

Linear Algebra

Gathering and Sharpening Tools

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To Students and Instructors

This book was first written for Washington & Jefferson College (W&J) students. At W&J, we teach Linear Algebra not only for its content but also (a) as a second course-length exposure to mathematical writing, and (b) as an introduction to the abstract and axiomatic thinking that is characteristic of higher mathematics. This initial audience and purpose has shaped this book.

As a text to be used in a *second* writing course, we have not included explicit writing instruction in the text. While training in logic and proof techniques is assumed from the beginning, we take care to identify proof strategies being used in these pages—especially in the early chapters. Along these lines, in most sections we have identified exercises as writing exercises which students should prioritize (and which instructors should consider requiring).

The fact that this text is written to introduce students to abstract concepts has influenced the choice and order of content. We introduce fields before vector spaces, and we introduce both structures earlier than many other texts. Theorems are then stated for general fields (instead of just for $\mathbb{R}$ or $\mathbb{C}$) throughout the text until the topic requires a restriction—such as inner products in Chapter 7, p. 177. To emphasize that not all fields should be thought of as $\mathbb{R}$, finite fields, the complex numbers, and other fields show up explicitly as examples. (Readers should note that care must be taken when using calculators/software/online aids to row-reduce matrices over fields which are not $\mathbb{R}$. Some pointers are provided in Section 2.2, p. 41.)

Other Features of this Book. Each section of this book contains *Reading Questions*. Ideally, instructors will assign and students will complete these questions before the class period in which material from that section of the book will be covered. The main goals of these questions is to force students to read the text and to guide a somewhat thoughtful first exposure to the content. We have found it helpful to emphasize to students that we do not expect them to fully understand the material upon this first reading, but that multiple exposures are often needed for comprehension and it is helpful if the classroom exposure is the second encounter, not the first. (If the instructor can skim the answers to these reading questions before class, then in-class examples and explanations can be customized accordingly.)

We have designated some problems at the end of each section as *Writing Exercises*. These are collected to help students practice and develop their writing. Some of these problems might be good to discuss in groups during class, and some might make useful homework exercises. As mentioned above, the separation of these problems is driven by one of the original purposes of this text—to help students improve their mathematical writing.

Many of the problems at the end of the sections have *answers or solutions* included in Appendix B, p. 203. Readers will notice that not all problems have

answers or solutions written out—some of this incompleteness is due to the fact that this book is still a work in progress, some is because we required students to turn in written solutions to specific problems and did not want those solutions staring our students in the face.

Finally, a word on the title. Virtually every Linear Algebra text is titled, *Linear Algebra*. That's fine; textbook titles are not, perhaps, where an author's greatest skill and creativity are deployed. The subtitle of this book was chosen with the metaphor of an apprentice handyman in mind. Such a worker needs to acquire some tools and improve their skill in using others. Additional implements may need sharpening or other maintenance. Linear Algebra is used widely in mathematics, and for a book pitched at intermediate-level undergraduate students of mathematics, this metaphor seemed appropriate. Plus, a good acronym which sounds somewhat French (LAGAST!) is hard to resist.

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Chapter 1

Solving Systems of Linear Equations

Linear Algebra is a remarkable area of mathematics. It is wide, deep, and has a huge variety of applications. One of the greatest parts of the subject is that it takes very little in terms of prerequisites to reap great rewards!

While the abstraction in this book begins in earnest in Chapter 2, p. 34, readers will find this first chapter imminently practical and full of calculations. At the heart of this chapter is an algorithm to solve systems of linear equations. Though it might seem easy to some students, it is important to understand this chapter before moving on, as it serves as an essential foundation for what follows.

1.1 Systems of Linear Equations

We will introduce the first set of big ideas in this book with an example.

Example 1.1.1 Alex reaches into his pocket and pulls out a handful of coins. He tells us that he's holding 90 cents in his hand, consisting of only nickels (worth five cents each) and quarters (worth 25 cents each). How many coins of each type is Alex holding?

We can model this information with a linear equation. Let x be the number of nickels in Alex's hand, and let y be the number of quarters. The equation that captures the information Alex shared is

$$5x + 25y = 90. \tag{1.1}$$

In this case, we don't have enough information to answer Alex's question. It could be that Alex is holding three nickels and three quarters ($15 + 75 = 90$) or that he is holding 13 nickels and one quarter ($65 + 25 = 90$). There are quite a few solutions to equation (1.1). (Note that in this example, since it would not make sense to have only part of a coin, we need our values of x and y to be non-negative integers.)

Now imagine that Alex gives us additional information by telling us that he is holding exactly ten coins. We can put this information into a second linear equation, and we now have what is called a **system of equations**. We need to know what values of x and y satisfy the following equations simultaneously:

$$5x + 25y = 90$$

$$x + y = 10.$$

A small amount of work shows us that the values $x = 8$ and $y = 2$ satisfy these equations simultaneously. (The reader should not worry right now about where those numbers came from. We'll get there soon.) This means that Alex is holding eight nickels and two quarters. ▲

The first major task of Linear Algebra is to learn how to handle systems of linear equations like the one given in Example 1.1.1, p. 1. We will learn a method to analyze such systems and determine their solutions (if solutions exist). We need a number of definitions as we get started.

Note 1.1.2 We will use the symbol $\mathbb{R}$ to denote the set of all real numbers. At the beginning of our work, we will be using $\mathbb{R}$ as our set of numbers for almost everything. However, when we get to Section 2.1, p. 34, we will move away from $\mathbb{R}$ to a more general description of a set of numbers that “works” for solving linear equations.

Definition 1.1.3 A **linear equation** in the variables $x_1, \dots, x_n$ is one which can be written in the form

$$a_1x_1 + \cdots + a_nx_n = b,$$

where the a_i and b are constants. The numbers $a_1, \dots, a_n, b$ all come from $\mathbb{R}$. In the special situation where $b = 0$, this is called a **homogeneous linear equation**. (Note that when there are only two variables in view we may use x and y instead of x_1 and x_2 ; similarly, when there are only three variables present, we may use x , y , and z .) ♦

The word “linear” in the phrase “linear equation” should make us think of a single power of a variable. In a linear equation, therefore, we will have no terms involving x_1^4 , $\sin(x_2)$, x_2x_3 , $\sqrt{x_4}$, or anything other than a single power of a variable.

Example 1.1.4 The following are linear equations:

$$4x + 2y = 98, \quad -2x_1 - 6x_2 + 5x_3 = 0, \quad 14x_1 - 15x_3 = -7.$$

These are all linear equations because every appearance of a variable term in these equations contains only a single variable raised to the first power.

The following are not linear equations:

$$4x^2 - y = 9, \quad \ln(x_1) + 3x_2 = 2, \quad x_1x_2 - \tan(x_3) = 0.$$

These are not linear equations because each equation has at least one variable term with something other than a single variable to a single power. ▲

Definition 1.1.5 A **system of linear equations** is a collection of one or more equations involving the same variables. (We sometimes shorten this and refer to a **linear system**.) When all of these linear equations are homogeneous linear equations, this may be called a **homogeneous linear system**. ♦

Example 1.1.6 Here is a system of linear equations involving the variables x_1 , x_2 , x_3 , and x_4 :

$$\begin{aligned} 2x_1 - x_3 + 8x_4 &= 10 \\ -x_1 + 3x_2 - 6x_4 &= -4. \end{aligned}$$

Note that not all variables need to be present in each equation. When a variable is missing, we consider that variable to have a zero coefficient in that equation.

It may be convenient (and preferred) to align the terms with the same variables vertically, but when an equation lacks one of the variables this creates a blank horizontal space. ▲

Definition 1.1.7 A **solution** to a system of linear equations in n variables is a list of n numbers, $(c_1, \dots, c_n)$, such that when the corresponding variables are assigned these numeric values (plug c_1 in everywhere for x_1 , plug c_2 in everywhere for x_2 , and so on), all the equations are true statements. The set of all solutions is called the **solution set** of the linear system. Two linear systems are said to be **equivalent** if their solution sets are equal. ◆

The language of *solution sets* and *equivalent linear systems* may seem unnecessarily complex. However, the method (see Algorithm 1.3.8, p. 20) for solving a linear system is much easier to describe with these terms firmly in hand.

Example 1.1.8 Consider the following linear system:

$$\begin{aligned} 2x + y &= -1 \\ x - 3y &= 17. \end{aligned}$$

A solution to this system is $(2, -5)$. (We use this notation to mean $x = 2$ and $y = -5$.) We verify this claim by plugging these numbers in for the variables and checking that both equations turn out to be true. (In fact, this is the *only* solution to this system.) ▲

The fact that we can write down a system of linear equations does not mean that it has a solution. Many linear systems have no solution at all. Others have one or an infinite number of solutions. This can be illustrated by some examples of lines graphed in the familiar setting of $\mathbb{R}^2$.

Example 1.1.9 We first consider the linear system consisting of two simple equations where the solution to each equation is a line in $\mathbb{R}^2$:

$$\begin{aligned} 2x + y &= 5 \\ -x + y &= -1. \end{aligned}$$

Readers will likely realize that the solution to this system happens when the graphs of these lines intersect. We see from the graph below that the intersection occurs at the point $(2, 1)$. Since this is the only intersection point, there is only one solution to this system.

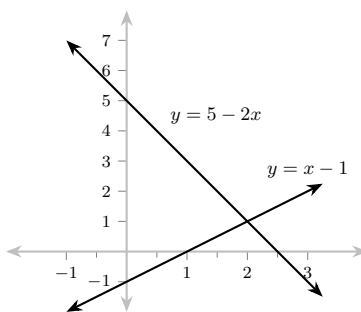


Figure 1.1.10 A graph of two intersecting lines

We now consider a second linear system:

$$\begin{aligned} 2x + y &= 5 \\ 2x + y &= 3. \end{aligned}$$

The graphs of these two lines appear below, but in this case the lines do not intersect at all because they are parallel. (The slope of both lines is -2 .)

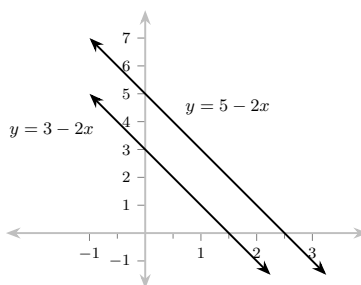


Figure 1.1.11 A graph of two parallel lines

Here is a third linear system in two variables:

$$\begin{aligned} 2x + y &= 5 \\ 6x + 3y &= 15. \end{aligned}$$

Each of these equations has a solution set whose graph is a line. In this particular case, we obtain *the same line* for both equations, which means that the linear system has an infinite number of solutions. Each one of the infinite solutions to the first equation is a solution to the second equation, and vice versa.

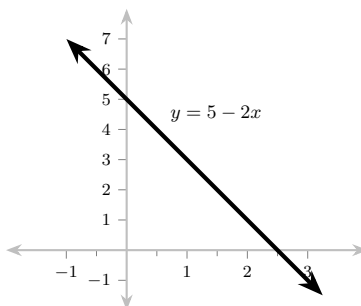


Figure 1.1.12 A graph of two identical lines

What we saw in Example 1.1.9, p. 3 is no accident. This simple collection of examples in $\mathbb{R}^2$ exposes our need for the following definitions. ▲

Definition 1.1.13 A system of linear equations is **consistent** if it has at least one solution. A system is **inconsistent** if it has no solutions. When a linear system has only one solution we say that the solution is **unique**. ◆

This definition and the principle illustrated in Example 1.1.9, p. 3 mean that whenever we encounter a system of linear equations we have two questions to ask. *Is this system consistent? If the system is consistent, is the solution unique?*

Reading Questions

1. This question is designed to help you understand linear equations.
 - (a) Write down an example of a linear equation involving the variables x , y , and z .
 - (b) Write down an example of an equation involving the variables x , y , and z which is *not* linear.

2. Find the solution set to the following linear system:

$$\begin{aligned} 3x + 7y &= 9 \\ -2x + 6y &= -2. \end{aligned}$$

Exercises

1. Carlos has ten coins totalling \$1.10. Each coin is either a nickel, a dime, or a quarter. He has twice as many nickels as he has quarters. How many of each coin does Carlos have?

Model this problem with a linear system. (You do not need to find the solution set of this linear system.)

2. Suppose that $f(x) = ax^2 + bx + c$ is a function whose graph passes through the points $(-1, 6)$, $(1, 0)$, and $(3, -2)$. What are the values of a , b , and c ?

Model this problem with a linear system. (You do not need to find the solution set of this linear system.)

3. For each part, determine whether the equations form a linear system.

(a)

$$\begin{aligned} 3y - 4z + x &= \sqrt{2} \\ -z + 2w + 7x &= 14 \end{aligned}$$

(b)

$$\begin{aligned} 2x - 10y + \frac{3}{z} &= 16 \\ 7x + 12z &= -1 \end{aligned}$$

(c)

$$\begin{aligned} x - e^4 z &= 9 \\ \ln(8)x + 13y &= -3 \end{aligned}$$

4. Determine whether or not each given list is a solution to this linear system.

$$\begin{aligned} x_1 - 2x_2 + 8x_3 &= -5 \\ -x_1 + 3x_2 - 10x_3 &= 6 \\ 2x_1 - 3x_2 + 14x_3 &= -9. \end{aligned}$$

(a) $(-7, 3, 1)$

(b) $(-5, 0, 0)$

(c) $(-4, 4, 1)$

(d) $(5, -3, -2)$

5. Determine whether or not each given list is a solution to this linear system.

$$\begin{aligned} x - y + 3z &= -6 \\ -3x + 4y - 10z &= 22 \\ -2x + 4y - 8z &= 21. \end{aligned}$$

(a) $(1, 1, -2)$

(b) $(-2, 3, 4)$

(c) $(0, 3, -1)$

6. For the following linear systems, graph the solution set of each equation. Then graph the solution set of the linear system.

(a)

$$-2x + y = -3$$

$$x - 5y = 7$$

(b)

$$2x - 2y = 5$$

$$3x + y = -2$$

$$-x - y = 1$$

(c)

$$2x - 3y = -2$$

$$-2x + 3y = -3$$

(d)

$$2x - 4y = -1$$

$$-x + 2y = \frac{1}{2}$$

(e)

$$x + 4y = 5$$

$$3x - y = 0$$

$$-2x - 8y = -10$$

Writing Exercises

7. For each of the following, write an example of a linear system with of two equations and two variables with the given property. (The system must be one you've not yet seen!) Explain why your example has the property.
- (a) The system has no solutions.
 - (b) The system has exactly one solution.
 - (c) The system has infinitely many solutions.
8. Consider a linear system of m equations and n variables and suppose that $(c_1, \dots, c_n)$ and $(d_1, \dots, d_n)$ are both solutions to this system. Under what conditions is $(c_1 + d_1, \dots, c_n + d_n)$ also a solution to the same system? Explain.
9. Consider a linear system of m equations and n variables and suppose that $(c_1, \dots, c_n)$ and $(d_1, \dots, d_n)$ are both solutions to this system. Let t be a real number. Prove that $(tc_1 + (1 - t)d_1, \dots, tc_n + (1 - t)d_n)$ is also a solution to this same linear system.

1.2 Matrices

In this section we will introduce matrices, one of the main computational tools in linear algebra. We will learn how to translate the information about a linear system to a matrix and then manipulate that matrix to solve the original system.

Some of the matrix manipulations later in this section may seem to come from nowhere. We intend this first example to motivate these upcoming operations.

Example 1.2.1 Consider the following system of two linear equations in two variables:

$$\begin{aligned} 3x + 3y &= -3 \\ 2x - y &= 7. \end{aligned}$$

We first multiply both sides of the top equation by $\frac{1}{3}$ in order to get “easier” coefficients on our variables. After taking this action, we have the following equivalent linear system:

$$\begin{aligned} x + y &= -1 \\ 2x - y &= 7. \end{aligned}$$

We can now use the first equation to eliminate one of the variables in the second equation. If we add -2 times the first equation to the second equation, we’ll only have the y variable left. The second equation is transformed, resulting in this equivalent linear system:

$$x + y = -1 \tag{1.2}$$

$$-3y = 9. \tag{1.3}$$

We can now multiply both sides of Equation (1.3) by $-\frac{1}{3}$ to find that $y = -3$. Plugging this value in for y in Equation (1.2) quickly gives us $x = 2$. We have solved the linear system at this point, and we have determined that the only solution to the system is $(2, -3)$. (Readers should check that this is in fact a solution by plugging these values into the original linear equations.) ▲

Example 1.2.1, p. 7 is much longer than it needs to be, and at this point some readers may wonder what they’ve gotten themselves into—the mathematics so far (such as it is) is far from advanced. The real point of this example is to make explicit the operations used to solve a simple linear system. Once we have better notation and terminology, solving linear systems will be much faster (especially with the aid of technology). This is where matrices come in so handy.

Definition 1.2.2 A **matrix** is a rectangular array of numbers. If m and n are natural numbers, then an $m \times n$ matrix is one with m rows and n columns. The entries in a matrix are referred to by their row and column numbers, so entry (i, j) is the number in the i th row and j th column. (Row numbers increase from the top of the matrix down, and column numbers increase from the left of the matrix to the right.)

The **main diagonal** of a matrix refers to those entries on the (i, i) -diagonal of the matrix—starting at the upper left and going down to the right. In other words, an entry is on the main diagonal if and only if its row and column numbers are the same.

A **submatrix** of a matrix refers to the matrix that remains after removing one or more rows and/or columns from a matrix. ♦

Every system of linear equations generates two important matrices—the **coefficient matrix** and the **augmented matrix**.

Definition 1.2.3 Given a system of m linear equations in n variables (hereafter, we will call this an $m \times n$ linear system),

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

the **coefficient matrix** of the system is

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix},$$

and the **augmented matrix** of the system is

$$\left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right].$$

We form the augmented matrix by “augmenting” the coefficient matrix with the column of constants. ♦

Note that the number of equations in the linear system corresponds to the number of rows of both the coefficient and augmented matrices, and the number of variables in the linear system corresponds to the number of columns in the coefficient matrix. (The augmented matrix of a linear system has one more column than the number of variables.)

Example 1.2.4 For the linear system

$$\begin{aligned} 2x_1 - 7x_2 + x_3 &= -8 \\ -x_1 + 4x_3 &= -2, \end{aligned}$$

the coefficient and augmented matrices are, respectively,

$$\begin{bmatrix} 2 & -7 & 1 \\ -1 & 0 & 4 \end{bmatrix} \quad \text{and} \quad \left[\begin{array}{ccc|c} 2 & -7 & 1 & -8 \\ -1 & 0 & 4 & -2 \end{array} \right].$$

Note the 0 in position (2, 2) as it corresponds to the absence of an x_2 term in the second equation of the linear system. ▲

We must get comfortable switching between linear systems and their associated matrices. In particular, we need to understand why specific forms of matrices are especially useful when solving linear systems.

Example 1.2.5 Consider the following matrix as the augmented matrix for a linear system:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & \frac{2}{3} \end{array} \right].$$

In one sense, this is the best possible augmented matrix we could have for a

3×3 linear system, because the solutions to the system are obvious. Translating this matrix back to equation form gives us $x = 7$, $y = -2$, and $z = \frac{2}{3}$. ▲

Almost no linear systems will come to us with an augmented matrix as simple as the one in Example 1.2.5, p. 8. (This is mostly because such a linear system is, well, boring. It takes no work to solve a system like this.) Our goal, however, is to take any given linear system and manipulate its augmented matrix to be *as close to* this sort of matrix as we can get.

As we work with augmented matrices, we are restricted in the arithmetic we perform on them because, most of all, we want to preserve the solution sets of the corresponding linear systems. In our next definition, we describe the three “legal” ways we have to manipulate a matrix in this fashion.

Definition 1.2.6 The following operations on a matrix are called **elementary row operations**.

1. Add a multiple of one row to another row, replacing that second row with the result. (We will call this the **replace** row operation.)
2. Multiply every entry in a row by a nonzero constant. (We will call this the **scale** row operation.)
3. Switch the location of any two rows in the matrix. (We will call this the **switch** row operation.)

Two matrices are called **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other. ◆

When defining “row equivalent” in the previous definition, careful readers will note one implied fact that must be checked. The word “equivalent” brings to mind an *equivalence relation*, which includes the property of the relation being symmetric. Therefore, the definition implies that all of the elementary row operations are “reversible”—that is, for each elementary row operation, there is an elementary row operation that reverses the change that was just made. This is something to prove! (See Exercise 1.2.9, p. 16.)

Theorem 1.2.7 Suppose that A and B are augmented matrices corresponding to systems of linear equations. Then if A and B are row equivalent, the linear systems to which they correspond are also equivalent.

Proof. We will prove this statement directly. Because elementary row operations only involve one or two rows of a matrix at a time, it is sufficient to prove this theorem for systems of two linear equations.

We suppose that we have the following $2 \times n$ linear system:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2, \end{aligned}$$

which produces the following augmented matrix:

$$A = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ a_{21} & \cdots & a_{2n} & b_2 \end{array} \right].$$

Further, we suppose that $(c_1, \dots, c_n)$ is a solution to the linear system. If we apply the *switch* row operation to A , this corresponds to writing the second equation above the first in the linear system. It is immediate that $(c_1, \dots, c_n)$ is still a solution to this system.

If we apply the *scale* row operation to A , multiplying row 1 (without loss

of generality) by a nonzero constant d , then we have the matrix

$$B = \left[\begin{array}{ccc|c} da_{11} & \cdots & da_{1n} & db_1 \\ a_{21} & \cdots & a_{2n} & b_2 \end{array} \right].$$

We must show that $(c_1, \dots, c_n)$ is a solution to the corresponding linear system.

It is obvious that $(c_1, \dots, c_n)$ satisfies the second equation in this new linear system since that equation is unchanged. If $(c_1, \dots, c_n)$ satisfied the first equation of the original system, then

$$a_{11}c_1 + \cdots + a_{1n}c_n = b_1.$$

We can now show that $(c_1, \dots, c_n)$ satisfies the first equation of the second system by substitution:

$$\begin{aligned} da_{11}c_1 + \cdots + da_{1n}c_n &= d(a_{11}c_1 + \cdots + a_{1n}c_n) \\ &= d(b_1). \end{aligned}$$

We must now show that the *replace* row operation preserves solutions. We let k be a nonzero constant and we replace (without loss of generality) the second row of A with the old second row plus k times the first row. Here is the resulting matrix C :

$$C = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ ka_{11} + a_{21} & \cdots & ka_{1n} + a_{2n} & kb_1 + b_2 \end{array} \right].$$

In the linear system which corresponds to C , the first equation is unchanged from the first equation in the linear system that corresponds to A , so we only need to be concerned with the second equation. By virtue of the fact that $(c_1, \dots, c_n)$ satisfied both equations of the first system, we know that

$$a_{11}c_1 + \cdots + a_{1n}c_n = b_1$$

and

$$a_{21}c_1 + \cdots + a_{2n}c_n = b_2.$$

We can now use this in the following calculation:

$$\begin{aligned} (ka_{11} + a_{21})c_1 + \cdots + (ka_{1n} + a_{2n})c_n &= (ka_{11}c_1 + \cdots + ka_{1n}c_n) + (a_{21}c_1 + \cdots + a_{2n}c_n) \\ &= k(a_{11}c_1 + \cdots + a_{1n}c_n) + (a_{21}c_1 + \cdots + a_{2n}c_n) \\ &= kb_1 + b_2. \end{aligned}$$

There is technically one more collection of facts to verify before this proof is complete. We have shown that, for all three elementary row operations, if $(c_1, \dots, c_n)$ is a solution to the linear system corresponding to A then it will also be a solution to the linear system corresponding to B (where B is the result of applying one elementary row operation to A). But “equivalent” linear systems means that the solution sets are equal *as sets*. This means that we must assume we have a solution for the linear system corresponding to the matrix B and show that it is a solution for the linear system corresponding to A . We claim that this concern can be dispensed with by invoking another result in this section. Connecting the last dots of this proof is left to the reader in Exercise 1.2.12, p. 17. ■

Roughly speaking, we want to use elementary row operations to transform the augmented matrix for a linear system into a matrix which has 1s along the

main diagonal and 0s as the other entries in those columns. This is not always possible, and we will describe the situation more precisely below, but here is an example to illustrate the process.

Example 1.2.8 We start by considering the following matrix as the augmented matrix of a linear system:

$$\left[\begin{array}{ccc|c} 2 & 2 & -1 & 8 \\ -3 & -2 & 2 & -12 \\ 5 & 0 & 4 & 11 \end{array} \right].$$

We first *scale* the first row by $\frac{1}{2}$ to produce a 1 in the (1, 1) entry:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ -3 & -2 & 2 & -12 \\ 5 & 0 & 4 & 11 \end{array} \right].$$

We then *replace* the second row with the sum of the second row and three times the first row:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ 0 & 1 & \frac{1}{2} & 0 \\ 5 & 0 & 4 & 11 \end{array} \right].$$

We will again use the 1 in the (1, 1) entry to “eliminate” the 5 in the (3, 1) entry. We *replace* the third row with -5 times the first row plus the third:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -5 & \frac{13}{2} & -9 \end{array} \right].$$

Now that we have “cleared out” the entries under the (1, 1) entry, we can do the same for the 1 in the (2, 2) entry. (In future examples we may need to *scale* first to have a 1 here.) We *replace* the third row with 5 times the second row plus the third:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 9 & -9 \end{array} \right].$$

We can now *scale* the third row by $\frac{1}{9}$ to produce 1s along the main diagonal:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 \end{array} \right].$$

We are halfway done, as we have 0s below the main diagonal. We now need to use elementary row operations to produce 0s above the main diagonal. We first *replace* the second row with $-\frac{1}{2}$ times the third row plus the second:

$$\left[\begin{array}{ccc|c} 1 & 1 & -\frac{1}{2} & 4 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 \end{array} \right].$$

We can again use the 1 in the (3, 3) position to produce a 0 in the (1, 3) position. We *replace* the first row with $\frac{1}{2}$ times the third row plus the first:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & \frac{7}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 \end{array} \right].$$

Finally, we use the 1 in the (2, 2) entry to produce a 0 in the (1, 2) entry. We *replace* the first row with -1 times the second row plus the first:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 \end{array} \right].$$

We now have the matrix in the form we wanted, because we can read off the solution: $x = 3$, $y = \frac{1}{2}$, and $z = -1$. ▲

We will now define and standardize this form of the augmented matrix that is so helpful in solving the related linear system. In what follows, by “nonzero row (column)” we mean a row (column) with at least one nonzero entry, and by the “leading entry” of a row we mean the row’s leftmost nonzero entry.

Definition 1.2.9 A matrix A is in **row-echelon form** (REF) if all of the following hold:

- all rows of all zeros are collected at the bottom of the matrix;
- each leading entry of a row is in a column to the right of the column of the leading entry for the row above it; and
- all entries in a column below the leading entry of a row are zeros.

If a matrix A is in row-echelon form and also satisfies the following two conditions, it is in **reduced row-echelon form** (RREF):

- each leading entry in a nonzero row is 1; and
- each leading 1 is the only nonzero entry in its column.



Note 1.2.10 The reader should observe that every matrix which is in RREF is automatically in REF, but the converse is not true.

These definitions are a mouthful, but they are useful. Here is an example showing some matrices that do and do not meet these criteria.

Example 1.2.11 The following two matrices are in REF but not RREF:

$$\left[\begin{array}{cccc} 2 & -1 & 3 & 0 \\ 0 & 5 & -3 & 7 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{ccc} -4 & 9 & 6 \\ 0 & 12 & -10 \end{array} \right].$$

The following two matrices are in RREF:

$$\left[\begin{array}{cccc} 1 & 0 & 4 & 5 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{cccc} 1 & -7 & 0 & 2 \\ 0 & 0 & 1 & -5 \end{array} \right].$$

Neither of the following matrices are in REF or RREF:

$$\left[\begin{array}{cccc} 0 & 2 & 5 & -8 \\ 0 & -1 & 7 & 7 \\ 4 & 2 & 1 & 0 \end{array} \right] \quad \left[\begin{array}{cccc} 2 & 3 & -2 & 5 \\ 0 & -1 & 3 & 6 \\ -4 & -3 & 0 & 12 \end{array} \right].$$



Definition 1.2.12 When a matrix A is in row-echelon form, then the leading entry in each row is called a **pivot**. The location of this entry within the matrix is called a **pivot position**. Finally, any column containing a pivot is called a **pivot column**. ♦

One of the reasons we have Definition 1.2.9, p. 12 is because (we will prove this below) *every* matrix can be put into RREF. What follows here is a description of the algorithm used to transform every matrix into RREF. This process is usually called “Gaussian elimination” or “Gauss-Jordan elimination.”

Algorithm 1.2.13 The Row Reduction Algorithm. *This row reduction algorithm consists of five steps. The first four (collectively) produce a matrix in row-echelon form; after the fifth step the matrix will be in reduced row-echelon form.*

1. *Start with the leftmost nonzero column. This will be a pivot column with the pivot position at the top.*
2. *Select a nonzero entry in this column and use the switch operation to move it to the top of the column (if necessary).*
3. *Use elementary row operations to create zeros below this pivot.*
4. *Ignore the row with the pivot just considered. Apply steps 1–3 to the submatrix that remains. Repeat this process until all nonzero rows have been handled.*
5. *Select the rightmost pivot and use the scale operation to make it a 1. Now use elementary row operations to create 0s above this pivot. Move upward and to the left, repeating this process for all remaining pivots.*

Example 1.2.14 We consider the following matrix A and put it into reduced row-echelon form:

$$A = \left[\begin{array}{cccc|c} -2 & -2 & 6 & 14 & 4 \\ 2 & 3 & -4 & -4 & -1 \\ -3 & -5 & 4 & -2 & -2 \end{array} \right].$$

The first column is nonzero, and we will leave the current top row in its place. Though it is not *necessary* to create a 1 in the pivot position at this point in the process, it is often useful to do so. (We are not violating Algorithm 1.2.13, p. 13 by producing a leading 1 this early in the process, but this is a step of row reduction that many, including this author, prefer to perform now to make future steps less painful.) We multiply the first row by $-\frac{1}{2}$ to achieve this:

$$\left[\begin{array}{cccc|c} 1 & 1 & -3 & -7 & -2 \\ 2 & 3 & -4 & -4 & -1 \\ -3 & -5 & 4 & -2 & -2 \end{array} \right].$$

We now use the 1 in the $(1, 1)$ position to create zeros in the column below it. We add -2 times row 1 to row 2 and we add 3 times row 1 to row 3. Since there is no “interaction” between these operations, we will perform them at the same time, though the reader should certainly take one operation at a time if this combination raises one’s blood pressure:

$$\left[\begin{array}{cccc|c} 1 & 1 & -3 & -7 & -2 \\ 0 & 1 & 2 & 10 & 3 \\ 0 & -2 & -5 & -23 & -8 \end{array} \right].$$

According to the algorithm, we now ignore row 1 and repeat the process for the remaining matrix. There is already a 1 at the “top” of (this portion

of) the second column, so we use that entry to create zeros below it. We add twice the second row to the third row:

$$\left[\begin{array}{cccc|c} 1 & 1 & -3 & -7 & -2 \\ 0 & 1 & 2 & 10 & 3 \\ 0 & 0 & -1 & -3 & -2 \end{array} \right].$$

The matrix is in row-echelon form now, so we proceed to step 5 of the algorithm. We multiply the third row by -1 to produce a 1 in the $(3, 3)$ position:

$$\left[\begin{array}{cccc|c} 1 & 1 & -3 & -7 & -2 \\ 0 & 1 & 2 & 10 & 3 \\ 0 & 0 & 1 & 3 & 2 \end{array} \right].$$

We now use the 1 we just created to produce zeros in the column above it. We add -2 times the third row to the second, and we add 3 times the third row to the first:

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 2 & 4 \\ 0 & 1 & 0 & 4 & -1 \\ 0 & 0 & 1 & 3 & 2 \end{array} \right].$$

The final step is to use the pivot in position $(2, 2)$ to create a 0 above it. We add -1 times the second row to the first:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 5 \\ 0 & 1 & 0 & 4 & -1 \\ 0 & 0 & 1 & 3 & 2 \end{array} \right]. \quad (1.4)$$

The process is now complete, so the matrix in (1.4) is the result of reducing A to RREF. ▲

We note that this example is different from Example 1.2.8, p.11 in an important way. The matrix in Example 1.2.14, p.13 corresponds to a system with three equations and four variables, while the matrix in Example 1.2.8, p.11 corresponds to a system with three equations and three variables. We only mention this to point out that the size of the original matrix puts some restrictions on the possibilities for its RREF, and the reader should be aware of this when completing the reading questions and the exercises at the end of this section.

Reading Questions

1. Write down the coefficient matrix and the augmented matrix for the following linear system:

$$\begin{aligned} 2x_1 - 3x_2 + \frac{1}{2}x_3 &= 7 \\ -5x_2 + x_3 - x_1 &= -4 \\ 6x_2 + 9x_1 &= -1. \end{aligned}$$

2. Although the reduced row-echelon form of a matrix is unique (we will prove this soon), the row-echelon form of a matrix is not unique. For the following matrix A , write two distinct row-echelon forms:

$$A = \left[\begin{array}{cc|c} 2 & -3 & 4 \\ -1 & 2 & 0 \end{array} \right].$$

3. Put the following matrix into reduced row-echelon form. Record each step along the way (both the elementary row operations used and the matrices

obtained as a result):

$$A = \left[\begin{array}{cc|c} 2 & -3 & 5 \\ -1 & 2 & -4 \\ 5 & -9 & 8 \end{array} \right].$$

Exercises

1. Write the augmented matrix that corresponds to the following linear system:

$$\begin{aligned} 2x_2 - 4x_1 + \sqrt{2}x_4 &= 8 \\ 3x_3 - 19x_2 &= -1 \\ 0.5x_4 - 3x_3 + x_1 &= 0. \end{aligned}$$

2. Write the linear system that corresponds to the following matrix, assuming this is the augmented matrix of the system:

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 4 \\ 0 & -2 & 1 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

3. For each of the following, describe the elementary row operation that was used to transform the matrix on the left into the matrix on the right. Be specific in your description of the operation. (You should not just answer “scale” or “switch”, etc.)

$$(a) \left[\begin{array}{ccc} 2 & 1 & -3 \\ -1 & 4 & 5 \\ -2 & 0 & 2 \end{array} \right] \longrightarrow \left[\begin{array}{ccc} 2 & 1 & -3 \\ 1 & 5 & 2 \\ -2 & 0 & 2 \end{array} \right]$$

$$(b) \left[\begin{array}{ccc} 2 & 1 & -3 \\ -1 & 4 & 5 \\ -2 & 0 & 2 \end{array} \right] \longrightarrow \left[\begin{array}{ccc} -1 & 4 & 5 \\ 2 & 1 & -3 \\ -2 & 0 & 2 \end{array} \right]$$

$$(c) \left[\begin{array}{ccc} 2 & 1 & -3 \\ -1 & 4 & 5 \\ -2 & 0 & 2 \end{array} \right] \longrightarrow \left[\begin{array}{ccc} 2 & 1 & -3 \\ -1 & 4 & 5 \\ -1 & 0 & 1 \end{array} \right]$$

4. For each of the following matrices, determine if the matrix is in row-echelon form, reduced row-echelon form, both, or neither.

$$(a) \left[\begin{array}{ccc} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{array} \right]$$

$$(b) \left[\begin{array}{ccc} 0 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$(c) \left[\begin{array}{cc} -1 & 2 \\ 0 & 2 \\ 0 & 0 \end{array} \right]$$

$$(d) \left[\begin{array}{ccc} 1 & 0 & 7 \\ 0 & 1 & -4 \end{array} \right]$$

5. Consider the following matrix A :

$$A = \begin{bmatrix} -2 & 3 \\ -3 & 5 \end{bmatrix}.$$

Find two distinct matrices B and C which are row equivalent to A and are in row echelon form. (There are many correct answers!)

6. Use Algorithm 1.2.13, p. 13 to put each of these matrices into RREF.

(a) $\begin{bmatrix} -3 & 4 \\ 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 4 & 5 \\ -1 & 3 & -4 \end{bmatrix}$

(c) $\begin{bmatrix} 4 & 8 & 8 \\ 4 & -1 & -8 \\ 0 & 8 & -7 \end{bmatrix}$

(d) $\begin{bmatrix} -3 & -8 & -1 \\ -4 & -3 & -3 \\ 2 & 13 & -1 \end{bmatrix}$

7. For each of the following matrix sizes, list all of the possible RREFs for matrices of that size. Use the symbols ■ for a pivot, * for an unspecified number, and 0 for a zero entry.

(a) 2×2

(b) 2×3

(c) 3×2

Writing Exercises

- 8.

(a) Suppose A is a 3×4 matrix. What is the maximum number of pivots in any RREF of A ? Explain.

(b) Suppose B is a 6×4 matrix. What is the minimum number of rows of zeros in any RREF of B ? Explain.

9. Prove that each of the elementary row operations is reversible. In other words, if an elementary row operation was used to transform a matrix A into a matrix B , prove that there is another elementary row operation (of the same type) which will transform B back into A .
10. Recall that an equivalence relation is reflexive, symmetric, and transitive. Now, fix the integers m and n and consider row equivalence as a relation on all $m \times n$ matrices. Prove that this is an equivalence relation. (Hint: another exercise in this section may be helpful in your argument.)
11. Prove or disprove: The following two matrices are row equivalent. (Hint: another exercise in this section may be helpful in your argument.)

$$A = \begin{bmatrix} 9 & -9 & -10 \\ -1 & -1 & 9 \\ -5 & -2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 3 & -2 \\ 7 & -2 & 1 \\ 7 & 6 & -8 \end{bmatrix}$$

12. Complete the proof of Theorem 1.2.7, p. 9. This can be accomplished by proving the statement in the final paragraph of the given proof.

1.3 Solving Linear Systems With Matrices

In the previous section, we learned how to write the augmented matrix of a linear system and Algorithm 1.2.13, p. 13 provided a process to reduce any matrix to RREF. In this section, we will learn how to use the RREF of a matrix to solve the corresponding linear system. We will also prove some important results related to these solutions.

We first introduce some terminology. These terms relate what we saw in Definition 1.2.12, p. 13 back to the corresponding linear systems.

Definition 1.3.1 Suppose that A is the coefficient matrix corresponding to a system of linear equations and that A is in REF. Then a variable corresponding to a pivot column in A is called a **basic variable** (or **pivot variable**), and a variable corresponding to a non-pivot column in A is called a **free variable**. ♦

Note 1.3.2 Note that in this definition A is the *coefficient matrix* (not the augmented matrix) of the linear system. We are using the coefficient matrix because we are making a definition concerning variables, and the rightmost column in the augmented matrix does not correspond to a variable in the linear system.

The augmented matrix of an $m \times n$ linear system is of size $m \times (n + 1)$. This puts some limitations on the different reduced row-echelon forms that we could see in this context. In the following two examples, we will consider specific reduced row-echelon forms and what they say about the linear systems to which they correspond.

Example 1.3.3 Consider the following as the augmented matrix corresponding to a system of linear equations:

$$A = \left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

This is a 3×4 matrix, so the original linear system has three equations and three variables. Hopefully the reader can see that this matrix is indeed in RREF. (Consult Definition 1.2.9, p. 12 for a refresher.)

We will now write the equations which correspond to each row of the matrix, and we will pay special attention to the final row:

$$x_1 + 0x_2 + 3x_3 = 0$$

$$0x_1 + x_2 + 2x_3 = 0$$

$$0x_1 + 0x_2 + 0x_3 = 1.$$

Usually we omit terms in linear equations when the coefficient is 0, but we are including those terms here to make a point. When the coefficient of a variable is 0, the entire linear term disappears since no value of the variable could make the product with the coefficient anything other than 0. This means that each of these equations can be written in a more simple form. In particular, the third equation can be written as $0 = 1$.

This may feel rather elementary, but there are no possible variable values to make $0 = 1$ a true statement. It is false *all of the time*. Since we are searching

for values of the variables which satisfy all the equations simultaneously, and since one of the equations has no solution, the linear system has no solution. This is an *inconsistent* linear system.

We proved in Theorem 1.2.7, p. 9 that row-equivalent matrices correspond to equivalent linear systems. Therefore, we can say that the original linear system for this example is inconsistent. ▲

What we saw in Example 1.3.3, p. 17 we will be able to generalize (see Theorem 1.3.5, p. 19) in our effort to categorize inconsistent linear systems. Before we do that, let's look at an augmented matrix which has a different RREF.

Example 1.3.4 Consider the following matrix as the augmented matrix of a linear system:

$$A = \left[\begin{array}{ccc|c} 1 & 0 & -3 & 5 \\ 0 & 1 & 2 & 7 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

(Again, the reader should verify that this matrix is in reduced row-echelon form.)

The final row of this matrix corresponds to the equation $0 = 0$. Since this equation is true all the time, for all values of the involved variables, we won't consider it any longer as it places no further restrictions on the solutions.

The first two rows of A correspond to the following two linear equations:

$$\begin{aligned} x_1 - 3x_3 &= 5 \\ x_2 + 2x_3 &= 7. \end{aligned}$$

From Definition 1.3.1, p. 17, we see that in this system x_1 and x_2 are basic variables and x_3 is a free variable. What does that mean for the solutions of this system? We call x_3 “free” because any element of $\mathbb{R}$ that we put into x_3 will produce a solution for this system. The variable x_3 is “free” to take on any value, and then the values of the basic variables x_1 and x_2 are determined by that value and the linear equations.

For example, in this system of equations, if $x_3 = 2$, then $x_1 = 11$ and $x_2 = 3$, giving $(11, 3, 2)$ as a solution to the linear system. If $x_3 = -1$, then $x_1 = 2$ and $x_2 = 9$, giving $(2, 9, -1)$ as a solution to the system. Since x_3 can take on any value in $\mathbb{R}$, and since we have a solution to the system for each value of x_3 , this means we have a solution to the system for each element of $\mathbb{R}$. We conclude that there are an infinite number of solutions to this system.

The solutions to this linear system can be written in a number of ways. We will prefer the following form:

$$\begin{cases} x_1 = 5 + 3x_3 \\ x_2 = 7 - 2x_3 \\ x_3 \text{ is free.} \end{cases}$$

This is called a **parametric description** of the solution set. Sometimes solutions like this are written with the letter t or s in place of x_3 to better match the usage of the word “parameter” elsewhere. We will follow the convention of using the free variables as parameters in our solutions. ▲

In Example 1.3.4, p. 18, we saw that having a free variable corresponded to having an infinite number of solutions. But we need to be careful about our conclusions, because there was also a free variable (x_3) in Example 1.3.3, p. 17, and in that case there were *no solutions* to the system. A free variable only indicates an infinite number of solutions when the system is consistent.

The two previous examples, combined with Example 1.2.5, p. 8, give us a

sense of what solutions to linear systems can look like given certain characteristics of the associated augmented matrices. We can now state in theorem form what we observed to be true in these examples.

Theorem 1.3.5 *Suppose A is the augmented matrix of a linear system, and suppose that A is in reduced row-echelon form. Then the linear system is consistent if and only if there is no pivot in the final column of A .*

Proof. We note that the pivot columns do not change when a matrix goes from row-echelon form to reduced row-echelon form (see Algorithm 1.2.13, p. 13), so we are not losing any generality with our assumption that A is in RREF.

This theorem is a biconditional statement, and we will prove one implication directly. We assume there is no pivot in the final column of A . Then when we consider the linear equations which correspond to the rows of A , we see that each of the basic variables can be written in terms of the free variables, if any free variables exist. If no free variables exist, then all basic variables have an assigned value and the system is consistent. In the case that there is at least one free variable, we can pick an element of $\mathbb{R}$ —let's say, 0—to assign to each of the free variables. This produces a solution to the linear system, and our system is consistent.

We will prove the contrapositive of the other implication. If there is a pivot in the final column of A , then the corresponding linear equation reduces to $0 = 1$. This means that there is no solution to the linear system, so the system is inconsistent. ■

Given that two major questions about the solutions to linear systems involve *consistency* and *uniqueness*, the next natural theorem to consider is related to this second concept.

Theorem 1.3.6 *Suppose that A is the augmented matrix corresponding to a consistent $m \times n$ linear system, and suppose that A is in reduced row-echelon form. Then the system has a unique solution if and only if there is a pivot in each of the first n columns of A .*

Proof. As with the proof of Theorem 1.3.5, p. 19, we are not losing any generality by assuming that A is in RREF.

We first suppose that there is a pivot in each of the first n columns of A ; this implies that $m \geq n$. We also recall that the linear system is assumed to be consistent, meaning that the first n rows of A have the following form:

$$\left[\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & b_1 \\ 0 & 1 & \cdots & 0 & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & b_n \end{array} \right].$$

(The matrix A may have rows of all zeros below the n rows here, but that will not affect our discussion.)

If the matrix A has the form we have just detailed, then the original linear system is equivalent to one with equations of the form $x_1 = b_1$, $x_2 = b_2$, ..., $x_n = b_n$. That is, the system has a unique solution.

We will prove the contrapositive of the other implication. We suppose that there is at least one free variable (call it x_j) in the linear system. We recall (see Example 1.3.4, p. 18) that when the matrix for a consistent system is in RREF, all solutions can be written by expressing basic variables in terms of any existing free variables. Therefore, this system has a solution in which all free variables are set equal to 0. Further, this system has a solution in which all free variables except x_j are set equal to 0 and $x_j = 1$. This may not change

the value of any of the basic variables, and there may not be any free variables aside from x_j , but these two solutions we have just described are not the same since x_j has a different value in each one. Therefore, this system has more than one solution, meaning that the system does not have a unique solution. ■

Corollary 1.3.7 *If m and n are natural numbers and $m < n$, then an $m \times n$ linear system cannot have a unique solution.*

Proof. Suppose A is the augmented matrix corresponding to an $m \times n$ linear system with $m < n$. The RREF of A can have at most m pivots, so by Theorem 1.3.6, p. 19 the system cannot have a unique solution. ■

The two previous theorems provide the last step to an algorithm for solving any linear system.

Algorithm 1.3.8 Algorithm for Solving Linear Systems. *Suppose that we have an $m \times n$ system of linear equations. Here are the steps to solve the system.*

1. *Form the augmented matrix of the linear system. We will call this matrix A .*
2. *Find the reduced row-echelon form of A . (If A is small, this can be done by hand; if A is not small, technology should be used to complete this step.)*
3. *Determine whether or not the system is consistent by observing the location of the pivots in the RREF of A . If there is a pivot in the rightmost column, the linear system is inconsistent and we need not proceed any further in the algorithm. If there is no pivot in the rightmost column, the system is consistent.*
4. *Determine whether or not the system has a unique solution. If there is a pivot in each of the first n columns, then the system has a unique solution which can be recorded. If there is not a pivot in each of the first n columns, then the system does not have a unique solution; in this case, a parametric description of the solution set can be recorded.*

The earlier examples in this section can be completed using this algorithm. We will include an additional example so the reader can practice using the algorithm once more.

Example 1.3.9 Consider the following linear system:

$$\begin{aligned} -7x - 4y + 7z &= -3 \\ -2x + 2y - 4z &= 2 \\ 5x + 4y + z &= 5. \end{aligned}$$

Does this system have a solution? If the system has a solution, write down the solution.

Solution. We follow Algorithm 1.3.8, p. 20 and form the augmented matrix A for this system. When we row reduce this matrix, we find

$$A = \left[\begin{array}{ccc|c} -7 & -4 & 7 & -3 \\ -2 & 2 & -4 & 2 \\ 5 & 4 & 1 & 5 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1/9 \\ 0 & 1 & 0 & 4/3 \\ 0 & 0 & 1 & 2/9 \end{array} \right].$$

We will refer to the RREF of A as B . Since B has no pivot in the rightmost column, our linear system is consistent. Secondly, since B has a pivot in each of the first three columns, the solution to our linear system is unique. We record this solution as $x = -1/9$, $y = 4/3$, and $z = 2/9$. ▲

The rest of this section will be devoted to proving Theorem 1.3.13, p. 22, which states that the RREF of a matrix is unique. (There are actually a couple of places so far in this section where we have been a bit sloppy and referred to *the* RREF of a matrix, but in each of these cases the uniqueness of the RREF was not essential to the argument.)

We will begin with a lemma. (Our approach to proving Theorem 1.3.13, p. 22 has been heavily influenced by Kuttler's treatment ([math.libretexts.org/Bookshelves/Linear_Algebra/A_First_Course_in_Linear_Algebra_\(Kuttler\)/](http://math.libretexts.org/Bookshelves/Linear_Algebra/A_First_Course_in_Linear_Algebra_(Kuttler)/)).)

Lemma 1.3.10 *In a consistent $m \times n$ linear system, all solutions can be expressed by writing the basic variables as linear functions of the free variables (if they exist). Further, each basic variable x_i can be written as a linear function of only those free variables x_j with $j > i$.*

Proof. The first sentence in this lemma has essentially been proved in the discussion within Example 1.3.4, p. 18. We will prove the second statement directly. We consider A as the augmented matrix of a consistent $m \times n$ linear system. Suppose A is row equivalent to B , where B is in RREF. We recall that part of the definition of RREF (Definition 1.2.9, p. 12) is that pivots are the leftmost non-zero number in their row.

We consider the linear equation corresponding to row d of B ; this equation will begin with a basic variable x_i and will possibly involve other variables x_j , with $j > i$, before the equals sign. However, all of these other variables x_j will be free variables, because any other basic variable x_k , with $k > i$, will correspond to a column in which that pivot is the only non-zero number. In other words, all entries b_{dk} along row d in B which correspond to pivot columns k , for $k > i$, will be zero. ■

The basic idea for the proof of Theorem 1.3.13, p. 22 is to prove the result for homogeneous linear systems first and then to obtain the proof for general linear systems as an extension. We turn to our first result about homogeneous linear systems. (For a refresher on homogeneous systems, see Definition 1.1.5, p. 2.)

Lemma 1.3.11 *If x_i is a basic variable of a homogeneous linear system, then any solution of the system with $x_j = 0$ for all free variables x_j with $j > i$, must have $x_i = 0$.*

Proof. We will prove this directly. Suppose that x_i is a basic variable of a homogeneous linear system and that in a solution of this system, $x_j = 0$ for all free variables x_j with $j > i$. From Lemma 1.3.10, p. 21 we know that in the description of the solution to this system, x_j will be written as a linear function of the free variables with larger indices. But the nature of a homogeneous linear system demands that such a linear function will involve *only* free variables and no constants (the constants are all 0). Therefore, if $x_j = 0$ for all free variables x_j with $j > i$, we must have $x_i = 0$ as well. ■

We will now prove the uniqueness result for augmented matrices of homogeneous systems. (We should note here, perhaps, that while we introduced the notions of REF and RREF for augmented matrices, the row reduction algorithm can be applied to any matrix at all.)

Proposition 1.3.12 *Let A be the augmented matrix corresponding to a homogeneous linear system. Suppose that A is row equivalent to matrices B and C , both of which are in reduced row-echelon form. Then $B = C$.*

Proof. We proceed by contradiction and assume that $B \neq C$. Since B and C are row equivalent and both are in RREF, they must have the same pivot

positions. (The reader is asked to prove this in Exercise 1.3.15, p. 24.) Since $B \neq C$, these matrices must differ in some row, call it row k . Since B and C have the same pivot positions, we assume there is a pivot in column i of row k in both matrices. There must be some position j , with $j > i$, such that $b_{kj} \neq c_{kj}$. The variable x_j must not be a basic variable in the linear system, because if so, we would have $b_{kj} = c_{kj} = 0$. So x_j is a free variable.

Homogeneous linear systems are always consistent. (The reader is asked to prove this in Exercise 1.3.10, p. 24.) There must exist a solution to the linear system where $x_j = 1$ and all other free variables take on the value of 0. In this solution, using the linear equations that correspond to the rows in B , we solve and obtain $x_i = -b_{kj}$. Using the linear equations that correspond to the rows in C , we find $x_i = -c_{kj}$. Since a solution is completely determined by the values of the free variables, this implies that $b_{kj} = c_{kj}$, which is a contradiction. ■

With this proposition in hand, we can state and prove our first large result.

Theorem 1.3.13 *Let A be an $m \times n$ matrix and let A be row equivalent to both B and C . If B and C are in reduced row-echelon form, then $B = C$.*

Proof. We first form the matrix A' by augmenting the matrix A with an additional column on the right consisting of all zeros. We similarly form the matrices B' and C' from B and C . We note that B' and C' are also in RREF and they are obtained from A' using the same row operations that reduced A to B and C .

We can consider A' , B' , and C' as augmented matrices corresponding to $m \times n$ homogeneous linear systems. By Proposition 1.3.12, p. 21, since A' is row equivalent to both B' and C' , where both B' and C' are in RREF, we must have $B' = C'$. By the construction of B' and C' , this implies $B = C$. ■

Reading Questions

1. Consider the following linear system:

$$\begin{aligned} 4x_1 + 7x_2 + 17x_3 &= 23 \\ -3x_1 - 5x_2 - 12x_3 &= -17. \end{aligned}$$

Determine which of the variables are basic variables and which are free variables. Explain your answer.

2. Consider the following linear system:

$$\begin{aligned} x_1 - 2x_2 + 2x_3 &= -1 \\ -x_1 + 2x_2 - x_3 &= -1 \\ 3x_1 - 6x_2 + 7x_3 &= -5. \end{aligned}$$

Write a parametric description of the solution set of this system. (Follow Example 1.3.4, p. 18.)

Exercises

1. In each of the following, suppose the augmented matrix for a linear system has been reduced to the following RREF. Write down the solution(s) to the system (if they exist).

$$(a) \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$(b) \left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$(c) \left[\begin{array}{ccc|c} 1 & 0 & -3 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$(d) \left[\begin{array}{ccccc|c} 1 & 0 & 2 & -4 & 0 & 7 \\ 0 & 1 & 9 & -1 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right]$$

2. Solve the following linear system.

$$2x - 3y + 5z = 0$$

$$-x + 2y - 3z = 0$$

$$x + 4y - 4z = 0$$

3. Solve the following linear systems.

(a)

$$x - 2y + 3z = 4$$

$$5x - 6y + 7z = 5$$

(b)

$$x - 2y + 2z - w = 1$$

$$3x - 7y - z + 5w = 2$$

(c)

$$2x - y + z = 3$$

$$-x + 3y + 2z = 1$$

$$3x + y + 4z = 7$$

(d)

$$4x - 2y = -4$$

$$2x + 5y + 6z = 2$$

$$3x - y + \frac{1}{2}z = 0$$

4. Determine the values of a for which the following linear system has no solutions, exactly one solution, or infinitely many solutions. Explain your answers.

$$x + 2y + 2z = -4$$

$$2x - 2y + 4z = 7$$

$$x + 2y - (a^2 - 5)z = a + 1$$

5. Under what conditions is the following linear system consistent? Your answer should be an equation or equations that must be satisfied by the b_i . Explain your answer.

$$\begin{aligned}x + 2y &= b_1 \\ -2x - 5y + 3z &= b_2 \\ x + 4y - 6z &= b_3\end{aligned}$$

6. Solve the following system for x , y , and z . (Hint: define new variables to produce a linear system.) Explain your solution.

$$\begin{aligned}-\frac{1}{x} + \frac{2}{y} - \frac{4}{z} &= 2 \\ -\frac{2}{x} + \frac{2}{y} - \frac{4}{z} &= 6 \\ -\frac{3}{x} + \frac{4}{y} + \frac{4}{z} &= -4\end{aligned}$$

7. Answer the question posed in Exercise 1.1.1, p.5 by solving the linear system that was created in that exercise.
8. Answer the question posed in Exercise 1.1.2, p.5 by solving the linear system that was created in that exercise.
9. Suppose that the graph of the function $f(x) = ax^3 + bx^2 + cx + d$ passes through the points $(-1, 2)$, $(-2, -9)$, $(1, 4)$, and $(2, 15)$. Determine the values of a , b , c , and d .

Writing Exercises

10. Prove that every homogeneous linear system is consistent.
- 11.

(a) Prove that if $ad - bc \neq 0$, then the reduced row-echelon form of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

(b) Use part (a) to prove that if $ad - bc \neq 0$, then the linear system

$$\begin{aligned}ax + by &= p \\ cx + dy &= q\end{aligned}$$

has exactly one solution.

12. Using only results from this section, explain why every linear system over $\mathbb{R}$ has either no solutions, exactly one solution, or an infinite number of solutions.
13. Suppose that a 3×4 coefficient matrix for a linear system has three pivot columns. Is the system consistent? Explain.
14. What would one have to know about the pivot columns in an augmented matrix in order to conclude that the linear system has exactly one solution? Explain.
15. Suppose that matrices A and B are row equivalent and both matrices are in REF. Prove that A and B have the same pivot positions.

1.4 Vectors

Up to this point in the chapter, we have been concerned with solving systems of linear equations, and we have used the tool of matrices to great effect. In this section we will connect linear systems to some basic geometric concepts, and this will result in alternate ways of writing linear systems which, at times, will be more helpful.

1.4.1 The Basics of Vectors

Although we will shortly connect “vectors” to geometric notions, at the beginning a vector will be strictly an algebraic object.

Definition 1.4.1 An n -dimensional vector over $\mathbb{R}$ is an ordered list of n real numbers. We will adopt the convention that, unless stated otherwise, vectors are column vectors written in this form:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

where $v_i \in \mathbb{R}$ for all i . (Column vectors are therefore matrices with a single column.) The set of all n -dimensional vectors over $\mathbb{R}$ is denoted $\mathbb{R}^n$. ♦

Note 1.4.2 We say that two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are equal if $u_i = v_i$ for each $i = 1, \dots, n$. This means that an equation involving vectors in $\mathbb{R}^n$ captures the same information as n equations involving real numbers.

As we have said, vectors should be thought of first as algebraic objects, and there are several ways to combine these objects.

Definition 1.4.3 We can combine and modify vectors through addition and a form of multiplication. We will describe the multiplication first.

Let $\mathbf{v} \in \mathbb{R}^n$ have the form

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

and let $c \in \mathbb{R}$. Then the **scalar multiple** of $\mathbf{v}$ by c is the vector $c\mathbf{v}$ in $\mathbb{R}^n$ obtained by multiplying each entry of $\mathbf{v}$ by c ; that is,

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}.$$

In this context, we will often refer to real numbers as **scalars**.

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ with the form

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

then the **sum** of $\mathbf{u}$ and $\mathbf{v}$ is the vector $\mathbf{u} + \mathbf{v}$ in $\mathbb{R}^n$ obtained by adding the corresponding entries in $\mathbf{u}$ and $\mathbf{v}$. That is,

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

◆

Note 1.4.4 We will use the notation $\mathbf{0}$ to denote the **zero vector**—that is, the vector in $\mathbb{R}^n$ whose entries are all 0. We will also use the notation $-\mathbf{u}$ to indicate the scalar multiple $(-1)\mathbf{u}$.

We present some quick calculations in the following example.

Example 1.4.5 Let $\mathbf{u} \in \mathbb{R}^2$ and let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ such that

$$\mathbf{u} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -6 \\ -3 \\ 3 \end{bmatrix}.$$

Then we can calculate $2\mathbf{u}$ and $\mathbf{v} + \mathbf{w}$ using Definition 1.4.3, p. 25:

$$2\mathbf{u} = \begin{bmatrix} -6 \\ 10 \end{bmatrix}, \quad \mathbf{v} + \mathbf{w} = \begin{bmatrix} -4 \\ -2 \\ -1 \end{bmatrix}.$$

▲

When combining vectors, we are limited to adding vectors of the same dimension—that is, vectors with the same number of entries. However, we are not limited to either addition *or* scalar multiplication; we can certainly do both at once. Nor are we limited to adding only two vectors at a time. The following definition provides the correct generalization.

Definition 1.4.6 Let $c_1, c_2, \dots, c_m$ be real numbers and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{R}^n$. Then the **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ with **weights** $c_1, c_2, \dots, c_m$ is

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_m\mathbf{v}_m.$$

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is the set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ and is written $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$. In other words, the span is defined to be

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\} = \left\{ \sum_{i=1}^m c_i \mathbf{v}_i \mid c_1, \dots, c_m \in \mathbb{R} \right\}.$$

◆

Example 1.4.7 Taking $\mathbf{v}$ and $\mathbf{w}$ from Example 1.4.5, p. 26, we can calculate the linear combination of these vectors with weights 2 and -1 :

$$2\mathbf{v} - \mathbf{w} = \begin{bmatrix} 4 \\ 2 \\ -8 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ -11 \end{bmatrix}.$$

▲

Note 1.4.8 We have defined addition and scalar multiplication for vectors here, but these concepts also make sense for matrices. We will set the stage briefly here for a return in Section 2.3, p. 46.

If we think of an $m \times n$ matrix A in terms of its columns, then the n columns of A are all vectors in $\mathbb{R}^m$. For two $m \times n$ matrices A and B , we define the sum $A + B$ in this way: column j of $A + B$ is the sum of the two vectors in $\mathbb{R}^m$ which are the j th columns of A and B . Similarly, if $c \in \mathbb{R}$, then we can define the scalar multiple cA in terms of its columns: column j of cA is the scalar multiple of the j th column of A by c . In this way these algebraic notions for matrices are built upon the corresponding notions for vectors.

We include two initial calculations as examples. If A and B are defined as

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 2 & 7 \\ 0 & -4 & 4 \end{bmatrix},$$

then we have

$$A + B = \begin{bmatrix} -1 & 1 & 7 \\ 3 & 0 & -1 \end{bmatrix} \quad \text{and} \quad -2A = \begin{bmatrix} -4 & 2 & 0 \\ -6 & -8 & 10 \end{bmatrix}.$$

We finally note that, as with vectors, the sum (and thus the linear combination) of two or more matrices only makes sense if all involved matrices are of the same size.

Before we explore the concept of span, we need to discuss the connection between systems of linear equations and vector equations. We saw a glimpse of this connection in Note 1.4.2, p. 25, and we will develop it further here.

Let us consider the following simple linear system:

$$\begin{aligned} 3x_1 - 8x_2 &= 9 \\ -2x_1 + 5x_2 &= -1. \end{aligned} \tag{1.5}$$

Solving this linear system involves (as always) investigating whether there are values of x_1 and x_2 which satisfy both of these equations simultaneously. We will now arrive at this same question from a different angle.

Let's consider the following three vectors in $\mathbb{R}^2$:

$$\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -8 \\ 5 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 9 \\ -1 \end{bmatrix}.$$

Since all of these vectors are in $\mathbb{R}^2$, we can ask this question: Can $\mathbf{w}$ be written as a linear combination of $\mathbf{u}$ and $\mathbf{v}$? (This question is equivalent to asking if $\mathbf{w}$ belongs to $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.) In other words, do there exist scalars x_1 and x_2 such that $x_1\mathbf{u} + x_2\mathbf{v} = \mathbf{w}$?

The equation in this final question is equivalent to other vector equations:

$$\begin{aligned} x_1\mathbf{u} + x_2\mathbf{v} &= \mathbf{w} \\ x_1 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} -8 \\ 5 \end{bmatrix} &= \begin{bmatrix} 9 \\ -1 \end{bmatrix} \\ \begin{bmatrix} 3x_1 \\ -2x_1 \end{bmatrix} + \begin{bmatrix} -8x_2 \\ 5x_2 \end{bmatrix} &= \begin{bmatrix} 9 \\ -1 \end{bmatrix} \\ \begin{bmatrix} 3x_1 - 8x_2 \\ -2x_1 + 5x_2 \end{bmatrix} &= \begin{bmatrix} 9 \\ -1 \end{bmatrix}. \end{aligned}$$

Because of Note 1.4.2, p. 25, this last vector equation is equivalent to the linear system in (1.5).

Example 1.4.9 Consider the following vectors in $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}, \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}.$$

Is $\mathbf{v}_3$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

We have written this question in terms of a vector equation, but by the previous discussion we should be able to translate this question to a more familiar one about solutions to linear systems and answer the question using Algorithm 1.3.8, p. 20.

The question about vectors is the same as asking if this linear system is consistent:

$$\begin{aligned} 3x_2 &= 5 \\ 2x_1 - 2x_2 &= -3 \\ -x_1 + 4x_2 &= 1. \end{aligned}$$

The augmented matrix for this system is

$$\left[\begin{array}{cc|c} 0 & 3 & 5 \\ 2 & -2 & -3 \\ -1 & 4 & 1 \end{array} \right],$$

and the RREF of this matrix is

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

Since there is a pivot in the final column of this matrix, we conclude (by invoking Theorem 1.3.5, p. 19) that the linear system is inconsistent. This means that the answer to the original question is no, $\mathbf{v}_3$ is not in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. ▲

Example 1.4.9, p. 28 illustrates a general workflow for questions like this. There is no need to explicitly write out the intermediate step involving linear systems; instead, we can form a matrix using the given vectors as the appropriate columns, row reduce the matrix, and answer the question.

Combining vectors in $\mathbb{R}^n$ is already (or soon will be) fairly natural for readers of this chapter. Under the operations of scalar multiplication and addition, vectors in $\mathbb{R}^n$ have some useful properties, which we record in the following fact. We will not spend time with these properties now, but we will look at them intently in Section 2.3, p. 46. These properties can all be verified using the corresponding properties of addition and multiplication of real numbers. (And the fact that real numbers have these properties is *essential*!)

Fact 1.4.10 For all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^n$, and for all real numbers c and d , the following properties hold.

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
4. $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$
5. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

6. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$

7. $c(d\mathbf{u}) = (cd)\mathbf{u}$

8. $1\mathbf{u} = \mathbf{u}$

Thinking about linear systems through the lens of vectors also brings some structure to the solutions to consistent linear systems. When a linear system is consistent and the solution is unique, we have recorded this in terms of the variables involved. For example, we recorded the solution to the linear system in Example 1.3.9, p.20 as $x = -1/9$, $y = 4/3$, and $z = 2/9$. However, if we think of these variables as forming a vector in $\mathbb{R}^3$, then we can record the solution this way:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1/9 \\ 4/3 \\ 2/9 \end{bmatrix}.$$

Similarly, when a consistent system does not have a unique solution, we can again use vector notation. (This would replace the parametric description of the solution set we presented in Example 1.3.4, p.18.) In Example 1.3.4, p.18 we considered such a linear system. The solution set can now be written this way:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 + 3x_3 \\ 7 - 2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}.$$

If we label the vectors $\mathbf{v}$ and $\mathbf{w}$ as

$$\mathbf{v} = \begin{bmatrix} 5 \\ 7 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix},$$

then all solutions to this system can be written as $\mathbf{v} + t\mathbf{w}$, where t can be any real number. As we learn more about the geometric interpretations of the solutions to linear systems, this phrasing in terms of vectors will be useful.

1.4.2 Vectors and Geometry

Before we leave this chapter, we need to introduce the connection between vectors and geometry. Readers who have taken multivariable calculus are likely aware of some of these concepts. For the sake of simplicity, we will restrict most of our discussion (and all of our illustrations) to $\mathbb{R}^2$ in this section.

We will visualize a vector $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$ by identifying it with the point (a, b) in the plane. We will draw an arrow from the origin to (a, b) to aid this visualization. So, for example, the vectors $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ would be visualized in the following picture.

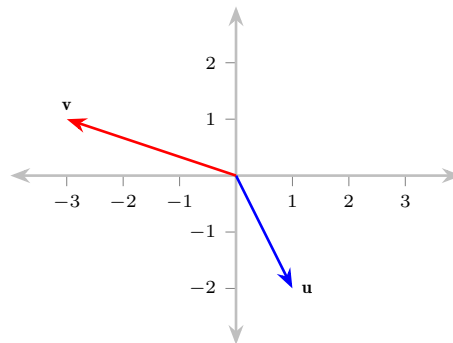


Figure 1.4.11 Illustration of two vectors in the plane

Scalar multiplication of a vector can be seen as stretching or contracting the vector, if the scalar is positive. If the scalar is negative, the vector will be pointing in the opposite direction, then stretched or contracted. Here is an example using the vector $\mathbf{v}$ previously defined.

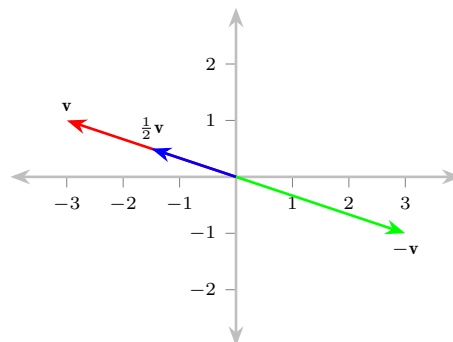


Figure 1.4.12 Illustration of scalar multiplication

Addition of vectors is also easy to visualize. To add two vectors, move the second vector from the origin so that its beginning coincides with the end of the first vector. The vector sum can be represented by the arrow from the origin to the end of this relocated second vector. (The dashed arrow in the following diagram is the relocated vector $\mathbf{v}$.)

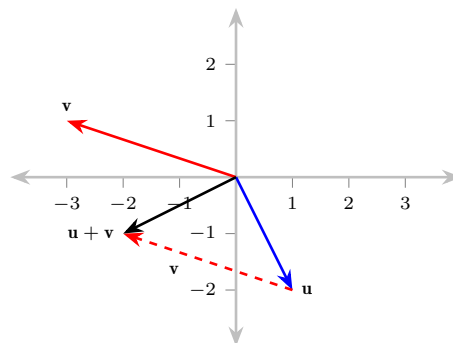


Figure 1.4.13 Illustration of vector addition

We can put these previous two ideas together to explain the way to visualize linear combinations (and spans). Let's consider the vectors $\mathbf{u}$ and $\mathbf{v}$ as previously defined, and we will also define $\mathbf{w}$ as $\mathbf{w} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

When we ask whether $\mathbf{w}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$, this is the same as asking whether we can form $\mathbf{w}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$. We answer this question

by forming a matrix with the appropriate columns and row reducing. We find the matrix

$$[\mathbf{u} \quad \mathbf{v} \mid \mathbf{w}] = \left[\begin{array}{cc|c} 1 & -3 & 2 \\ -2 & 1 & 2 \end{array} \right],$$

and the RREF of this matrix is

$$\left[\begin{array}{cc|c} 1 & 0 & -1.6 \\ 0 & 1 & -1.2 \end{array} \right].$$

This tells us that $\mathbf{w}$ is in the span of $\mathbf{u}$ and $\mathbf{v}$, and that

$$\mathbf{w} = -1.6\mathbf{u} - 1.2\mathbf{v}.$$

Here is the picture illustrating this.

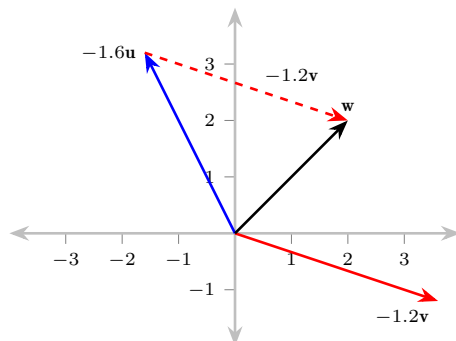


Figure 1.4.14 Illustration of a linear combination

The reader may be able to see that no matter what vector $\mathbf{w}$ in $\mathbb{R}^2$ was chosen in this last instance, that vector would be in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$. (This means that $\text{Span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2$.) This has everything to do with the relationship between $\mathbf{u}$ and $\mathbf{v}$.

We will discuss this more carefully in Chapter 5, p.107, but this brief comment will suffice for now. As long as $\mathbf{u} \neq \mathbf{0}$ in $\mathbb{R}^2$, then $\text{Span}\{\mathbf{u}\}$ is a line through the origin. (Remember that the span in this case consists of *all* multiples of $\mathbf{u}$.) Similarly, as long as neither $\mathbf{u}$ nor $\mathbf{v}$ is a multiple of the other vector, then $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is *all of* $\mathbb{R}^2$. (If we were talking about vectors in $\mathbb{R}^3$ instead of $\mathbb{R}^2$ this span would be a plane through the origin.)

1.4.3 Reading Questions

- Let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 2 \\ -7 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

Calculate each of the following vectors.

- $-2\mathbf{u}$
- $\mathbf{u} - \mathbf{v}$
- $3\mathbf{u} - 4\mathbf{v}$

2. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}.$$

Is the vector

$$\begin{bmatrix} 7 \\ -1 \\ 8 \end{bmatrix}$$

in $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$? Explain your answer using techniques from this section. (See especially Example 1.4.9, p. 28.)

1.4.4 Exercises

1. Write a linear system that is equivalent to the following vector equation:

$$x \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ -1 \end{bmatrix}.$$

2. Write a vector equation that is equivalent to the following linear system:

$$\begin{aligned} 2x - 5z &= 9 \\ x + 3y + z &= -1 \\ -y - 7z &= 0. \end{aligned}$$

3. Let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}.$$

List five vectors in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$. For each vector, write down the specific linear combination of $\mathbf{u}$ and $\mathbf{v}$ used to generate the vector.

4. Describe all possible ways of writing $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$.
5. For each of the following, determine whether $\mathbf{b}$ is a linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. Explain your answer.

(a)

$$\mathbf{u} = \begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -3 \\ -3 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -4 \\ 0 \\ -4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 1.5 \end{bmatrix}.$$

(b)

$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 5 \\ -1 \\ -2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 10 \\ -1 \\ 4 \end{bmatrix}$$

6. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{b}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -2 \\ -6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ -1 \\ h \end{bmatrix}.$$

For what value(s) of h is $\mathbf{b}$ in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$? Explain your answer.

7. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{b}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Under what conditions is $\mathbf{b}$ in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$? This will be an equation (or equations) satisfied by the b_i . Explain your answer.

Writing Exercises

8. Let $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Show that $\begin{bmatrix} h \\ k \end{bmatrix}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ for all h and k . Explain your answer.
9. Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are called *collinear* if $\mathbf{u} = k\mathbf{v}$ or $\mathbf{v} = k\mathbf{u}$ for some $k \in \mathbb{R}$. Show that the span of any two nonzero vectors in $\mathbb{R}^2$ which are not collinear is all of $\mathbb{R}^2$.
10. Give an example of three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^3$ such that no two of them are collinear but $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \neq \mathbb{R}^3$. Explain why your example works.
11. State criteria on vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in $\mathbb{R}^3$ such that $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \mathbb{R}^3$. Explain your answer.
12. Prove property 5 from Fact 1.4.10, p.28. Note which properties of the real numbers you use in this proof.

Chapter 2

Fields and Vector Spaces

2.1 Fields

In this section we will consider the real numbers and study their most important properties in a general setting. In the process, we will learn how to handle axioms and abstract algebraic concepts.

In Chapter 1, p. 1, we used the real numbers as the familiar world within which all of our calculations took place. Before we offer any definitions or results in this chapter, we will ponder exactly what properties of the real numbers were essential to these calculations. (The reader is likely so familiar with the real numbers that its important properties have been internalized and are not in the conscious mind. In this section we will make those properties explicit.)

The most basic concept in this book is the linear equation. What properties of the real numbers are used to solve a simple linear equation? In what follows we will solve the equation $2x + 7 = 12$ and draw to the surface some of these important properties:

$$2x + 7 = 12 \tag{2.1}$$

$$(2x + 7) - 7 = 12 - 7 \tag{2.2}$$

$$2x + (7 - 7) = 5 \tag{2.3}$$

$$2x + 0 = 5 \tag{2.4}$$

$$2x = 5 \tag{2.5}$$

$$\frac{1}{2}(2x) = \frac{1}{2}5 \tag{2.6}$$

$$\left(\frac{1}{2}2\right)x = \frac{5}{2} \tag{2.7}$$

$$(1)x = \frac{5}{2} \tag{2.8}$$

$$x = \frac{5}{2}. \tag{2.9}$$

No student wants to write out all of these steps, and no instructor wants to grade such a solution! But the point is to notice the properties of the real numbers which are vital to solving an equation like this and which we usually ignore.

- In (2.3) we used the *associativity* of addition in $\mathbb{R}$. That is, we can move the parentheses around in addition and still have an equivalent expression. (The reader should see subtraction as a form of addition.)

- In (2.4) we used the fact that 7 has an *additive inverse*—a number we can add to 7 to get 0. (The additive inverse of 7 is -7 .)
- In (2.5) we used the fact that 0 is an *additive identity* in $\mathbb{R}$ —when we add 0 to any real number r , we get r again.
- In (2.7) we used the *associativity* of multiplication in $\mathbb{R}$. We can move the parentheses around in multiplication just like we can in addition.
- In (2.8) we used the fact that $\frac{1}{2}$ is the *multiplicative inverse* of 2 in $\mathbb{R}$. In other words, we can multiply 2 by $\frac{1}{2}$ to get 1.
- Finally, in (2.9) we used the fact that 1 is a *multiplicative identity* in $\mathbb{R}$ —when we multiply any real number by 1, that real number is unchanged.

By identifying these properties, our goal is to envision other mathematical realms in which solving linear equations would work the same way it does within $\mathbb{R}$. Toward this end, we now define an algebraic object called a “field” which has all of the properties used above (plus a few we haven’t yet mentioned).

Definition 2.1.1 A **field** is a set $\mathbb{F}$ with operations $+$ and $\cdot$ and distinct elements $0, 1 \in \mathbb{F}$ such that all of the following properties hold.

1. For all $a, b \in \mathbb{F}$, $a + b \in \mathbb{F}$. (We say that $\mathbb{F}$ is **closed under addition**.)
2. For all $a, b \in \mathbb{F}$, $a \cdot b \in \mathbb{F}$. (We say that $\mathbb{F}$ is **closed under multiplication**.)
3. For all $a, b \in \mathbb{F}$, $a + b = b + a$. (We say that addition in $\mathbb{F}$ is **commutative**.)
4. For all $a, b, c \in \mathbb{F}$, $a + (b + c) = (a + b) + c$. (We say that addition in $\mathbb{F}$ is **associative**.)
5. For each $a \in \mathbb{F}$, $a + 0 = 0 + a = a$. (We say that 0 is an **additive identity** in $\mathbb{F}$.)
6. For each $a \in \mathbb{F}$, there exists an element $b \in \mathbb{F}$, such that $a + b = b + a = 0$. (We say that each a has an **additive inverse** in $\mathbb{F}$.)
7. For each $a, b \in \mathbb{F}$, $a \cdot b = b \cdot a$. (We say that multiplication is **commutative** in $\mathbb{F}$.)
8. For each $a, b, c \in \mathbb{F}$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$. (We say that multiplication is **associative** in $\mathbb{F}$.)
9. For each $a \in \mathbb{F}$, $a \cdot 1 = 1 \cdot a = a$. (We say that 1 is a **multiplicative identity** in $\mathbb{F}$.)
10. For each $a \in \mathbb{F}$ with $a \neq 0$, there exists an element $c \in \mathbb{F}$ such that $a \cdot c = c \cdot a = 1$. (We say that every nonzero element a in $\mathbb{F}$ has a **multiplicative inverse**.)
11. For each $a, b, c \in \mathbb{F}$, $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$. (We say that addition and multiplication in $\mathbb{F}$ satisfy the **distributive law**.)



Note 2.1.2 Axioms 5 and 6 could perhaps be stated with a bit more precision. Some texts state axiom 5 like this: the set $\mathbb{F}$ must have an additive identity, that is, an element e such that $x + e = e + x = x$ for all $x \in \mathbb{F}$. Those same

texts may also state axiom 6 without reference to the symbol 0: for each $a \in \mathbb{F}$, there exists an element $b \in \mathbb{F}$ such that $a + b = b + a = e$, where e is the additive identity of the field.

For some students, the use of the symbol 0 might be helpful, reminding them of the essential property of the additive identity through familiarity. But this may be confusing when handling a set which contains the symbol 0 when that symbol does *not* function as the relevant additive identity.

Our point here is that the use of 0 in the axioms for a field (and some of the subsequent theorems) is intended to stand in for a generic symbol for the additive identity. Similarly, the symbol 1 stands in as the generic symbol for the multiplicative identity. (A reader could imagine a restatement of axioms 9 and 10 above using a letter as the multiplicative identity in the way we have demonstrated axioms 5 and 6 can be restated.)

The reader should also note that in a generic field the term “nonzero” means “not the additive identity.”

A note about axioms. What are presented in Definition 2.1.1, p. 35 are known as the **axioms** of a field. This may be the reader’s first exposure to axioms in mathematics, and this is worthy of a comment or two.

Much of theoretical mathematics is built upon axiomatic reasoning. The thinking goes like this: If we assume a limited number of properties are true, and we assume nothing beyond those properties, what else follows *necessarily*? In this way we can ask what is true of a field, not just what is true of the real numbers. While the real numbers may have specific properties that a general field does not, anything that is true of a general field must be true of the real numbers.

Working through some examples (and non-examples) will help us make sense of this definition.

Example 2.1.3 The set of real numbers $\mathbb{R}$ is a field. (If the real numbers were not a field, then we wouldn’t have done a very good job of abstracting the properties of the real numbers for this definition!) ▲

Example 2.1.4 The complex numbers are defined in this way:

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\},$$

where $i^2 = -1$. Addition and multiplication are defined in this way:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i \\ (a + bi) \cdot (c + di) &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Notice that the addition and multiplication occurring within the parentheses on the right side of these equations are happening within $\mathbb{R}$. In this way, showing that some of the field axioms hold for $\mathbb{C}$ depends on $\mathbb{R}$ being a field.

The elements 0 and 1 in $\mathbb{C}$ are

$$\begin{aligned}0 &= 0 + 0i \\ 1 &= 1 + 0i,\end{aligned}$$

and these elements are not equal. (We recall that part of the definition of a field is that the elements 0 and 1 are distinct.)

With these definitions, one can check that $\mathbb{C}$ satisfies the properties of a field. We will prove a few of these properties, and we will assign a few of these proofs in the exercises.

The definitions of $+$ and $\cdot$ in $\mathbb{C}$ show that $\mathbb{C}$ is closed under addition and multiplication. (Notably, this relies on $\mathbb{R}$ being closed under addition and multiplication!) To prove that the addition in $\mathbb{C}$ is commutative, we consider two complex numbers $a + bi$ and $c + di$. Then

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad (2.10)$$

$$= (c + a) + (d + b)i \quad (2.11)$$

$$= (c + di) + (a + bi). \quad (2.12)$$

(We note that line (2.11) used the fact that addition of real numbers is commutative.) This proves that addition in $\mathbb{C}$ is commutative.

We will also prove that 0 is an additive identity in $\mathbb{C}$. If $a + bi \in \mathbb{C}$, then

$$(a + bi) + (0 + 0i) = (a + 0) + (b + 0)i = a + bi.$$

(This uses the fact that $0 \in \mathbb{R}$ is an additive identity.) We note that although the definition of a field requires us to examine addition by 0 “on both sides,” since we just proved that addition is commutative, what we have already shown is sufficient. $\blacktriangle$

Example 2.1.5 The set of non-negative real numbers $\mathbb{R}^{\geq 0}$ is not a field. (When thinking about a subset of the real numbers, we will assume that the usual addition and multiplication are in view unless otherwise stated.) While this set is closed under addition and multiplication, it does not contain *additive inverses* for positive real numbers. For example, the number 9 has no additive inverse in this set. $\blacktriangle$

Example 2.1.6 The set of **rational numbers** $\mathbb{Q}$ is a field. The rational numbers are defined as the set of all quotients (hence the symbol $\mathbb{Q}$) of integers; more formally,

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

We do not need to show that all of the field axioms hold in order to prove that $\mathbb{Q}$ is a field. Since $\mathbb{Q}$ is a subset of $\mathbb{R}$, and since it therefore inherits its operations from $\mathbb{R}$, some of the field axioms hold automatically. (These would include the associativity and commutativity of addition. What else holds “by inheritance?”) Checking some of these details is left to the exercises. $\blacktriangle$

Note 2.1.7 This notion of “inherited” properties comes up with some frequency, so we will undertake a brief explanation. Suppose we have $A \subseteq B$ for two sets A and B , and we have an operation $\oplus$ defined on B . Any property that is intrinsic to the operation (rather than to the set) will hold in both B and A . For example, if $\oplus$ is commutative in B , then $\oplus$ will also be commutative in A since $A \subseteq B$.

Not all field properties are “inherited” in this same way with subsets because some properties depend on the elements of the set A and not just the operation. Example 2.1.5, p. 37 is a good example of this; $\mathbb{R}^{\geq 0}$ is a subset of $\mathbb{R}$ with the same operations, but it is not a field because (for example) the property of additive inverses is not an inherited property in the same way that the commutativity or associativity of addition is inherited.

We refer the reader to Appendix A, p. 200 for the definitions of $\mathbb{Z}_n$ and $\mathbb{F}_p$, related notational conventions, and a refresher on modular arithmetic. All of this is necessary for the following example.

Example 2.1.8 When p is a prime integer, then $\mathbb{F}_p$ is a field.

Working within $\mathbb{F}_p$ can be a bit destabilizing at first, as the calculations

take some practice. However, this effort pays off because the smallest finite fields offer some of the most tangible sandboxes in which to play. We will be using $\mathbb{F}_p$ for small p throughout this chapter to develop some interesting examples.

Part of the definition of a field is that one can divide by any nonzero element. But because we work within $\mathbb{R}$ so often, division within $\mathbb{F}_p$ is strange. In $\mathbb{F}_3$, for instance, the multiplicative inverse of 2 is 2. (Another way to say this is that, within $\mathbb{F}_3$, the element that acts the most like $\frac{1}{2}$ is 2.)

The field $\mathbb{F}_2$ contains only the elements 0 and 1, and it therefore is a model for anything that is *binary*. For this reason, working in $\mathbb{F}_2$ is often very useful in computer science. $\blacktriangle$

Having defined fields, we now turn to the consequences of this definition. In other words, if a set with two operations meets the definition of a field, what else must also be true? The following theorem presents some basic results that flow from the axioms of a field. Some of these will look familiar because they are true in $\mathbb{R}$ and that's where most people are comfortable working.

Theorem 2.1.9 *For any field $\mathbb{F}$, the following are true.*

1. *The additive identity in $\mathbb{F}$ is unique.*
2. *The multiplicative identity in $\mathbb{F}$ is unique.*
3. *Additive inverses in $\mathbb{F}$ are unique. (This means that for every element $x \in \mathbb{F}$, there exists a unique element $y \in \mathbb{F}$ which is the additive inverse of x . The truth of this statement justifies our use of the notation $-x$ for the additive inverse of an element $x \in \mathbb{F}$.)*
4. *Multiplicative inverses in $\mathbb{F}$ are unique. (The truth of this statement justifies our use of the notation x^{-1} for the multiplicative inverse of a nonzero element $x \in \mathbb{F}$.)*
5. *For every $x \in \mathbb{F}$, $-(-x) = x$.*
6. *For every nonzero $x \in \mathbb{F}$, $(x^{-1})^{-1} = x$.*
7. *For every $x \in \mathbb{F}$, $0 \cdot x = 0$.*
8. *For every $x \in \mathbb{F}$, $(-1) \cdot x = -x$.*
9. *If $x, y \in \mathbb{F}$ and $x \cdot y = 0$, then either $x = 0$ or $y = 0$.*

Proof. We will prove a few of these and leave the rest as exercises.

To prove (Item 1, p. 38) that the additive identity is unique, we must prove that 0 is the only element in $\mathbb{F}$ which has the properties of an additive identity. We suppose that $a \in \mathbb{F}$ is such that $a + x = x + a = x$ for every element $x \in \mathbb{F}$. Since this must be true for every x , it must be true for $x = 0$. The previous equation then becomes

$$a + 0 = 0 + a = 0.$$

Since 0 is an additive identity, $a + 0 = a$, which combined with the fact that $a + 0 = 0$ means that $a = 0$. This proves that the additive identity, 0, is unique.

To prove (Item 3, p. 38) that additive inverses are unique, we must prove that for any element $x \in \mathbb{F}$, there is a unique element that behaves like an additive inverse of x . We let $x \in \mathbb{F}$ and we suppose that $y, z \in \mathbb{F}$ are both additive inverses of x . We wish to show that $y = z$.

Our assumptions mean that

$$x + y = y + x = 0$$

$$x + z = z + x = 0.$$

We then use these assumptions, along with some of the properties of fields (the associativity of addition and the properties of the additive identity) in this calculation:

$$\begin{aligned} x + y &= 0 \\ z + (x + y) &= z + 0 \\ (z + x) + y &= z \\ 0 + y &= z \\ y &= z. \end{aligned}$$

This proves that additive inverses are unique. ■

We offer one final note on notation. We will often use juxtaposition to denote multiplication within a field. That is, we may write xy instead of $x \cdot y$ to indicate the product of two elements in a field. We trust the reader will adjust quickly to this seismic shift.

Reading Questions

1. Carry out the following calculations within $\mathbb{F}_7$. (All answers should be a number between 0 and 6.)
 - (a) $4 + 3$
 - (b) $5 \cdot 6$
 - (c) $2 \cdot (3 + 6)$ (Complete this calculation in the two different ways present in the distributive law and verify that they are equal.)
2. What is the additive inverse of 3 in $\mathbb{F}_5$? What is the multiplicative inverse of 3 in $\mathbb{F}_7$? Explain your answers.
3. Theorem A.0.6, p. 201 says that $\mathbb{Z}_9$ is not a field because 9 is not prime. Identify a nonzero element of $\mathbb{Z}_9$ that does not have a multiplicative inverse and explain why it does not have an inverse.

Exercises

1. Carry out the following calculations in $\mathbb{Z}_8$. (Remember that your answer for calculations in $\mathbb{Z}_n$ should be a number between 0 and $n - 1$.)
 - (a) $6 + 7$
 - (b) $3 + 7 + 5$
 - (c) $7 \cdot 3$
 - (d) $6 \cdot 4 \cdot 3$
 - (e) $(-3) \cdot (5 + 3 + (-4))$
2. We know that since 12 is not a prime, $\mathbb{Z}_{12}$ is not a field. In particular, the axiom about multiplicative inverses does not hold. For each nonzero element of $\mathbb{Z}_{12}$, determine whether or not it has a multiplicative inverse. If the element has a multiplicative inverse, state that inverse.
3. For each nonzero element of $\mathbb{F}_{11}$, find the multiplicative inverse.

Writing Exercises

4. Finish Example 2.1.4, p. 36. In other words, complete the proof begun in Example 2.1.4, p. 36 that $\mathbb{C}$ is a field.
5. Consider Example 2.1.6, p. 37. Which of the field axioms for $\mathbb{Q}$ hold “by inheritance” and for which of the axioms is there something that needs to be proved? Put each of the field axioms into one of these two categories. For each axiom that doesn’t hold merely “by inheritance,” prove that it holds for $\mathbb{Q}$.
6. Let A be a set and let n be a nonnegative integer. Define a polynomial $p(x)$ of degree n over A by

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

where $a_i \in A$ for all i . Define $A[x]$ as the set of polynomials of any degree over A .

- (a) Is $\mathbb{Z}[x]$ a field? Justify your answer.
- (b) Is $\mathbb{R}[x]$ a field? Justify your answer.
7. In this problem we consider “adding” an element to $\mathbb{F}_3$. If $\mathbb{F}_3[\alpha]$ is defined by

$$\mathbb{F}_3[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{F}_3, \alpha^2 = 2\},$$

is $\mathbb{F}_3[\alpha]$ a field? Justify your answer.

8. In this problem we consider “adding” an element to $\mathbb{F}_5$.

- (a) If $\mathbb{F}_5[\alpha]$ is defined by

$$\mathbb{F}_5[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{F}_5, \alpha^2 = 4\},$$

is $\mathbb{F}_5[\alpha]$ a field? Justify your answer.

- (b) If $\mathbb{F}_5[\alpha]$ is defined by

$$\mathbb{F}_5[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{F}_5, \alpha^2 = 2\},$$

is $\mathbb{F}_5[\alpha]$ a field? Justify your answer.

9. Define $\mathbb{Q}[\sqrt{2}]$ by

$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}.$$

Is $\mathbb{Q}[\sqrt{2}]$ a field? Justify your answer.

10. Consider the set $\mathbb{R}^2$ with operations defined as follows:

$$\begin{aligned}(a, b) \oplus (c, d) &= (a + c, b + d) \\ (a, b) \otimes (c, d) &= (ac - 2bd, ad + bc).\end{aligned}$$

Is $\mathbb{R}^2$ with these operations a field? Justify your answer.

11. Prove Item 2, p. 38 of Theorem 2.1.9, p. 38.
12. Prove Item 4, p. 38 of Theorem 2.1.9, p. 38.
13. Prove Item 7, p. 38 of Theorem 2.1.9, p. 38.
14. Prove Item 8, p. 38 of Theorem 2.1.9, p. 38.
15. Prove Item 9, p. 38 of Theorem 2.1.9, p. 38.

2.2 Solving Linear Systems Over Fields

Having now defined a field, in this section we will show how the problems of chapter 1 can be solved in this general setting. We have laid the groundwork for reproducing the results of the first three sections of Chapter 1, p. 1 for a general field $\mathbb{F}$ instead of the real numbers.

We now return to where we began in Section 1.1, p. 1: the humble linear equation. If fields are generalizations of the real numbers, and if we can solve linear equations when everything in sight comes from the real numbers, we should be able to solve linear equations when everything in sight comes from a general field.

Example 2.2.1 Consider the following equation where all variable values, constants, and coefficients are drawn from $\mathbb{F}_5$:

$$3x + 2 = 1.$$

Solving this equation in $\mathbb{R}$ would be easy; let's solve it in $\mathbb{F}_5$.

We first note that the additive inverse of 2 in $\mathbb{F}_5$ is 3, so our first step is to add 3 to both sides of the equation:

$$\begin{aligned} 3x + 2 + 3 &= 1 + 3 \\ 3x &= 4. \end{aligned}$$

We now need the multiplicative inverse of 3 in $\mathbb{F}_5$, which is 2. We multiply both sides by 2 to get our answer:

$$\begin{aligned} 2(3x) &= 2(4) \\ x &= 3. \end{aligned}$$

When we check that our solution works (we plug $x = 3$ back into the original equation and perform the computations in $\mathbb{F}_5$), we find that it does:

$$3x + 2 = 3(3) + 2 = 9 + 2 = 11 = 1.$$



The point of this section is that the algorithm for solving linear systems (Algorithm 1.3.8, p. 20) which worked for $\mathbb{R}$ also works for a general field $\mathbb{F}$. In order to be comfortable with this notion, we need to talk quickly through the development of that algorithm in this general setting.

Because a general field contains both 0 and 1 (or rather, an additive and a multiplicative identity), and because within a field we can perform all of the operations needed to solve linear systems, everything we want to do is legitimate. The definitions of the coefficient and augmented matrices, the elementary row operations, the echelon forms, pivots, and the row reduction algorithm (all found in Section 1.2, p. 7 and Section 1.3, p. 17) are the same once we move away from $\mathbb{R}$. Similarly, the three important theorems we have encountered so far (which we will reproduce below) all hold over a general field $\mathbb{F}$. We will omit the proofs of these theorems because the earlier proofs, when translated from $\mathbb{R}$ to $\mathbb{F}$, are still valid.

Theorem 2.2.2 *Suppose A is the augmented matrix of a linear system over a field $\mathbb{F}$, and suppose that A is in reduced row-echelon form. Then the linear system is consistent if and only if there is no pivot in the final column of A .*

Theorem 2.2.3 Suppose that A is the augmented matrix corresponding to a consistent $m \times n$ linear system over a field $\mathbb{F}$, and suppose that A is in reduced row-echelon form. Then the system has a unique solution if and only if there is a pivot in each of the first n columns of A .

Theorem 2.2.4 Let A be an $m \times n$ matrix with entries in a field $\mathbb{F}$ and let A be row equivalent to both B and C . If B and C are in reduced row-echelon form, then $B = C$.

The algorithm for solving a linear system, supported by these theorems, remains the same as in Algorithm 1.3.8, p. 20. We will finish this section with two examples where we go through this algorithm carefully.

Example 2.2.5 The following is a linear system over $\mathbb{F}_3$:

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 1 \\2x_1 + 2x_2 &= 1 \\x_1 + 2x_3 &= 0.\end{aligned}$$

We will begin to solve this system by forming the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \end{array} \right].$$

Since working with fields other than $\mathbb{R}$ is still new, we will explain all of the steps needed to reduce this matrix to its RREF. We first add the first row to the second row to produce a 0 in the $(2, 1)$ position. (Remember that 1 is the additive inverse of 2 in $\mathbb{F}_3$!) This matrix is the result:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 \end{array} \right].$$

We now add twice the first row to the third to produce a 0 in the $(3, 1)$ position. Here is that matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array} \right].$$

We now notice that the second and third rows are the same. This means the third row will end up being a row of zeros, and we can achieve this by adding twice the second row to the third (in $\mathbb{F}_3$, this is the same as subtracting the second row from the third):

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The final step in reducing this matrix is to take care of the entry which is above the pivot in the $(2, 2)$ position. We add the second row to the first, and this is the matrix which results:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Our matrix is now in RREF, and from Theorem 2.2.2, p. 41 we can conclude that this linear system is consistent. Further, from Theorem 2.2.3, p. 42 we can

see that there is not a unique solution to this system. We can write the solutions to this system, however:

$$\begin{cases} x_1 = x_3 \\ x_2 = 2 + 2x_3 \\ x_3 \text{ is free.} \end{cases}$$

Since $\mathbb{F}_3$ has three elements, there are three possible values for x_3 , meaning that there are three solutions to this linear system. $\blacktriangle$

Note 2.2.6 A linear system over a field $\mathbb{F}$ will have either no solutions, a unique solution, or a number of solutions related to the size of the field and the number of free variables. When the field is infinite, we have what is sometimes known as the “trichotomy law”, namely that the number of solutions to a linear system over such a field will either be zero, one, or infinite. This law needs some modification when the field is finite. (See Exercise 1.3.12, p. 24 and compare it to Exercise 2.2.11, p. 46.)

Example 2.2.7 The following is a linear system over $\mathbb{C}$:

$$\begin{aligned} (2 - i)x_1 + (2 + 4i)x_2 + (-7 + 6i)x_3 &= 2 - 6i \\ (1 - 2i)x_1 + (5 + 2i)x_2 + (-2 + 12i)x_3 &= -3 - 8i \\ -3x_1 + (1 - 5i)x_2 + 9x_3 &= -4 + 3i. \end{aligned}$$

This example may seem a bit intimidating at first, especially for readers who have not dealt much with $\mathbb{C}$. But when we follow our step-by-step approach, we should arrive at a solution with minimal problems.

First, we write down the augmented matrix of this system:

$$\left[\begin{array}{ccc|c} 2 - i & 2 + 4i & -7 + 6i & 2 - 6i \\ 1 - 2i & 5 + 2i & -2 + 12i & -3 - 8i \\ -3 & 1 - 5i & 9 & -4 + 3i \end{array} \right].$$

To start row reducing this matrix, we need a 1 in the (1, 1) entry. Instead of following Algorithm 1.3.8, p. 20 rigidly by exchanging rows and then dividing (or multiplying by an inverse), we will skip the first step and handle the rows as they are.

For a nonzero element $a + bi$ of $\mathbb{C}$, the multiplicative inverse is

$$\frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

This means that the inverse of $2 - i$ is $\frac{2}{5} + \frac{1}{5}i$. So, in order to get a 1 in the (1, 1) position, we multiply the first row of the matrix by $\frac{2}{5} + \frac{1}{5}i$. Here is the result:

$$\left[\begin{array}{ccc|c} 1 & 2i & -4 + i & 2 - 2i \\ 1 - 2i & 5 + 2i & -2 + 12i & -3 - 8i \\ -3 & 1 - 5i & 9 & -4 + 3i \end{array} \right].$$

We now work to clear out the other entries in the first column. We add $(-1 + 2i)$ times the first row to the second and we add 3 times the first row to the third. (We are taking care of two steps at once here.) This is the result:

$$\left[\begin{array}{ccc|c} 1 & 2i & -4 + i & 2 - 2i \\ 0 & 1 & 3i & -1 - 2i \\ 0 & 1 + i & -3 + 3i & 2 - 3i \end{array} \right].$$

(For readers who are new to $\mathbb{C}$, verifying these calculations would be an excellent exercise!)

Since we already have a 1 in the $(2, 2)$ entry, we can use that to produce a zero below it in that column. We add $(-1 - i)$ times the second row to the third row, and we get this:

$$\left[\begin{array}{ccc|c} 1 & 2i & -4+i & 2-2i \\ 0 & 1 & 3i & -1-2i \\ 0 & 0 & 0 & 1 \end{array} \right].$$

Even though this matrix is not yet in RREF, we do not need to continue with our algorithm. Theorem 2.2.2, p. 41 tells us that the original system is inconsistent because of the pivot in the final column. Therefore, this system has no solution. $\blacktriangle$

Note 2.2.8 Readers will be able to put a matrix over $\mathbb{R}$ into RREF without difficulty using various computational means—handheld calculators, smartphone applications, and any number of online matrix manipulators. However, working with matrices over fields which are not $\mathbb{R}$ presents some difficulties for most of these applications.

Students can find similar online calculators for matrices over the complex numbers through online searches. Finite fields are a bit trickier. David Augustat has created the website Matrixer (matrixer.davidaugustat.com/) for this purpose, and we encourage readers to take advantage of this resource *after* they have first mastered the mechanics themselves. (That is, tools like this are best used to check work that is done by hand and then, when the by-hand calculations are no longer the main point, to streamline the process.)

Finally, we have written an R package called `matrixmodp`, which can be found on CRAN (cran.r-project.org/web/packages/matrixmodp/index.html). This package can handle matrices over finite, prime fields, but it does not have the capability to handle matrices over more exotic finite fields. (We are aware that R is not the best vehicle for matrix algebra, but this book was largely written when we were also learning R, and this was a good excuse to learn how to author a package in that language.)

Reading Questions

1. Solve the following linear equation over $\mathbb{C}$. List the steps you take in solving the equation in terms of the axioms of a field.

$$(1 - 3i)x + (4 + 2i) = -1 - 3i$$

2. The following is a matrix with entries from $\mathbb{F}_5$. Reduce this matrix to REF. (It is not necessary to reduce the matrix to RREF.) Describe each step you take.

$$\begin{bmatrix} 4 & 2 & 0 & 1 \\ 3 & 1 & 1 & 3 \\ 2 & 0 & 4 & 4 \end{bmatrix}$$

Exercises

1. Solve the following linear system over $\mathbb{F}_2$:

$$\begin{aligned} x + z &= 1 \\ y + z &= 0 \\ x + y + z &= 1. \end{aligned}$$

2. Solve the following linear system over $\mathbb{F}_3$:

$$\begin{aligned} 2x + y + z &= 0 \\ x + 2z &= 2 \\ y + z &= 1 \\ 2x + 2y &= 1. \end{aligned}$$

3. Solve the following linear system over $\mathbb{F}_7$:

$$\begin{aligned} 2x_1 + 3x_2 + x_4 &= 4 \\ 3x_1 + x_3 + 5x_4 &= 6 \\ 4x_1 + x_2 + 2x_3 &= 5. \end{aligned}$$

4. The following matrix A is defined over $\mathbb{C}$:

$$A = \begin{bmatrix} i & 1+i & 3 \\ 1-i & -3+2i & 6+10i \end{bmatrix}.$$

Put this matrix into RREF.

5. Solve the following linear system over $\mathbb{C}$:

$$\begin{aligned} (-i)x + (3-i)z &= -1 \\ (3+2i)x + y + (-1+9i)z &= 6-3i \\ (1+i)x + (2-i)y - (2i)z &= 10-5i. \end{aligned}$$

6. Solve the following linear system over $\mathbb{F}_5$:

$$\begin{aligned} 4x + y &= 2 \\ 3x + 2y + 2z &= 1 \\ x + 3y + z &= 0. \end{aligned}$$

7. The following matrix A is defined over $\mathbb{Q}[\sqrt{2}]$ (see Exercise 2.1.9, p. 40):

$$A = \begin{bmatrix} 1+\sqrt{2} & 2 & 3 & -1+2\sqrt{2} \\ 3\sqrt{2} & 1-\sqrt{2} & 0 & 2+\sqrt{2} \\ 4 & -5+2\sqrt{2} & 3+\sqrt{2} & -1 \end{bmatrix}.$$

Put this matrix into RREF.

8. The following matrix A is defined over $\mathbb{F}_3[\alpha]$ (see Exercise 2.1.7, p. 40):

$$A = \begin{bmatrix} 2+\alpha & 2\alpha & 1 \\ 1+2\alpha & \alpha & 2 \\ 0 & 1+\alpha & 2+2\alpha \end{bmatrix}.$$

Put this matrix into RREF.

Writing Exercises

- 9.

- (a) Suppose that the following is a linear system over $\mathbb{F}_3$:

$$\begin{aligned} ax + by &= e \\ cx + dy &= f. \end{aligned}$$

Show that if $ad + 2bc \neq 0$, then this linear system has a unique solution. (For reference, see Exercise 1.3.11, p. 24.)

- (b) Suppose that the system stated in part (a) of this problem is a system over $\mathbb{F}_p$. What is the correct inequality the coefficients a , b , c , and d must satisfy in order for the system to have a unique solution? State your answer and prove it is correct.

10. Consider a linear system

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

with all coefficients and constants in the integers.

- (a) Show that if the system has a solution $\mathbf{x} \in \mathbb{R}^n$ then it must have a solution $\mathbf{y} \in \mathbb{Q}^n$.
- (b) Show that if the system has a *unique* solution $\mathbf{x} \in \mathbb{Q}^n$ then $\mathbf{x}$ is also the *unique* solution in $\mathbb{R}^n$.
11. Give descriptions of linear systems with each of the following properties, or state why such a system is impossible. Specific numbers and equations are not necessary, but your description should include what field is involved, the number of equations and variables, the number of free variables, etc. Explain your answers.
- (a) A consistent system with exactly 8 solutions
- (b) A consistent system with exactly 10 solutions
- (c) A consistent system with exactly 9 solutions
- (d) A consistent system with exactly 17 solutions

2.3 Vector Spaces

We now move to our next task of abstraction. We have generalized the real numbers and introduced the idea of a field, and we will now generalize the set and structure of the vectors $\mathbb{R}^n$ (see Section 1.4, p. 25) to a vector space.

Definition 2.3.1 A **vector space** over a field $\mathbb{F}$ is a set V on which are defined the operations of *addition* and *scalar multiplication* such that all of the following properties hold.

1. For all $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} \in V$. (The sum of two vectors is a vector.)
2. For all $c \in \mathbb{F}$ and all $\mathbf{v} \in V$, $c\mathbf{v} \in V$. (The scalar multiple of a vector is a vector.)
3. For all $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. (Vector addition is commutative.)
4. For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$. (Vector addition is associative.)
5. There is a vector $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$. (There is a vector which is the identity for vector addition. We call this the *zero vector*.)
6. For each $\mathbf{u} \in V$ there exists a vector $\mathbf{v} \in V$ such that $\mathbf{u} + \mathbf{v} = \mathbf{0}$. (Each vector has an additive inverse.)

7. For each $\mathbf{v} \in V$, $1\mathbf{v} = \mathbf{v}$. (Scalar multiplication by 1 is an identity.)
8. For each $\mathbf{v} \in V$ and all $c, d \in \mathbb{F}$, $c(d\mathbf{v}) = (cd)\mathbf{v}$. (Scalar multiplication of a vector is associative.)
9. For all $\mathbf{u}, \mathbf{v} \in V$ and each $c \in \mathbb{F}$, $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$. (Scalar multiplication distributes over the sum of vectors.)
10. For each $\mathbf{v} \in V$ and all $c, d \in \mathbb{F}$, $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$. (Scalar multiplication distributes over the sum of field elements.)

The elements of V are called *vectors* and the elements of $\mathbb{F}$ are called *scalars*. ◆

Note 2.3.2 The reader will want to note when the multiplication of field elements is in view and when scalar multiplication (of a scalar times a vector) is in view. The context should help, as should our practice of using bold notation for vectors. The product of vectors (in the way that we take the product of field elements) is not a defined construction in a general vector space.

Faced with this rather abstract definition, some examples are in order.

Example 2.3.3 The set $\mathbb{R}^n$, with the operations of scalar multiplication and vector addition defined in Definition 1.4.3, p. 25, is a vector space over $\mathbb{R}$. As with our definition of a field, if $\mathbb{R}^n$ is not a vector space then we have carried out the enterprise of abstraction rather poorly. (See Fact 1.4.10, p. 28 where we stated most of the properties of a vector space for $\mathbb{R}^n$, though we did not use that language.) ▲

Example 2.3.4 Let $\mathbb{F}$ be a field and let $\mathbb{F}^n$ be defined in the following way:

$$\mathbb{F}^n = \left\{ \left[\begin{array}{c} a_1 \\ a_2 \\ \vdots \\ a_n \end{array} \right] \mid a_i \in \mathbb{F} \right\}.$$

The operations of scalar multiplication and vector addition are defined for $\mathbb{F}^n$ over $\mathbb{F}$ in the same way that the operations of scalar multiplication and vector addition are defined for $\mathbb{R}^n$ over $\mathbb{R}$. Then $\mathbb{F}^n$ is a vector space.

This should be relatively easy for the reader to believe, but checking the details could be a helpful exercise in the definitions of both fields and vector spaces. ▲

Example 2.3.5 Let A be a subset of the real numbers. Define the set V as the set of all functions $A \rightarrow \mathbb{R}$. Given appropriate operations, this is a vector space.

If f and g are elements of V , then we define the sum as

$$(f + g)(t) = f(t) + g(t)$$

for all $t \in A$. (The vector $f + g$ must be a function $A \rightarrow \mathbb{R}$, so we here define how to calculate the value of this function for every $t \in A$.) Additionally, if $c \in \mathbb{R}$ and $f \in V$, then we define scalar multiplication as

$$(cf)(t) = cf(t)$$

for all $t \in A$. (Again, we are defining the value of the function cf at each element of A .) We will now argue that V is a vector space.

Looking back at Definition 2.3.1, p. 46, the first two axioms hold by the way we defined the operations. Axioms three and four hold because addition in $\mathbb{R}$

is both commutative and associative. (Since these vectors in V are functions which take values in $\mathbb{R}$, the properties of $\mathbb{R}$ are once again crucial.)

The zero vector in V is the function $f(t) = 0$, the function which is uniformly zero for each value $t \in A$. This function has the properties of the zero vector mandated by axiom five.

The additive inverse of a function $g \in V$ is the function $(-1)g = -g$ since

$$(g + (-g))(t) = g(t) + (-g)(t) = g(t) - g(t) = 0$$

for all $t \in A$. This shows that axiom six holds.

The rest of the axioms hold because of the definitions of vector addition and scalar multiplication and the properties of the real numbers. $\blacktriangle$

Note 2.3.6 The previous example can be difficult to digest, because we are considering functions to be vectors. It may take some adjustment to think of functions—rather than single real numbers (or even ordered lists of real numbers)—as the objects of study.

Example 2.3.7 Let P_n denote the set of all real-valued polynomials of degree less than or equal to n with real coefficients. (Recall that the *degree* of a polynomial is the largest exponent of the variable that has a nonzero coefficient.) This means that the polynomial $p(x) = 7x^{10} - \frac{1}{2}x^6 + 9$ has degree 10 and is an element of P_n for all $n \geq 10$.

We consider vector addition to be the usual algebraic sum of polynomials and scalar multiplication to be the usual product of a polynomial by a constant. Then P_n is a vector space over $\mathbb{R}$. $\blacktriangle$

Example 2.3.8 Consider the set $V = \mathbb{R}^2$ but with non-standard operations. (In this example and the next, we will use horizontal instead of vertical notation for $\mathbb{R}^2$. This is purely for ease of notation in these limited instances.) We define the sum of two vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ to be

$$\mathbf{u} \oplus \mathbf{v} = (u_1 v_1, 1),$$

and we define scalar multiplication of a vector $\mathbf{u} = (u_1, u_2)$ by a real number c to be

$$c \odot \mathbf{u} = (cu_1, 1).$$

It is a good exercise to grapple with the axioms and determine whether or not V with these operations is a vector space.

(We are using the symbols $\oplus$ and $\odot$ because these operations we are defining are not familiar ones, and most readers likely have no previous associations with these symbols. For our purposes, these symbols have no overarching meaning; they will be redefined for the purposes of specific examples and exercises.)

With some work, the reader will find that this is *not* a vector space. All we need to do in order to show this is not a vector space is to find one axiom that does not hold. We will show that the fifth axiom (regarding the zero vector) does not hold.

We must be careful here, because the terminology can become confusing. By “zero vector” we do not always mean in $\mathbb{R}^2$ the ordered pair $(0, 0)$. Depending on the operations that are in view, the zero vector may exist in a form other than $(0, 0)$. To show this axiom does not hold, we must argue that *no* element of $\mathbb{R}^2$ can have the properties of a zero vector given these operations.

We will argue by contradiction. Suppose there is a vector $\mathbf{u} = (u_1, u_2)$ which has the properties of a zero vector. This means that, for any $\mathbf{v} \in V$, we have $\mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u} = \mathbf{v}$. We consider the vector $\mathbf{v} = (2, 2)$. If $\mathbf{u}$ is the zero

vector, then we must have

$$(u_1, u_2) \oplus (2, 2) = (2, 2),$$

but the addition on this set means

$$(u_1, u_2) \oplus (2, 2) = (2u_1, 1).$$

Since it is impossible to have $(2, 2) = (2u_1, 1)$ as the second coordinates are not equal as elements of $\mathbb{R}$, we have a contradiction, and therefore there can be no zero vector in V with these operations. $\blacktriangle$

Example 2.3.9 We again consider the set $V = \mathbb{R}^2$. Vector addition will be the usual sum in $\mathbb{R}^2$, but scalar multiplication will be different. For any real number c and any $\mathbf{u} = (u_1, u_2) \in \mathbb{R}^2$, we define $c \odot \mathbf{u}$ by

$$c \odot \mathbf{u} = (cu_1, \frac{1}{2}cu_2).$$

Once again we pose the question: Is this a vector space?

While it is good practice to check all of the axioms, again we only need to find one axiom which does not hold in order to show this is not a vector space. (If it were a vector space, we would need to prove that all of the axioms hold.) We will show that scalar multiplication by 1 is not an identity (axiom 7).

We will examine the scalar product $1 \odot (1, 1)$:

$$1 \odot (1, 1) = (1, \frac{1}{2}).$$

If axiom 7 were to hold, we would have $1 \odot (1, 1) = (1, 1)$. Since $(1, 1) \neq (1, \frac{1}{2})$, we have shown that axiom 7 does not hold. Therefore, this is not a vector space. $\blacktriangle$

Example 2.3.10 If we fix positive integers m and n and a field $\mathbb{F}$, then $M_{m,n}(\mathbb{F})$ is the set of all $m \times n$ matrices with entries in $\mathbb{F}$. When $m = n$, we use the notation $M_n(\mathbb{F})$.

With the standard addition and scalar multiplication of matrices (see Note 1.4.8, p. 27 for a description), the set $M_{m,n}(\mathbb{F})$ is a vector space over $\mathbb{F}$ for a fixed m and n . The quick explanation here is that because of the way scalar multiplication and addition of matrices are defined, once a reader believes that $\mathbb{F}^n$ is a vector space (see Example 2.3.4, p. 47), believing that $M_{m,n}(\mathbb{F})$ is a vector space just amounts to considering several columns at the same time. But since these columns have no interaction with each other as far as these operations are concerned, none of the vector space axioms will be violated. The zero vector for this vector space is the zero matrix. $\blacktriangle$

As evidence of the fact that $\mathbb{R}^n$ is the model for our general definition of a vector space, we now repeat the definitions of *linear combination* and *span*. These definitions in the general setting will be necessary for the following section. (See the original definitions in Definition 1.4.6, p. 26.)

Definition 2.3.11 Let $\mathbb{F}$ be a field and let $c_1, c_2, \dots, c_m \in \mathbb{F}$. Let V be a vector space over $\mathbb{F}$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in V$. Then the **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ with **weights** $c_1, c_2, \dots, c_m$ is

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_m\mathbf{v}_m.$$

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is the set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ and is written $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$. In other words, the

span is defined to be

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\} = \left\{ \sum_{i=1}^m c_i \mathbf{v}_i \mid c_1, \dots, c_m \in \mathbb{F} \right\}.$$

◆

The final result in this section is similar to Theorem 2.1.9, p.38 in which we summarize some of the “obvious” facts which are true in any vector space. We will prove a few of these facts here and leave some others as exercises.

Theorem 2.3.12 *Let V be a vector space over a field $\mathbb{F}$. Then all of the following hold.*

1. *The zero vector in V is unique.*
2. *Additive inverses of vectors in V are unique.*
3. *For every $\mathbf{v} \in V$, $-(-\mathbf{v}) = \mathbf{v}$.*
4. *For every $\mathbf{v} \in V$, $0\mathbf{v} = \mathbf{0}$.*
5. *For every $c \in \mathbb{F}$, $c\mathbf{0} = \mathbf{0}$.*
6. *For every $\mathbf{v} \in V$, $(-1)\mathbf{v} = -\mathbf{v}$.*
7. *If $c \in \mathbb{F}$, $\mathbf{v} \in V$, and $c\mathbf{v} = \mathbf{0}$, then either $c = 0$ or $\mathbf{v} = \mathbf{0}$.*

Proof. We will first prove property 2. Suppose that $\mathbf{u} \in V$ and that both $\mathbf{v}, \mathbf{w} \in V$ have the properties of being additive inverses of $\mathbf{u}$. We will show that $\mathbf{v} = \mathbf{w}$.

Since $\mathbf{u} + \mathbf{v} = \mathbf{0}$ and $\mathbf{u} + \mathbf{w} = \mathbf{0}$, we have

$$\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}.$$

Adding $\mathbf{v}$ to both sides (and using both the associative law and the properties of the zero vector) we have

$$\begin{aligned} (\mathbf{v} + \mathbf{u}) + \mathbf{v} &= (\mathbf{v} + \mathbf{u}) + \mathbf{w} \\ \mathbf{0} + \mathbf{v} &= \mathbf{0} + \mathbf{w} \\ \mathbf{v} &= \mathbf{w}. \end{aligned}$$

We will now show that property 4 holds. Since $0 = 0 + 0$ in $\mathbb{F}$, we can use the distributive property (axiom 10 in Definition 2.3.1, p.46) to find the following:

$$0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}.$$

We have $0\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$, and subtracting $0\mathbf{v}$ from both sides gives us $\mathbf{0} = 0\mathbf{v}$. (The reader should recall that by “subtracting” we mean adding the additive inverse, which every vector in a vector space possesses.)

To prove property 6, we must show that $(-1)\mathbf{v}$ has the properties of the additive inverse of $\mathbf{v}$. (Then, by property 2, we can conclude what we want.) We will use the distributive law (axiom 10 again) as well as the just-proved property 4:

$$\begin{aligned} \mathbf{v} + (-1)\mathbf{v} &= 1\mathbf{v} + (-1)\mathbf{v} \\ &= (1 - 1)\mathbf{v} \\ &= 0\mathbf{v} = \mathbf{0}. \end{aligned}$$

■

Reading Questions

- In this question we will explore $\mathbb{F}_5^3$. (See Example 2.3.4, p. 47. The vector space $\mathbb{F}_5^3$ should be thought of as being the same as $\mathbb{R}^3$ except that the entries and the operations are coming from $\mathbb{F}_5$.)
 - Write down two nonzero elements of $\mathbb{F}_5^3$ and compute their sum.
 - Write down one nonzero element of $\mathbb{F}_5^3$ and compute the scalar multiple of this vector by the element 3 of $\mathbb{F}_5$.
- Write down one element of P_4 . (See Example 2.3.7, p. 48.)
- Consider the vector $\mathbf{u} = (1, 3)$ in $\mathbb{F}_5^2$. (Again, we are writing the vector horizontally for convenience.) How many elements does $\text{Span}\{\mathbf{u}\}$ contain? Write all of them down.

Exercises

- For each of the following, determine whether the given subset $V \subseteq \mathbb{R}^2$ is a real vector space, using the usual operations of vector addition and scalar multiplication in $\mathbb{R}^2$. (By a “real vector space” we simply mean a vector space over the field $\mathbb{R}$.)
 - $V = \{(x, y) \mid x, y \geq 0\}$
 - $V = \{(x, y) \mid xy \geq 0\}$
 - $V = \{(x, y) \mid y = x\}$
- Consider the set $V = \mathbb{F}_5^2$ with the usual scalar multiplication but with vector addition defined this way:

$$(u_1, v_1) \oplus (u_2, v_2) = (u_1 + u_2, v_1 + v_2 + 1).$$

This is not a vector space over $\mathbb{F}_5$. Determine which of the vector space axioms fail, and give an explanation for each such axiom.

- Consider the set $V = \mathbb{Q}^2$ with the usual vector addition but with scalar multiplication defined this way:

$$c \odot (x, y) = (0, cy).$$

This is not a vector space over $\mathbb{Q}$. Determine which of the vector space axioms fail, and give an explanation for each such axiom.

- Consider the following three elements of P_3 :

$$p_1 = 2t^2 - t + 6$$

$$p_2 = -t^3 - t^2 + 4t$$

$$p_3 = 4t^3 + \frac{1}{2}t^2 + 2t + 1.$$

Calculate the linear combination of p_1 , p_2 , and p_3 with weights 2, -3 , and -2 , respectively.

- Consider the following three elements of $\mathbb{F}_3^2$:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Calculate the linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ with weights 1, 2, and 2,

respectively.

6. Consider the vector space $\mathbb{F}_2^3$.

- (a) How many distinct vectors are in $\mathbb{F}_2^3$?
 (b) If $\mathbf{u}$ and $\mathbf{v}$ are the following vectors,

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

how many distinct vectors are in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$?

7. Consider the vector space V of all functions $\mathbb{R} \rightarrow \mathbb{R}$. (This general setting was introduced in Example 2.3.5, p. 47.) Let f and g be the following vectors in V :

$$f(t) = \sin^2(t), \quad g(t) = \cos^2(t).$$

Is the constant function $h(t) = 4$ an element of $\text{Span}\{f, g\}$? Explain your answer.

8. Consider the following vectors in $\mathbb{F}_5^3$:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Is $\mathbf{w} \in \text{Span}\{\mathbf{u}, \mathbf{v}\}$? Explain your answer.

9. Consider the following vectors in $\mathbb{C}^3$:

$$\mathbf{u} = \begin{bmatrix} 2i \\ 1-i \\ 2-i \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1+2i \\ 0 \\ 4+i \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2+2i \\ 2-i \\ 3i \end{bmatrix}.$$

Is $\mathbf{w} \in \text{Span}\{\mathbf{u}, \mathbf{v}\}$? Explain your answer.

Writing Exercises

10. Consider the following subset of $\mathbb{R}^2$:

$$V = \{(x, y) \mid y = 4x\}.$$

Prove that V , with the usual operations of vector addition and scalar multiplication in $\mathbb{R}^2$, is a vector space over $\mathbb{R}$. (Hint: Since V is a subset of $\mathbb{R}^2$ with the same operations as $\mathbb{R}^2$, some of the axioms may not need to be proved from scratch.)

11. Is $\mathbb{R}$ a vector space over $\mathbb{Q}$? (Addition and scalar multiplication should be understood as the obvious operations in $\mathbb{R}$.) Explain your answer.
 12. Is $\mathbb{Q}$ a vector space over $\mathbb{R}$? (Addition and scalar multiplication should be understood as the obvious operations in $\mathbb{Q}$.) Explain your answer.
 13. Prove that $\mathbb{C}$ is a vector space over $\mathbb{R}$. (Addition and scalar multiplication should be understood as the obvious operations in $\mathbb{C}$.)
 14. Prove that every vector space over $\mathbb{C}$ is a vector space over $\mathbb{R}$.
 15. Prove Item 1, p. 50 in Theorem 2.3.12, p. 50.
 16. Prove Item 5, p. 50 in Theorem 2.3.12, p. 50.

17. Prove Item 7, p. 50 in Theorem 2.3.12, p. 50.

2.4 Subspaces

Whenever we begin to deal with abstraction in mathematics, we also consider the relationship between these abstract concepts. We have already made it clear how the notions of fields and vector spaces are related—a vector space *requires* that we know what a field is! But mathematicians also consider how these abstract objects might fit inside or contain each other.

The most basic example of this is something with which all readers are likely familiar. When we learn about *sets*, we soon also learn about *subsets*. For two sets A and B , we say that A is a subset of B when all of the elements of A are contained in B . We denote this by $A \subseteq B$.

A vector space is a set with a lot more structure. So a “sub-vector space” must be a subset with the properties of a vector space. Here is the formal definition.

Definition 2.4.1 Suppose that V is a vector space over the field $\mathbb{F}$. Then a subset $W \subseteq V$ is a **subspace** of V if W is also a vector space over $\mathbb{F}$ with the same operations of vector addition and scalar multiplication that are used for V . ♦

To put this less formally: A subspace is a subset which is a vector space over the same field with the operations *inherited* from the larger space.

Since the operations of a subspace are inherited, we need not check the full list of properties (from Definition 2.3.1, p. 46) to prove that a subset is a subspace. In fact, when we consult that definition, we see that axioms 3, 4, and 7–10 will automatically be satisfied—these are properties of the operations, not of the set on which the operation is taking place. This leads to the following fact.

Fact 2.4.2 Let V be a vector space over the field $\mathbb{F}$ and let W be a subset of V . Then W is a subspace of V if the following properties hold.

- For all $\mathbf{u}, \mathbf{v} \in W$, $\mathbf{u} + \mathbf{v} \in W$. (The subset W is closed under addition.)
- For all $c \in \mathbb{F}$ and all $\mathbf{v} \in W$, $c\mathbf{v} \in W$. (The subset W is closed under scalar multiplication.)
- The zero vector $\mathbf{0}$ of V belongs to W .
- For each $\mathbf{u} \in W$ there exists a vector $\mathbf{v} \in W$ such that $\mathbf{u} + \mathbf{v} = \mathbf{0}$. (The subset W contains all additive inverses.)

From this fact, there appear to be four conditions to check in order to prove that a subset is a subspace. However, we will refer the reader back to Theorem 2.3.12, p. 50. By part 6 of that theorem, we know that if a subset of a vector space is closed under scalar multiplication it must also contain all additive inverses. This means that there are only three conditions to check to prove that a subset is a subspace. We summarize this as a theorem.

Theorem 2.4.3 Let V be a vector space over the field $\mathbb{F}$ and let W be a subset of V . Then W is a subspace of V if the following properties hold.

1. For all $\mathbf{u}, \mathbf{v} \in W$, $\mathbf{u} + \mathbf{v} \in W$.
2. For all $c \in \mathbb{F}$ and all $\mathbf{v} \in W$, $c\mathbf{v} \in W$.
3. The zero vector $\mathbf{0}$ of V belongs to W .

As always, some examples are in order!

Example 2.4.4 Every vector space V has a subspace consisting of a single vector—the **zero subspace**: $\{0\}$. In a trivial way, this set meets all of the conditions listed in Theorem 2.4.3, p. 53. ▲

Example 2.4.5 Let V be the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$ as defined in Example 2.3.5, p. 47. Let W be the subset of all polynomials. (This means we include polynomials of any degree.) Then W is a subspace of V .

Since the sum of two polynomials is another polynomial and the scalar multiple of a polynomial is a polynomial, the first two conditions in Theorem 2.4.3, p. 53 hold. Additionally, the zero function in V is the same as the zero polynomial in W . This proves that W is a subspace of V . ▲

Example 2.4.6 The vector space $\mathbb{R}^2$ is *not* a subspace of the vector space $\mathbb{R}^3$. This may seem surprising, as the operations for these spaces are clearly compatible, and we often think of $\mathbb{R}^2$ as “living inside” of $\mathbb{R}^3$.

However, this common way of thinking is wrong because the space $\mathbb{R}^2$ is not even a subset of $\mathbb{R}^3$; thus, it is impossible for $\mathbb{R}^2$ to be a subspace of $\mathbb{R}^3$. (The set $\mathbb{R}^2$ consists of ordered pairs of real numbers while $\mathbb{R}^3$ consists of ordered triples of real numbers.)

We can make a slight adjustment to match the way that many people think of $\mathbb{R}^2$ as “living inside” of $\mathbb{R}^3$. We define the set A in the following way:

$$A = \{(x, y, 0) \mid x, y \in \mathbb{R}\}.$$

Then A is what we usually call the “ xy -plane” in $\mathbb{R}^3$. This subset A is a subspace of $\mathbb{R}^3$. ▲

Example 2.4.7 Here is a collection of examples that generalize Example 2.4.6, p. 54. Every line through the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$, and every line or plane through the origin in $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$. On the other hand, lines in $\mathbb{R}^2$ which do *not* pass through the origin are *not* subspaces of $\mathbb{R}^2$, and lines and planes which do *not* pass through the origin in $\mathbb{R}^3$ are *not* subspaces of $\mathbb{R}^3$. The details are left for the exercises. ▲

The notion of the span of a set of vectors gives us another angle through which we can identify subspaces. We begin with a result.

Theorem 2.4.8 Let V be a vector space over a field $\mathbb{F}$. If $\mathbf{v}_1, \dots, \mathbf{v}_n$ are vectors in V , then $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a subspace of V .

Proof. We let $W = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. By the definition of the span of a set of vectors, every element $\mathbf{w}$ of W can be written in the following form:

$$\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n,$$

where $c_1, \dots, c_n \in \mathbb{F}$. We first observe that $\mathbf{0} \in W$ by taking $c_1 = \cdots = c_n = 0$. Next, if $\mathbf{u}$ and $\mathbf{w}$ are elements of W , we can write these vectors as

$$\mathbf{u} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$$

$$\mathbf{w} = d_1\mathbf{v}_1 + \cdots + d_n\mathbf{v}_n,$$

for some scalars c_i and d_i . The sum of these elements is

$$\begin{aligned} \mathbf{u} + \mathbf{w} &= (c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n) + (d_1\mathbf{v}_1 + \cdots + d_n\mathbf{v}_n) \\ &= (c_1 + d_1)\mathbf{v}_1 + \cdots + (c_n + d_n)\mathbf{v}_n. \end{aligned}$$

(We are using some of the properties of a vector space from Definition 2.3.1, p. 46 in order to carry out this algebraic manipulation.) Since $c_i + d_i \in \mathbb{F}$ for each i , this proves that $\mathbf{u} + \mathbf{w} \in W$.

Finally, we let $c \in \mathbb{F}$ and $\mathbf{w} \in W$. We want to show that $c\mathbf{w} \in W$. We can assume that $\mathbf{w}$ has the form

$$\mathbf{w} = d_1\mathbf{v}_1 + \cdots + d_n\mathbf{v}_n,$$

where $d_1, \dots, d_n \in \mathbb{F}$. Then we have

$$\begin{aligned} c\mathbf{w} &= c(d_1\mathbf{v}_1 + \cdots + d_n\mathbf{v}_n) \\ &= c(d_1\mathbf{v}_1) + \cdots + (cd_n\mathbf{v}_n) \\ &= (cd_1)\mathbf{v}_1 + \cdots + (cd_n)\mathbf{v}_n. \end{aligned}$$

(Again, we are using properties of the vector space V here.) Since $cd_i \in \mathbb{F}$ for each i , this proves that W is closed under scalar multiplication.

Since W has all of the properties from Theorem 2.4.3, p. 53, we have shown that W is a subspace of V . ■

In the final example of this section, we will use Theorem 2.4.8, p. 54 to prove that a set is a subspace by realizing it as the span of a set of vectors.

Example 2.4.9 Consider the following subset of $\mathbb{R}^3$:

$$A = \left\{ \begin{bmatrix} 2a+b \\ 5b-\frac{1}{2}a \\ 6a \end{bmatrix} \mid a, b \in \mathbb{R} \right\}.$$

This notation means that every given pair of real numbers a and b specifies an element of A . For example, when $a = 1$ and $b = -1$, we have $(1, -\frac{11}{2}, 6) \in A$.

We will use Theorem 2.4.8, p. 54 to prove that A is a subspace of $\mathbb{R}^3$. Any element $\mathbf{v} \in A$ can be written in the following way:

$$\mathbf{v} = \begin{bmatrix} 2a+b \\ 5b-\frac{1}{2}a \\ 6a \end{bmatrix} = a \begin{bmatrix} 2 \\ -\frac{1}{2} \\ 6 \end{bmatrix} + b \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}.$$

This proves that if

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ -\frac{1}{2} \\ 6 \end{bmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix},$$

then $A = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. We conclude that A is a subspace of $\mathbb{R}^3$ by Theorem 2.4.8, p. 54. ▲

Reading Questions

1. Let V be the vector space $\mathbb{R}^2$ with the usual addition and scalar multiplication. Let A be the following set:

$$A = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\}.$$

(We can identify A with the first quadrant in $\mathbb{R}^2$.) Determine whether or not A is a subspace of V . Explain your answer.

2. Consider the vector space P_4 and let A be the following set:

$$A = \{\text{polynomials in } P_4 \text{ with even degree}\}.$$

Determine whether or not A is a subspace of P_4 . Explain your answer.

3. Let V be the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$. Let f and g be the following functions:

$$f(t) = 2t^2 \qquad g(t) = \cos(3t).$$

Let $A = \text{Span}\{f, g\}$. We know that A is a subspace of V by Theorem 2.4.8, p. 54. Write down four distinct vectors in A .

Exercises

1. For each of the following, determine whether or not the subset W of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$. Explain your answers. (We are writing vectors horizontally for convenience.)

(a) $W = \{(a, 0, 0) \mid a \in \mathbb{R}\}$

(b) $W = \{(a, b, c) \mid b = a + c\}$

(c) $W = \{(a, 1, c) \mid a, c \in \mathbb{R}\}$

(d) $W = \{(a, 0, c) \mid a, c \in \mathbb{R}\}$

2. For each of the following, determine whether or not the subset W of P_3 is a subspace of P_3 . Explain your answers.

(a) $W = \{at^3 + at \mid a \in \mathbb{R}\}$

(b) $W = \{at^2 + bt + c \mid a, b, c \in \mathbb{R}\}$

(c) $W = \{p \in P_3 \mid p \text{ has rational coefficients}\}$

(d) $W = \{p \in P_3 \mid p(0) = 0\}$

(e) $W = \{a_3t^3 + a_2t^2 + a_1t + a_0 \mid a_3 + a_2 + a_1 + a_0 = 0\}$

3. Let $C[0, 10]$ be the vector space of continuous functions defined $[0, 10] \rightarrow \mathbb{R}$, and let $D[0, 10]$ be the set of differentiable functions defined $[0, 10] \rightarrow \mathbb{R}$. (The fact that $C[0, 10]$ is a vector space is established in Exercise 2.4.5, p. 57.)

For each of the following, determine whether or not the subset W of $C[0, 10]$ is a subspace of $C[0, 10]$. Explain your answers.

(a) $W = \{f \in C[0, 10] \mid f(1) = 0\}$

(b) $W = \{f \in C[0, 10] \mid f(1) = 1\}$

(c) $W = \{f \in D[0, 10] \mid f'(1) = 0\}$

(d) $W = \{f \in D[0, 10] \mid f'(1) = 1\}$

(e) $W = \{f \in D[0, 10] \mid f'(t) \text{ is constant}\}$

4. For each of the following, determine whether or not the subset W of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$. Explain your answers. (We are writing vectors horizontally for convenience.)

(a) $W = \{(2a - b, a + 5b, -a) \mid a, b \in \mathbb{R}\}$

(b) $W = \{(a + b, 2b, ab) \mid a, b \in \mathbb{R}\}$

(c) $W = \{(a + b, b - c, a + 2d) \mid a, b, c, d \in \mathbb{R}\}$

(d) $W = \{(2a - 4b, 5, 3c + b) \mid a, b, c \in \mathbb{R}\}$

Writing Exercises

5. Let V denote the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$. (See Example 2.3.5, p. 47 for the relevant definitions and operations.) Let a and b be real numbers with $a < b$, and let $C[a, b]$ denote the set of all continuous functions $[a, b] \rightarrow \mathbb{R}$. Prove that $C[a, b]$ is a subspace of V .
6. Let a and b be real numbers with $a < b$. Show that the set of all functions $f \in C[a, b]$ such that

$$\int_a^b f(x) \, dx = 0$$

is a subspace of $C[a, b]$. (See Exercise 2.4.5, p. 57 for the definition of $C[a, b]$.)

7. Prove that a line in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$ if and only if the line passes through the origin. (Note that all lines in $\mathbb{R}^2$ can be written in the form $ax + by + c = 0$.)
8. Prove that a plane in $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$ if and only if the plane passes through the origin. (Note that all planes in $\mathbb{R}^3$ can be written in the form $ax + by + cz + d = 0$.)
9. Let V be a vector space and let W_1 and W_2 be subspaces of V . Must $W_1 \cup W_2$ be a subspace of V ? Justify your answer.
10. Let V be a vector space and let W_1 and W_2 be subspaces of V . Must $W_1 \cap W_2$ be a subspace of V ? Justify your answer.
11. Let V be a vector space and let W_1 and W_2 be subspaces of V . Define the *sum* of W_1 and W_2 like this:

$$W_1 + W_2 = \{\mathbf{v} \mid \mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2 \text{ for some } \mathbf{w}_1 \in W_1 \text{ and some } \mathbf{w}_2 \in W_2\}.$$

Must $W_1 + W_2$ be a subspace of V ? Justify your answer.

Chapter 3

Linear Transformations

3.1 Linear Transformations

Speaking broadly, mathematicians are often concerned about (mathematical) objects and the right sort of functions between those objects. The structure of specific objects can be illuminated by a look at the functions to and from those objects. In linear algebra, the objects in view are vector spaces (see Definition 2.3.1, p.46), and the functions between these objects are called *linear transformations*.

3.1.1 Introduction to Linear Transformations

Definition 3.1.1 If V and W are vector spaces over a field $\mathbb{F}$, then a function $T : V \rightarrow W$ is called a **linear transformation** if both of the following properties hold.

- For all $\mathbf{u}, \mathbf{v} \in V$, we have $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
- For all $\mathbf{v} \in V$ and all $c \in \mathbb{F}$, we have $T(c\mathbf{v}) = cT(\mathbf{v})$.

These functions are sometimes referred to as *linear maps* or *linear operators*.

If $T : V \rightarrow W$ is a linear transformation, then V is the **domain** of T and W is the **codomain** of T . ♦

Note 3.1.2 Many readers will be more familiar with the idea of the *range* of a function than the *codomain* of a function. The **range** of a linear transformation $T : V \rightarrow W$ is the set $\{T(\mathbf{v}) \in W \mid \mathbf{v} \in V\}$. In words, the range is the subset of the codomain consisting of the “outputs” of the function for all elements of the domain. We will often use the term **image** when discussing the range of a linear transformation.

Linear transformations are the “right” types of functions to study between vector spaces because they preserve the primary vector space operations. The first property of linear transformations means that such a function respects vector addition, and the second property means that such a function respects scalar multiplication.

Example 3.1.3 We consider the real vector spaces P_5 and P_4 , along with the function $D : P_5 \rightarrow P_4$ which takes the derivative. That is, $D(p) = p'$ for all $p \in P_5$. So if $p = 3t^5 - 2t^3 + 10t$, then $D(p) = 15t^4 - 6t^2 + 10$. We note that $p \in P_5$ and $D(p) \in P_4$.

The fact that our function D is a linear transformation between these vector

spaces is a consequence of calculus. For all differentiable functions f and g , and all real numbers c , it is true that

$$\begin{aligned}[f + g]' &= f' + g' \\ [cf]' &= cf'.\end{aligned}$$

(If the reader doubts or has forgotten these facts, the closest textbook on single-variable calculus should be consulted *posthaste*.)

These calculus facts confirm that $D(p+q) = D(p)+D(q)$ and $D(cp) = cD(p)$ for all $p, q \in P_5$ and all $c \in \mathbb{R}$. This proves that $D : P_5 \rightarrow P_4$ is a linear transformation. $\blacktriangle$

Example 3.1.4 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the function which reflects a vector in the Cartesian plane across the x -axis. So $T(x, y) = (x, -y)$. Additionally, let $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the function which rotates a vector counter-clockwise around the origin by $\frac{\pi}{2}$ radians. So $S(x, y) = (-y, x)$. Then both T and S are linear transformations.

We will supply two calculations here to give the sense of these functions. The reader should note that T takes the vector $(-3, 2)$ in the second quadrant and reflects it across the x -axis to the vector $(-3, -2)$ in the third quadrant. Also, S rotates the vector $(-3, 2)$ counter-clockwise around the origin by $\frac{\pi}{2}$ radians to the vector $(-2, -3)$. (It is fairly obvious that the length of the vectors $(-3, 2)$ and $(-2, -3)$ are the same. To check the claim about the angles, one would calculate the angles between the positive x -axis and both the vectors $(-3, 2)$ and $(-2, -3)$. The first angle is roughly 2.55 radians and the second is 4.12, giving a difference of 1.57 radians, or roughly $\frac{\pi}{2}$.)

We first check the additivity condition. Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$. Then we have

$$\begin{aligned}T((x_1, y_1) + (x_2, y_2)) &= T(x_1 + x_2, y_1 + y_2) = (x_1 + x_2, -(y_1 + y_2)) \\ T(x_1, y_1) + T(x_2, y_2) &= (x_1, -y_1) + (x_2, -y_2) = (x_1 + x_2, -y_1 - y_2).\end{aligned}$$

From the distributive property of the real numbers (in the second coordinate of these calculations), we can see that the additive property holds for T . (The calculation for S is similar.)

We now check the scalar multiplication property. (Again, the calculations for T and S are similar, so we will only show one of them.) Let $c \in \mathbb{R}$ and let $(x, y) \in \mathbb{R}^2$. Then we have

$$\begin{aligned}S(c(x, y)) &= S(cx, cy) = (-cy, cx) \\ cS(x, y) &= c(-y, x) = (-cy, cx).\end{aligned}$$

Note that we used the commutativity of multiplication in $\mathbb{R}$ in this calculation.

These brief calculations show that both T and S are linear transformations. $\blacktriangle$

3.1.2 Linear Transformations and Matrices

While linear algebra is not *only* about matrices, matrices are valuable tools and provide a rich source of examples in this subject. In fact, matrices are so central to the notion of linear transformations that we will devote this subsection to their discussion.

Example 3.1.5 Let $\mathbb{F}$ be a field and let A be an $m \times n$ matrix with entries from $\mathbb{F}$. (We will refer to this in what follows as “a matrix over $\mathbb{F}$.”) Then multiplication by A is a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$. (We will denote

the function which is multiplication by A by $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$.)

To justify this claim we must first explain what we mean by “multiplication by A .” We will let $\mathbf{v} \in \mathbb{F}^n$ and denote entry (i, j) in A by a_{ij} . We will further denote the entries of $\mathbf{v}$ by

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

Then the **matrix-vector product** $A\mathbf{v}$ is defined to be the following vector in $\mathbb{F}^m$:

$$A\mathbf{v} = \begin{bmatrix} a_{11}v_1 + \cdots + a_{1n}v_n \\ \vdots \\ a_{m1}v_1 + \cdots + a_{mn}v_n \end{bmatrix}. \quad (3.1)$$

One way to state this is that entry j in $A\mathbf{v}$ is the sum of the entry-wise product of row j in A with $\mathbf{v}$. Since $A\mathbf{v}$ is an element of $\mathbb{F}^m$, the domain and codomain of T_A are correct.

What we have defined is the product of a matrix and a vector. However, an alternate description of this product will be more useful in proving that T_A is a linear transformation.

If the columns of A are thought of as vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$, then the product $A\mathbf{v}$ is also

$$A\mathbf{v} = v_1\mathbf{a}_1 + \cdots + v_n\mathbf{a}_n = \sum_{i=1}^n v_i\mathbf{a}_i. \quad (3.2)$$

In words, $A\mathbf{v}$ is a linear combination of the columns of A with weights coming from the entries of $\mathbf{v}$. (We have reserved proving the equivalence of these two formulations to Exercise 3.1.7.16, p. 68.)

With this equivalent definition, proving that T_A is a linear transformation is a snap. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{F}^n$ and let $c \in \mathbb{F}$. We will further denote the entries of $\mathbf{u}$ and $\mathbf{v}$ by

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

Then we have the following:

$$\begin{aligned} T_A(\mathbf{u} + \mathbf{v}) &= A(\mathbf{u} + \mathbf{v}) = \sum_{i=1}^n (u_i + v_i)\mathbf{a}_i \\ T_A(\mathbf{u}) + T_A(\mathbf{v}) &= A\mathbf{u} + A\mathbf{v} = \sum_{i=1}^n u_i\mathbf{a}_i + \sum_{i=1}^n v_i\mathbf{a}_i. \end{aligned}$$

These two expressions are equal due to the fact that $\mathbb{F}^m$ is a vector space.

We have one final calculation to prove that T_A is a linear transformation. Let $\mathbf{v}$ be a vector in $\mathbb{F}^n$ and let c be in $\mathbb{F}$. Then we have

$$\begin{aligned} T_A(c\mathbf{v}) &= A(c\mathbf{v}) = \sum_{i=1}^n (cv_i)\mathbf{a}_i \\ cT_A(\mathbf{v}) &= cA(\mathbf{v}) = c \sum_{i=1}^n v_i\mathbf{a}_i. \end{aligned}$$

Once again, these expressions are equal because $\mathbb{F}^m$ is a vector space.

These calculations prove that T_A is a linear transformation. ▲

Note 3.1.6 To summarize, when $\mathbb{F}$ is a field, multiplication by an $m \times n$ matrix A is a linear transformation $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$.

General matrices are rectangular, not necessarily square. When a matrix is square, however, we have additional properties to discuss.

Definition 3.1.7 Let A be an $n \times n$ matrix. (So A is *square*.) We say that A is a **diagonal matrix** if $a_{ij} = 0$ for all (i, j) such that $i \neq j$. If A is diagonal and $a_{ii} = 1$ for all $i = 1, \dots, n$, then A is called an **identity matrix**. ♦

The next example shows what the linear transformation T_A is like when A is a diagonal or identity matrix.

Example 3.1.8 Let A and B be the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We note that both A and B are diagonal matrices and that B is an identity matrix.

Building on Example 3.1.5, p. 59, both T_A and T_B are linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$. Let $\mathbf{x}$ be an element of $\mathbb{R}^3$ where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Then we have

$$T_A(\mathbf{x}) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ -x_2 \\ -3x_3 \end{bmatrix}.$$

We also have

$$T_B(\mathbf{x}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{x}.$$

Since $T_B(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$, the linear transformation T_B acts as the identity function on $\mathbb{R}^3$. The action of T_A on $\mathbb{R}^3$ is slightly more complex, but only by a bit. Because of all the zeros in A , the effect of the transformation T_A is just multiplication of each coordinate of a vector in $\mathbb{R}^3$ by the corresponding diagonal entry in A . What we observe for multiplication by a diagonal or identity matrix in $M_3(\mathbb{R})$ can easily be extended to $M_n(\mathbb{F})$ for any $n \in \mathbb{N}$ and any field $\mathbb{F}$. ▲

Note 3.1.9 We often use the notation $\mathbf{e}_1, \dots, \mathbf{e}_n$ to refer to the columns of the $n \times n$ identity matrix. In other words, $\mathbf{e}_j$ is the vector with a 1 in entry j and zeros elsewhere.

3.1.3 Properties of Linear Transformations

Recall that while linear transformations must have special properties, they are first of all *functions*. And, as functions, we can discuss the composition of linear transformations as well as properties like injectivity and surjectivity.

Definition 3.1.10 If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear transformations between vector spaces, then the function $S \circ T : U \rightarrow W$ defined by $(S \circ T)(\mathbf{u}) = S(T(\mathbf{u}))$ for each $\mathbf{u} \in U$ is the **composition** of the transformations T and S . ♦

Note 3.1.11 We note that Definition 3.1.10, p. 61 is only possible when the *codomain* of T is the same as the *domain* of S . It is not difficult to show that $S \circ T$ is a linear transformation. (See Exercise 3.1.7.12, p. 68.)

Definition 3.1.12 Let $T : V \rightarrow W$ be a linear transformation between vector spaces. We say that T is **injective** if $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ implies $\mathbf{v}_1 = \mathbf{v}_2$ for all $\mathbf{v}_1, \mathbf{v}_2 \in V$. Injective linear transformations are also referred to as *one-to-one* since no two distinct elements of the domain may correspond to the same element of the image.¹

A linear transformation T is called **surjective** if for every $\mathbf{w} \in W$ there exists a vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. For surjective functions, the image is the same as the codomain. (The image is a subset of the codomain for every function, but these sets are equal if and only if the function is surjective.) Sometimes surjective functions are referred to as *onto* functions.

If a linear transformation is both injective and surjective, we say that it is **bijective**. ♦

Example 3.1.13 Let's reconsider the linear transformation $D : P_5 \rightarrow P_4$ which appeared in Example 3.1.3, p. 58. We observe that D is surjective but not injective.

The transformation is surjective because we know about the antiderivative. Let $q \in P_4$ have the form

$$q(t) = a_4t^4 + a_3t^3 + a_2t^2 + a_1t + a_0.$$

This is a generic element of P_4 , so we only need to supply an element $p \in P_5$ such that $D(p) = q$, and this will prove that D is surjective. Consider the element p defined as

$$p(t) = \frac{1}{5}a_4t^5 + \frac{1}{4}a_3t^4 + \frac{1}{3}a_2t^3 + \frac{1}{2}a_1t^2 + a_0t.$$

It is but the work of a Calculus I student to verify that $D(p) = q$, thus showing that D is surjective. (We note that we could have chosen p to have any constant term at all; we used the constant term of 0.)

Finally, we will show that D is not injective by looking at an example of two elements of P_5 which have the same image under D in P_4 . Let $p_1 = t^2 + 10$ and $p_2 = t^2 + 20$. Then we see that even though $p_1 \neq p_2$, we have $D(p_1) = D(p_2) = 2t$, and this proves that D is not injective. ▲

We will define one more property of linear transformations here that will resurface in Section 3.2, p. 68.

Definition 3.1.14 Let $T : V \rightarrow W$ be a linear transformation between vector spaces. The **identity transformation** on V is the linear transformation $I_V : V \rightarrow V$ which is $I_V(\mathbf{v}) = \mathbf{v}$ for each $\mathbf{v} \in V$. (If the vector space we have in mind is clear, we will drop the subscript and use the notation I .)

We say that the linear transformation T is **invertible** if there exists a linear transformation $S : W \rightarrow V$ such that $S \circ T = I_V$. ♦

Note 3.1.15 In Definition 3.1.14, p. 62, we usually use the notation T^{-1} in place of S . Considered only as a function, S is unique and therefore using T^{-1} is not ambiguous.

¹Recall that by “image” we mean “range”.

3.1.4 Isomorphisms

Bijjective functions are important in almost all settings, and the linear algebra setting is no exception. We have a specific name for bijective linear transformations.

Definition 3.1.16 A bijective linear transformation T between vector spaces V and W is called an **isomorphism**. If there exists an isomorphism between vector spaces V and W , then these spaces are said to be **isomorphic**. ♦

The reader should think of isomorphic vector spaces as *essentially the same*. Such spaces will not be exactly the same, of course, in the same way that two finite sets of the same size are not necessarily identical. But the presence of an isomorphism means that the vector space operations are compatible in such a way that such spaces share many of the same properties.

Example 3.1.17 The vector spaces $\mathbb{R}^2$ and P_1 are isomorphic. To prove this, we consider the function $T : \mathbb{R}^2 \rightarrow P_1$ defined by

$$T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = a + bt.$$

To justify the claim that $\mathbb{R}^2$ and P_1 are isomorphic, we must prove that T is a bijective linear transformation. We will prove that T is a linear transformation and ask the reader to prove bijectivity in the exercises.

We let $\mathbf{x}_1$ and $\mathbf{x}_2$ be elements of $\mathbb{R}^2$ where

$$\mathbf{x}_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}.$$

Then we have

$$T(\mathbf{x}_1 + \mathbf{x}_2) = T\left(\begin{bmatrix} a_1 + a_2 \\ b_1 + b_2 \end{bmatrix}\right) = (a_1 + a_2) + (b_1 + b_2)t,$$

and

$$T(\mathbf{x}_1) + T(\mathbf{x}_2) = (a_1 + b_1t) + (a_2 + b_2t).$$

The properties P_1 possesses as a vector space show that

$$T(\mathbf{x}_1 + \mathbf{x}_2) = T(\mathbf{x}_1) + T(\mathbf{x}_2).$$

We now suppose that $\mathbf{x} \in \mathbb{R}^2$ and $c \in \mathbb{R}$, where

$$\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Then

$$\begin{aligned} T(c\mathbf{x}) &= T\left(\begin{bmatrix} ca \\ cb \end{bmatrix}\right) = ca + cbt \\ cT(\mathbf{x}) &= c(a + bt) = ca + cbt. \end{aligned}$$

Again, we see that $T(c\mathbf{x}) = cT(\mathbf{x})$ using the properties of P_1 as a vector space.

We have shown that T is a linear transformation. It is also bijective (see Exercise 3.1.7.18, p. 68), so it is an isomorphism, meaning that $\mathbb{R}^2$ and P_1 are isomorphic. ▲

Note 3.1.18 If V and W are vector spaces, then the set of all linear transformations from $V \rightarrow W$ is denoted $L(V, W)$. When $W = V$, we will write $L(V)$ instead of $L(V, V)$.

We can now prove that two concepts we have defined in this section are one and the same for linear transformations.

Proposition 3.1.19 *Let $T \in L(V, W)$. If T is bijective, meaning $T^{-1} : W \rightarrow V$ exists, then T^{-1} is also a linear transformation.*

Proof. We will check the two properties of a linear transformation. (See Definition 3.1.1, p. 58.) Suppose that $\mathbf{w}_1, \mathbf{w}_2 \in W$. Since $T \circ T^{-1} = I_W$, we have

$$\mathbf{w}_1 + \mathbf{w}_2 = T(T^{-1}(\mathbf{w}_1)) + T(T^{-1}(\mathbf{w}_2)) = T(T^{-1}(\mathbf{w}_1) + T^{-1}(\mathbf{w}_2)).$$

When we apply T^{-1} to the beginning and end of this equality, using $T^{-1} \circ T = I_V$, we get

$$T^{-1}(\mathbf{w}_1 + \mathbf{w}_2) = T^{-1}(\mathbf{w}_1) + T^{-1}(\mathbf{w}_2).$$

We will now check the scalar multiple property in a similar fashion. Let $\mathbf{w} \in W$ and let $c \in \mathbb{F}$. Then we have

$$c\mathbf{w} = cT(T^{-1}(\mathbf{w})) = T(cT^{-1}(\mathbf{w})).$$

Applying T^{-1} to both sides again we get

$$T^{-1}(c\mathbf{w}) = cT^{-1}(\mathbf{w}).$$

This proves that $T^{-1} \in L(W, V)$. ■

Proposition 3.1.20 *Let V and W be vector spaces over $\mathbb{F}$, and let $T \in L(V, W)$. Then T is an isomorphism if and only if T is invertible.*

Proof. Most of this result is true without worrying about any of the linear transformation properties. (A function is bijective if and only if it has an inverse.)

Here, then, is the brief argument in full. If T is an isomorphism, then T^{-1} exists because T is bijective. Then Proposition 3.1.19, p. 64 tells us that T^{-1} is a linear transformation. This shows that T is invertible (see Definition 3.1.14, p. 62). On the other hand, if T is invertible, this implies both the existence of T^{-1} and the fact that T^{-1} is a linear transformation. This proves that T is an isomorphism. ■

Before we leave this subsection, it is worth pointing out that when V and W are vector spaces, the set $L(V, W)$ itself has some important structure.

When V and W are vector spaces over $\mathbb{F}$, we can define the sum and scalar multiple of linear transformations since both of these operations happen on the level of elements. If $S, T \in L(V, W)$ and $c \in \mathbb{F}$, then we define $S + T$ and cT in the following way. For all $\mathbf{v} \in V$,

- $(S + T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$, and
- $(cT)(\mathbf{v}) = cT(\mathbf{v})$.

These operations as defined make $L(V, W)$ into its own vector space. We will leave the proof of this theorem for the exercises.

Theorem 3.1.21 *Let V and W be vector spaces over a field $\mathbb{F}$, and let $S, T \in L(V, W)$ and $c \in \mathbb{F}$. Then $S + T$ and cT are both linear transformations*

from V to W , and $L(V, W)$ is a vector space with these operations.

3.1.5 The Matrix-Vector Form of a Linear System

Having defined the product of a matrix and a vector in Example 3.1.5, p. 59, we can reformulate one of the foundational (and introductory) matters of this book. We will now put the notion of a linear system—in particular, the solutions to linear systems—in a different context.

Let's consider the following system of linear equations over a field $\mathbb{F}$, as we saw in Section 2.2, p. 41:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

If we let A be the matrix $A = [a_{ij}]$, $\mathbf{x}$ be the vector of variables $\mathbf{x} = [x_j]$, and $\mathbf{b}$ be the vector of constants $\mathbf{b} = [b_j]$, then this linear system can be written efficiently as $A\mathbf{x} = \mathbf{b}$.

With this reformulation, the questions of the existence and uniqueness of solutions to a system of equations (see the end of Section 1.1, p. 1) can now be stated in the language of the injectivity and surjectivity of linear transformations.

Example 3.1.22 Consider the linear transformation $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which is multiplication by this matrix:

$$A = \begin{bmatrix} 2 & 3 & 2 \\ 1 & -2 & 8 \\ -1 & 4 & -12 \end{bmatrix}.$$

We will show that T_A is neither injective nor surjective.

Let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors in $\mathbb{R}^3$:

$$\mathbf{u} = \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 7 \\ 0 \\ 2 \end{bmatrix}.$$

By forming and row-reducing the augmented matrices $[A \mid \mathbf{u}]$ and $[A \mid \mathbf{v}]$, we can determine how many solutions there are to the equations $T_A(\mathbf{x}) = \mathbf{u}$ and $T_A(\mathbf{x}) = \mathbf{v}$, respectively. Here are the calculations:

$$[A \mid \mathbf{u}] \sim \left[\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right], \quad (3.3)$$

$$[A \mid \mathbf{v}] \sim \left[\begin{array}{ccc|c} 1 & 0 & 4 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]. \quad (3.4)$$

From (3.3), since there is a pivot in the final column of the RREF of $[A \mid \mathbf{u}]$, we see that $\mathbf{u}$ is not in the image of T_A . This means that the matrix equation $A\mathbf{x} = \mathbf{u}$ has no solution, so T_A is not surjective; equivalently, the linear system which corresponds to the augmented matrix $[A \mid \mathbf{u}]$ is inconsistent.

From (3.4), we see that $\mathbf{v}$ is in the image of T_A . Since there is no pivot in the final column of the RREF of $[A \mid \mathbf{v}]$, and since there is a free variable in

that same RREF, this means that the matrix equation $A\mathbf{x} = \mathbf{v}$ has multiple solutions, so T_A is not injective. Specifically, if

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{x}_2 = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix},$$

then we have both $T_A(\mathbf{x}_1) = \mathbf{v}$ and $T_A(\mathbf{x}_2) = \mathbf{v}$. (The vector $\mathbf{x}_1$ results from setting the free variable equal to 0, and we obtain $\mathbf{x}_2$ by setting the free variable equal to 1.) Finally, we note that the linear system which corresponds to the augmented matrix $[A \mid \mathbf{v}]$ is consistent with many solutions—that is, a solution is not unique. $\blacktriangle$

3.1.6 Reading Questions

- For each of the following, determine the number of rows and columns that a matrix would have if multiplication by that matrix is a linear transformation with the given domain and codomain.

(a) domain: $\mathbb{R}^2$, codomain: $\mathbb{R}^3$

(b) domain: $\mathbb{Q}^4$, codomain: $\mathbb{Q}^2$

- Let A , $\mathbf{u}$, $\mathbf{b}$, and $\mathbf{c}$ be defined as follows:

$$A = \begin{bmatrix} 2 & 0 & 6 \\ 1 & 3 & 6 \\ -1 & 5 & 2 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix}.$$

Define a linear transformation $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ to be multiplication by A .

- Find $T_A(\mathbf{u})$.
- Find an $\mathbf{x}$ in $\mathbb{R}^3$ such that $T_A(\mathbf{x}) = \mathbf{b}$.
- Is there more than one $\mathbf{x}$ whose image under T_A is $\mathbf{b}$? How do you know?
- Determine whether or not $\mathbf{c}$ is in the image of T_A . (Reminder: by “image” here we mean “range”.)

3.1.7 Exercises

- Consider the function $T : P_2 \rightarrow P_3$ defined by $T(p) = tp$. (So, for example, $T(2 + t) = 2t + t^2$.) Is T a linear transformation? Justify your answer.
- Consider the function $T : P_2 \rightarrow P_2$ defined by $T(p) = p(0) + p(1)t + p(2)t^2$. Is T a linear transformation? Justify your answer.
- Consider the function $T : P_2 \rightarrow P_1$ defined by $T(p) = p(0) + p'(0)t$. Is T a linear transformation? Justify your answer.
- Let $T : \mathbb{F}_5^3 \rightarrow \mathbb{F}_5^2$ be the function defined by $T(x, y, z) = (3x + y - 2z, -xy)$. Is T a linear transformation? Justify your answer.
- Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the function defined by $T(x, y) = (2x, x - 3y, 0)$. Is T a linear transformation? Justify your answer.

6. Consider the following matrix over $\mathbb{F}_3$:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix}.$$

For each of the following vectors $\mathbf{v}$, calculate the matrix-vector product $A\mathbf{v}$.

- (a) $\mathbf{v} = (1, 1, 0)$
- (b) $\mathbf{v} = (2, 1, 2)$
- (c) $\mathbf{v} = (0, 2, 1)$

7. Consider the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} -1 & 2 & 3 \\ -2 & 5 & 0 \end{bmatrix}.$$

- (a) If T is the linear transformation which is multiplication by A , what are the domain and codomain of T ?
- (b) Calculate the image of the vector $\mathbf{v} = (-3, 1, 4)$ under the linear transformation T .
- (c) Is the vector $\mathbf{w} = (-2, -1)$ in the image of T ? Explain your answer.

8. Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 4 \\ -1 & 0 \end{bmatrix}.$$

Let T be the linear transformation which is multiplication by A .

- (a) Is the vector $(1, 1, 1)$ in the image of T ? Explain your answer.
- (b) Is T surjective? Explain your answer.

9. Let A be the following matrix over $\mathbb{F}_7$:

$$A = \begin{bmatrix} 2 & 0 & 4 \\ 4 & 3 & 5 \\ 5 & 1 & 2 \end{bmatrix}.$$

Let T be the linear transformation which is multiplication by A .

- (a) Is the vector $(3, 1, 1)$ in the image of T ? Explain your answer.
- (b) The vector $\mathbf{w} = (5, 4, 0)$ is in the image of T . Find one $\mathbf{x} \in \mathbb{F}_7^3$ such that $T(\mathbf{x}) = \mathbf{w}$.
- (c) Is there more than one $\mathbf{x} \in \mathbb{F}_7^3$ such that $T(\mathbf{x}) = \mathbf{w}$? How do you know?
- (d) Is T injective? Is T surjective? Explain your answers.

10. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation which is multiplication by the following matrix:

$$A = \begin{bmatrix} 4 & -2 & 0 \\ 3 & 2 & 3 \end{bmatrix}.$$

Give a description of all vectors $\mathbf{x} \in \mathbb{R}^3$ such that $T(\mathbf{x}) = \mathbf{0}$.

Writing Exercises

11. Define the function $T : C[0, \infty) \rightarrow C[0, \infty)$ to be the following:

$$(T(f))(x) = \int_0^x f(y) \, dy.$$

Prove that T is a linear transformation. (See Exercise 2.4.5, p. 57 for a reminder of the definition of $C[0, \infty)$.)

12. Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations between vector spaces over a field $\mathbb{F}$. Prove that $S \circ T$ is also a linear transformation.
13. Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations between vector spaces over a field $\mathbb{F}$.
- Prove that if $S \circ T$ is injective, then T must be injective.
 - Prove that if $S \circ T$ is surjective, then S must be surjective.
14. Let $T : V \rightarrow W$ be a function between vector spaces over $\mathbb{F}$.
- If T is a linear transformation, must it be true that $T(\mathbf{0}_V) = \mathbf{0}_W$? Either prove this is true or produce a counterexample.
 - If $T(\mathbf{0}_V) = \mathbf{0}_W$, must T be a linear transformation? Either prove this is true or produce a counterexample.
15. Let $\mathbf{v}_1, \dots, \mathbf{v}_m$ be vectors which span a vector space V . If $T : V \rightarrow V$ is a linear transformation for which $T(\mathbf{v}_i) = \mathbf{0}$ for all $i = 1, \dots, m$, prove that T is the zero transformation. (In other words, prove that $T(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x} \in V$.)
16. Let A be an $m \times n$ matrix over a field $\mathbb{F}$, and let $\mathbf{v}$ be a vector in $\mathbb{F}^n$. Prove that the formulations of the matrix-vector product given in (3.1) and (3.2) are equivalent.
17. Prove Theorem 3.1.21, p. 64.
18. Finish the proof in Example 3.1.17, p. 63 that $\mathbb{R}^2$ and P_1 are isomorphic. Specifically, prove that the linear transformation given in this example is bijective.

3.2 The Matrix of a Linear Transformation

As we saw in the previous section, linear transformations can be defined using matrices and they can also be defined with no matrices in sight. In this section we will see that, for a certain class of linear transformations, there is *always* a matrix in sight.

3.2.1 Constructing the Matrix

Our claim might seem fanciful at first. *Can every linear transformation be realized using a matrix?* The surprising answer is *yes*, for a specific kind of linear transformation.

We first make an observation related to the definition of the matrix-vector product in Example 3.1.5, p. 59.

Note 3.2.1 If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if we recall the definition of $\mathbf{e}_j$ from Note 3.1.9, p. 61, then

$$A(\mathbf{e}_j) = \mathbf{a}_j.$$

The truth of this equality comes by thinking of $A(\mathbf{e}_j)$ in the way expressed in (3.2), as a linear combination of the columns of A with weights from the entries in $\mathbf{e}_j$.

We now suppose that $\mathbb{F}$ is a field and that $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is a linear transformation. We claim that there is a unique $m \times n$ matrix A such that for every $\mathbf{v} \in \mathbb{F}^n$, $T(\mathbf{v}) = A\mathbf{v}$. In other words, we claim that the work of the linear transformation T can be carried out through multiplication by A .

We will define the matrix A which does the job. For each $j = 1, \dots, n$, define the vector $\mathbf{a}_j$ by $\mathbf{a}_j = T(\mathbf{e}_j)$. We then define A as the matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Since any vector $\mathbf{v} \in \mathbb{F}^n$, written as $\mathbf{v} = [v_i]$, has the property that

$$\mathbf{v} = \sum_{j=1}^n v_j \mathbf{e}_j,$$

we can verify that the action of T is the same as the action of multiplication by A :

$$T(\mathbf{v}) = T\left(\sum_{j=1}^n v_j \mathbf{e}_j\right) = \sum_{j=1}^n v_j T(\mathbf{e}_j) = \sum_{j=1}^n v_j \mathbf{a}_j = A\mathbf{v}.$$

Note that we used the fact that T is a linear transformation in this last string of equalities.

We have just proved the following theorem.

Theorem 3.2.2 *If $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is a linear transformation, then there exists a unique $m \times n$ matrix A over $\mathbb{F}$ such that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{F}^n$.*

A scrupulous reader may protest our use of the word “unique” in the statement of this theorem. Here is the argument concerning uniqueness. If the theorem is true, then (for this theorem) there is only one way it could possibly work. If a matrix A exists, it must have the property that $A\mathbf{e}_j = T(\mathbf{e}_j)$ for all j . Since we have shown that such a construction does work, the matrix A we obtain must be unique.

This theorem is quite powerful. We will demonstrate that power through two examples that find their origin in Section 3.1, p. 58.

Example 3.2.3 We take our notation from Example 3.1.4, p. 59. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which reflects a vector in the Cartesian plane across the x -axis, and let $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which rotates a vector counter-clockwise around the origin by $\frac{\pi}{2}$ radians. In this example we will find the 2×2 matrices A and B such that $T(\mathbf{v}) = A\mathbf{v}$ and $S(\mathbf{v}) = B\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^2$.

In the proof of Theorem 3.2.2, p. 69, we saw that the way to form the matrix of a linear transformation is to calculate the image of the vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$. In this context, we need to calculate the image of $\mathbf{e}_1$ and $\mathbf{e}_2$ under T and S .

The calculations we seek are below:

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad S(\mathbf{e}_1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad S(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

This tells us that the matrices A and B are as follows:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Any curious reader can check that these matrices are correct by choosing a vector in $\mathbb{R}^2$ and multiplying by A and by B separately. The results should align with the actions of T and S , respectively. $\blacktriangle$

3.2.2 Composition and Matrix Multiplication

Since linear transformations are functions, we can *compose* them with other linear transformations. In order for this to make sense, we need to have the codomains and domains match up correctly. (The reader should recall a brief introduction to this idea in Definition 3.1.10, p. 61.)

If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear transformations between vector spaces, then the linear transformation $S \circ T : U \rightarrow W$ is defined. If $U = \mathbb{F}^n$, $V = \mathbb{F}^m$, and $W = \mathbb{F}^p$, then the linear transformation $S \circ T$ is defined from $\mathbb{F}^n$ to $\mathbb{F}^p$, and Theorem 3.2.2, p. 69 says that there is a unique matrix over $\mathbb{F}$ which carries out this linear transformation. What is that matrix?

Theorem 3.2.2, p. 69 tells us that there are matrices A and B such that the transformations T and S are multiplication by B and A , respectively. The matrix B is $m \times n$ and A is $p \times m$. We will *define* the product of A and B so that the matrix of $S \circ T$ is the matrix product AB .

Definition 3.2.4 Let A be a $p \times m$ matrix over a field $\mathbb{F}$ and let B be an $m \times n$ matrix over $\mathbb{F}$. Then the **matrix product** AB is the unique $p \times n$ matrix over $\mathbb{F}$ such that for all $\mathbf{u} \in \mathbb{F}^n$,

$$A(B\mathbf{u}) = (AB)\mathbf{u}.$$

◆

Note 3.2.5 When we take the matrix product AB , the number of columns of A must match the number of rows of B . The matrix product makes no sense (and cannot be computed) otherwise. The matrix AB then has the same number of rows as A and the same number of columns as B .

Though we have defined matrix multiplication in terms of the composition of linear transformations, we can multiply matrices of the correct dimensions even when we have no specific linear transformations in mind. This is similar to our understanding of row-reducing a matrix—row reduction arose in the context of solving linear systems, but the process can be carried out on any matrix.

There is one alternate, useful way to think about matrix multiplication—in terms of the columns of the matrix.

Proposition 3.2.6 Let $A \in M_{m,n}(\mathbb{F})$, $B \in M_{n,p}(\mathbb{F})$, and let the columns of B be $\mathbf{b}_1, \dots, \mathbf{b}_p$. Then the columns of AB are $A\mathbf{b}_1, \dots, A\mathbf{b}_p$.

Proof. By our definition of the matrix product, for each $j = 1, \dots, p$ we have

$$(AB)\mathbf{e}_j = A(B\mathbf{e}_j).$$

The observation in Note 3.2.1, p. 69 means that $B\mathbf{e}_j = \mathbf{b}_j$, so we have

$$(AB)\mathbf{e}_j = A\mathbf{b}_j.$$

Since $(AB)\mathbf{e}_j$ is the j th column of AB , this proves the proposition. $\blacksquare$

From the understanding we developed in Example 3.1.5, p. 59, this proposition means that every column of the matrix product AB is a linear combination of the columns of A .

We have defined matrix multiplication, but we have not specified how the entries in the matrix product are calculated. Fear not; the wait is over.

We will use the definition of matrix multiplication and the formula we have for the product of a matrix and a vector (see formula (3.1)). Since $(AB)\mathbf{u}$ is a vector, we will record a formula for entry i in this vector. In what follows, we assume $A = [a_{ij}]$ and the entries of $B = [b_{ij}]$; we also assume $\mathbf{u} = [u_i]$:

$$\begin{aligned} [(AB)\mathbf{u}]_i &= [A(B\mathbf{u})]_i = \sum_{k=1}^m a_{ik} [B\mathbf{u}]_k \\ &= \sum_{k=1}^m a_{ik} \sum_{j=1}^n b_{kj} u_j = \sum_{j=1}^n \left(\sum_{k=1}^m a_{ik} b_{kj} \right) u_j. \end{aligned}$$

When we look again at the formula in (3.1) for the product of a matrix and a vector, we see that

$$[AB]_{ij} = \sum_{k=1}^m a_{ik} b_{kj} \quad (3.5)$$

for all $1 \leq i \leq p$ and all $1 \leq j \leq n$. In words, this means that the (i, j) -entry of AB is the entry-wise product of row i in A with column j in B . (In Subsection 7.1.1, p. 177 we will acknowledge this as the *dot product* of two vectors in $\mathbb{F}^m$.)

We will try to make this concrete with some examples.

Example 3.2.7 Let A and B be the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix}.$$

Note that the product AB makes sense since the number of columns of A is the same as the number of rows of B . Here is the matrix product:

$$AB = \begin{bmatrix} 2(-2) - 1(1) & 2(0) - 1(-3) \\ 3(-2) + 4(1) & 3(0) + 4(-3) \end{bmatrix} = \begin{bmatrix} -5 & 3 \\ -2 & -12 \end{bmatrix}.$$

Since the sizes of A and B allow it, we can also calculate BA in this example:

$$BA = \begin{bmatrix} -4 & 2 \\ -7 & -13 \end{bmatrix}.$$

Finally, we observe that $AB \neq BA$. ▲

Example 3.2.8 Let A and B be the following matrices over $\mathbb{F}_5$:

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 4 \\ 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 3 \\ 4 & 2 \end{bmatrix}.$$

Since A is 3×2 and B is 2×2 , we can calculate AB , which will be 3×2 . (In this example we cannot calculate BA .) Here is the matrix product:

$$AB = \begin{bmatrix} 4(3) + 0(4) & 4(3) + 0(2) \\ 1(3) + 4(4) & 1(3) + 4(2) \\ 3(3) + 0(4) & 3(3) + 0(2) \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 4 & 1 \\ 4 & 4 \end{bmatrix}.$$

To obtain the last equality, we remember that we are working in $\mathbb{F}_5$. ▲

Since we defined matrix multiplication in the context of the composition of linear transformations, our next example picks up on this theme.

Example 3.2.9 We return to Example 3.2.3, p.69 and consider the linear transformations $S, T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where T reflects a vector in the Cartesian plane across the x -axis and S rotates a vector counter-clockwise around the origin by $\frac{\pi}{2}$ radians. In the previous example, we calculated the matrices A and B for T and S , respectively. What is the matrix for $S \circ T$?

We have defined matrix multiplication to answer questions exactly like this. We only need to multiply the matrices in the proper order. The matrix for $S \circ T$ is

$$BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

A related question in this context is whether or not linear transformations *commute*. In other words, is $S \circ T = T \circ S$? For this example, answering that question boils down to comparing the matrix product AB with the product BA which we have just calculated:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

From this we can see that $S \circ T$ and $T \circ S$ are distinct linear transformations. ▲

As we start to deal more regularly with matrices in the context of linear transformations, we need to recall the notation $M_{m,n}(\mathbb{F})$ and $M_n(\mathbb{F})$ from Example 2.3.10, p. 49.

The next theorem records some facts about matrix multiplication which will be useful later in the text. We will walk the reader through the proof of this theorem in the exercises at the end of this section.

Theorem 3.2.10 Let $A, A_1, A_2 \in M_{m,n}(\mathbb{F})$, $B, B_1, B_2 \in M_{n,p}(\mathbb{F})$, and $C \in M_{p,q}(\mathbb{F})$. Then

1. $A(BC) = (AB)C$,
2. $A(B_1 + B_2) = AB_1 + AB_2$, and
3. $(A_1 + A_2)B = A_1B + A_2B$.

This theorem says that, if all of the matrix products make sense, matrix multiplication is associative and obeys both of the distributive laws.

3.2.3 The Transpose of a Matrix

The *transpose* of a matrix is useful notation for some formulas that will appear later.

Definition 3.2.11 If $A \in M_{m,n}(\mathbb{F})$, then the **transpose** of A , denoted A^T , is the element of $M_{n,m}(\mathbb{F})$ whose rows are the columns of A . In other words,

$$[A^T]_{ij} = [A]_{ji}$$

for all $1 \leq i \leq n$ and all $1 \leq j \leq m$. ◆

Note 3.2.12 The transpose is an easy way to turn a column vector into a row vector and vice versa.

Example 3.2.13 If A is the 2×3 matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -2 & 4 & 5 \end{bmatrix},$$

then A^T is the 3×2 matrix

$$A^T = \begin{bmatrix} 2 & -2 \\ -1 & 4 \\ 0 & 5 \end{bmatrix}.$$

Some matrices are unaffected by taking the transpose. These deserve a special designation! ▲

Definition 3.2.14 A matrix which is equal to its own transpose is called a **symmetric matrix**. (All symmetric matrices must be square by necessity.) ◆

The following theorem collects some properties related to the transpose of a matrix.

Theorem 3.2.15 Let $A, C \in M_{m,n}(\mathbb{F})$, let $B \in M_{n,p}(\mathbb{F})$, and let $k \in \mathbb{F}$. Then the following properties hold:

1. $(A^T)^T = A$;
2. $(A + C)^T = A^T + C^T$;
3. $(kA)^T = kA^T$; and
4. $(AB)^T = B^T A^T$.

Proof. The first three parts of this theorem are immediate from the definitions and require no proof. To prove the fourth part, we will compare the (i, j) -entry of both $(AB)^T$ and $B^T A^T$. First, from the definition of the transpose and (3.5) we see that

$$[(AB)^T]_{ij} = [AB]_{ji} = \sum_{k=1}^n a_{jk} b_{ki}.$$

To compare, entry (i, j) of $B^T A^T$ is

$$[B^T A^T]_{ij} = \sum_{k=1}^n [B^T]_{ik} [A^T]_{kj} = \sum_{k=1}^n b_{ki} a_{jk}.$$

Since multiplication is commutative in fields, these two expressions are equal. ■

Note 3.2.16 While it might be more aesthetically pleasing if we did not have to switch the order of the multiplication when taking the transpose of a product, this type of formula makes sense when considering the dimensions of the matrices involved. If A is $m \times n$ and B is $n \times p$, then the expression $A^T B^T$ wouldn't even make sense unless $m = p$.

3.2.4 Reading Questions

1. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which is rotation *clockwise* around the origin by $\frac{\pi}{2}$ radians. Find the matrix for T . (Refer to Example 3.2.3, p. 69.) Explain your process.

2. Consider the following two matrices A and B over $\mathbb{R}$:

$$A = \begin{bmatrix} 0 & 3 \\ 5 & -1 \\ -1 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & -2 & 4 \\ 0 & 2 & -1 \end{bmatrix}.$$

Calculate both AB and BA .

3. Write down a 3×3 matrix over $\mathbb{F}_5$ which is symmetric. (See Definition 3.2.14, p. 73.)

3.2.5 Exercises

1. Let A , B , and C be the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & -2 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -3 \\ 2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 4 & 1 \\ 3 & 2 & -2 \\ 4 & -3 & 3 \end{bmatrix}.$$

For each of the following, determine whether the given calculation makes sense. If it does, find the requested matrix. (Do this by hand, without technology.) If it doesn't make sense, explain why it doesn't.

- (a) A^2
 - (b) AB
 - (c) AC
 - (d) BC
 - (e) BA
 - (f) B^2
2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which reflects a vector across the line $y = x$. Find the matrix for T .
3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which projects a vector onto the line $y = x$. Find the matrix for T .
4. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which projects a vector onto the line $y = -x$. Find the matrix for T .
5. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which rotates a vector counter-clockwise around the origin by an angle of θ radians. Find the matrix for T . (Each entry in the matrix should be an expression involving θ .)

Writing Exercises

6. In fields, we have the *cancellation law* for multiplication. If $ab = ac$ and $a \neq 0$, then $b = c$. Does matrix multiplication have this property?
Let A , B , and C be matrices over $\mathbb{F}$ such that AB and AC make sense and are the same size and A is not the zero matrix. If $AB = AC$, must it be true that $B = C$? Either prove this is true or provide a counterexample.
7. In fields, multiplication has the *no zero divisors* property. If $xy = 0$, then either $x = 0$ or $y = 0$. Does matrix multiplication have this property?
Let A and B be matrices over $\mathbb{F}$ such that AB makes sense. Let Z

be the matrix of the same size as AB consisting of all zeros. If $AB = Z$, must it be true that either A or B is a matrix of all zeros? Either prove this is true or provide a counter-example.

8. Let $A \in M_2(\mathbb{F}_5)$ be of the form

$$A = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}.$$

- (a) What conditions must a , b , and c satisfy so that $A^2 = I_2$?
- (b) How many matrices in $M_2(\mathbb{F}_5)$ of this form have the property that $A^2 = I_2$?

3.3 Inverting a Matrix

In this section we will introduce the definition of an *invertible matrix* and discuss some properties of the *inverse* of a matrix. We will then introduce a method to determine whether or not a matrix is invertible. Additionally, when a matrix is invertible we will be able to calculate its inverse.

3.3.1 Invertible Matrices

With matrix multiplication defined in terms of the composition of linear transformations, we turn to a specific composition in this subsection. Specifically, we will think about matrices for linear transformations S and T when $S \circ T$ is the identity transformation.

In Definition 3.1.14, p. 62, we called such linear transformations *invertible*. When such a linear transformation can be accomplished by matrix multiplication, we will refer to the connected matrix using this same term.

Definition 3.3.1 Let $A \in M_n(\mathbb{F})$. The matrix B is an **inverse** matrix for A if

$$AB = BA = I_n.$$

If a matrix A has an inverse, we say that A is **invertible** or **non-singular**. If A is a matrix for which no inverse matrix exists, we say that A is **singular** or not invertible. ♦

It may strike the reader as strange that only square matrices have a chance at being invertible. (That is, we have only defined invertibility for square matrices.) There is a good reason for this, which we will explore in the exercises.

If we know an inverse B of a matrix A , then we can solve matrix-vector equations with ease:

$$\begin{aligned} A\mathbf{x} &= \mathbf{y} \\ B(A\mathbf{x}) &= B\mathbf{y} \\ (BA)\mathbf{x} &= B\mathbf{y} \\ I\mathbf{x} &= B\mathbf{y} \\ \mathbf{x} &= B\mathbf{y}. \end{aligned}$$

The next three propositions present some properties of matrix inverses.

Proposition 3.3.2 *If a matrix $A \in M_n(\mathbb{F})$ has an inverse, that inverse is unique.*

Proof. Let $A \in M_n(\mathbb{F})$ and suppose that both $B, C \in M_n(\mathbb{F})$ are inverses for A . Then we have

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

■

This proposition allows us to refer to *the inverse* of a matrix A and to use the notation A^{-1} for that matrix.

Proposition 3.3.3 *Suppose that $A, B \in M_n(\mathbb{F})$ are both invertible. Then AB is invertible as well and*

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Proof. If A and B are both invertible, then both A^{-1} and B^{-1} exist. Since matrix multiplication is associative, the following calculations show that the matrix $B^{-1}A^{-1}$ satisfies the properties of the inverse of AB , thereby making AB invertible:

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} = A(I_n)A^{-1} = AA^{-1} = I_n; \\ (B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B = B^{-1}(I_n)B = B^{-1}B = I_n.\end{aligned}$$

■

This final proposition states what may seem like an obvious fact, but which should still be justified. That justification is left to the exercises.

Proposition 3.3.4 *Let $A \in M_n(\mathbb{F})$ be invertible. Then A^{-1} is also invertible and $(A^{-1})^{-1} = A$.*

While we are not yet ready to calculate the inverse of a matrix (stay tuned!), we can provide examples of invertible matrices and their inverses.

Example 3.3.5 Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}.$$

We can verify that A^{-1} is

$$A^{-1} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$$

by computing the product in both orders and verifying that the result is I_2 in both cases.

Similarly, here is a 3×3 matrix over $\mathbb{F}_3$ which is invertible:

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

The reader is encouraged to verify that the following matrix satisfies the properties of the inverse of B :

$$B^{-1} = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

▲

3.3.2 Elementary Matrices

The method of determining the invertibility of a matrix starts with a simple definition.

Definition 3.3.6 An **elementary matrix** is one that is formed by performing a single elementary row operation on an identity matrix. $\blacklozenge$

Because elementary matrices are related to elementary row operations, there are three types of elementary matrices. The following example provides one elementary matrix (over $\mathbb{R}$) of each type.

Example 3.3.7 Our first elementary matrix results from switching the second and third rows in I_3 :

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Next we will look at a matrix which comes about by adding 4 times the first row of I_2 to the second row:

$$\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}. \quad (3.6)$$

Finally, we have a matrix which is formed by multiplying the second row of I_4 by 7:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

▲
Multiplying by an elementary matrix has the effect of carrying out an elementary row operation. In other words, if the $n \times n$ matrix E results from applying an elementary row operation to I_n , and if A is another $n \times n$ matrix, then EA is the matrix A after this same elementary row operation has been applied. We will demonstrate this in an example before stating the relevant theorem. (The proof of the theorem is saved for the exercises.)

Example 3.3.8 Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}.$$

If we label as E the matrix in (3.6), then we can calculate EA :

$$EA = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -5 & 6 \end{bmatrix}.$$

The reader can verify that EA is the result of adding four times the first row of A to the second row of A . $\blacktriangle$

Theorem 3.3.9 *If the elementary matrix E results from performing an elementary row operation on I_n , and if A is an $n \times n$ matrix, then EA is the matrix that results from applying this same elementary row operation to A .*

Each elementary row operation is “reversible” in the sense that there is another elementary row operation that reverses the work done by the first. (This appears as Exercise 1.2.9, p. 16.) We can use this fact to establish the following useful proposition.

Proposition 3.3.10 *Every elementary matrix is invertible.*

Proof. Let E be an elementary matrix and let E' denote the elementary matrix which represents the reverse elementary row operation from E . By the definition of these matrices and Theorem 3.3.9, p. 77, we see that

$$EE' = I_n, \text{ and } E'E = I_n.$$

This shows that E and E' are inverses of each other and, in particular, this proves that E is invertible. ■

We will now connect elementary matrices to the RREF of a matrix in the following proposition. This is largely a restatement of Algorithm 1.2.13, p. 13 using elementary matrices.

Proposition 3.3.11 *If $A \in M_{m,n}(\mathbb{F})$, then there exists $B \in M_{m,n}(\mathbb{F})$ in RREF and elementary matrices $E_1, \dots, E_k \in M_m(\mathbb{F})$ such that*

$$A = E_1 \cdots E_k B.$$

Proof. Since each matrix can be reduced to a matrix in RREF, and since elementary row operations are accomplished by multiplying by elementary matrices, there exist elementary matrices $D_1, \dots, D_k \in M_m(\mathbb{F})$ such that

$$B = D_k \cdots D_1 A.$$

Since elementary matrices are invertible, by repeated application of Proposition 3.3.3, p. 76 we see that $D_k \cdots D_1$ is invertible and

$$(D_k \cdots D_1)^{-1} = D_1^{-1} \cdots D_k^{-1}.$$

Then we have

$$\begin{aligned} B &= (D_k \cdots D_1) A \\ (D_k \cdots D_1)^{-1} B &= (D_k \cdots D_1)^{-1} (D_k \cdots D_1) A \\ D_1^{-1} \cdots D_k^{-1} B &= A. \end{aligned}$$

We note that each D_i^{-1} is an elementary matrix, and if we define $E_i = D_i^{-1}$ for each $i = 1, \dots, k$, we have our result. ■

3.3.3 Finding the Inverse of a Matrix

We will now move on to develop an algorithm for finding the inverse of a matrix (when one exists). We need a lemma before stating our most important result of the section.

Lemma 3.3.12 *If $A \in M_n(\mathbb{F})$ is invertible, then for every $\mathbf{b} \in \mathbb{F}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.*

Proof. Let $\mathbf{b}$ be any vector in $\mathbb{F}^n$. Since A is invertible, A^{-1} exists, and we can show that $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution to the equation $A\mathbf{x} = \mathbf{b}$:

$$A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I_n\mathbf{b} = \mathbf{b}.$$

To show that this solution is unique, suppose $\mathbf{v}$ is another solution to this equation, so $A\mathbf{v} = \mathbf{b}$. Then we have

$$\begin{aligned} A\mathbf{v} &= \mathbf{b} \\ A^{-1}(A\mathbf{v}) &= A^{-1}\mathbf{b} \\ (A^{-1}A)\mathbf{v} &= A^{-1}\mathbf{b} \end{aligned}$$

$$\begin{aligned} I_n \mathbf{v} &= A^{-1} \mathbf{b} \\ \mathbf{v} &= A^{-1} \mathbf{b}. \end{aligned}$$

■

Theorem 3.3.13 *A matrix $A \in M_n(\mathbb{F})$ is invertible if and only if A is row equivalent to I_n . When A is invertible, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .*

Proof. If A is invertible, then by Lemma 3.3.12, p. 78 the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{F}^n$. Now Theorem 1.3.6, p. 19 means that the RREF of A has a pivot in each of its n columns. Since A is square, this means the RREF has a pivot in each row as well, meaning that the RREF of A must be I_n .

Conversely, suppose that A is row equivalent to I_n . By Proposition 3.3.11, p. 78, there exist elementary matrices $E_1, \dots, E_k$ such that

$$A = E_1 \cdots E_k I_n. \quad (3.7)$$

This means that $A = E_1 \cdots E_k$, and since the product of invertible matrices is invertible, this proves that A is invertible.

If we multiply both sides of (3.7) by $(E_1 \cdots E_k)^{-1}$, we get

$$E_k^{-1} \cdots E_1^{-1} A = I_n,$$

which shows the sequence of elementary row operations (through multiplication by elementary matrices) used to transform A into I_n . On the other hand, if we take the equation $A = E_1 \cdots E_k$ from the previous paragraph and invert both sides, we get

$$A^{-1} = E_k^{-1} \cdots E_1^{-1},$$

which we can easily adjust to

$$A^{-1} = E_k^{-1} \cdots E_1^{-1} I_n.$$

This establishes the final claim in the theorem. ■

This theorem provides an algorithm for us to determine when a matrix is invertible and, in the case it is invertible, to calculate its inverse.

Algorithm 3.3.14 *In order to determine whether or not a matrix $A \in M_n(\mathbb{F})$ is invertible, follow these steps.*

1. Reduce the matrix $[A \mid I_n]$ to its RREF.
2. If the RREF has the form $[I_n \mid B]$, then A is invertible and $B = A^{-1}$.
3. If the RREF does not have I_n in its left n columns, then A is not invertible.

We will end this section with several examples in which we work through this algorithm.

Example 3.3.15 Consider the following matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -2 \\ 0 & 3 & 2 \end{bmatrix}.$$

To determine whether or not this matrix is invertible, we row reduce the matrix

$[A \mid I_3]$:

$$\left[\begin{array}{ccc|ccc} 2 & 0 & -3 & 1 & 0 & 0 \\ 1 & -1 & -2 & 0 & 1 & 0 \\ 0 & 3 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -4 & 9 & 3 \\ 0 & 1 & 0 & 2 & -4 & -1 \\ 0 & 0 & 1 & -3 & 6 & 2 \end{array} \right].$$

We see that A is invertible and that

$$A^{-1} = \begin{bmatrix} -4 & 9 & 3 \\ 2 & -4 & -1 \\ -3 & 6 & 2 \end{bmatrix}.$$

▲

Example 3.3.16 Consider the following matrix $A \in M_4(\mathbb{Q})$:

$$A = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 1 & 1 & -3 & 7 \\ 0 & -2 & 1 & 0 \\ -1 & -4 & 2 & -2 \end{bmatrix}.$$

We now row reduce $[A \mid I_4]$ to determine whether or not A is invertible. We find that $[A \mid I_4]$ is row equivalent to

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 2 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & -1 & 0 & -1/5 & -1/5 & -1/5 \\ 0 & 0 & 1 & -2 & 0 & -2/5 & 3/5 & -2/5 \\ 0 & 0 & 0 & 0 & 1 & 3/5 & -17/5 & 13/5 \end{array} \right].$$

This calculation shows that A is not invertible.

▲

Example 3.3.17 Consider the following matrix $A \in M_2(\mathbb{F}_3)$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}.$$

We now row reduce $[A \mid I_2]$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 \end{array} \right].$$

This proves that A is invertible and that

$$A^{-1} = \begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}.$$

▲

3.3.4 Reading Questions

1. Consider the following matrix defined over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}.$$

Write down elementary matrices $E_1, \dots, E_n$ which reduce A to I_2 . In other words, find elementary matrices $E_1, \dots, E_n$ such that $I_2 = E_n \cdots E_1 A$.

2. Consider the following matrix defined over $\mathbb{F}_5$:

$$A = \begin{bmatrix} 4 & 0 & 4 \\ 3 & 3 & 1 \\ 1 & 0 & 3 \end{bmatrix}.$$

Determine whether or not A is invertible. Explain your answer. If A is invertible, find the inverse.

3.3.5 Exercises

1. For each of the following matrices A , find the RREF of A (call it B), and elementary matrices $E_1, \dots, E_k$ such that $B = E_k \cdots E_1 A$.

- (a) $A \in M_{2,3}(\mathbb{R})$:

$$A = \begin{bmatrix} -4 & 6 & 1 \\ -5 & 2 & -1 \end{bmatrix}$$

- (b) $A \in M_2(\mathbb{F}_5)$:

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 0 \end{bmatrix}$$

- (c) $A \in M_2(\mathbb{C})$:

$$A = \begin{bmatrix} -1-i & -3 \\ -3 & 2-3i \end{bmatrix}$$

2. For each of the following matrices A , find the RREF of A (call it B), and elementary matrices $E_1, \dots, E_k$ such that $B = E_k \cdots E_1 A$.

- (a) $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 3 & 2 & -2 \\ 2 & -3 & 1 \\ -2 & -2 & 0 \end{bmatrix}$$

- (b) $A \in M_3(\mathbb{F}_3)$:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

3. For each of the following matrices A in $M_3(\mathbb{R})$, determine whether or not A is invertible using the algorithm from this section. In the cases where A is invertible, find the inverse.

(a) $A = \begin{bmatrix} -5 & -5 & 20 \\ 4 & -4 & 32 \\ 0 & -1 & 6 \end{bmatrix}$

(b) $A = \begin{bmatrix} -1 & -1 & -6 \\ -3 & 0 & -5 \\ 3 & 6 & 5 \end{bmatrix}$

4. For each of the following matrices A in $M_3(\mathbb{F}_5)$, determine whether or not A is invertible using the algorithm from this section. In the cases where A is invertible, find the inverse.

(a) $A = \begin{bmatrix} 4 & 3 & 0 \\ 2 & 3 & 1 \\ 2 & 4 & 3 \end{bmatrix}$

$$(b) \ A = \begin{bmatrix} 0 & 3 & 4 \\ 4 & 0 & 4 \\ 4 & 3 & 3 \end{bmatrix}$$

5. Use an inverse matrix to solve the following linear system over $\mathbb{F}_3$:

$$x + 2y = 1$$

$$x + y = 2.$$

6. Use an inverse matrix to solve the following linear system over $\mathbb{F}_5$:

$$2x + 4y = 1$$

$$x + 3y = 4.$$

7. Use an inverse matrix to solve the following linear system over $\mathbb{R}$:

$$-4x + 2y + 4z = 1$$

$$-2x + y - 6z = -2$$

$$-3x - y + 2z = 3.$$

Writing Exercises

8. Let $A \in M_n(\mathbb{F})$ be invertible. Prove that A^T is invertible and that $(A^T)^{-1} = (A^{-1})^T$.
9. Let $A \in M_n(\mathbb{F})$. Prove that A is invertible if and only if A^T is invertible.
10. Let $E \in M_n(\mathbb{F})$ be an elementary matrix. Prove that E^T is also an elementary matrix.
11. Let $A \in M_{m,n}(\mathbb{F})$.
- Suppose that A is **left-invertible**, meaning that there is an $n \times m$ matrix B such that $BA = I_n$. Prove that $m \geq n$.
 - Suppose that A is **right-invertible**, meaning that there is an $n \times m$ matrix B such that $AB = I_m$. Prove that $m \leq n$.
 - Prove that any A which is invertible must be a square matrix.
12. Prove Proposition 3.3.4, p. 76.
13. Suppose $AB = AC$, where A is an $n \times n$ matrix and B and C are $n \times p$ matrices.
- Show that if A is invertible, then $B = C$.
 - Provide an example where $AB = AC$ but A is not invertible and $B \neq C$.
14. Suppose that $(B - C)D = 0$, where B and C are $m \times n$ matrices, D is an $n \times n$ matrix, and 0 is the $m \times n$ zero matrix.
- Show that if D is invertible, then $B = C$.
 - Provide an example where $(B - C)D = 0$ but D is not invertible and $B \neq C$.
15. Suppose that A and B are $n \times n$ matrices and that AB is invertible. Prove that A is invertible.

16. Suppose that $A \in M_n(\mathbb{F})$ is upper triangular and invertible. (A matrix is **upper triangular** when all entries below the main diagonal are 0.) Prove that A^{-1} is also upper triangular.
17. How many matrices in $M_2(\mathbb{F}_2)$ are invertible? What proportion of the matrices in $M_2(\mathbb{F}_2)$ are invertible?
18. How many matrices in $M_2(\mathbb{F}_3)$ are invertible? What proportion of the matrices in $M_2(\mathbb{F}_3)$ are invertible?
19. Prove Theorem 3.3.9, p. 77. (You will need three cases.)

3.4 Subspaces and Linear Transformations

Every linear transformation between vector spaces brings with it some descriptions of related subspaces of the domain and codomain. We will explore some of these subspaces in this section.

3.4.1 The Kernel of a Linear Transformation

The *kernel* of a linear transformation T is the set of all vectors that T sends to the zero vector.

Definition 3.4.1 If $T : V \rightarrow W$ is a linear transformation between vector spaces, then the **kernel** of T is the set

$$\ker(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}.$$



While it may seem strange to single out the vectors that are sent to $\mathbf{0}$, this set reveals a lot about T .

Theorem 3.4.2 Let $T : V \rightarrow W$ be a linear transformation. Then $\ker(T)$ is a subspace of V .

Proof. We will prove this theorem using the criteria for a subspace spelled out in Theorem 2.4.3, p. 53. Since $T(\mathbf{0}_V) = \mathbf{0}_W$, we have $\mathbf{0}_V \in \ker(T)$. (The fact that $T(\mathbf{0}_V) = \mathbf{0}_W$ is found in Exercise 3.1.7.14, p. 68.)

Let $\mathbf{v}_1, \mathbf{v}_2 \in \ker(T)$. Using the additive property of T and the fact that these vectors are in the kernel, we have

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W.$$

Finally, we let $\mathbf{v} \in \ker(T)$ and $k \in \mathbb{F}$. Then we have

$$T(k\mathbf{v}) = kT(\mathbf{v}) = k\mathbf{0}_W = \mathbf{0}_W.$$

This calculation used the fact that $\mathbf{v}$ was assumed to be in the kernel and the scalar multiplication property of T . ■

Example 3.4.3 Let $V = D[a, b]$ be the set of all differentiable functions from $[a, b] \rightarrow \mathbb{R}$. Let $T : V \rightarrow V$ be the linear transformation which takes the derivative. (We proved a very similar function was a linear transformation in Example 3.1.3, p. 58.) What is the kernel of T ?

We recall from calculus that a function f has $f'(x) = 0$ for all x in an interval if and only if f is a constant function. This proves that $\ker(T)$ is the set of all constant functions, and it further establishes that set as a subspace of $D[a, b]$. ▲

While not all linear transformations are linked to matrices, some are. The kernel has an alternate name in those situations.

Definition 3.4.4 If $A \in M_{m,n}(\mathbb{F})$, then the **null space** of A is

$$\text{null}(A) = \{\mathbf{v} \in \mathbb{F}^n \mid A\mathbf{v} = \mathbf{0}\}.$$

◆

Since we have shown that the kernel is a subspace, the word “space” in the term *null space* is justified.

We also recall the link between matrix-vector equations like $A\mathbf{x} = \mathbf{b}$ and linear systems. The definition of the null space shows that the set of solutions to a *homogeneous* linear system can be described as the null space of a matrix.

Example 3.4.5 Let A be the matrix

$$A = \begin{bmatrix} 3 & 1 & -3 \\ -1 & 0 & 2 \\ 2 & -2 & -10 \end{bmatrix}.$$

We can find $\text{null}(A)$ by row reducing the matrix $[A \mid \mathbf{0}]$. Here is the RREF:

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

From this we see that $x_1 = 2x_3$, $x_2 = -3x_3$, and x_3 is free. In other words, any vector $\mathbf{x}$ in $\text{null}(A)$ looks like

$$\mathbf{x} = \begin{bmatrix} 2x_3 \\ -3x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

So we have

$$\text{null}(A) = \text{Span}\{\mathbf{v}\}$$

where

$$\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

▲

Note 3.4.6 There is an important fact contained in this last example. When we have a homogeneous system, we can always pay attention to just the coefficient matrix instead of the augmented matrix. No elementary row operation can produce a non-zero entry in a column of zeros.

The following theorem is one of the reasons that the kernel is so useful.

Theorem 3.4.7 Let $T : V \rightarrow W$ be a linear transformation. Then T is injective if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof. We first suppose that T is injective. (We recall the definition of injectivity from Definition 3.1.12, p. 62.) Since $T(\mathbf{0}_V) = \mathbf{0}_W$, the injectivity of T implies that $\mathbf{v} = \mathbf{0}_V$ if $T(\mathbf{v}) = \mathbf{0}_W$ for any $\mathbf{v} \in V$. Therefore, $\ker(T) = \{\mathbf{0}\}$.

Next, we suppose that $\ker(T) = \{\mathbf{0}\}$. We want to prove that T is injective, so we let $\mathbf{v}_1, \mathbf{v}_2 \in V$ with $T(\mathbf{v}_1) = T(\mathbf{v}_2)$. We want to show that $\mathbf{v}_1 = \mathbf{v}_2$. By the linearity of T we have

$$T(\mathbf{v}_1) - T(\mathbf{v}_2) = \mathbf{0}$$

$$T(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}.$$

Since $\ker(T) = \{\mathbf{0}\}$, we must have $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$, meaning that $\mathbf{v}_1 = \mathbf{v}_2$. This proves that T is injective. ■

Example 3.4.8 Consider the linear transformation $T : P_1 \rightarrow \mathbb{R}^2$ given by

$$T(p) = \begin{bmatrix} p(0) \\ p'(0) \end{bmatrix}.$$

To examine $\ker(T)$, we need to look at polynomials $p \in P_1$ such that $p(0) = 0$ and $p'(0) = 0$. If $p(t) = a + bt$, then $a = p(0)$ and $b = p'(0)$, so if $T(p) = \mathbf{0}$, we must have $a = 0$ and $b = 0$. This means that the only polynomial in $\ker(T)$ is the zero polynomial. Therefore, T is injective. ▲

We now present one final fact related to the kernel.

Corollary 3.4.9 *Suppose that the linear system represented by the equation $A\mathbf{x} = \mathbf{b}$ is consistent. Then this system has a unique solution if and only if $\text{null}(A) = \{\mathbf{0}\}$.*

Proof. We first assume that the system has a unique solution. Since the linear system is consistent, then there exists a vector $\mathbf{v}$ such that $A\mathbf{v} = \mathbf{b}$. If $\mathbf{w} \in \text{null}(A)$, then $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, so $\mathbf{v} + \mathbf{w}$ is also a solution. But since there is a unique solution, we must have $\mathbf{v} = \mathbf{v} + \mathbf{w}$, so $\mathbf{w} = \mathbf{0}$. This shows that $\text{null}(A) = \{\mathbf{0}\}$.

We now assume that $\text{null}(A) = \{\mathbf{0}\}$. From Theorem 3.4.7, p. 84 we know that the associated linear transformation is injective. Since the system is consistent, there must be only one vector that the transformation sends to $\mathbf{b}$ so the system has a unique solution. ■

3.4.2 The Image as a Subspace

We have examined the kernel as a subspace of the domain of a linear transformation. We now turn our attention to a well-known subset of the codomain. The reader will be familiar with the image (or range) of a linear transformation. We can now prove that this is a subspace.

Note 3.4.10 We will use the notation $\text{im}(T)$ for the image of a linear transformation T .

Theorem 3.4.11 *Let $T : V \rightarrow W$ be a linear transformation between vector spaces over $\mathbb{F}$. Then $\text{im}(T)$ is a subspace of W .*

Proof. Since we know that $T(\mathbf{0}_V) = \mathbf{0}_W$, it follows that $\mathbf{0}_W \in \text{im}(T)$. We now need to show the other properties demanded by Theorem 2.4.3, p. 53.

If $\mathbf{w}_1, \mathbf{w}_2 \in \text{im}(T)$, then there exist vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ with $T(\mathbf{v}_1) = \mathbf{w}_1$ and $T(\mathbf{v}_2) = \mathbf{w}_2$. Then, using the linearity of T , we have

$$\mathbf{w}_1 + \mathbf{w}_2 = T(\mathbf{v}_1) + T(\mathbf{v}_2) = T(\mathbf{v}_1 + \mathbf{v}_2).$$

This proves that $\mathbf{w}_1 + \mathbf{w}_2 \in \text{im}(T)$, so $\text{im}(T)$ is closed under addition.

Finally, we let $k \in \mathbb{F}$ and $\mathbf{w} \in \text{im}(T)$, so there exists a vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. Then, using the fact that T is a linear transformation, we have

$$k\mathbf{w} = kT(\mathbf{v}) = T(k\mathbf{v}),$$

which proves that $k\mathbf{w} \in \text{im}(T)$. Thus, $\text{im}(T)$ is closed under scalar multiplication. We conclude that $\text{im}(T)$ is a subspace of W . ■

As usual, when our linear transformation is linked to a matrix, we have more to say.

Definition 3.4.12 If $A \in M_{m,n}(\mathbb{F})$, then the **column space** of A , written $\text{col}(A)$, is the set of all linear combinations of the columns of A . If $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$, then

$$\text{col}(A) = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

◆

When we introduced the matrix-vector product in Example 3.1.5, p. 59, we noted that $A\mathbf{v}$ is a linear combination of the columns of A with weights coming from the entries in $\mathbf{v}$. So, a vector in $\text{col}(A)$ can be written as $A\mathbf{x}$ for some $\mathbf{x}$. Therefore, another way to write the column space is

$$\text{col}(A) = \{\mathbf{w} \in \mathbb{F}^m \mid \mathbf{w} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{F}^n\}.$$

From this description, we can see that the *column space* of a matrix and the *image* of a linear transformation are the same.

Fact 3.4.13 If $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is a linear transformation given by multiplication by a matrix A , then

$$\text{im}(T) = \text{col}(A).$$

We can also restate the consistency of linear systems using the language of the column space.

Fact 3.4.14 If $A \in M_{m,n}(\mathbb{F})$, then the linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \text{col}(A)$.

Since the column space of a matrix is a subspace of the codomain of the associated linear transformation, there will be some occasions when that subspace is as large as it could be. The next theorem gives conditions for just that situation.

Theorem 3.4.15 A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans $\mathbb{F}^m$ if and only if the RREF of the matrix $A = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ has a pivot in every row.

Proof. The set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans $\mathbb{F}^m$ if and only if $\mathbf{b} \in \text{col}(A)$ for every $\mathbf{b} \in \mathbb{F}^m$. This happens when the linear system with augmented matrix $[A \mid \mathbf{b}]$ is consistent for each $\mathbf{b}$.

We know from Theorem 2.2.2, p. 41 that a linear system over $\mathbb{F}$ is consistent if and only if there is no pivot in the final column of the augmented matrix. If the RREF of A has a pivot in every row, then there cannot be a pivot in the final column of the RREF of $[A \mid \mathbf{b}]$ since each row already contains one pivot.

We will prove the contrapositive of the other implication. Suppose that the RREF of A does not have a pivot in every row. We will create a vector in $\mathbb{F}^m$ which is not in the span of this set of vectors. Since the RREF of A does not have a pivot in every row, let the smallest row number with no pivot be k . Form the augmented matrix with the RREF of A and the vector $\mathbf{e}_k$. Now reverse the elementary row operations that were taken to reduce A to its RREF. The result will be an augmented matrix $[A \mid \mathbf{b}]$ which is related to an inconsistent system. (There will be a pivot in the final column of the RREF of this matrix. We constructed it this way!) This proves that the set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ does not span $\mathbb{F}^m$. ■

Example 3.4.16 Define the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{F}_5^3$ as

$$\mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix}.$$

How large is $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$?

We form the matrix $[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ and find the RREF:

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & 4 & 4 \\ 3 & 4 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since there is a pivot in each row, the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ spans $\mathbb{F}_5^3$. ▲

Example 3.4.17 Consider the following vectors in $\mathbb{R}^3$:

$$\mathbf{u} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -5 \\ 6 \\ 2 \end{bmatrix}.$$

When we row reduce the matrix $[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$, we get

$$\begin{bmatrix} -2 & 1 & -5 \\ 3 & -2 & 6 \\ -1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since there is not a pivot in every row, the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ does not span $\mathbb{R}^3$. ▲

3.4.3 Reading Questions

- Let A be the following matrix:

$$A = \begin{bmatrix} 2 & 2 \\ 3 & -2 \\ 3 & -5 \end{bmatrix}.$$

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation which is multiplication by A .

- Calculate $\text{null}(A)$.
 - Is T injective? Explain.
- Consider the following three vectors in $\mathbb{F}_3^2$:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Does the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ span $\mathbb{F}_3^2$? Explain.

3.4.4 Exercises

- Consider the following matrix A in $M_{3,4}(\mathbb{F}_7)$:

$$A = \begin{bmatrix} 6 & 3 & 3 & 6 \\ 0 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{bmatrix}.$$

For each of the following vectors $\mathbf{x} \in \mathbb{F}_7^4$, determine whether or not $\mathbf{x} \in \text{null}(A)$.

$$(a) \mathbf{x} = \begin{bmatrix} 6 \\ 1 \\ 4 \\ 2 \end{bmatrix}$$

$$(b) \mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ 1 \\ 2 \end{bmatrix}$$

2. For each of the following matrices A over $\mathbb{R}$, find $\text{null}(A)$ by producing a set of vectors that spans $\text{null}(A)$.

$$(a) A = \begin{bmatrix} 2 & 3 & 4 & 2 \\ -5 & -1 & 5 & -3 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 2 & -4 & -5 & -3 & 1 \\ 1 & 4 & -5 & 4 & 5 \\ 0 & -2 & 5 & 0 & 5 \end{bmatrix}$$

3. Find a matrix A so that the following set is $\text{col}(A)$:

$$\left\{ \begin{bmatrix} 2r + 4s \\ 3s - 7t \\ -r - s + 5t \end{bmatrix} \mid r, s, t \in \mathbb{R} \right\}.$$

4. Let A and $\mathbf{x}$ be the following, with entries from $\mathbb{R}$:

$$A = \begin{bmatrix} 3 & -1 \\ -9 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

- (a) Show that $\mathbf{x} \in \text{null}(A)$ and $\mathbf{x} \notin \text{col}(A)$.
- (b) Explain why it is not possible, for this particular matrix A , to find a non-zero vector in $\text{null}(A) \cap \text{col}(A)$.
- (c) Is it possible to find a matrix $A \in M_2(\mathbb{R})$ such that there exists a non-zero vector in $\text{null}(A) \cap \text{col}(A)$? Justify your answer thoroughly.
5. Consider the following linear transformation $T : \mathbb{F}_7^4 \rightarrow \mathbb{F}_7^3$:

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 + 4x_2 + 6x_3 + x_4 \\ 4x_1 + 6x_2 + 5x_3 \\ 4x_1 + 2x_2 + 3x_3 \end{bmatrix}.$$

This T is not injective. Find distinct vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\mathbb{F}_7^4$ such that $T(\mathbf{x}_1) = T(\mathbf{x}_2)$.

6. Consider the following linear transformation $T : \mathbb{F}_7^3 \rightarrow \mathbb{F}_7^4$:

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 3x_2 + 3x_3 \\ 2x_1 + 6x_2 + 3x_3 \\ x_2 + 3x_3 \\ 3x_1 + 5x_2 + 6x_3 \end{bmatrix}.$$

This T is not surjective. Find a vector $\mathbf{y} \in \mathbb{F}_7^4$ such that $\mathbf{y}$ is not in the image of T .

7. Consider the following linear transformation $T : \mathbb{R}^3 \rightarrow P_2$:

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = (a+b) + (b+c)t + (a+b+c)t^2.$$

- (a) Either prove or disprove that T is injective.
 (b) Either prove or disprove that T is surjective.

8. Consider the following linear transformation $T : \mathbb{R}^2 \rightarrow P_2$:

$$T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = a + bt + bt^2.$$

- (a) Either prove or disprove that T is injective.
 (b) Either prove or disprove that T is surjective.

9. Consider the following linear transformation $T : P_2 \rightarrow \mathbb{R}^2$:

$$T(a + bt + ct^2) = \begin{bmatrix} a - 2b \\ 3b + c \end{bmatrix}.$$

- (a) Either prove or disprove that T is injective.
 (b) Either prove or disprove that T is surjective.

10. For each of the following, consider a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ which is multiplication by the given $m \times n$ matrix A . In each case, determine whether or not T is injective and whether or not T is surjective. Explain your answers.

(a) $A = \begin{bmatrix} 2 & -4 & -14 \\ -4 & -1 & 10 \\ -2 & 2 & 10 \end{bmatrix}$

(b) $A = \begin{bmatrix} -3 & -5 & -2 & 3 \\ 1 & 1 & -5 & -5 \\ 1 & -3 & 1 & 2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 2 & -2 & 2 \\ 4 & -1 & 7 \\ -3 & -1 & -7 \\ 0 & 5 & 5 \end{bmatrix}$

(d) $A = \begin{bmatrix} -2 & -3 & -1 \\ 5 & -5 & -2 \\ 0 & 4 & -4 \end{bmatrix}$

Writing Exercises

11. Suppose that $T : U \rightarrow V$ is a linear transformation between vector spaces over $\mathbb{F}$. If U' is a subspace of U , prove that $T(U')$ is a subspace of V , where

$$T(U') = \{T(\mathbf{u}) \in V \mid \mathbf{u} \in U'\}.$$

12. Suppose that $T : U \rightarrow V$ is a linear transformation between vector spaces

over $\mathbb{F}$. If V' is a subspace of V , define the set U_1 by

$$U_1 = \{\mathbf{u} \in U \mid T(\mathbf{u}) \in V'\}.$$

Prove that U_1 is a subspace of U .

- 13.** Let $A \in M_{m,n}(\mathbb{F})$ and let $B \in M_{n,p}(\mathbb{F})$. Prove that $\text{col}(AB) \subseteq \text{col}(A)$.
- 14.** Prove that if $n < m$, then no set of n vectors can span $\mathbb{F}^m$.

Chapter 4

Determinants

4.1 Defining the Determinant

In the previous chapter, we learned an algorithm for finding the inverse of an invertible $n \times n$ matrix A . This algorithm also told us when a matrix is *not invertible*—that is, when it is singular. The centerpiece of the algorithm was the row reduction of an $n \times 2n$ matrix.

In this section, we will learn about the useful and powerful function called *the determinant*. The determinant by itself will not give us the inverse of a matrix, but it will reveal whether or not a matrix is invertible.

4.1.1 The Definition of the Determinant

In Exercise 1.3.11, p.24, we saw that when a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has the property that $ad - bc \neq 0$, A is row equivalent to I_2 . (In that context we were working over $\mathbb{R}$, but this is true over any field.) The converse is also true, though we haven't yet established it.

Connecting this fact with Theorem 3.3.13, p.79, when $ad - bc \neq 0$, we see that the 2×2 matrix A is invertible. (We will see that the converse of this statement is also true.)

This quantity $ad - bc$ for A is special—it is called the *determinant* of A , and this relationship between the determinant and invertibility also holds for larger matrices. The goal of this section is to define a number for any $n \times n$ matrix which functions the same way that $ad - bc$ does for a 2×2 matrix.

Definition 4.1.1 For an $m \times n$ matrix A , we define A_{ij} to be the submatrix of A that results from deleting row i and column j from A . ♦

Example 4.1.2 If A is the following matrix,

$$A = \begin{bmatrix} 3 & -7 & -4 & -7 \\ 2 & 0 & -8 & -2 \\ 8 & 6 & 2 & 2 \end{bmatrix},$$

then the submatrices A_{13} and A_{22} are

$$A_{13} = \begin{bmatrix} 2 & 0 & -2 \\ 8 & 6 & 2 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 3 & -4 & -7 \\ 8 & 2 & 2 \end{bmatrix}.$$



Perhaps surprisingly, this bit of notation is all we need to define the determinant. We note that this is a *recursive* definition, which means that the calculation happens in stages by reducing the size of the matrix.

Definition 4.1.3 Suppose that $A \in M_n(\mathbb{F})$ with $n \geq 1$, and let the entries of A be denoted a_{ij} . Then the **determinant** of A , denoted $\det(A)$ or $|A|$, is defined as follows. If $n = 1$, then $\det(A) = a_{11}$. If $n \geq 2$, then

$$\begin{aligned}\det(A) &= a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{1+n} a_{1n} \det(A_{1n}) \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).\end{aligned}$$

◆

To calculate the determinant of an $n \times n$ matrix, we need to know the determinant of a lot of $(n-1) \times (n-1)$ submatrices—in fact, we need to know the determinant of n submatrices. The reader will appreciate that when we begin with a small matrix, this is a manageable calculation; when the matrix is larger, carrying out this calculation by hand would be a decidedly less pleasant task.

Note 4.1.4 When $n = 2$, the formula for the determinant of A reduces to the familiar expression with which we opened this section. Note that, in this case, $A_{11} = [a_{22}]$ and $A_{12} = [a_{21}]$ are both 1×1 matrices. We have

$$\begin{aligned}\det(A) &= a_{11} \det(A_{11}) - a_{12} \det(A_{12}) \\ &= a_{11}a_{22} - a_{12}a_{21}.\end{aligned}$$

This last expression is precisely $ad - bc$ with different symbols.

We will put our new definition to use in the following example.

Example 4.1.5 Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & -4 & -1 \\ 6 & -6 & 0 \\ -5 & -3 & -5 \end{bmatrix}.$$

We will find $\det(A)$ using the formula:

$$\begin{aligned}\det(A) &= 2 \begin{vmatrix} -6 & 0 \\ -3 & -5 \end{vmatrix} - (-4) \begin{vmatrix} 6 & 0 \\ -5 & -5 \end{vmatrix} + (-1) \begin{vmatrix} 6 & -6 \\ -5 & -3 \end{vmatrix} \\ &= 2(30 - 0) + 4(-30 - 0) - (-18 - 30) \\ &= 60 - 120 + 48 = -12.\end{aligned}$$

We see that $\det(A) = -12$.

▲

Note 4.1.6 For the purposes of what follows, we will introduce another bit of notation. If $A = [a_{ij}]$, then the (i, j) -**cofactor** of A , denoted by C_{ij} , is

$$C_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Our first determinant formula could then be written

$$\det(A) = \sum_{j=1}^n a_{1j} C_{1j}.$$

4.1.2 Additional Ways to Calculate the Determinant

The definition of the determinant uses the first row of the matrix as its “home base” for calculations. For this reason, the original formula is sometimes referred to as *cofactor expansion along the first row of A* . A perhaps surprising result, and one we will not prove here because of its complicated nature, asserts that the determinant can be calculated using expansion along any row, not just the first.

Theorem 4.1.7 *Suppose that $A = [a_{ij}]$ and that $A \in M_n(\mathbb{F})$ with $n \geq 2$. Then $\det(A)$ can be calculated using cofactor expansion across any row of A . In other words, for any $1 \leq i \leq n$, we have*

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Because we can target any row of a matrix in our calculation of the determinant, rows which contain several zero entries are particularly attractive.

Example 4.1.8 We consider the following matrix A :

$$A = \begin{bmatrix} 2 & 1 & -3 \\ 0 & 4 & 0 \\ -1 & -5 & -2 \end{bmatrix}.$$

Since we can expand along any row to calculate the determinant, and since the second row contains two zeros, we will expand along the second row. The zeros make it unnecessary to calculate the cofactors C_{21} and C_{23} :

$$\begin{aligned} \det(A) &= 0 \cdot C_{21} + 4 \cdot C_{22} + 0 \cdot C_{23} \\ &= (-1)^{2+2}(4) \begin{vmatrix} 2 & -3 \\ -1 & -2 \end{vmatrix} \\ &= 4(-4 - 3) = -28. \end{aligned}$$

▲

In addition to expanding along any row of a matrix to calculate the determinant, we can also use any column. Instead of proving this directly using the definition of the determinant, we will show that a matrix and its transpose have the same determinant. The result about using columns to calculate the determinant will follow.

Theorem 4.1.9 *Let $A \in M_n(\mathbb{F})$. Then $\det(A) = \det(A^T)$.*

Note 4.1.10 While we have the machinery to prove this result here, the proof we could offer is clever but not exactly intuitive for some students. (See Beezer (linear.ups.edu) for a good example of such a proof.) Instead, we will take advantage of results in the next section and prove this result in Subsection 4.2.3, p. 102. The reader should be aware that some results between the statement and proof of Theorem 4.1.9, p. 93 depend on the truth of this theorem. We have resolved the possible inconsistency for ourselves knowing that we *could* prove this result now but are choosing to offer a *clearer* proof later.

Corollary 4.1.11 *Suppose that $A = [a_{ij}]$ and that $A \in M_n(\mathbb{F})$ with $n \geq 2$. Then $\det(A)$ can be calculated using cofactor expansion along any column of A .*

In other words, for any $1 \leq i \leq n$, we have

$$\det(A) = \sum_{j=1}^n a_{ji} C_{ji}.$$

Proof. We note that column i of A is the same as row i of A^T . So, calculating $\det(A)$ using expansion along column i is the same as calculating $\det(A^T)$ using expansion along row i . But since $\det(A^T) = \det(A)$, this calculation will result in $\det(A)$. ■

The following example shows how we can use this flexibility in calculating the determinant.

Example 4.1.12 We will calculate the determinant of the following matrix:

$$A = \begin{bmatrix} 3 & 1 & 0 & -1 \\ -2 & -1 & 2 & 4 \\ 1 & 3 & 0 & -3 \\ 2 & -2 & 0 & 0 \end{bmatrix}.$$

Since the third column of A has three zeros, we will expand along that column:

$$\det(A) = \sum_{i=1}^4 a_{i3} C_{i3} = 2 \cdot C_{23}.$$

Just for this example, we will rename $A_{23} = B$, so $\det(A) = -2 \det(B)$. Here is B :

$$B = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -3 \\ 2 & -2 & 0 \end{bmatrix}.$$

We will take advantage of the zero in the $(3, 3)$ -entry of B by expanding along the third row to calculate $\det(B)$:

$$\begin{aligned} \det(B) &= (-1)^{3+1}(2) \begin{vmatrix} 1 & -1 \\ 3 & -3 \end{vmatrix} + (-1)^{3+2}(-2) \begin{vmatrix} 3 & -1 \\ 1 & -3 \end{vmatrix} \\ &= 2(0) + 2(-8) = -16. \end{aligned}$$

Since we determined that $\det(A) = -2 \det(B)$, this means that $\det(A) = 32$. ▲

We will close this section with one final result, the proof of which will be Exercise 4.1.4.10, p. 96. The truth of this proposition also relies on rows/columns with many zeros. By a “triangular” matrix we mean either an upper triangular or a lower triangular matrix.

Proposition 4.1.13 *Let $A \in M_n(\mathbb{F})$ be a triangular matrix. Then $\det(A)$ is the product of the entries along the main diagonal of A .*

4.1.3 Reading Questions

1. Compute the determinant of the matrix A by cofactor expansion across the first row. Write out all of your calculations.

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \\ 1 & -1 & 5 \end{bmatrix}.$$

2. Consider the following matrix B :

$$B = \begin{bmatrix} 2 & 3 & 1 \\ 6 & -2 & -1 \\ 0 & 0 & 4 \end{bmatrix}.$$

- Compute the determinant of B by cofactor expansion down the third column. Write out all of your calculations.
- There is a better row or column to use for this calculation. Which one is it? Choose that row or column and compute the determinant of B by cofactor expansion along that row or column. Write out all of your calculations. Why is this easier?

4.1.4 Exercises

1. For each of the following matrices over $\mathbb{R}$, find the determinant by hand using cofactor expansion along a row or column of your choice.

$$(a) A = \begin{bmatrix} 3 & 3 & -0.5 \\ 2 & -3 & 0 \\ -1 & 2 & 2 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 2 & -4 & 4 & -1 \\ -1 & 0 & -4 & 3 \\ 2 & 2 & -4 & 0 \\ 4 & 0 & -2 & 1 \end{bmatrix}$$

2. For each of the following matrices over $\mathbb{R}$, find the determinant by hand using cofactor expansion along a row or column of your choice.

$$(a) A = \begin{bmatrix} -3 & 3 & 0 \\ 0 & 4 & -2 \\ -1 & -4 & 2 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} -2 & 4 & 0 & 0 \\ 0.5 & -3.5 & -2 & -4 \\ 0.5 & 0 & 0 & -2 \\ 1 & -2 & 3 & -3 \end{bmatrix}$$

3. Consider the following matrix A with entries in $\mathbb{F}_5$:

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 4 & 1 \\ 4 & 2 & 4 \end{bmatrix}.$$

Find $\det(A)$ by hand using cofactor expansion along a row or column of your choice. (The answer should be a number in $\mathbb{F}_5$.)

4. For each of the following, write down the elementary matrix that performs the given elementary row operation on a 3×3 matrix. Then, calculate the determinant of that elementary matrix.
- Multiply row 2 by -3
 - Switch rows 2 and 3
 - Replace the third row with the sum of the third row and seven times the first row

5. Let $A = \begin{bmatrix} 3 & 1 \\ 2 & -2 \end{bmatrix}$. Write down the matrix $5A$. How are $\det(A)$ and $\det(5A)$ related?
6. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a matrix over $\mathbb{R}$ and let $k \in \mathbb{R}$. Find a formula that relates $\det(kA)$ to k and $\det(A)$.
7. Let $A = \begin{bmatrix} 1 & -2 \\ 2.5 & c \end{bmatrix}$ be a matrix over $\mathbb{R}$. Find a value of c so that $\det(A) = 0$ or explain why this is not possible.
8. Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & -2 & 1 \\ 3 & -2 & 0 \\ 2 & 3 & c \end{bmatrix}.$$

Find a value of c so that $\det(A) = 0$ or explain why this is not possible.

9. Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} x-5 & 1 \\ 4 & x-2 \end{bmatrix}.$$

Find all values of x such that $\det(A) = 0$.

Writing Exercises

10. Prove that the determinant of a triangular matrix is the product of the entries along the main diagonal.
11. Prove that if A is a 2×2 matrix and if B is the result of switching the rows in A , then $\det(B) = -\det(A)$.
12. Suppose that A is a 2×2 matrix and that B is the result of applying the *replace* row operation to A . Prove that $\det(B) = \det(A)$.

4.2 Properties of the Determinant

We have introduced the determinant, but we have not yet backed up our assertion that the determinant is *useful* or *powerful*. Our goal in this section is to establish just that. In particular, by the end of this section we will be able to conclude that the determinant gives a characterization of the invertibility of a square matrix.

4.2.1 The Determinant and Elementary Row Operations

In this subsection we will discover how elementary row operations affect the determinant of a matrix. These will be essential facts for proving the big theorems of this chapter. We begin with a result that is obvious in light of Theorem 4.1.7, p. 93.

Proposition 4.2.1 *Let $A \in M_n(\mathbb{F})$. If A has a row of zeros, then $\det(A) = 0$.*

Proof. To calculate $\det(A)$, we use cofactor expansion along the row of zeros. This immediately shows that $\det(A) = 0$. ■

Note 4.2.2 We observe that Proposition 4.2.1, p. 96 is also true if the word “row” is replaced by “column” since a matrix and its transpose have equal determinants. The reader should consider each result in this section and reflect on whether the statement would still hold after making the same word exchange.

Now, we examine the effect of the *switch* elementary row operation.

Theorem 4.2.3 *Let $A \in M_n(\mathbb{F})$ and let B be the result of switching two rows in A . Then $\det(B) = -\det(A)$.*

Proof. We will proceed by induction on n . This result only makes sense for $n \geq 2$, and the base case of $n = 2$ was covered in Exercise 4.1.4.11, p. 96.

We let k be an integer such that $k \geq 2$ and we assume the result is true for all $k \times k$ matrices. Let A be a $(k+1) \times (k+1)$ matrix and let B be the result of switching two rows in A . We want to show that $\det(B) = -\det(A)$.

Since $k \geq 2$, we have $k+1 \geq 3$, which means that we can calculate $\det(B)$ by expansion along a row that is not involved in the row exchange. Suppose that B was produced by switching rows p and q . We will calculate $\det(B)$ by expanding along row i , where i is distinct from both p and q . We have

$$\det(B) = \sum_{j=1}^{k+1} (-1)^{i+j} [B]_{ij} \det(B_{ij}).$$

We note that since i will never be p or q , $[B]_{ij} = [A]_{ij}$ for all j . Additionally, for all j , B_{ij} can be obtained by performing a switch row operation on A_{ij} . This means that, by the inductive hypothesis, we have $\det(B_{ij}) = -\det(A_{ij})$ for all j since these matrices are $k \times k$. So, we have

$$\begin{aligned} \det(B) &= \sum_{j=1}^{k+1} (-1)^{i+j} [A]_{ij} (-1) \det(A_{ij}) \\ &= - \sum_{j=1}^{k+1} (-1)^{i+j} [A]_{ij} \det(A_{ij}) \\ &= -\det(A). \end{aligned}$$

This completes the inductive step.

We have shown that the result holds for all $n \geq 2$ by the Principle of Mathematical Induction. ■

The second elementary row operation we will consider is the *scale* operation. How is the determinant of a matrix affected if one row is multiplied by a non-zero element of the field?

Theorem 4.2.4 *Let $A \in M_n(\mathbb{F})$ and let B be the result of multiplying a row in A by a non-zero $c \in \mathbb{F}$. Then $\det(B) = c \det(A)$.*

Proof. We will not need induction for this argument. Suppose that B is formed by multiplying row i in A by $c \in \mathbb{F}$ where $c \neq 0$. We will calculate $\det(B)$ by expanding along row i . Note that since row i is the only row affected by this operation, $B_{ij} = A_{ij}$ for all $1 \leq j \leq n$. Additionally, we note that $[B]_{ij} = c[A]_{ij}$ for all $1 \leq j \leq n$. Now we have

$$\begin{aligned} \det(B) &= \sum_{j=1}^n (-1)^{i+j} [B]_{ij} \det(B_{ij}) \\ &= \sum_{j=1}^n (-1)^{i+j} c [A]_{ij} \det(A_{ij}) \end{aligned}$$

$$\begin{aligned}
&= c \sum_{j=1}^n (-1)^{i+j} [A]_{ij} \det(A_{ij}) \\
&= c \det(A).
\end{aligned}$$

■

We now present the result related to the remaining elementary row operation, the *replace* operation.

Theorem 4.2.5 *Let $A \in M_n(\mathbb{F})$ and let B be the result of replacing a row in A with the sum of that row and c times another row in A . Then $\det(B) = \det(A)$.*

Proof. We proceed by induction on n . This result only makes sense when $n \geq 2$, and the base case of $n = 2$ is covered in Exercise 4.1.4.12, p. 96.

We let k be an integer such that $k \geq 2$ and we assume the result is true for all $k \times k$ matrices. Let A be a $(k+1) \times (k+1)$ matrix and let B be the result of replacing row q in A with the sum of row q and c times row p in A . We want to show that $\det(B) = \det(A)$.

We observe that k is large enough that we can calculate $\det(B)$ by expanding along a row which is not row q ; we will call that row i . Since $i \neq q$, we have $[B]_{ij} = [A]_{ij}$ for all $1 \leq j \leq n$. Additionally, for each j , B_{ij} is a $k \times k$ matrix which has been obtained from A_{ij} by a *replace* row operation. The inductive hypothesis means that $\det(B_{ij}) = \det(A_{ij})$ for all $1 \leq j \leq n$. Therefore, we have the following:

$$\begin{aligned}
\det(B) &= \sum_{j=1}^n (-1)^{i+j} [B]_{ij} \det(B_{ij}) \\
&= \sum_{j=1}^n (-1)^{i+j} [A]_{ij} \det(A_{ij}) \\
&= \det(A).
\end{aligned}$$

This completes the inductive step.

We have shown that the result holds for all $n \geq 2$ by the Principle of Mathematical Induction. ■

The following example shows how these three theorems can be used to calculate the determinant of a matrix using row reduction.

Example 4.2.6 Let A be the following matrix:

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & 2 \\ -2 & 3 & 0 \end{bmatrix}.$$

We will find $\det(A)$ using row reduction. We first switch rows 1 and 2, which introduces a negative sign:

$$\det(A) = - \begin{vmatrix} 1 & -1 & 2 \\ 2 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}.$$

Once we reduce the matrix to a triangular form, we can use Proposition 4.1.13, p. 94, so we do not need to reduce the matrix to RREF, only to REF. This means that the rest of the row reduction can be performed using only the

replace operation, which does not change the determinant:

$$\det(A) = - \begin{vmatrix} 1 & -1 & 2 \\ 2 & 0 & -3 \\ 0 & 3 & -3 \end{vmatrix} = - \begin{vmatrix} 1 & -1 & 2 \\ 0 & 2 & -7 \\ 0 & 3 & -3 \end{vmatrix} = - \begin{vmatrix} 1 & -1 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & \frac{15}{2} \end{vmatrix}.$$

We have reduced the matrix far enough so that we can calculate its determinant using the product of the entries along the main diagonal:

$$\det(A) = (-1)(1)(2)(\frac{15}{2}) = -15.$$



4.2.2 Invertibility and the Determinant

We will use the results that have accumulated thus far in this section to prove two major results. First, we need to record an easy fact.

Lemma 4.2.7 *For any $n \geq 1$, we have $\det(I_n) = 1$.*

Proof. Since the identity matrix is, among other things, a triangular matrix, Proposition 4.1.13, p.94 applies. The entries along the main diagonal are all 1, so $\det(I_n) = 1$. ■

We will now apply this lemma to record the determinant of any elementary matrix.

Proposition 4.2.8 *Let $E \in M_n(\mathbb{F})$ be an elementary matrix.*

1. *If E performs the switch elementary row operation, then $\det(E) = -1$.*
2. *If E performs the scale elementary row operation, and if the scaling is by $c \in \mathbb{F}$, then $\det(E) = c$.*
3. *If E performs the replace elementary row operation, then $\det(E) = 1$.*

Proof. Every elementary matrix in $M_n(\mathbb{F})$ is the result of performing a single elementary row operation on I_n . We have theorems in this section which tell us how these elementary row operations affect the determinant of a matrix, and since from Lemma 4.2.7, p.99 we know that $\det(I_n) = 1$, we will be able to arrive at our result.

If E performs a *switch* row operation, then by Theorem 4.2.3, p.97 we have $\det(E) = \det(EI_n) = -\det(I_n) = -1$.

If E scales one row of a matrix by a non-zero $c \in \mathbb{F}$, then by Theorem 4.2.4, p.97 we have $\det(E) = \det(EI_n) = c \det(I_n) = c$.

Finally, if E performs a *replace* row operation, then by Theorem 4.2.5, p.98 we have $\det(E) = \det(EI_n) = \det(I_n) = 1$, which completes the proof. ■

Example 4.2.9 Sometimes, the easiest way to find a determinant by hand is to use a combination of cofactor expansion and row reduction techniques. Let $A \in M_4(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 0 & 1 & -1 & 2 \\ 1 & 3 & 0 & -2 \\ 2 & 4 & 1 & -1 \\ -2 & 0 & -1 & -3 \end{bmatrix}.$$

To find $\det(A)$, we first use the *replace* row operation, using the 1 in the (2,1)

position to put zeros in the column below it:

$$A \sim \begin{bmatrix} 0 & 1 & -1 & 2 \\ 1 & 3 & 0 & -2 \\ 0 & -2 & 1 & 3 \\ 0 & 6 & -1 & -7 \end{bmatrix} = B.$$

Since the *replace* row operation doesn't change the determinant, we have $\det(A) = \det(B)$. We now use cofactor expansion along the first column to calculate $\det(B)$. Since there is only one non-zero entry in that column, we have

$$\det(B) = - \begin{vmatrix} 1 & -1 & 2 \\ -2 & 1 & 3 \\ 6 & -1 & -7 \end{vmatrix}.$$

We can now use the *replace* row operation three more times, to produce zeros in the $(2, 1)$, $(3, 1)$, and $(3, 2)$ positions of this 3×3 matrix:

$$\det(B) = - \begin{vmatrix} 1 & -1 & 2 \\ 0 & -1 & 7 \\ 0 & 5 & -19 \end{vmatrix} = - \begin{vmatrix} 1 & -1 & 2 \\ 0 & -1 & 7 \\ 0 & 0 & 16 \end{vmatrix}.$$

We now invoke Proposition 4.1.13, p. 94 to see that $\det(B) = -(-1)(16) = 16$. Since $\det(B) = \det(A)$, we have $\det(A) = 16$. $\blacktriangle$

In this next result, we use Proposition 4.2.8, p. 99 to show that the determinant respects matrix multiplication, at least when one of the factors is an elementary matrix.

Theorem 4.2.10 *Let $A, E \in M_n(\mathbb{F})$, and let E be an elementary matrix. Then*

$$\det(EA) = \det(E) \det(A).$$

Proof. This argument uses Proposition 4.2.8, p. 99 and requires three cases. If E performs a *switch* row operation, then we know from Theorem 4.2.3, p. 97 that $\det(EA) = -\det(A)$. Since we now know that $\det(E) = -1$, we have

$$\det(EA) = -\det(A) = \det(E) \det(A).$$

If E performs a *scale* row operation, and if the scaling is by a non-zero $c \in \mathbb{F}$, then we know from Theorem 4.2.4, p. 97 that $\det(EA) = c \det(A)$. Since $\det(E) = c$, we have

$$\det(EA) = c \det(A) = \det(E) \det(A).$$

Finally, if E performs a *replace* row operation, then we know from Theorem 4.2.5, p. 98 that $\det(EA) = \det(A)$. We know that $\det(E) = 1$, so

$$\det(EA) = \det(A) = 1 \cdot \det(A) = \det(E) \det(A).$$

$\blacksquare$

Armed with this result, we can now prove one of the most useful facts about determinants.

Theorem 4.2.11 *For any $n \times n$ matrix A over $\mathbb{F}$, A is invertible if and only if $\det(A) \neq 0$.*

Proof. For $A \in M_n(\mathbb{F})$, let $B \in M_n(\mathbb{F})$ be the unique RREF of A . From Proposition 3.3.11, p. 78, we know there exist elementary matrices $E_1, \dots, E_k$

such that

$$A = E_1 \cdots E_k B.$$

We can apply Theorem 4.2.10, p. 100 repeatedly to see that

$$\det(A) = \det(E_1 \cdots E_k B) = \det(E_1) \cdots \det(E_k) \det(B).$$

Since $\det(E_i) \neq 0$ for each i by Proposition 4.2.8, p. 99, we conclude that $\det(A) \neq 0$ if and only if $\det(B) \neq 0$.

We now assume that A is invertible. Theorem 3.3.13, p. 79 tells us that $B = I_n$, so $\det(B) \neq 0$. This proves one direction of the theorem.

We will prove the contrapositive of the other direction of the theorem. We now assume that A is not invertible, which (again by Theorem 3.3.13, p. 79) means that $B \neq I_n$. Specifically, B must have fewer than n pivots, which means that B must have at least one row of zeros. By Proposition 4.2.1, p. 96 we have $\det(B) = 0$. Therefore, we must also have $\det(A) = 0$. ■

If a casual math student spends some time away from linear algebra, this previous theorem might be the one and only fact they remember about the determinant. It is powerful and used *frequently*.

Example 4.2.12 Using this theorem, if $A \in M_3(\mathbb{R})$ is

$$A = \begin{bmatrix} 2 & -4 & 2 \\ 1 & 0 & 3 \\ 3.5 & 2 & 12.5 \end{bmatrix},$$

then we can say that A is not invertible since $\det(A) = 0$.

We can also analyze the invertibility of matrices over other fields. Consider the matrix $B \in M_3(\mathbb{F}_5)$ given by

$$B = \begin{bmatrix} 3 & 4 & 1 \\ 1 & 4 & 0 \\ 1 & 2 & 4 \end{bmatrix}.$$

We find that $\det(B) = 0$, so B is not invertible. (If B were a matrix over $\mathbb{R}$, we would have $\det(B) = 30$. But this means that, in $\mathbb{F}_5$, $\det(B) = 0$.)

Finally, we consider another matrix $C \in M_3(\mathbb{F}_5)$:

$$C = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & 2 & 1 \end{bmatrix}.$$

Since $\det(C) = 1$ in $\mathbb{F}_5$, C is invertible. ▲

We present one final, important result about determinants in the last theorem of this chapter.

Theorem 4.2.13 Let $A, B \in M_n(\mathbb{F})$. Then $\det(AB) = \det(A) \det(B)$.

Proof. We will prove this in two cases. First, if A is not invertible, then neither is AB , by Exercise 3.3.5.15, p. 82. This means that $\det(AB) = \det(A) \det(B)$ since, by Theorem 4.2.11, p. 100, both sides of the equation are zero.

If A is invertible, then A is row equivalent to I_n , and there exist elementary matrices $E_1, \dots, E_k$ such that

$$A = E_1 \cdots E_k I_n = E_1 \cdots E_k.$$

In the calculation that follows, we use this factorization as well as repeated application of Theorem 4.2.10, p. 100. We first use Theorem 4.2.10, p. 100 to

peel the determinant of elementary matrices away from $\det(B)$; we then use the same result to put those determinants back together to form $\det(A)$. Here is the argument:

$$\begin{aligned}
 \det(AB) &= \det(E_1 \cdots E_k B) \\
 &= \det(E_1) \det(E_2 \cdots E_k B) \\
 &= \det(E_1) \det(E_2) \det(E_3 \cdots E_k B) = \cdots \\
 &= \det(E_1) \cdots \det(E_k) \det(B) \\
 &= \det(E_1 E_2) \cdots \det(E_k) \det(B) = \cdots \\
 &= \det(E_1 \cdots E_k) \det(B) = \det(A) \det(B).
 \end{aligned}$$

This completes the proof. ■

We take a step back for a moment to marvel at this theorem. We defined matrix multiplication in the context of the composition of linear transformations (see Subsection 3.2.2, p. 70), and the calculations were quite involved. The definition of the determinant was also complicated, but in a different way, so the fact that these two notions fit together so nicely is worthy of our admiration.

Example 4.2.14 In this example, we will verify Theorem 4.2.13, p. 101 for a specific example. Let A and B be the following matrices:

$$A = \begin{bmatrix} -1 & -2 \\ -3 & -4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -2 & 4 \\ 4 & 2 \end{bmatrix}.$$

We calculate AB as

$$AB = \begin{bmatrix} -6 & -8 \\ -10 & -20 \end{bmatrix}.$$

We see that $\det(A) = -2$, $\det(B) = -20$, and $\det(AB) = 40$, so the relationship $\det(AB) = \det(A) \det(B)$ holds. ▲

4.2.3 Proving Theorem 4.1.9

We will devote the final subsection to the proof of Theorem 4.1.9, p. 93. The reader holding their breath since the statement of this theorem will soon be able to exhale.

Results about the multiplicative property of the determinant and the behavior of the transpose under matrix multiplication make this result easy to prove with but a single lemma.

Lemma 4.2.15 *If E is an elementary $n \times n$ matrix over a field $\mathbb{F}$, then $\det(E) = \det(E^T)$.*

Proof. We have three cases to consider, but two of the cases are immediate. If E is an elementary matrix that performs the *scale* or *switch* row operation, then $E = E^T$, so the result follows easily. (We ask the reader to prove this fact in Exercise 4.2.5.15, p. 106.)

We now suppose that E performs the *replace* row operation. We assume that E performs the row operation of adding k times row i to row j , where $i \neq j$. This means that E is the matrix I_n with the extra feature of containing the entry k in position (j, i) . From Proposition 4.2.8, p. 99, we know that $\det(E) = 1$, so we only need to prove that $\det(E^T) = 1$.

The matrix E^T is I_n except for the fact that it contains the element k in position (i, j) . We will calculate $\det(E^T)$ by using cofactor expansion along

row j :

$$\det(E^T) = \sum_{q=1}^n a_{jq} C_{jq} = 1 \cdot C_{jj}.$$

This calculation reduces to one term because there is only one nonzero element in row j of E^T . (Choosing row j for expansion means the element k in position (i, j) is removed when calculating the determinant of the submatrix.) Since $(E^T)_{jj} = I_{n-1}$, we have

$$\det(E^T) = (-1)^{j+j} \det((E^T)_{jj}) = (-1)^{2j} \det(I_{n-1}) = 1 \cdot 1 = 1.$$

Since $\det(E^T) = 1$ and $\det(E) = 1$, this concludes our final case. $\blacksquare$

We are now ready for the long-promised proof of Theorem 4.1.9, p. 93.

Proof of Theorem 4.1.9. We assume that $A \in M_n(\mathbb{F})$. If A is not invertible, then A^T is also not invertible (see Exercise 3.3.5.9, p. 82), meaning that both $\det(A) = 0$ and $\det(A^T) = 0$ by Theorem 4.2.11, p. 100. This proves that $\det(A) = \det(A^T)$.

We now assume that A is invertible. Using Proposition 3.3.11, p. 78 and Theorem 3.3.13, p. 79, we know that there exist elementary matrices $E_1, \dots, E_k$ such that

$$A = E_1 \cdots E_k I_n = E_1 \cdots E_k. \quad (4.1)$$

By repeated use of Theorem 4.2.10, p. 100, we know that

$$\det(A) = \det(E_1) \cdots \det(E_k).$$

We can take the transpose of both sides of (4.1), and using Theorem 3.2.15, p. 73 (part 4) repeatedly we have

$$A^T = (E_1 \cdots E_k)^T = E_k^T \cdots E_1^T.$$

We again use Theorem 4.2.10, p. 100 repeatedly (the transpose of an elementary matrix is an elementary matrix, see Exercise 3.3.5.10, p. 82), and we have

$$\det(A^T) = \det(E_k^T) \cdots \det(E_1^T).$$

Finally, using Lemma 4.2.15, p. 102 and the fact that multiplication within $\mathbb{F}$ is commutative, we conclude that $\det(A) = \det(A^T)$. $\blacksquare$

4.2.4 Reading Questions

1. Consider the following three matrices:

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 1 & 2 & 0 \\ 1 & 1 & 2 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 3 & -1 & -2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & -1 & -2 \\ 1 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}.$$

- Calculate $\det(A)$ using cofactor expansion along some row or column. Show your work.
- The matrix A_1 was obtained from A by a single elementary row operation. Which one?
- Knowing $\det(A)$ and given your answer to (b), what do you predict $\det(A_1)$ to be? (Consult Theorem 4.2.3, p. 97.)
- Calculate $\det(A_1)$ using cofactor expansion along some row or column. Show your work.

- (e) The matrix A_2 was obtained from A by a single elementary row operation. Which one?
- (f) Knowing $\det(A)$ and given your answer to (e), what do you predict $\det(A_2)$ to be? (Consult Theorem 4.2.5, p. 98.)
- (g) Calculate $\det(A_2)$ using cofactor expansion along some row or column. Show your work.
2. Verify Theorem 4.2.13, p. 101 for the following two matrices A and B :

$$A = \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 2 \\ 1 & -3 \end{bmatrix}.$$

(You should follow Example 4.2.14, p. 102)

4.2.5 Exercises

1. Find the determinant of the matrix using row reduction.

(a) $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -4 & -2 \\ -4 & -3 & 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} -1 & -2 & 0 & 3 \\ -2 & -2 & 0 & -2 \\ 0 & 2 & 1 & 0 \\ 3 & 8 & 3 & 7 \end{bmatrix}$

2. Find the determinant of the matrix using row reduction.

(a) $A = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 0 & 2 \\ 3 & -3 & 0 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 0 & -2 & 1 \\ 3 & 1 & 1 & -2 \\ -1 & -2 & -1 & 3 \end{bmatrix}$

3. Find the determinant using a combination of row reduction and cofactor expansion:

$$A = \begin{bmatrix} 2 & 1 & -3 & 1 \\ 4 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ -2 & 1 & 2 & 1 \end{bmatrix}.$$

4. Find the determinant using a combination of row reduction and cofactor expansion:

$$A = \begin{bmatrix} -1 & 2 & 1 & 4 \\ 3 & -4 & 1 & -3 \\ 4 & -10 & -1 & 0 \\ -1 & 4 & 2 & 3 \end{bmatrix}.$$

5. Use the determinant to determine whether or not the matrix is invertible. (Note that not all fields are $\mathbb{R}$!)

(a) $A \in M_3(\mathbb{F}_3)$, $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$

$$(b) \ A \in M_3(\mathbb{R}), \ A = \begin{bmatrix} -3 & -1 & -1 \\ 0 & -3 & -3 \\ 2 & -3 & 3 \end{bmatrix}$$

$$(c) \ A \in M_3(\mathbb{F}_5), \ A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

$$(d) \ A \in M_2(\mathbb{C}), \ A = \begin{bmatrix} 2+i & 2-3i \\ 4-i & -2+4i \end{bmatrix}$$

$$(e) \ A \in M_3(\mathbb{C}), \ A = \begin{bmatrix} 0 & 3-2i & -2-4i \\ -2 & 2+4i & 0 \\ 3+i & -1+i & 0 \end{bmatrix}$$

6. Calculate $\det(A^3)$ if

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

7. Construct an invertible matrix $A \in M_3(\mathbb{R})$. For each entry of A , compute the corresponding cofactor. Create a new 3×3 matrix with these cofactors in the same position as the entry of A on which they were based; call this matrix C . Calculate AC^T . What do you observe?

Writing Exercises

8. Suppose that A is a square matrix with two identical columns. Prove that $\det(A) = 0$.
9. Suppose that $A \in M_n(\mathbb{F})$ is invertible. Prove that $\det(A^{-1}) = \frac{1}{\det(A)}$.
- 10.
- (a) Suppose that $A \in M_n(\mathbb{R})$ and that $\det(A^4) = 0$. Prove that A is not invertible.
- (b) Does the result or your argument in part (a) change if $\mathbb{R}$ is replaced with $\mathbb{F}_7$? Explain.
11. Suppose that $A, B \in M_n(\mathbb{F})$. Show that $\det(AB) = \det(BA)$ regardless of whether or not $AB = BA$.
12. Let $A \in M_n(\mathbb{F})$ and let $k \in \mathbb{F}$. Find a formula for $\det(kA)$ and prove that your formula is correct.
- 13.

- (a) Verify that $\det(A) = \det(B) + \det(C)$ where A , B , and C are

$$A = \begin{bmatrix} a+e & b+f \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad C = \begin{bmatrix} e & f \\ c & d \end{bmatrix}.$$

- (b) Let A and B be

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Show that $\det(A+B) = \det(A) + \det(B)$ if and only if $a+d=0$.

- (c) Provide an example where $A, B \in M_3(\mathbb{R})$ to prove that $\det(A+B) = \det(A) + \det(B)$ is not always true.

14. Consider the following matrix (called a *Vandermond* matrix):

$$V = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}.$$

- (a) Use row operations to explain why $\det(V) = (b-a)(c-a)(c-b)$.
 - (b) Explain why V is invertible if and only if a , b , and c are all distinct real numbers.
15. Suppose that $\mathbb{F}$ is a field and that $E \in M_n(\mathbb{F})$ is an elementary matrix which performs the *scale* or *switch* row operation. Prove that E is a symmetric matrix.

Chapter 5

The Dimension of a Vector Space

Thus far in this text, the only way we have related vector spaces to each other is through linear transformations between those spaces. But we have not had any *intrinsic* quality of a vector space that enables comparison between spaces.

The notion of the *dimension* of a vector space allows just such a comparison. In this chapter we will develop the necessary machinery for defining dimension, and we will relate this concept to matrices, linear transformations, and more.

5.1 Linear Independence

Linear independence—or, rather, its opposite—is related to the idea of *redundancy*. If there is a *linear dependence* among a set of vectors, then we don't need all of those vectors to produce the same span.

Definition 5.1.1 Consider a set of vectors $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V . When $n \geq 2$, we say that V' is **linearly dependent** if, for some i , $1 \leq i \leq n$, $\mathbf{v}_i$ is a linear combination of the other vectors in the set. When $n = 1$, the set V' is **linearly dependent** if and only if $\mathbf{v}_1 = \mathbf{0}$. $\blacklozenge$

Example 5.1.2 Consider the following three vectors in $\mathbb{F}_3^3$:

$$\mathbf{u} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent since $\mathbf{v} = 2\mathbf{u} + \mathbf{w}$. $\blacktriangle$

The definition of *linear dependence* we have given is the intuitive one, but it is not the one most widely used. The following result provides an equivalent definition of linear dependence that is much easier to deploy.

Proposition 5.1.3 A set of vectors $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V is linearly dependent if and only if there exist $c_1, \dots, c_n \in \mathbb{F}$, not all of which are 0, such that

$$\mathbf{0} = c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n.$$

Proof. We will first dispatch with the case where $n = 1$. If $n = 1$ and V' is linearly dependent, then $\mathbf{v}_1 = \mathbf{0}$. Then the equation $1\mathbf{v}_1 = \mathbf{0}$ is satisfied. Conversely, if $c_1\mathbf{v}_1 = \mathbf{0}$ for some $c_1 \neq 0$, then by Theorem 2.3.12, p. 50, Item 7,

p. 50, we must have $\mathbf{v}_1 = \mathbf{0}$, meaning V' is linearly dependent.

We now consider the case where $n \geq 2$. If V' is linearly dependent, then some vector in V' , call it $\mathbf{v}_j$, is a linear combination of the other vectors in V' . This means we have

$$\mathbf{v}_j = c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} + c_{j+1}\mathbf{v}_{j+1} + \cdots + c_n\mathbf{v}_n.$$

If we subtract $\mathbf{v}_j$ from both sides, we have

$$\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} - \mathbf{v}_j + c_{j+1}\mathbf{v}_{j+1} + \cdots + c_n\mathbf{v}_n.$$

Since we now have written $\mathbf{0}$ as a non-trivial linear combination of the vectors in V' —that is, the coefficients in the linear combination are not all zero—we have completed half of the proof.

We now suppose that there is a linear combination of the vectors in V' ,

$$\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n,$$

where not all of the coefficients are zero. If $c_j \neq 0$, then we can write

$$\begin{aligned} -c_j\mathbf{v}_j &= c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} + c_{j+1}\mathbf{v}_{j+1} + \cdots + c_n\mathbf{v}_n \\ \mathbf{v}_j &= \left(-\frac{c_1}{c_j}\right)\mathbf{v}_1 + \cdots + \left(-\frac{c_{j-1}}{c_j}\right)\mathbf{v}_{j-1} \\ &\quad + \left(-\frac{c_{j+1}}{c_j}\right)\mathbf{v}_{j+1} + \cdots + \left(-\frac{c_n}{c_j}\right)\mathbf{v}_n. \end{aligned}$$

Thus we have written $\mathbf{v}_j$ as a linear combination of the other vectors in V' , so V' is linearly dependent. $\blacksquare$

We will often use this statement in Proposition 5.1.3, p. 107 as our definition of linear dependence.

Definition 5.1.4 A set of vectors $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V is **linearly independent** if it is not linearly dependent. $\blacklozenge$

Note 5.1.5 In practice, we will think about linear independence in the following way. A set $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent if the vector equation

$$x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n = \mathbf{0}$$

has only the trivial solution.

Further, when a set V' is linearly dependent, then we will call a non-trivial linear combination

$$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

a **linear dependence relation** for the vectors in V' .

We will try to make the notions of linear dependence and linear independence more concrete with some examples.

Example 5.1.6 Consider the set $V' = \{p_1, p_2\}$ in P_2 , where

$$p_1 = 1 + t \text{ and } p_2 = 3t^2.$$

We can see that the set V' is linearly independent, because the only way to produce the zero polynomial from the combination $c_1p_1 + c_2p_2$ is if $c_1 = c_2 = 0$. This is relatively easy to see in this example, since the degrees of t are not at all shared between p_1 and p_2 . If the coefficient of t^2 must be zero in the sum $c_1p_1 + c_2p_2$, then we must have $c_2 = 0$. And if the coefficient of t must be zero in the sum $c_1p_1 + c_2p_2$, then we must have $c_1 = 0$. $\blacktriangle$

Example 5.1.7 Consider the following vectors in $\mathbb{R}^2$:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ -2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 5 \\ -5 \end{bmatrix}.$$

We can show that $\mathbf{v}_3 \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ by a familiar matrix reduction:

$$\begin{bmatrix} 2 & -4 & 5 \\ 4 & -2 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2.5 \\ 0 & 1 & -2.5 \end{bmatrix}.$$

This shows us that $\mathbf{v}_3 = -2.5\mathbf{v}_1 - 2.5\mathbf{v}_2$, which proves that the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent. Further, we can conclude that the following is a linear dependence relation for the vectors in the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$:

$$\mathbf{0} = 2.5\mathbf{v}_1 + 2.5\mathbf{v}_2 + \mathbf{v}_3.$$

▲
We will see now that sets of *two vectors* are particularly nice when it comes to determining linear independence. (This means that there was an easier way to complete Example 5.1.6, p. 108.)

Consider a set $V' = \{\mathbf{v}, \mathbf{w}\}$ in a vector space V . If V' is linearly dependent, then we can write $\mathbf{v} = c\mathbf{w}$ or $\mathbf{w} = d\mathbf{v}$ for some $c, d \in \mathbb{F}$. Conversely, if V' is linearly independent, then we cannot have either $\mathbf{v} = c\mathbf{w}$ or $\mathbf{w} = d\mathbf{v}$. This means that we have a handy characterization of linear dependence for sets of two vectors.

Fact 5.1.8 *A set of two vectors $\{\mathbf{v}, \mathbf{w}\}$ is linearly dependent if and only if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither vector is a multiple of the other.*

Example 5.1.9 If $\mathbf{v}$ and $\mathbf{w}$ are the following vectors in $\mathbb{R}^3$,

$$\mathbf{v} = \begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix},$$

then the set $\{\mathbf{v}, \mathbf{w}\}$ is linearly independent since neither $\mathbf{v}$ nor $\mathbf{w}$ is a multiple of the other. ▲

There is one other notable fact that will allow us to determine whether particular sets of vectors are linearly dependent.

Fact 5.1.10 *Any set of vectors that contains the zero vector is linearly dependent. This is true because a linear dependence relation is easy to construct. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of vectors in a vector space V , and if $\mathbf{v}_i = \mathbf{0}$, then*

$$\mathbf{0} = 0\mathbf{v}_1 + \dots + 0\mathbf{v}_{i-1} + 1\mathbf{v}_i + 0\mathbf{v}_{i+1} + \dots + 0\mathbf{v}_n$$

is a non-trivial linear combination of the vectors in the set which produces the zero vector.

A reader may guess that we will occasionally need to figure out whether or not a given set of vectors is linearly independent. As with questions about span, this turns out to be easier when the vector space is $\mathbb{F}^n$ for some field $\mathbb{F}$. For other sorts of vector spaces, we will need different methods.

Proposition 5.1.11 *Let $A \in M_{m,n}(\mathbb{F})$. The columns of A are linearly independent if and only if $\mathbf{x} = \mathbf{0}$ is the only solution to the linear system represented by the equation $A\mathbf{x} = \mathbf{0}$. That is, the columns of A are linearly independent if and only if $\text{null}(A) = \{\mathbf{0}\}$.*

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{F}^m$ be the columns of A . Then the vector form of the equation $A\mathbf{x} = \mathbf{0}$ is

$$x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n = \mathbf{0}.$$

If the columns of A are linearly independent, then the only solution to this equation is $\mathbf{x} = \mathbf{0}$, which means that $\text{null}(A) = \{\mathbf{0}\}$. Alternatively, if the columns of A are linearly dependent, then $\text{null}(A)$ contains a non-zero vector—namely, the vector of scalars which provides a linear dependence relation for these vectors. ■

Note 5.1.12 The reader may note the slight abuse of terminology in the previous proof. We referred to the columns of a matrix being linearly dependent or independent instead of the set containing the columns of the matrix. We trust that the reader will forgive and overlook this misstep since the meaning is clear and the verbal gymnastics needed to be precise at all times can prove tiresome.

The following proposition provides another test of the linear dependence of a set of vectors.

Corollary 5.1.13 *If $n > m$, then every set of n vectors in $\mathbb{F}^m$ is linearly dependent.*

Proof. Let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a set of vectors in $\mathbb{F}^m$ and let A be the matrix with the elements of V' as its columns. By Corollary 1.3.7, p. 20 (or, technically, by the version of this result generalized to any field $\mathbb{F}$), we know that the $m \times n$ linear system represented by $A\mathbf{x} = \mathbf{0}$ cannot have a unique solution. Since $\mathbf{x} = \mathbf{0}$ is a known solution, the presence of another solution means that the columns of A must be linearly dependent by Proposition 5.1.11, p. 109. ■

Proposition 5.1.11, p. 109 provides us with a convenient algorithm to determine whether or not a set of vectors in $\mathbb{F}^m$ is linearly independent.

Algorithm 5.1.14 *Suppose $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of vectors in a vector space $\mathbb{F}^m$. In order to determine whether or not V' is linearly independent, we follow these steps.*

1. Form the matrix $A = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ and row reduce it to REF. Call this matrix B .
2. The set V' is linearly independent if and only if B has a pivot in every column.

Example 5.1.15 Consider the following vectors in $\mathbb{F}_5^3$:

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent because the matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ has I_3 as its RREF. ▲

Example 5.1.16 We consider the following vectors in $\mathbb{R}^4$:

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 4.5 \\ 3.5 \\ -4.5 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3.5 \\ 5 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -5 \\ 28 \\ 22 \\ -14 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 5 \\ 0 \\ 0.5 \\ -4 \end{bmatrix}.$$

The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is linearly dependent, because the matrix A which has

the vectors $\mathbf{v}_i$ as its columns has the following RREF:

$$\begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

▲

Algorithm 5.1.14, p. 110 only covers the situations when our vectors come from some $\mathbb{F}^m$. In the case of other vector spaces, we will need to do more work.

Example 5.1.17 Consider the following three elements of P_2 :

$$p_1 = 1 + t, \quad p_2 = t - t^2, \quad p_3 = 2 + 2t + t^2.$$

To determine whether or not the set $\{p_1, p_2, p_3\}$ is linearly dependent, we need to return to the definition. Suppose that we have

$$c_1 p_1 + c_2 p_2 + c_3 p_3 = 0$$

for some $c_1, c_2, c_3 \in \mathbb{R}$. In other words, this linear combination is the zero polynomial, so we have

$$c_1 p_1 + c_2 p_2 + c_3 p_3 = 0 + 0t + 0t^2.$$

For these specific polynomials, this means we have

$$\begin{aligned} c_1(1 + t) + c_2(t - t^2) + c_3(2 + 2t + t^2) &= 0 + 0t + 0t^2 \\ (c_1 + 2c_3)1 + (c_1 + c_2 + 2c_3)t + (-c_2 + c_3)t^2 &= 0 + 0t + 0t^2. \end{aligned}$$

Since the coefficients of the corresponding powers of t must be equal on both sides of this equation, we have a linear system to solve:

$$\begin{aligned} c_1 + 2c_3 &= 0 \\ c_1 + c_2 + 2c_3 &= 0 \\ -c_2 + c_3 &= 0. \end{aligned}$$

Our variables in this system are c_1 , c_2 , and c_3 , and we solve the system using the techniques from Section 1.3, p. 17. We find that

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This shows that the only solution to this linear system is the trivial one: $c_1 = c_2 = c_3 = 0$. That means that the set $\{p_1, p_2, p_3\}$ is linearly independent.

▲

We end this section with two additional results.

Corollary 5.1.18 *A set of n vectors in $\mathbb{F}^n$ is linearly independent if and only if that set spans $\mathbb{F}^n$.*

Proof. Let $A = [\mathbf{v}_1 \cdots \mathbf{v}_n] \in M_n(\mathbb{F})$. By Theorem 3.4.15, p. 86, we know that $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans $\mathbb{F}^n$ if and only if the RREF of A has a pivot in every row. Further, Algorithm 5.1.14, p. 110 says that V' is linearly independent if and only if the RREF of A has a pivot in every column. Since A is a square matrix, each of these happen exactly when the RREF of A is I_n . ■

The following result appears to be little more than a slight restatement of the definition of linear dependence. However, the precise wording used in this theorem turns out to be quite useful in proving some results later in the text.

Theorem 5.1.19 The Linear Dependence Lemma. *Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly dependent set in a vector space V and that $\mathbf{v}_1 \neq \mathbf{0}$. Then there exists $j \in \{2, \dots, n\}$ such that $\mathbf{v}_j \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\}$.*

Proof. Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a linearly dependent set of vectors in a vector space V , and suppose that $\mathbf{v}_1 \neq \mathbf{0}$. Then there exist scalars $c_1, \dots, c_n$, not all of which are zero, such that

$$\mathbf{0} = c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n.$$

Let k be the largest index such that $c_k \neq 0$. It must be that $k \geq 2$ since we assumed $\mathbf{v}_1 \neq \mathbf{0}$. Then

$$c_k\mathbf{v}_k = -c_1\mathbf{v}_1 - \dots - c_{k-1}\mathbf{v}_{k-1}.$$

Since $c_k \neq 0$, we have

$$\mathbf{v}_k = \left(-\frac{c_1}{c_k}\right)\mathbf{v}_1 + \dots + \left(-\frac{c_{k-1}}{c_k}\right)\mathbf{v}_{k-1}.$$

This shows that $\mathbf{v}_k \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}$ and completes the proof. ■

Note 5.1.20 It is important to record that Theorem 5.1.19, p. 112 doesn't say that in linearly dependent sets *every* vector is a combination of the vectors with smaller subscripts. We merely have the *existence* of a vector with that property.

Reading Questions

1. For each of the following, determine whether the given set of vectors in $\mathbb{R}^3$ is linearly dependent or linearly independent. (You should NOT need to do any matrix row reduction to figure this out.) Refer to a fact or theorem from the section when you are giving your answer.

(a) $\{\mathbf{v}_1, \mathbf{v}_2\}$ where

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ 4 \\ -10 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -6 \\ 15 \end{bmatrix}$$

(b) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -8 \\ -9 \\ 3 \end{bmatrix}$$

(c) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ 5 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -8 \\ -9 \\ 3 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 10 \\ 7 \\ -2 \end{bmatrix}$$

(d) $\{\mathbf{v}_1, \mathbf{v}_2\}$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -7 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

2. Determine whether the following sets in P_3 are linearly independent. Explain your answers. You should not need to do any calculations.

(a) $\{1 + t^2, t^3\}$

(b) $\{1, 2t^2, -7 + 6t^2\}$

(c) $\{2 - 5t^2, -4 + 10t^2\}$

Exercises

1. For each of the following, determine by inspection (without doing any calculation) whether the given set is linearly dependent or linearly independent. Explain your answers.

(a) $\{\mathbf{v}_1, \mathbf{v}_2\}$ in $\mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 8 \\ -4 \end{bmatrix}$$

(b) $\{p_1, p_2\}$ in P_2 where

$$p_1 = 2t - 4t^2, \quad p_2 = -t + 2t^2$$

(c) $\{\mathbf{v}_1, \mathbf{v}_2\}$ in $\mathbb{F}_5^2$ where

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

(d) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{F}_7^2$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

(e) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -3 \\ 9 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -2 \\ -4 \end{bmatrix}$$

2. Determine the value(s) of c , if any, that will make the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent in $\mathbb{R}^3$.

(a) $\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ -4 \\ -3 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 9 \\ -11 \\ c \end{bmatrix}$

(b) $\mathbf{v}_1 = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -6 \\ -4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ -2 \\ c \end{bmatrix}$

3. Determine the value(s) of c , if any, that will make the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly dependent in $\mathbb{R}^3$, if

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -4 \\ -4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ c \end{bmatrix}.$$

4. How many pivot columns must a 6×4 matrix have if its columns are linearly independent? Explain.
5. Determine whether the following statements are true or false. Justify your answer either way.

(a) If $V' = \{\mathbf{v}_1, \mathbf{v}_2\}$ is a subset of a vector space V and $\mathbf{v}_2$ is not a scalar multiple of $\mathbf{v}_1$, then V' is linearly independent.

(b) If $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is a subset of a vector space V and $\mathbf{v}_3$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_4$, then V' is linearly independent.

(c) If $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is a subset of a vector space V and both $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ are linearly independent, then V' is linearly independent.

(d) If $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is a subset of a vector space V and V' is linearly independent, then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is also linearly independent.

6. Determine whether or not the following set of vectors is linearly independent in the given vector space.

(a) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^4$ if

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -2.5 \\ -5 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 1 \\ -1 \\ -4.5 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4 \\ -1 \\ 3 \\ 2.5 \end{bmatrix}$$

(b) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{F}_5^3$ if

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

(c) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{F}_3^3$ if

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

7. For each of the following subsets $\{p_1, p_2, p_3\}$ of P_2 , determine whether the set is linearly dependent or linearly independent. Explain your answers.

(a) $p_1 = 3 + 5t^2$, $p_2 = -5 - 3t + 2t^2$, $p_3 = -4 - 5t - 2t^2$

(b) $p_1 = 2 - t + t^2$, $p_2 = -3 + 5t - 12t^2$, $p_3 = -2 - 2t + 8t^2$

Writing Exercises

8. Let $T : V \rightarrow W$ be a linear transformation between vector spaces.

(a) Prove that if $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly dependent set in V , then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a linearly dependent set in W .

(b) Prove that if T is injective and if $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a linearly dependent set in W , then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly dependent set in V .

9. Suppose that V_1 and V_2 are subsets of a vector space V . Prove that if $V_1 \subseteq V_2$ and V_1 is linearly dependent, then V_2 is linearly dependent.

10. Let $A \in M_n(\mathbb{F})$ and let $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$ be a linear transformation.

(a) Prove that $\text{null}(A) = \{\mathbf{0}\}$ if and only if $\text{col}(A) = \mathbb{F}^n$.

(b) Prove that T is injective if and only if it is surjective.

11. Let V be a vector space over $\mathbb{Q}$ and let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a subset of V . Prove that V' is linearly dependent if and only if there exist *integers* $c_1, \dots, c_n$, not all of which are zero, such that

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0}.$$

12. Let $A \in M_{m,n}(\mathbb{F})$ and suppose that $\text{null}(A) \neq \{\mathbf{0}\}$. Prove that the set of vectors that spans $\text{null}(A)$ is linearly independent.

5.2 Basis of a Vector Space

We have previously examined when a set of vectors spans a vector space. In this section, we will learn how to work with the most efficient spanning set possible.

5.2.1 The Definition of a Basis

We begin with the notion of finite- and infinite-dimensional vector spaces.

Definition 5.2.1 A vector space V is **finite-dimensional** if there is a finite set of vectors that spans V . A vector space is **infinite-dimensional** if it is not finite-dimensional. $\blacklozenge$

We recall that *linear independence* in Section 5.1, p. 107 was introduced as a way to eliminate redundancy. We pick up on this idea in the next definition.

Definition 5.2.2 Let V be a finite-dimensional vector space. Then a set $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **basis** for V if B is a linearly independent set and if $V = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. $\blacklozenge$

Note 5.2.3 The notion of a basis exists for infinite-dimensional vector spaces, but since the overwhelming majority of our work will be with finite-dimensional spaces, we have only given the definition in that setting.

Example 5.2.4 We recall that $\mathbf{e}_i$ is the vector in $\mathbb{F}^n$ with 1 in the i th coordinate and zeros elsewhere. Then the set $E = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis for $\mathbb{F}^n$. If we form the $n \times n$ matrix with these vectors as columns, we see that it is the $n \times n$ identity matrix. Since there is a pivot in every column, E is linearly independent according to Algorithm 5.1.14, p. 110. Then Corollary 5.1.18, p. 111 tells us that E also spans $\mathbb{F}^n$. This proves that E is a basis for $\mathbb{F}^n$.

We call this basis the **standard basis** for $\mathbb{F}^n$. $\blacktriangle$

Example 5.2.5 We now consider the subset $B = \{1, t, t^2\}$ of the vector space P_2 . Since any vector in P_2 can be written as $a(1) + b(t) + c(t^2)$, it is clear that B spans P_2 . It is also true that B is linearly independent: the set $\{1, t\}$ is linearly independent since neither vector is a scalar multiple of the other. And then since t^2 is not a linear combination of 1 and t , we conclude that B is linearly independent by (the contrapositive of) the Linear Dependence Lemma (Theorem 5.1.19, p. 112). This proves that B is a basis for P_2 .

The analogous basis for P_n , $\{1, t, \dots, t^n\}$, is often called the **standard basis** for P_n . $\blacktriangle$

Example 5.2.6 Consider the following matrix $A \in M_{3,5}(\mathbb{R})$:

$$A = \begin{bmatrix} 3 & -2 & -4 & -4 & 3 \\ 1 & -2 & 1 & 1 & 2 \\ 0 & 0 & 4 & 0 & -4 \end{bmatrix}.$$

We will find a basis for $\text{null}(A)$.

Following the procedure we first encountered in Example 3.4.5, p. 84, we start by finding the RREF of A :

$$A \sim \begin{bmatrix} 1 & 0 & 0 & -5/2 & -2 \\ 0 & 1 & 0 & -7/4 & -5/2 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}.$$

We see that x_4 and x_5 are free variables, and that any vector $\mathbf{x}$ in $\text{null}(A)$ can be written as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} (5/2)x_4 + 2x_5 \\ (7/4)x_4 + (5/2)x_5 \\ x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_4 \begin{bmatrix} 5/2 \\ 7/4 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 2 \\ 5/2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

If we label the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 5/2 \\ 7/4 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 5/2 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

then we can see that $\text{null}(A) = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Further, we see that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent (neither vector is a scalar multiple of the other), so $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\text{null}(A)$. $\blacktriangle$

Note 5.2.7 What we observed in Example 5.2.6, p. 116 is true more generally. Since the method we use to find a spanning set for $\text{null}(A)$ always produces a linearly independent set (see Exercise 5.1.12, p. 115), this method will always produce a basis for $\text{null}(A)$.

Here is an example where we are looking at whether a set of two vectors is a basis.

Example 5.2.8 It turns out that it is fairly easy to tell whether a set of two vectors in $\mathbb{R}^2$ forms a basis for $\mathbb{R}^2$. Since linear independence is easy to check with two vectors—is either vector a scalar multiple of the other?—we can focus on this characteristic. This means that the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

is a basis for $\mathbb{R}^2$. Neither vector is a scalar multiple of the other, so the set is linearly independent. And then Corollary 5.1.18, p. 111 tells us that this set must also span $\mathbb{R}^2$. (We could also easily see this by row reducing the matrix $[\mathbf{v}_1 \ \mathbf{v}_2]$.)

On the other hand, the set $W' = \{\mathbf{w}_1, \mathbf{w}_2\}$, where

$$\mathbf{w}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \text{and} \quad \mathbf{w}_2 = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$$

is not a basis for $\mathbb{R}^2$. Since $\mathbf{w}_2 = -2\mathbf{w}_1$, W' is not linearly independent, so it cannot be a basis. ▲

Putting some facts together, there is a fairly straightforward condition for when a set of vectors in $\mathbb{F}^n$ is a basis for that space.

Proposition 5.2.9 *The set $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is a basis for $\mathbb{F}^n$ if and only if the RREF of the matrix $[\mathbf{v}_1 \cdots \mathbf{v}_m]$ is I_n .*

Proof. From Theorem 3.4.15, p. 86 we know that the set $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ spans $\mathbb{F}^m$ if and only if the RREF of $A = [\mathbf{v}_1 \cdots \mathbf{v}_m]$ has a pivot in every row. Additionally, Algorithm 5.1.14, p. 110 tells us that V' is linearly independent if and only if the RREF of A has a pivot in each column. The only way a matrix in RREF can have a pivot in every row and every column is if that RREF is the identity matrix. ■

We put this proposition into action in the following example.

Example 5.2.10 Let $A \in M_3(\mathbb{F}_5)$ be the following matrix:

$$A = \begin{bmatrix} 3 & 4 & 4 \\ 3 & 0 & 1 \\ 4 & 3 & 4 \end{bmatrix}.$$

We will label column i in A as the vector $\mathbf{v}_i \in \mathbb{F}_5^3$.

Since the RREF of A is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix},$$

the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is not a basis for $\mathbb{F}_5^3$.

On the other hand, if $B \in M_3(\mathbb{F}_5)$ is the matrix

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 3 \\ 0 & 3 & 0 \end{bmatrix},$$

then the columns of B form a basis for $\mathbb{F}_5^3$ since the RREF of B is I_3 . $\blacktriangle$

5.2.2 The Properties of a Basis

Having a basis is a powerful tool. In particular, it guarantees a uniqueness that is quite useful.

Theorem 5.2.11 The Unique Representation Theorem. *A set of vectors $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V is a basis for V if and only if each element of V can be uniquely represented as a linear combination of the vectors in V' .*

Proof. We will prove the forward direction of this biconditional statement directly. Suppose that $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis of V . Since $V = \text{Span}(V')$, every vector in V can be written as a linear combination of the vectors in V' . Let $\mathbf{v}$ be a vector in V , and suppose that $\mathbf{v}$ can be written as a linear combination of the vectors in V' in two ways:

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i \quad \text{and} \quad \mathbf{v} = \sum_{i=1}^n b_i \mathbf{v}_i.$$

We want to show that $a_i = b_i$ for each i , $1 \leq i \leq n$. Since both of these representations are equal to $\mathbf{v}$, they are equal to each other, so we have

$$\mathbf{0} = \sum_{i=1}^n a_i \mathbf{v}_i - \sum_{i=1}^n b_i \mathbf{v}_i = \sum_{i=1}^n (a_i - b_i) \mathbf{v}_i.$$

Since V' is a linearly independent set (since we are assuming it is a basis), it must be that $a_i - b_i = 0$ for each i . Therefore, $a_i = b_i$ and the representation of $\mathbf{v}$ is unique.

For the other direction, we suppose that every element of V can be uniquely represented as a linear combination of the vectors in V' . Since every element of V can be represented as a linear combination of the vectors in V' , we see that V' spans V . Since every element in V can be represented *uniquely* as a linear combination of the vectors in V' , and since $\mathbf{0} \in V$ can be represented as the trivial linear combination of the vectors in V' , this means that V' is linearly independent. (The trivial linear combination of vectors in V' is the *only* way to obtain $\mathbf{0}$ as a linear combination of the vectors in V' .) Since V' is linearly independent and spans V , this proves that V' is a basis for V . $\blacksquare$

This next result shows us how to trim a spanning set down until we reach a basis.

Theorem 5.2.12 The Spanning Set Theorem. *Suppose that V is a nonzero, finite-dimensional vector space and that $V = \text{Span}(B)$ for some set of vectors $B \subseteq V$.*

1. *If B is a linearly dependent set and a vector $\mathbf{w} \in B$ can be written as a linear combination of the rest of the vectors in B , then $\text{Span}(B) = \text{Span}(B - \{\mathbf{w}\})$.*
2. *A subset of B is a basis for V .*

Proof. We suppose that $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. If B is linearly dependent, then by the Linear Dependence Lemma (Theorem 5.1.19, p. 112), there exists a vector $\mathbf{v}_k \in B$ such that $\mathbf{v}_k$ can be written as a linear combination of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$. We suppose this combination is

$$\mathbf{v}_k = a_1 \mathbf{v}_1 + \dots + a_{k-1} \mathbf{v}_{k-1}. \quad (5.1)$$

Now suppose $\mathbf{v}$ is a vector in V . We have

$$\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_{k-1}\mathbf{v}_{k-1} + c_k\mathbf{v}_k + c_{k+1}\mathbf{v}_{k+1} + \cdots + c_n\mathbf{v}_n. \quad (5.2)$$

Using (5.1), we can substitute this expression in for $\mathbf{v}_k$ in (5.2) and, once the algebraic dust settles, we will have written $\mathbf{v}$ as a combination of the vectors in $B - \{\mathbf{v}_k\}$. This shows that $\text{Span}(B) = \text{Span}(B - \{\mathbf{v}_k\})$. (Since $B - \{\mathbf{v}_k\} \subseteq B$, it is true that $\text{Span}(B - \{\mathbf{v}_k\}) \subseteq \text{Span}(B)$. The argument thus far in this proof has established the other subset containment.)

If B is linearly independent, then it is already a basis for V . If it is linearly dependent, then we can remove a vector according to the above procedure to obtain a set $B_1 = B - \{\mathbf{w}\}$ which still spans V . As long as there are two or more vectors in the spanning set, we can repeat this process until we are left with a linearly independent set and thus a basis. If the spanning set is eventually reduced to a single vector, that vector will be nonzero since V is nonzero, and therefore that set will be linearly independent and therefore a basis. ■

Corollary 5.2.13 *Every finite-dimensional vector space has a basis.*

Proof. Since a finite-dimensional vector space by definition has a finite spanning set B , Theorem 5.2.12, p. 118 tells us that a subset of B will be a basis for the vector space. ■

While the proof of Theorem 5.2.12, p. 118 provides a way to trim a spanning set down to a basis, it does not offer a practical method for this process. The following algorithm provides such a method for certain vector spaces.

Algorithm 5.2.14 *Let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a set of vectors in $\mathbb{F}^m$. The following steps result in a basis B for $\text{Span}(V')$.*

1. Put the matrix $A = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ in RREF.
2. If column i in the RREF contains a pivot, then include $\mathbf{v}_i$ in B .

Proof. We form the matrix $A = [\mathbf{v}_1 \cdots \mathbf{v}_n]$. If A is already in RREF, then the non-pivot columns are linear combinations of the pivot columns that precede them (when reading from left to right). So, those can be discarded and the pivot columns will be a basis, according to Theorem 5.2.12, p. 118.

We will complete the proof with a reminder about the effect of elementary row operations on the columns of a matrix. If a column $\mathbf{v}_k$ of A is a linear combination of the columns that precede it, then

$$\mathbf{v}_k = \sum_{i=1}^{k-1} c_i \mathbf{v}_i$$

for some scalars c_i . This means that the column vector $[c_i]$ is a solution to the linear system represented by the augmented matrix $[\mathbf{v}_1 \cdots \mathbf{v}_{k-1} \mid \mathbf{v}_k]$. One of the earliest facts we learned about elementary row operations is that they preserve the solution sets of linear systems, so the same vector $[c_i]$ will be a solution to the linear system represented by the RREF of $[\mathbf{v}_1 \cdots \mathbf{v}_{k-1} \mid \mathbf{v}_k]$. This proves that the relationships between the columns of a matrix are the same as the relationships between the columns of the RREF of that matrix.

So, if A is not in RREF, we can find the RREF of A , call it C . The non-pivot columns of C indicate that the corresponding columns of A should not be included in the basis. In other words the pivot columns of C indicate that the corresponding columns of A are the ones that should remain to form the basis. ■

Note 5.2.15 We emphasize here that the pivot columns in the reduced matrix do not provide the vectors for the basis! The pivot columns merely provide the *instructions* for which of the original vectors should be kept to form the basis.

Example 5.2.16 Consider the following matrix $A \in M_{4,5}(\mathbb{F}_5)$:

$$A = \begin{bmatrix} 3 & 2 & 3 & 3 & 3 \\ 0 & 0 & 1 & 4 & 0 \\ 3 & 2 & 0 & 1 & 2 \\ 0 & 0 & 2 & 3 & 1 \end{bmatrix}.$$

We will find a basis for $\text{col}(A)$ using Algorithm 5.2.14, p. 119. When we put A into RREF, we find

$$A \sim \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivots are in columns 1, 3, and 5, so a basis for $\text{col}(A)$ is $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{v}_5 = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \quad .$$



We arrive at the end of this section with two helpful perspectives on a basis. A basis can be formed by trimming a spanning set down until it is linearly independent. Thus, a basis is a spanning set that is as small as possible. On the other hand, a linearly independent set can always be enlarged until it spans. Therefore, a basis is also a linearly independent set that is as large as possible.

5.2.3 Reading Questions

1. Consider the set $V' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ in $\mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 4 \\ -1 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 2 \\ -2 \\ 7 \end{bmatrix}.$$

Find a basis for $\text{Span}(V')$. Follow Example 5.2.16, p. 120 and explain your answer.

2. Determine whether or not the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ forms a basis for $\mathbb{F}_7^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 5 \\ 6 \\ 2 \end{bmatrix}.$$

Explain your answer.

5.2.4 Exercises

1. For each of the following, determine whether the given set of vectors forms a basis for the indicated vector space.

- (a)
- $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- in
- $\mathbb{R}^3$
- if

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ 0.5 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4.5 \\ -2.5 \\ 4.5 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4.5 \\ 1.5 \\ -4.5 \end{bmatrix}$$

- (b)
- $\{\mathbf{v}_1, \mathbf{v}_2\}$
- in
- $\mathbb{F}_5^2$
- if

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

- (c)
- $\{p_1, p_2, p_3\}$
- in
- P_2
- if

$$p_1 = 4 + 2t + 4t^2, \quad p_2 = -3 + 4t, \quad p_3 = 2 - 2t - 4t^2$$

2. For each of the following, determine whether the given set of vectors forms a basis for the indicated vector space.

- (a)
- $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$
- in
- $\mathbb{R}^4$
- if

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \\ -3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 2 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -4 \\ -2 \\ 5 \\ -13 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

- (b)
- $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- in
- $\mathbb{F}_3^3$
- if

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

- (c)
- $\{p_1, p_2, p_3\}$
- in
- P_2
- if

$$p_1 = 3 + 3t - 4t^2, \quad p_2 = 3 - 3t, \quad p_3 = 12 + 6t - 12t^2$$

3. For each matrix A , find a basis for $\text{null}(A)$ and $\text{col}(A)$.

- (a)
- $A \in M_{3,4}(\mathbb{R})$
- ,

$$A = \begin{bmatrix} -3 & 2 & -2 & 0.5 \\ 2.5 & 5 & 15 & 5 \\ -5 & -1.5 & -13 & 0.5 \end{bmatrix}$$

- (b)
- $A \in M_{4,6}(\mathbb{F}_5)$
- ,

$$A = \begin{bmatrix} 3 & 2 & 0 & 4 & 3 & 4 \\ 3 & 3 & 3 & 1 & 3 & 1 \\ 4 & 0 & 2 & 0 & 3 & 3 \\ 4 & 4 & 1 & 2 & 0 & 2 \end{bmatrix}$$

4. Produce a matrix $A \in M_{3,4}(\mathbb{F}_5)$ which has two vectors in a basis for $\text{null}(A)$ and two vectors in a basis for $\text{col}(A)$.

5. Find a basis for $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\} \subseteq \mathbb{R}^4$, if

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ -3 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 0 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 9 \\ 10 \\ 1 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 3 \\ 2 \\ -4 \\ 9 \end{bmatrix}, \quad \mathbf{v}_5 = \begin{bmatrix} 12 \\ -16 \\ 5 \\ 1 \end{bmatrix}.$$

6. Find a basis for $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\} \subseteq \mathbb{F}_5^4$, if

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{v}_5 = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 3 \end{bmatrix}.$$

7. Let V be the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$. Find a basis for the subspace H , if

$$H = \text{Span}\{\sin(t), \sin(2t), \sin(t)\cos(t)\}.$$

8. Find a matrix $A \in M_2(\mathbb{R})$ such that

$$A \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Is A unique? Explain.

9. Suppose that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation which satisfies the following:

$$T \left(\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad T \left(\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T \left(\begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Calculate

$$T \left(\begin{bmatrix} -1 \\ 4 \\ 6 \end{bmatrix} \right).$$

10. Find a subset of the following set which is a basis for P_2 :

$$\{t - 1, t^2 - 2t, t^2 - 2, t^2 + 1\}.$$

Writing Exercises

11. Let $T : V \rightarrow W$ be a linear transformation between vector spaces, and let B be a basis for V .

- Produce an example to show that $T(B)$ does not need to be a basis of W .
- Suppose that T is injective. Must $T(B)$ be a basis for W ? If so, prove it. If not, produce a counter-example.
- Suppose that T is surjective. Must $T(B)$ be a basis for W ? If so, prove it. If not, produce a counter-example.

12. Suppose that $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for a vector space V . Prove that

$$\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_{n-1} + \mathbf{v}_n, \mathbf{v}_n\}$$

is also a basis for V .

- Prove or disprove: Every basis of P_2 must contain a polynomial of degree 2, a polynomial of degree 1, and a constant polynomial.
- Write down a basis for $M_2(\mathbb{R})$. Prove that your set is a basis. (There is no need to prove that $M_2(\mathbb{R})$ is a vector space as this was covered in Example 2.3.10, p. 49.)

5.3 Dimension

In this section we will define the *dimension* of a vector space, finally delivering on the promise made in the introduction to this chapter to describe an intrinsic quality of vector spaces that allows a comparison between spaces.

5.3.1 The Dimension of a Vector Space

We are on the threshold of the definition of dimension. We will first present a result that connects (for a finite-dimensional space) the number of vectors needed for a spanning set to the concept of linear independence. We will omit the proof.

Lemma 5.3.1 *Suppose that V is a vector space and that $V \subseteq \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$. If $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a linearly independent subset of V , then $m \geq n$.*

Proof. Since $\mathbf{w}_j \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ for each $j = 1, \dots, n$, there exist scalars $c_{ij} \in \mathbb{F}$ such that

$$\mathbf{w}_j = c_{1j}\mathbf{v}_1 + \dots + c_{mj}\mathbf{v}_m.$$

We now consider the homogeneous linear system represented by the equation $A\mathbf{x} = \mathbf{0}$, where A is the $m \times n$ matrix $A = [c_{ij}]$. If $\mathbf{x}$ is a solution to this system, then we have

$$0 = c_{i1}x_1 + \dots + c_{in}x_n$$

for all $i = 1, \dots, m$. This means that we have

$$\begin{aligned} \mathbf{0} &= \sum_{i=1}^m \left(\sum_{j=1}^n c_{ij}x_j \right) \mathbf{v}_i \\ &= \sum_{j=1}^n x_j \left(\sum_{i=1}^m c_{ij}\mathbf{v}_i \right) \\ &= \sum_{j=1}^n x_j \mathbf{w}_j. \end{aligned}$$

Since $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a linearly independent set, we must have $x_j = 0$ for all j , meaning that $\mathbf{x} = \mathbf{0}$ is the only solution to the equation $A\mathbf{x} = \mathbf{0}$. By Corollary 1.3.7, p. 20 (or, rather, the corresponding corollary to Theorem 2.2.3, p. 42), this means that $m \geq n$, as desired. ■

We will now use this lemma to prove a result related to dimension.

Theorem 5.3.2 *Suppose that, for every $n \geq 1$, a vector space V contains a linearly independent subset of size n . Then V is infinite-dimensional.*

Proof. We will prove the contrapositive. If V is finite-dimensional, then there exists a set $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ such that $V = \text{Span}(V')$. By Lemma 5.3.1, p. 123, for any $n > m$, V cannot contain a linearly independent subset of n vectors. Thus, there exists a natural number n such that V does not contain a linearly independent subset of that size. This completes the proof of the contrapositive. ■

This theorem gives us an introduction to our first infinite-dimensional example.

Example 5.3.3 Let P be the vector space of all polynomials with real coefficients. (We do not restrict the degree of polynomials in P .) For any $n \geq 1$, P

contains the linearly independent set

$$\{1, t, \dots, t^n\}.$$

Therefore, by Theorem 5.3.2, p. 123, P is infinite-dimensional. $\blacktriangle$

We now come to the bedrock result of this section, the result that makes the definition of *dimension* possible.

Theorem 5.3.4 Suppose that $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ are both bases for a vector space V . Then $m = n$.

Proof. Since $V \subseteq \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a linearly independent set, Lemma 5.3.1, p. 123 implies that $m \geq n$. However, since $V \subseteq \text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is a linearly independent set, Lemma 5.3.1, p. 123 also implies that $n \geq m$. Therefore, $m = n$. $\blacksquare$

Even though a vector space may have a huge number of bases, all of those bases have the same size. This is a number intrinsic to the vector space, not to any specific basis of that vector space. This is what we mean by the *dimension* of a vector space.

Definition 5.3.5 Let V be a finite-dimensional vector space. If $V \neq \{\mathbf{0}\}$, then the **dimension** of V , written $\dim(V)$, is the size of any basis of V . If $\dim(V) = n$, we say that V is **n -dimensional**.

If $V = \{\mathbf{0}\}$, then we define the dimension of V to be 0. $\blacklozenge$

Two of the families of vector spaces we frequently discuss have easy-to-determine dimensions, as the next two examples illustrate.

Example 5.3.6 Since $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis for $\mathbb{F}^n$, then $\dim(\mathbb{F}^n) = n$. $\blacktriangle$

Example 5.3.7 Since $\{1, t, \dots, t^n\}$ is a basis for P_n , then $\dim(P_n) = n + 1$. $\blacktriangle$

The proofs of the next two results are a consequence of Lemma 5.3.1, p. 123 and will appear in the exercises.

Proposition 5.3.8 The dimension of any vector space is less than or equal to the size of any spanning set.

Proposition 5.3.9 If a vector space V is finite-dimensional and $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a linearly independent set in V , then $\dim(V) \geq n$.

We will now begin to discuss dimension as a tool to compare vector spaces. Linear transformations are the main way we relate vector spaces to each other, so these next results rely on that machinery.

Theorem 5.3.10 Suppose that $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for a vector space V . Let $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be a subset of a vector space W . Then there is a unique linear transformation $T : V \rightarrow W$ such that $T(\mathbf{v}_i) = \mathbf{w}_i$ for each i , $1 \leq i \leq n$.

Proof. Given $\mathbf{v} \in V$, there exists a unique linear combination

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i$$

by Theorem 5.2.11, p. 118. We define the function T by

$$T(\mathbf{v}) = \sum_{i=1}^n c_i \mathbf{w}_i.$$

In words, we send a vector $\mathbf{v}$ to the linear combination of the $\mathbf{w}_i$ vectors using the same weights as those needed to form $\mathbf{v}$ from the basis V' . This gives $T(\mathbf{v}_i) = \mathbf{w}_i$ for each i , so we only need to show that T is a linear transformation.

Suppose that $\mathbf{u}, \mathbf{v} \in V$ with

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i \quad \text{and} \quad \mathbf{u} = \sum_{i=1}^n d_i \mathbf{v}_i.$$

Then we have

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T\left(\sum_{i=1}^n (c_i + d_i) \mathbf{v}_i\right) \\ &= \sum_{i=1}^n (c_i + d_i) \mathbf{w}_i \\ &= \sum_{i=1}^n c_i \mathbf{w}_i + \sum_{i=1}^n d_i \mathbf{w}_i \\ &= T(\mathbf{u}) + T(\mathbf{v}). \end{aligned}$$

Now we let $\mathbf{v} \in V$ and $d \in \mathbb{F}$. Then, if

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i,$$

we have

$$\begin{aligned} T(d\mathbf{v}) &= T\left(\sum_{i=1}^n (dc_i) \mathbf{v}_i\right) \\ &= \sum_{i=1}^n (dc_i) \mathbf{w}_i \\ &= d \sum_{i=1}^n c_i \mathbf{w}_i \\ &= dT(\mathbf{v}). \end{aligned}$$

We will complete the proof by justifying the claim that T is unique. Suppose that $T' \in L(V, W)$ with $T'(\mathbf{v}_i) = \mathbf{w}_i$ for each i . Then, if $\mathbf{v} \in V$ with

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i,$$

we have

$$T'(\mathbf{v}) = T'\left(\sum_{i=1}^n c_i \mathbf{v}_i\right) = \sum_{i=1}^n c_i T'(\mathbf{v}_i) = \sum_{i=1}^n c_i \mathbf{w}_i.$$

This shows that $T'(\mathbf{v}) = T(\mathbf{v})$ for every $\mathbf{v} \in V$, so $T' = T$ and T is unique. ■

The notion of *alikehood* that we use in linear algebra is when two vector spaces are isomorphic. The reader may wish to consult Definition 3.1.16, p. 63 for a refresher.

Theorem 5.3.11 *Let $T \in L(V, W)$ and let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then T is an isomorphism if and only if $T(B) = \{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a basis for W .*

Proof. We first suppose that T is an isomorphism. We want to show that $T(B)$ is a basis for W , so we begin with linear independence. Suppose that

$c_1, \dots, c_n \in \mathbb{F}$ such that

$$\mathbf{0} = \sum_{i=1}^n c_i T(\mathbf{v}_i).$$

Then we have

$$\mathbf{0} = \sum_{i=1}^n T(c_i \mathbf{v}_i) = T\left(\sum_{i=1}^n c_i \mathbf{v}_i\right).$$

Since T is injective, by Theorem 3.4.7, p. 84 we must have

$$\mathbf{0} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

But since B is a linearly independent set, we have $c_i = 0$ for all i . This proves that $T(B)$ is linearly independent.

We now prove that $T(B)$ spans W . Let $\mathbf{w} \in W$. Since T is surjective, there exists $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. Since B is a basis for V , we have

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

Then

$$\mathbf{w} = T(\mathbf{v}) = T\left(\sum_{i=1}^n c_i \mathbf{v}_i\right) = \sum_{i=1}^n c_i T(\mathbf{v}_i).$$

This proves that $W = \text{Span}(T(B))$, so $T(B)$ is a basis for W .

We now need to prove the other implication, and we assume that $T(B)$ is a basis for W . We need to show that T is an isomorphism. To show that T is injective, suppose that $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{0}$. We have

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i,$$

so

$$\mathbf{0} = T(\mathbf{v}) = T\left(\sum_{i=1}^n c_i \mathbf{v}_i\right) = \sum_{i=1}^n c_i T(\mathbf{v}_i).$$

But since $T(B)$ is a linearly independent set by assumption, this implies that $c_i = 0$ for all i . This means that $\mathbf{v} = \mathbf{0}$, so T is injective.

To prove that T is surjective, we assume that $\mathbf{w} \in W$. Since $T(B)$ spans W , we have

$$\mathbf{w} = \sum_{i=1}^n d_i T(\mathbf{v}_i)$$

for some $d_i \in \mathbb{F}$. We claim that if

$$\mathbf{v} = \sum_{i=1}^n d_i \mathbf{v}_i,$$

then $T(\mathbf{v}) = \mathbf{w}$. Here is the justification:

$$T(\mathbf{v}) = T\left(\sum_{i=1}^n d_i \mathbf{v}_i\right) = \sum_{i=1}^n d_i T(\mathbf{v}_i) = \mathbf{w}.$$

This proves that T is surjective and is thus an isomorphism. ■

When we view dimension as an intrinsic quality of a vector space that allows comparison between spaces, we find something surprising about vector spaces with the same dimension. They are essentially the same!

Theorem 5.3.12 *Let V and W be finite-dimensional vector spaces over $\mathbb{F}$. Then $\dim(V) = \dim(W)$ if and only if V and W are isomorphic.*

Proof. Suppose that $\dim(V) = \dim(W) = n$. Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V and let $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be a basis for W . By Theorem 5.3.10, p. 124, we can find $T \in L(V, W)$ such that $T(\mathbf{v}_i) = \mathbf{w}_i$ for each i , $1 \leq i \leq n$. Then by Theorem 5.3.11, p. 125, T is an isomorphism.

To prove the claim in the other direction, suppose that $T \in L(V, W)$ is an isomorphism. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V , then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a basis for W by Theorem 5.3.11, p. 125. Thus $\dim(V) = \dim(W)$. ■

Here is an immediate consequence of this result.

Corollary 5.3.13 *Every finite-dimensional vector space over $\mathbb{F}$ of dimension n is isomorphic to $\mathbb{F}^n$.*

Example 5.3.14 Since P_2 is a three-dimensional vector space over $\mathbb{R}$, $\mathbb{R}^3$ and P_2 are isomorphic. ▲

5.3.2 Dimension and Subspaces

If we know the dimension of a vector space, then we sometimes have a quicker path to finding a basis for that space. This next result says that if we have a spanning set of the same size as a basis, then it must be a basis.

Theorem 5.3.15 *Suppose that V is a vector space with $\dim(V) = n > 0$. If $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is such that $V = \text{Span}(B)$, then B is a basis for V .*

Proof. By The Spanning Set Theorem (Theorem 5.2.12, p. 118), we know that a subset B' of B will be a basis for V . But since $\dim(V) = n$, the size of B' must be n . Therefore, $B' = B$ and B is a basis for V . ■

What is true in Theorem 5.3.15, p. 127 for a spanning set is also true for a linearly independent set. To prove that, however, we first need the analog to The Spanning Set Theorem for linearly independent sets.

Theorem 5.3.16 *Suppose that V is a finite-dimensional vector space and that V' is a linearly independent set of vectors in V . Then there is a basis of V which contains V' .*

Proof. Let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a linearly independent set of vectors in V . If $V = \text{Span}(V')$, then V' is a basis and we are done. If $V \neq \text{Span}(V')$, then there exists some vector $\mathbf{v}_{n+1} \in V - \text{Span}(V')$. By the Linear Dependence Lemma (Theorem 5.1.19, p. 112), the set $V_1 = V' \cup \{\mathbf{v}_{n+1}\}$ is linearly independent.

We can repeat this process. If $V = \text{Span}(V_1)$, we are done; otherwise, we create $V_2 = V_1 \cup \{\mathbf{v}_{n+2}\}$ in the same fashion that we created V_1 . We can continue doing this, adding one vector at a time to this set and maintaining linear independence. Eventually we must reach the point where $V = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}, \dots, \mathbf{v}_{n+k}\}$, since otherwise Lemma 5.3.1, p. 123 would imply that V is infinite-dimensional. ■

We now have the machinery necessary to state the following theorem. The proof will appear in the exercises.

Theorem 5.3.17 *Suppose that V is a vector space with $\dim(V) = n$ and that B is a linearly independent subset of V of size n . Then B is a basis for V .*

The final result of this section collects some facts about dimension and subspaces which we will use in some of the sections that follow.

Theorem 5.3.18 *Suppose that V is a finite-dimensional vector space and that U is a subspace of V . Then the following hold.*

1. *The subspace U is finite-dimensional.*
2. *We have $\dim(U) \leq \dim(V)$.*
3. *We have $\dim(U) = \dim(V)$ if and only if $U = V$.*

Proof. We will prove these facts in order. If the subspace U is $\{\mathbf{0}\}$, then we have nothing to prove. If not, then there is some non-zero vector $\mathbf{u}_1 \in U$. If $U = \text{Span}\{\mathbf{u}_1\}$, we are done; if not, then there exists $\mathbf{u}_2 \in U - \text{Span}\{\mathbf{u}_1\}$. By Theorem 5.1.19, p.112, the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ is linearly independent. We can continue to repeat this process. At each stage we have a linearly independent set $U_k = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$, and this cannot continue indefinitely since U is a subspace of V , which is finite-dimensional. Thus this process must eventually stop when $U = \text{Span}(U_j)$ for some j , and that proves that U is finite dimensional.

The space U is finite dimensional, so it has a basis B . This is a linearly independent set of vectors in V , so Theorem 5.3.16, p.127 says that B can be extended to a basis B' of V . This means that B' will have at least as many vectors in it as B , so $\dim(V) \geq \dim(U)$.

If $U = V$ it is obvious that $\dim(U) = \dim(V)$, so we only need to prove the claim in the other direction. We will prove the contrapositive, so we assume $U \neq V$. Let $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis for U . Since $U \neq V$, there exists a vector $\mathbf{v}_1 \in V - \text{Span}(B)$. By Theorem 5.1.19, p.112 the set $\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{v}_1\}$ is linearly independent in V , implying that $\dim(V) \geq n + 1$. Therefore $\dim(U) \neq \dim(V)$. ■

Example 5.3.19 We can apply this latest result to the vector space $\mathbb{R}^3$. The familiar subspaces of $\mathbb{R}^3$ are *all* of the subspaces of $\mathbb{R}^3$.

1. The only subspace of dimension 0 in $\mathbb{R}^3$ is the zero subspace $\{\mathbf{0}\}$.
2. One-dimensional subspaces of $\mathbb{R}^3$ are lines through the origin. These can all be written as the span of a single (non-zero) vector.
3. Two-dimensional subspaces of $\mathbb{R}^3$ are planes through the origin. These are all spanned by sets of two linearly independent vectors.
4. The only three-dimensional subspace of $\mathbb{R}^3$ is $\mathbb{R}^3$ itself.



5.3.3 Reading Questions

1. Consider the following vectors in $\mathbb{R}^2$:

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

By inspection, why is the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ a basis for $\mathbb{R}^2$? Explain your answer.

2. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, and $\mathbf{v}_4$ be vectors in $\mathbb{R}^3$.

- (a) The set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is not a basis for $\mathbb{R}^3$, and there's a very short argument why. What is that argument?

- (b) Must there be a subset of S which is a basis of $\mathbb{R}^3$? Why or why not?

5.3.4 Exercises

- Find the dimension of the subspace of $\mathbb{R}^3$ consisting of all vectors whose first and third coordinates are equal.
- For each of the following sets of vectors in the given vector space, find the dimension of the subspace spanned by that set of vectors.

- (a) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ in $\mathbb{R}^3$ if

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 5 \\ -3 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 9 \\ -4 \\ 6 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 6.5 \\ -5 \\ 6 \end{bmatrix}$$

- (b) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ in $\mathbb{F}_5^3$ if

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

- For each of the following matrices A , determine the dimensions of $\text{null}(A)$ and $\text{col}(A)$.

- (a) $A \in M_{4,5}(\mathbb{R})$,

$$A = \begin{bmatrix} 1 & -2.5 & 4 & 1 & 2.5 \\ -2.5 & 6 & -9.5 & -4.5 & -8 \\ 5 & 3 & -11 & 3 & -5 \\ 2 & -1 & 0 & -4 & -5 \end{bmatrix}$$

- (b) $A \in M_{2,4}(\mathbb{F}_5)$,

$$A = \begin{bmatrix} 3 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

- Determine whether the following statements are true or false. Justify your answer either way.
 - If a set $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ spans a finite-dimensional space V , and if V' is a set of more than m vectors in V , then V' is linearly dependent.
 - The vector space $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$.
 - A vector space is infinite-dimensional if it is spanned by an infinite set.
- Determine whether the following statements are true or false. Justify your answer either way.
 - If $\dim(V) = m$, then there exists a spanning set of $m + 1$ vectors in V .
 - If every set of m vectors in V fails to span V , then $\dim(V) > m$.
 - If $m \geq 2$ and $\dim(V) = m$, then every set of $m - 1$ non-zero vectors in V is linearly independent.

6. The first four Hermite polynomials are 1 , $2t$, $-2 + 4t^2$, and $-12t + 8t^3$. Show that the set of these polynomials is a basis for P_3 .
7. The first four Laguerre polynomials are 1 , $1 - t$, $2 - 4t + t^2$, and $6 - 18t + 9t^2 - t^3$. Show that the set of these polynomials is a basis for P_3 .

Writing Exercises

8. Let A be a matrix.
 - (a) Prove that $\dim(\text{null}(A))$ is the number of non-pivot columns in A .
 - (b) Prove that $\dim(\text{col}(A))$ is the number of pivot columns of A .
9. Let V be the set of all functions $\mathbb{R} \rightarrow \mathbb{R}$. Prove that V is infinite-dimensional.
10. Suppose that $T : V \rightarrow W$ is a linear transformation between vector spaces and that V is finite-dimensional. Prove that $\dim(\text{range}(T)) \leq \dim(V)$.
11. Prove that $\mathbb{C}^2$ is two-dimensional as a vector space over $\mathbb{C}$ but four-dimensional as a vector space over $\mathbb{R}$.
12. Prove Proposition 5.3.8, p. 124.
13. Prove Proposition 5.3.9, p. 124.
14. Prove Theorem 5.3.17, p. 127.

5.4 Rank and Nullity

In this section we will connect dimension with the subspaces associated with linear transformations (see Section 3.4, p. 83).

5.4.1 Defining Rank and Nullity

We begin by defining the dimension of the image of a linear transformation. (Readers should recall that *image* and *range* have the same meaning, but we prefer to use *image*.)

Definition 5.4.1 Let T be a linear transformation. Then the **rank** of T , denoted $\text{rank}(T)$, is the dimension of the image of T :

$$\text{rank}(T) = \dim(\text{im}(T)).$$

The **rank** of a matrix A is the dimension of the column space of A :

$$\text{rank}(A) = \dim(\text{col}(A)).$$

It may seem strange to define the same word in two ways. However, since the image of T is exactly the column space of A when T is multiplication by A , these two definitions coincide. ◆

When A is an $m \times n$ matrix over $\mathbb{F}$, its rows are vectors in $\mathbb{F}^n$ and its columns are vectors in $\mathbb{F}^m$. This is why the column space of A is a subspace of $\mathbb{F}^m$. We can also examine the analogous space for the rows.

Definition 5.4.2 The set of all linear combinations of the rows of a matrix A is called the **row space** of A . We denote this by $\text{row}(A)$. ◆

Note 5.4.3 Since the rows of A are the columns of A^T , it is immediate that $\text{row}(A) = \text{col}(A^T)$.

With the definition of the row space it is natural to wonder how the sizes of the row and column spaces compare to each other. The following results will help us settle this matter.

Proposition 5.4.4 *If A and B are row equivalent matrices, then we have $\text{row}(A) = \text{row}(B)$. Further, if B is in REF, then the non-zero rows of B form a basis for $\text{row}(B)$ (and $\text{row}(A)$).*

Proof. We will first show that $\text{row}(A) = \text{row}(B)$ as sets. If A is reduced to B , then the rows of B are linear combinations of the rows of A . (The elementary row operations produce linear combinations of the original rows.) Therefore, any linear combination of the rows of B can be written as a linear combination of the rows of A . This proves that $\text{row}(B) \subseteq \text{row}(A)$. Since all row operations are reversible, we can use row operations to produce A from B , and this same argument shows that $\text{row}(A) \subseteq \text{row}(B)$. This proves that $\text{row}(A) = \text{row}(B)$.

If the matrix B is in REF, the nonzero rows are linearly independent because no nonzero row is a linear combination of the rows below it. Here we are applying the Linear Dependence Lemma (Theorem 5.1.19, p. 112) to the nonzero rows from bottom to top. Since the rows of B span $\text{row}(B)$ by definition, the fact that they are linearly independent means that they form a basis for $\text{row}(B)$. ■

Theorem 5.4.5 *Let $A \in M_{m,n}(\mathbb{F})$. Then $\text{rank}(A) = \dim(\text{row}(A))$.*

Proof. If we put A into REF, then Proposition 5.4.4, p. 131 tells us that the number of pivots is $\dim(\text{row}(A))$ since that is the number of vectors in a basis of $\text{row}(A)$. However, the number of pivots in a REF (or the RREF) of A is also $\text{rank}(A)$. (See Exercise 5.3.4.8, p. 130.) This proves that $\text{rank}(A) = \dim(\text{row}(A))$. ■

Note 5.4.6 This theorem says that $\text{rank}(A) = \text{rank}(A^T)$. This theorem also answers the question about the relative sizes of $\text{col}(A)$ and $\text{row}(A)$ —they are the same!

Example 5.4.7 Consider the following matrix $A \in M_{4,5}(\mathbb{F}_5)$:

$$A = \begin{bmatrix} 0 & 2 & 2 & 1 & 2 \\ 4 & 3 & 4 & 1 & 3 \\ 0 & 0 & 3 & 3 & 2 \\ 4 & 2 & 0 & 0 & 0 \end{bmatrix}.$$

We will find a basis for $\text{row}(A)$. Here is the RREF of A :

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 4 & 4 \\ 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

We have used the RREF of a matrix in the past to find bases for the null space and column space of a matrix. Now, we will use it to find a basis for the row space. Proposition 5.4.4, p. 131 tells us that the nonzero rows of this RREF are the basis we seek, therefore a basis for $\text{row}(A)$ is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\begin{aligned} \mathbf{v}_1 &= [1 \ 0 \ 0 \ 4 \ 4] \\ \mathbf{v}_2 &= [0 \ 1 \ 0 \ 2 \ 2] \\ \mathbf{v}_3 &= [0 \ 0 \ 1 \ 1 \ 4]. \end{aligned}$$



We have defined the dimension of the image of a linear transformation T , so we now turn to the kernel.

Definition 5.4.8 If T is a linear transformation, then the **nullity** of T is the dimension of the kernel of T . If A is a matrix, then the **nullity** of A is the dimension of the null space of A . ♦

Note 5.4.9 As we saw with rank, these two uses of “nullity” coincide for the situation when T is multiplication by A .

Some other texts use the notation $\text{null}(A)$ to indicate nullity instead of null space, as we have. We will not introduce any additional notation for the nullity, but we will use $\dim(\ker(T))$ or $\dim(\text{null}(A))$ as appropriate.

5.4.2 The Rank-Nullity Theorem

The following theorem brings together the rank and nullity of a matrix/linear transformation.

Theorem 5.4.10 The Rank-Nullity Theorem. *If $A \in M_{m,n}(\mathbb{F})$, then*

$$\text{rank}(A) + \dim(\text{null}(A)) = n.$$

If $T \in L(V, W)$, then

$$\text{rank}(T) + \dim(\ker(T)) = \dim(V).$$

Proof. We will prove the result for matrices. The proof for linear transformations is a bit more technical. (The reader should note that the result for linear transformations implies the result for matrices!)

If $A \in M_{m,n}(\mathbb{F})$, let B be the RREF of A . Then $\text{rank}(A)$ is the number of pivot columns in B . Further, $\dim(\text{null}(A))$ is the number of non-pivot columns in B . (See Exercise 5.3.4.8, p. 130.) Since each of the n columns of B must be either a pivot or a non-pivot column, and since A and B have the same number of columns, this proves the theorem. ■

Example 5.4.11 If A is a 5×6 matrix with a three-dimensional null space, this theorem tells us that the rank of A is $6 - 3 = 3$.

Let us consider an additional scenario: Could a 6×8 matrix A have a one-dimensional null space? If such a matrix existed, it would have a rank of $8 - 1 = 7$, according to Theorem 5.4.10, p. 132. But the largest rank that a 6×8 matrix can have is 6, since there cannot be more pivots than there are rows. So the answer is no, a 6×8 matrix cannot have a one-dimensional null space. ▲

When the dimensions of the domain and codomain of a linear transformation are equal, some properties of such a transformation coincide.

Corollary 5.4.12 *If $T \in L(V, W)$ and $\dim(V) = \dim(W)$, then the following are equivalent.*

1. *The transformation T is injective.*
2. *The transformation T is surjective.*
3. *The transformation T is an isomorphism.*

Proof. By Theorem 3.4.7, p. 84, T is injective if and only if $\ker(T) = \{\mathbf{0}\}$. In other words, T is injective if and only if $\dim(\ker(T)) = 0$. By Theorem 5.4.10, p. 132, this happens if and only if $\text{rank}(T) = \dim(V)$, and if $\dim(V) = \dim(W)$, $\text{rank}(T) = \dim(W)$ if and only if T is surjective. This proves that T is injective

if and only if T is surjective. Since a bijective linear transformation is an isomorphism, our proof is complete. ■

To close out this section, we present a long theorem with many equivalent statements. We will omit a proof, because the equivalence of most of these statements has been already established at various places in this text. (In other books, this theorem forms the central focus of the text. It is certainly important, but we have chosen a different emphasis.)

Theorem 5.4.13 The Invertible Matrix Theorem. *Let $A \in M_n(\mathbb{F})$. Then the following statements are equivalent.*

1. *The matrix A is invertible.*
2. *The matrix A is row equivalent to I_n .*
3. *The matrix A has n pivots.*
4. *The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.*
5. *The columns of A form a linearly independent set.*
6. *If $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$ is multiplication by A , then T is injective.*
7. *The equation $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b} \in \mathbb{F}^n$.*
8. *The columns of A span $\mathbb{F}^n$.*
9. *The linear transformation T_A is surjective.*
10. *There is an $n \times n$ matrix B such that $AB = BA = I_n$.*
11. *The matrix A^T is invertible.*
12. *We have $\det(A) \neq 0$.*
13. *The columns of A form a basis for $\mathbb{F}^n$.*
14. *We have $\text{col}(A) = \mathbb{F}^n$.*
15. *We have $\text{rank}(A) = n$.*
16. *We have $\text{null}(A) = \{\mathbf{0}\}$.*
17. *We have $\dim(\text{null}(A)) = 0$.*

This theorem ties together threads from almost every section we've covered, which is quite an achievement! The reader should note that this result only applies to *square* matrices.

5.4.3 Reading Questions

1. Consider the following matrix A :

$$A = \begin{bmatrix} 2 & 0 & 2 & 4 & 0 \\ -1 & 1 & -4 & 6 & -7 \\ 6 & 3 & -3 & 2 & 13 \end{bmatrix}.$$

Find a basis for $\text{row}(A)$. Explain your answer.

2. Suppose that $T : V \rightarrow W$ is a linear transformation and that $\dim(V) = 4$ and $\dim(W) = 5$. What are the possible values for $\dim(\ker(T))$? Explain.

5.4.4 Exercises

1. Find the rank and nullity of each of the following matrices.

(a) $A \in M_{3,4}(\mathbb{R})$,

$$A = \begin{bmatrix} 2 & -3 & 4 & 1 \\ -1 & 2 & -3 & 1 \\ 4 & -3 & 2 & 11 \end{bmatrix}$$

(b) $A \in M_{4,5}(\mathbb{F}_3)$,

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 0 \\ 1 & 2 & 2 & 1 & 1 \\ 2 & 1 & 2 & 0 & 1 \\ 2 & 1 & 2 & 0 & 2 \end{bmatrix}$$

2. Let $D : P_n \rightarrow P_{n-1}$ be the differentiation linear transformation. Calculate $\text{rank}(D)$ and $\dim(\ker(D))$.
- 3.
- (a) If a 5×3 matrix A has rank 3, find $\dim(\text{null}(A))$, $\dim(\text{row}(A))$, and $\text{rank}(A^T)$.
- (b) If the null space of a 7×6 matrix A is 5-dimensional, what is the dimension of the column space of A ?
- (c) If A is a 7×9 matrix, what is the smallest possible dimension of $\text{null}(A)$?
4. Suppose a nonhomogeneous linear system of nine equations and ten variables has a solution for all possible constants on the right side of the equations. Is it possible to find two nonzero solutions of the associated homogeneous system that are not multiples of each other? Explain.
5. Suppose $A \in M_{m,n}(\mathbb{F})$ and $\mathbf{b} \in \mathbb{F}^m$. What has to be true about the two numbers $\text{rank}([A \ \mathbf{b}])$ and $\text{rank}(A)$ in order for the equation $A\mathbf{x} = \mathbf{b}$ to be consistent? Explain.

Writing Exercises

6. If A and B are matrices, prove that $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
7. Suppose that $T \in L(V, W)$ and that V and W are both finite-dimensional.
- (a) Prove that T is surjective if and only if $\text{rank}(T) = \dim(W)$.
- (b) Prove that T is injective if and only if $\text{rank}(T) = \dim(V)$.
8. Suppose that $A\mathbf{x} = \mathbf{b}$ is a 6×6 linear system which is consistent but which does *not* have a unique solution. Prove that there must be a vector $\mathbf{c} \in \mathbb{F}^6$ such that the system $A\mathbf{x} = \mathbf{c}$ is inconsistent.
9. Prove that if $T \in L(V, W)$, then $\text{rank}(T) \leq \min\{\dim(V), \dim(W)\}$.
10. Prove that if $A \in M_{m,n}(\mathbb{F})$, then $\text{rank}(A) \leq \min\{m, n\}$.
11. Let $T \in L(V, W)$.
- (a) Prove that if $\dim(V) > \dim(W)$, then T cannot be injective.
- (b) Prove that if $\dim(W) > \dim(V)$, then T cannot be surjective.

5.5 Coordinates

We have recently shown (Corollary 5.3.13, p.127) that all n -dimensional vector spaces over $\mathbb{F}$ are isomorphic to $\mathbb{F}^n$. In this section we explore the vast implications of this isomorphism.

5.5.1 Coordinates of Vectors

If V is a finite-dimensional vector space over $\mathbb{F}$, then it has a basis $\mathcal{B}$. We have seen (Theorem 5.2.11, p.118) that each vector in V then has a *unique* representation as a linear combination of these basis vectors. In the definition that follows, we focus on the coefficients in these linear combinations.

Definition 5.5.1 The **coordinates** of a vector $\mathbf{v} \in V$ with respect to a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are the unique scalars $c_1, \dots, c_n$ such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n.$$

The **coordinate vector of $\mathbf{v}$ with respect to $\mathcal{B}$** is the vector $[\mathbf{v}]_{\mathcal{B}} \in \mathbb{F}^n$,

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

◆

Note 5.5.2 When the basis $\mathcal{B}$ we are using is unambiguous, we may drop a bit of the cumbersome terminology contained in the phrase “coordinate vector of $\mathbf{v}$ with respect to $\mathcal{B}$ ” and simply refer to the “coordinate vector of $\mathbf{v}$.”

This process of assigning to a vector $\mathbf{v} \in V$ a vector $[\mathbf{v}]_{\mathcal{B}} \in \mathbb{F}^n$ is sometimes called a **coordinate mapping**, and it defines a function $V \rightarrow \mathbb{F}^n$. This function is actually an isomorphism of vector spaces.

Theorem 5.5.3 Let V be an n -dimensional vector space over $\mathbb{F}$, and let $\mathcal{B}$ be a basis for V . Consider the coordinate map $C_{\mathcal{B}} : V \rightarrow \mathbb{F}^n$ given by

$$C_{\mathcal{B}}(\mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}.$$

Then $C_{\mathcal{B}}$ is an isomorphism.

Proof. The function $C_{\mathcal{B}}$ is a linear transformation. (We ask the reader to verify this in the exercises.) We note that $C_{\mathcal{B}}$ maps the basis vectors in $\mathcal{B}$ to the standard basis in $\mathbb{F}^n$. So, by Theorem 5.3.11, p.125, $C_{\mathcal{B}}$ is an isomorphism. ■

The existence of coordinate vectors means that just about everything for finite-dimensional vector spaces can be accomplished with vectors and matrices over $\mathbb{F}$. We explore this in the following examples.

Example 5.5.4 Let $\mathcal{B} = \{1, t, t^2\}$ be the standard basis of the vector space P_2 . If p_1 and p_2 are

$$p_1 = 2 - t + 4t^2 \quad \text{and} \quad p_2 = -3t^2 + 10,$$

then the coordinate vectors of p_1 and p_2 are

$$[p_1]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \quad \text{and} \quad [p_2]_{\mathcal{B}} = \begin{bmatrix} 10 \\ 0 \\ -3 \end{bmatrix}.$$

Note that the order of the coordinates really matters, so in this case the terms in p_2 had to be reordered (in increasing powers of t) before the coefficients were entered as the coordinate vector. $\blacktriangle$

Example 5.5.5 Within $\mathbb{F}_5^3$, consider $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}.$$

Since neither of these vectors is a scalar multiple of the other, $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly independent set and therefore a basis for W . If we let $\mathbf{v}_3$ be

$$\mathbf{v}_3 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix},$$

we can verify that $\mathbf{v}_3 \in W$ by row-reducing the appropriate matrix:

$$\begin{bmatrix} 2 & 1 & 0 \\ 3 & 0 & 3 \\ 1 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since there is no pivot in the final column, we see that $\mathbf{v}_3 \in W$. Further, we can write down the coordinate vector of $\mathbf{v}_3$ with respect to $\mathcal{B}$ by studying this row-reduced matrix. We see that

$$[\mathbf{v}_3]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

It may seem strange for a vector in the three-dimensional space $\mathbb{F}_5^3$ to have a coordinate vector with only two entries, but this is due to the fact that W is two-dimensional. (It has a basis of only two vectors!) The coordinate mapping in this case says that W is isomorphic to $\mathbb{F}_5^2$, and this is why the coordinate vector for any vector in W has only two entries. $\blacktriangle$

There are some consequences of Theorem 5.5.3, p. 135 that we want to spell out explicitly because of their usefulness. The proof of the following proposition can be found as part of the proof of Theorem 5.3.11, p. 125.

Proposition 5.5.6 *Let V be a vector space of dimension m over $\mathbb{F}$. Since the coordinate mapping $C_{\mathcal{B}} : V \rightarrow \mathbb{F}^m$ is an isomorphism, then the following statements are true.*

1. *A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in V is linearly independent if and only if the set of coordinate vectors $\{C_{\mathcal{B}}(\mathbf{v}_1), \dots, C_{\mathcal{B}}(\mathbf{v}_n)\}$ is linearly independent in $\mathbb{F}^m$.*
2. *A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans V if and only if the set of coordinate vectors $\{C_{\mathcal{B}}(\mathbf{v}_1), \dots, C_{\mathcal{B}}(\mathbf{v}_n)\}$ spans $\mathbb{F}^m$.*

Hopefully the reader can now see exactly how helpful the coordinate mapping isomorphism is. The following example should help to connect the dots.

Example 5.5.7 Consider the set of vectors $Y = \{p_1, p_2, p_3\}$ in P_3 , where

$$\begin{aligned} p_1 &= 1 - t - 3t^2 + 2t^3 \\ p_2 &= -5 + 4t + 2t^2 - t^3 \\ p_3 &= 1 + 3t + 4t^2 - 3t^3. \end{aligned}$$

With respect to the standard basis $\mathcal{B}$ of P_3 , these are the coordinate vectors:

$$[p_1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ -1 \\ -3 \\ 2 \end{bmatrix}, \quad [p_2]_{\mathcal{B}} = \begin{bmatrix} -5 \\ 4 \\ 2 \\ -1 \end{bmatrix}, \quad [p_3]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ -3 \end{bmatrix}.$$

By row-reducing the matrix which has these coordinate vectors as its columns, we can see that the set of coordinate vectors $\{[p_1]_{\mathcal{B}}, [p_2]_{\mathcal{B}}, [p_3]_{\mathcal{B}}\}$ is linearly independent in $\mathbb{R}^4$:

$$\begin{bmatrix} 1 & -5 & 1 \\ -1 & 4 & 3 \\ -3 & 2 & 4 \\ 2 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Using Proposition 5.5.6, p. 136, we conclude that the set Y is linearly independent in P_3 .

For dimension reasons, we already knew that the set Y cannot span P_3 , however this row-reduced matrix confirms it. Since there is not a pivot in each row, the set of coordinate vectors does not span $\mathbb{R}^4$, and this means that Y does not span P_3 . $\blacktriangle$

5.5.2 Coordinates and Linear Transformations

Back in Section 3.2, p. 68, we showed how every linear transformation $\mathbb{F}^n \rightarrow \mathbb{F}^m$ could be realized as multiplication by a matrix over $\mathbb{F}$. We now bring that understanding into contact with coordinate vectors. While not every linear transformation between vector spaces is multiplication by a matrix, every linear transformation between finite-dimensional vector spaces can be represented as multiplication by a matrix when considering the relevant coordinate vectors.

Definition 5.5.8 Let V and W be n - and m -dimensional vector spaces over $\mathbb{F}$, respectively, and let $T : V \rightarrow W$ be a linear transformation. Further, suppose that $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V and $\mathcal{C} = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ is a basis for W . If, for each j , $1 \leq j \leq n$, we have $a_{1j}, \dots, a_{mj}$ as the coordinates of $T(\mathbf{v}_j)$ with respect to $\mathcal{C}$, then the **matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$** is the matrix $A = [a_{ij}]$. (In other words, column j of this matrix is the coordinate vector $[T(\mathbf{v}_j)]_{\mathcal{C}}$.) We denote this matrix as $[T]_{\mathcal{B}, \mathcal{C}}$.

When $V = W$ and $\mathcal{B} = \mathcal{C}$, then we use the notation $[T]_{\mathcal{B}}$ and refer to the **matrix of T with respect to $\mathcal{B}$** .

Finally, when the basis/bases we are using are unambiguous, we may refer to $[T]_{\mathcal{B}}$ or $[T]_{\mathcal{B}, \mathcal{C}}$ as the **coordinate matrix** of T . $\blacklozenge$

The point of this rather long (and cumbersome!) definition is that we can represent a linear transformation T as multiplication by a matrix. That's what the following proposition shows.

Proposition 5.5.9 Let $T : V \rightarrow W$ be a linear transformation between finite-dimensional vector spaces. Suppose that V and W have bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $A = [T]_{\mathcal{B}, \mathcal{C}}$. Then, for any $\mathbf{v} \in V$,

$$[T(\mathbf{v})]_{\mathcal{C}} = A[\mathbf{v}]_{\mathcal{B}}.$$

Proof. Let the bases $\mathcal{B}$ and $\mathcal{C}$ be $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\mathcal{C} = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$. For $\mathbf{v} \in V$, suppose that

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n,$$

or, in other words,

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

We also assume that, for each j , $1 \leq j \leq n$, the coordinates of $T(\mathbf{v}_j)$ with respect to $\mathcal{C}$ are $a_{1j}, \dots, a_{mj}$.

Then, using the linearity of T , we have

$$T(\mathbf{v}) = \sum_{j=1}^n c_j T(\mathbf{v}_j) = \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \mathbf{w}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \mathbf{w}_i.$$

This says that the i th coordinate of $T(\mathbf{v})$ with respect to $\mathcal{C}$ is $\sum a_{ij} c_j$, which is the same as the i th entry of $A[\mathbf{v}]_{\mathcal{B}}$. ■

Note 5.5.10 According to this proposition, here is the way to realize a linear transformation as a matrix. Form $[T]_{\mathcal{B}, \mathcal{C}}$ by calculating the coordinate vector $[T(\mathbf{v}_j)]_{\mathcal{C}}$ for every vector $\mathbf{v}_j \in \mathcal{B}$. Then, to use this matrix to determine what happens to a vector $\mathbf{v} \in V$, find the coordinate vector $[\mathbf{v}]_{\mathcal{B}}$. After multiplying this vector by $[T]_{\mathcal{B}, \mathcal{C}}$, the result will be the coordinate vector $[T(\mathbf{v})]_{\mathcal{C}}$. In order to recover the value of $T(\mathbf{v})$, use the basis vectors in $\mathcal{C}$ and this coordinate vector to find the correct linear combination.

Note 5.5.11 There are several linear transformations involved in Proposition 5.5.9, p. 137—the transformation T , multiplication by the matrix A , and the two coordinate mappings. Because it is easy to confuse the roles of these transformations, mathematicians frequently employ a tool called a *commutative diagram* to keep their ideas and symbols organized.

We will produce here the diagram that summarizes the conclusion of Proposition 5.5.9, p. 137.

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ M_{\mathcal{B}} \downarrow & & \downarrow M_{\mathcal{C}} \\ \mathbb{F}^n & \xrightarrow{T_A} & \mathbb{F}^m \end{array}$$

Figure 5.5.12 A commutative diagram illustrating the relationship between a linear transformation and multiplication by a matrix.

We have labeled the coordinate mappings from V to $\mathbb{F}^n$ and from W to $\mathbb{F}^m$ by $M_{\mathcal{B}}$ and $M_{\mathcal{C}}$, respectively. Further, we have labeled the linear transformation which is multiplication by A , from $\mathbb{F}^n$ to $\mathbb{F}^m$, with T_A .

The proposition claims that this diagram *commutes*. In other words, for any vector $\mathbf{v} \in V$, going around the rectangle in either of the possible directions (right then down or down then right) will produce the same result. This would mean that $M_{\mathcal{C}}(T(\mathbf{v})) = T_A(M_{\mathcal{B}}(\mathbf{v}))$. With different notation, this is exactly the conclusion of Proposition 5.5.9, p. 137.

Example 5.5.13 Let $D : P_3 \rightarrow P_2$ be the differentiation function. (We proved that a very similar function was a linear transformation in Example 3.1.3, p. 58.) Let $\mathcal{B}$ be the standard basis for P_3 , and let $\mathcal{C}$ be the standard basis for P_2 . Here we calculate the coordinate vectors for the derivative of each of the

polynomials in $\mathcal{B}$:

$$[D(1)]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad [D(t)]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [D(t^2)]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad [D(t^3)]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}.$$

These coordinate vectors form the columns of the matrix $[D]_{\mathcal{B},\mathcal{C}}$.

We will now use this matrix to carry out the action of D . Let's take the derivative of $p = -2 - 4t - t^2 - t^3$. Since the coordinate vector of p with respect to $\mathcal{B}$ is

$$[p]_{\mathcal{B}} = \begin{bmatrix} -2 \\ -4 \\ -1 \\ -1 \end{bmatrix},$$

we can multiply this vector by $[D]_{\mathcal{B},\mathcal{C}}$ to get $[D(p)]_{\mathcal{C}}$:

$$[D(p)]_{\mathcal{C}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -4 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -3 \end{bmatrix}.$$

This tells us that the coordinates for $D(p)$ with respect to $\mathcal{C}$ are -4 , -2 , and -3 . In other words,

$$D(p) = -4(1) - 2(t) - 3(t^2),$$

and this matches what we know to be the derivative of p . ▲

Example 5.5.14 We consider a linear transformation $T : \mathbb{R}^3 \rightarrow P_2$ defined by

$$T \left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = (a + 2b) + (-3a + 4b - c)t + (2a - 4c)t^2.$$

We let $\mathcal{B}$ be the standard basis for $\mathbb{R}^3$ and $\mathcal{C}$ be the standard basis for P_2 . We now write $[T(\mathbf{e}_i)]_{\mathcal{C}}$ for each $\mathbf{e}_i \in \mathcal{B}$:

$$[T(\mathbf{e}_1)]_{\mathcal{C}} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \quad [T(\mathbf{e}_2)]_{\mathcal{C}} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}, \quad [T(\mathbf{e}_3)]_{\mathcal{C}} = \begin{bmatrix} 0 \\ -1 \\ -4 \end{bmatrix}.$$

These coordinate vectors make up the columns of the matrix $[T]_{\mathcal{B},\mathcal{C}}$. If we wanted to calculate $T(\mathbf{v})$, where

$$\mathbf{v} = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix},$$

we could do so using coordinate vectors and the matrix $[T]_{\mathcal{B},\mathcal{C}}$. Since the coordinate vector of $\mathbf{v}$ with respect to $\mathcal{B}$ is fairly obvious—it is $\mathbf{v}$ itself—we can proceed with this calculation:

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{B},\mathcal{C}}[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 1 & 2 & 0 \\ -3 & 4 & -1 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ -19 \\ -14 \end{bmatrix}.$$

This tells us that $T(\mathbf{v}) = -5 - 19t - 14t^2$. ▲

We will end this section with two results related to coordinate matrices.

This first result says that the composition of linear transformation matches the multiplication of matrices in the expected way.

Theorem 5.5.15 *Let U , V , and W be finite-dimensional vector spaces with bases $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$, respectively. Suppose that $T \in L(U, V)$ and that $S \in L(V, W)$. Then*

$$[ST]_{\mathcal{B}, \mathcal{D}} = [S]_{\mathcal{C}, \mathcal{D}}[T]_{\mathcal{B}, \mathcal{C}}.$$

Proof. We assume that $\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$, $\mathcal{C} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, and $\mathcal{D} = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$. Further, we let $A = [S]_{\mathcal{C}, \mathcal{D}}$, $B = [T]_{\mathcal{B}, \mathcal{C}}$, and $C = [ST]_{\mathcal{B}, \mathcal{D}}$. We want to show that $AB = C$.

The definition of A tells us that, for each $k = 1, \dots, n$, the k th column of A is the coordinate vector of $S(\mathbf{v}_k)$ with respect to $\mathcal{D}$. So

$$S(\mathbf{v}_k) = \sum_{i=1}^m a_{ik} \mathbf{w}_i.$$

Also, for each $j = 1, \dots, p$,

$$T(\mathbf{u}_j) = \sum_{k=1}^n b_{kj} \mathbf{v}_k.$$

Using the linearity of these transformations, we have

$$\begin{aligned} (ST)(\mathbf{u}_j) &= \sum_{k=1}^n b_{kj} S(\mathbf{v}_k) \\ &= \sum_{k=1}^n b_{kj} \left(\sum_{i=1}^m a_{ik} \mathbf{w}_i \right) \\ &= \sum_{i=1}^m \left(\sum_{k=1}^n a_{ik} b_{kj} \right) \mathbf{w}_i. \end{aligned}$$

This means that

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

for all i and j . Since this is how the (i, j) -entry of the matrix product AB is formed, this proves that $C = AB$, as desired.² ■

This final result states that the invertibility of a linear transformation and the invertibility of its coordinate matrix are tied together in the predictable way.

Corollary 5.5.16 *Let V and W be finite-dimensional vector spaces with bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Suppose that $T \in L(V, W)$ and that $A = [T]_{\mathcal{B}, \mathcal{C}}$. Then A is invertible if and only if T is invertible, and in that case, $[T^{-1}]_{\mathcal{C}, \mathcal{B}} = A^{-1}$.*

5.5.3 Reading Questions

1. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

- (a) The set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^2$. Without doing any calculations, explain why this is so. (I'm not looking for the definition

²This proof was taken from *Linear Algebra* (2018), Meckes and Meckes, pages 193-194.

of a basis, I want an explanation as to why this set satisfies that definition.)

(b) Let $\mathbf{w} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$. What is the coordinate vector of $\mathbf{w}$ with respect to $\mathcal{B}$?

2. Let $T : P_2 \rightarrow P_2$ be the following function:

$$T(p) = p(-1) + p(0)t + p(1)t^2.$$

Let $\mathcal{B}$ be the standard basis for P_2 .

(a) Find the coordinate matrix $[T]_{\mathcal{B}}$ for T .

(b) Use this coordinate matrix to calculate $T(q)$, if

$$q = -3 - 5t + 3t^2.$$

5.5.4 Exercises

1. For the given basis $\mathcal{B}$ of $\mathbb{R}^2$ and the given coordinate vector $[\mathbf{v}]_{\mathcal{B}}$, find $\mathbf{v}$.

(a) $[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$

(b) $[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \end{bmatrix} \right\}$

2. For the basis $\mathcal{B} = \{p_1, p_2, p_3\}$ of P_2 and the coordinate vector $[p]_{\mathcal{B}}$, find p if

$$p_1 = 2 - 4t^2, \quad p_2 = -1 - t, \quad p_3 = 3t + 2t^2$$

and

$$[p]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}.$$

3. Find the coordinate vectors $[\mathbf{v}]_{\mathcal{B}}$ for each of the following vectors $\mathbf{v}$ with respect to the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of $\mathbb{R}^3$, if

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ 5 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ 4 \\ -4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ -7 \\ 2 \end{bmatrix}.$$

(a) $\mathbf{v} = \begin{bmatrix} 16 \\ 5 \\ -16 \end{bmatrix}$

(b) $\mathbf{v} = \begin{bmatrix} 6 \\ -23 \\ 11 \end{bmatrix}$

4. Find the coordinate vectors $[p]_{\mathcal{B}}$ for each of the following polynomials p with respect to the basis $\mathcal{B} = \{p_1, p_2, p_3\}$ of P_2 , if

$$p_1 = 8 + 4t - 4t^2, \quad p_2 = 5 + 8t + 3t^2, \quad p_3 = -6 - 2t - 5t^2.$$

(a) $p = -2 + 2t - 23t^2$

(b) $p = 23 + 28t + 5t^2$

5. Use coordinate vectors to test the linear independence of the following sets of polynomials in P_3 .

(a) $\{p_1, p_2, p_3\}$ if

$$\begin{aligned} p_1 &= -6 + 7t + 6t^2 + 3t^3 \\ p_2 &= 2t - 4t^2 + 7t^3 \\ p_3 &= 2 + 6t - t^2 - 5t^3 \end{aligned}$$

(b) $\{p_1, p_2, p_3\}$ if

$$\begin{aligned} p_1 &= 6 + 7t - t^2 - 2t^3 \\ p_2 &= -5 - 7t - 6t^2 + 8t^3 \\ p_3 &= 7 + 7t - 8t^2 + 4t^3 \end{aligned}$$

6. Use coordinate vectors to test whether the following sets of vectors span P_2 .

(a) $\{p_1, p_2, p_3, p_4\}$ if

$$\begin{aligned} p_1 &= -4 + t + t^2 \\ p_2 &= 3 + 5t + t^2 \\ p_3 &= -2 - 4t + 2t^2 \\ p_4 &= 2 - 4t - t^2 \end{aligned}$$

(b) $\{p_1, p_2, p_3, p_4\}$ if

$$\begin{aligned} p_1 &= 4 + 6t + 5t^2 \\ p_2 &= -3t^2 \\ p_3 &= 4 + 6t - 4t^2 \\ p_4 &= 8 + 12t + t^2 \end{aligned}$$

7. Let $T : P_2 \rightarrow \mathbb{R}^2$ be the linear transformation

$$T(p) = \begin{bmatrix} p(0) + p(1) \\ p(1) - p(2) \end{bmatrix}.$$

Let $\mathcal{B}$ be the standard basis for P_2 and let $\mathcal{E}$ be the standard basis for $\mathbb{R}^2$.

(a) Find the coordinate matrix $[T]_{\mathcal{B}, \mathcal{E}}$.

(b) Use this coordinate matrix to calculate $T(-10 + 3t^2)$.

8. Let $T : \mathbb{R}^3 \rightarrow P_2$ be the linear transformation

$$T \left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = (2a - b) + (b - 3c)t + (a - b + c)t^2.$$

Let $\mathcal{E}$ be the standard basis for $\mathbb{R}^3$ and let $\mathcal{B}$ be the standard basis for P_2 .

(a) Find the coordinate matrix $[T]_{\mathcal{E}, \mathcal{B}}$.

- (b) Use this coordinate matrix to calculate $T(\mathbf{v})$ for

$$\mathbf{v} = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}.$$

9. Let $T : P_2 \rightarrow P_2$ be the linear transformation

$$T(p) = p' + p(1)t^2.$$

Let $\mathcal{B}$ be the standard basis for P_2 .

- (a) Choose a basis $\mathcal{C}$ for P_2 which is *not* the standard basis. Prove that your set of polynomials is a basis.
 - (b) Find the coordinate matrix $[T]_{\mathcal{C}, \mathcal{B}}$.
 - (c) Use this coordinate matrix to calculate $T(2 + t - 4t^2)$.
10. Let $D : P_3 \rightarrow P_2$ be the derivative and let $T : P_2 \rightarrow P_3$ be the linear transformation which is multiplication by t . Let $\mathcal{B}$ be the standard basis for P_2 and let $\mathcal{C}$ be the standard basis for P_3 .
- (a) Find the coordinate matrix $[T]_{\mathcal{B}, \mathcal{C}}$.
 - (b) Find the coordinate matrix $[DT]_{\mathcal{B}}$.
 - (c) Find the coordinate matrix $[TD]_{\mathcal{C}}$.
11. Consider the plane P in $\mathbb{R}^3$ defined by $x - 2y + 3z = 0$.

- (a) Find a basis for P .
- (b) Determine whether each of the following vectors is in P , and for each one that is, find its coordinate vector in terms of the basis you gave in part a.
 - i. $\mathbf{v}_1 = (1, -1, -1)$
 - ii. $\mathbf{v}_2 = (2, 3, 1)$
 - iii. $\mathbf{v}_3 = (5, -2, -3)$

Writing Exercises

12. Prove that the coordinate mapping in Theorem 5.5.3, p.135 is a linear transformation.
13. Without using Theorem 5.3.11, p. 125, prove that the coordinate mapping in Theorem 5.5.3, p. 135 is injective.
14. Without using Theorem 5.3.11, p. 125, prove that the coordinate mapping in Theorem 5.5.3, p. 135 is surjective.
15. Let $T : V \rightarrow W$ be a linear transformation between finite-dimensional vector spaces, and let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W , respectively. Prove that $\text{rank}(T)$ (the rank of T as a linear transformation) is the same as $\text{rank}([T]_{\mathcal{B}, \mathcal{C}})$ (the rank of the coordinate matrix of T).
16. Prove Corollary 5.5.16, p. 140.

5.6 Change of Basis

Every basis for a vector space gives a different angle on that space—we get a different coordinate system for each basis. Since any finite-dimensional vector space has many bases, in this section we explain how to move between bases.

5.6.1 The Change-of-Basis Matrix

We will first describe a situation in which this technique will be useful. Consider the following two bases for $\mathbb{R}^2$: $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad \mathbf{w}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

We can verify that $\mathcal{B}$ and $\mathcal{C}$ are bases for $\mathbb{R}^2$ since they are both linearly independent sets of two vectors in a two-dimensional space.

If we have a vector $\mathbf{v} \in \mathbb{R}^2$, it is straightforward to calculate both $[\mathbf{v}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{C}}$. The question for us is this: How do these two coordinate vectors relate to each other? Specifically, how might we calculate one coordinate vector from the other one?

It turns out that we already have the necessary machinery for this calculation. We summarize the process in the following proposition.

Proposition 5.6.1 *Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for a finite-dimensional vector space V . Then, for any $\mathbf{v} \in V$, we have*

$$[\mathbf{v}]_{\mathcal{C}} = [I]_{\mathcal{B},\mathcal{C}}[\mathbf{v}]_{\mathcal{B}},$$

where $I : V \rightarrow V$ is the identity transformation.

Proof. This is a simple application of Proposition 5.5.9, p. 137 to the identity transformation I :

$$[\mathbf{v}]_{\mathcal{C}} = [I(\mathbf{v})]_{\mathcal{C}} = [I]_{\mathcal{B},\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}.$$

■

Definition 5.6.2 If $\mathcal{B}$ and $\mathcal{C}$ are two bases for a finite-dimensional vector space V , then the matrix $P_{\mathcal{B},\mathcal{C}} = [I]_{\mathcal{B},\mathcal{C}}$ is called the **change-of-basis matrix** from $\mathcal{B}$ to $\mathcal{C}$. ♦

Example 5.6.3 We will continue the example begun earlier in this section. If $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2\}$, then we can calculate $P_{\mathcal{B},\mathcal{C}}$ by determining the coordinate vectors $[\mathbf{v}_1]_{\mathcal{C}}$ and $[\mathbf{v}_2]_{\mathcal{C}}$. We need only to row-reduce two matrices:

$$\begin{bmatrix} -2 & -1 & 1 \\ 1 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2/5 \\ 0 & 1 & -1/5 \end{bmatrix}, \quad \begin{bmatrix} -2 & -1 & 0 \\ 1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2/5 \\ 0 & 1 & 4/5 \end{bmatrix}.$$

From these calculations, we can see how to write the $\mathcal{B}$ -basis vectors in terms of the vectors in $\mathcal{C}$, and these form the columns of our change-of-basis matrix:

$$P_{\mathcal{B},\mathcal{C}} = \begin{bmatrix} -2/5 & -2/5 \\ -1/5 & 4/5 \end{bmatrix}.$$

We now consider a vector $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ in $\mathbb{R}^2$. We can calculate $[\mathbf{v}]_{\mathcal{B}}$ in this way:

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5/2 \end{bmatrix}.$$

Now that we have $[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 5/2 \end{bmatrix}$, we can use the change-of-basis matrix to find $[\mathbf{v}]_{\mathcal{C}}$:

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{B},\mathcal{C}}[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -2/5 & -2/5 \\ -1/5 & 4/5 \end{bmatrix} \begin{bmatrix} 2 \\ 5/2 \end{bmatrix} = \begin{bmatrix} -9/5 \\ 8/5 \end{bmatrix}.$$

We can verify that this is the correct $\mathcal{C}$ -coordinate vector for $\mathbf{v}$ by calculating it directly:

$$\begin{bmatrix} -2 & -1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -9/5 \\ 0 & 1 & 8/5 \end{bmatrix}.$$

▲

Example 5.6.4 We consider the vector space P_2 . Let $\mathcal{C}$ be the standard basis for P_2 and let $\mathcal{B}$ be the set $\{p_1, p_2, p_3\}$, where

$$p_1 = 2 + t, \quad p_2 = -1 - t + 2t^2, \quad p_3 = -2t + 3t^2.$$

In order to find the change-of-basis matrix, we need to write the coordinate vectors of the basis vectors of $\mathcal{B}$ with respect to $\mathcal{C}$. But since $\mathcal{C}$ is the standard basis of P_2 , this is an easy task to complete. Here is the change-of-basis matrix:

$$P_{\mathcal{B},\mathcal{C}} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & -2 \\ 0 & 2 & 3 \end{bmatrix}.$$

▲

What we saw in Example 5.6.4, p. 145 is an indication that some change-of-basis matrices are easier to calculate than others. In particular, when the standard basis is the *target* (not the *source*) basis, the result is almost immediate.

Lemma 5.6.5 Let $\mathcal{E}$ be the standard basis of $\mathbb{F}^n$, and let $\mathcal{B}$ be any other basis of $\mathbb{F}^n$. Then the columns of $P_{\mathcal{B},\mathcal{E}}$ are the vectors of $\mathcal{B}$, in order.

Proof. If $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, then column j of $P_{\mathcal{B},\mathcal{E}}$ is $[\mathbf{v}_j]_{\mathcal{E}}$. But since $\mathcal{E}$ is the standard basis, then $[\mathbf{v}_j]_{\mathcal{E}} = \mathbf{v}_j$. ■

The next lemma also shows that the change-of-basis matrices from one basis to another and back again have the inverse relationship we might expect.

Lemma 5.6.6 Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for a finite-dimensional vector space V . Then the relationship between the two change-of-basis matrices is

$$P_{\mathcal{C},\mathcal{B}} = (P_{\mathcal{B},\mathcal{C}})^{-1}.$$

Proof. Since $P_{\mathcal{B},\mathcal{C}} = [I]_{\mathcal{B},\mathcal{C}}$, by Theorem 5.5.15, p. 140 we have

$$P_{\mathcal{B},\mathcal{C}}[I]_{\mathcal{C},\mathcal{B}} = [I]_{\mathcal{B},\mathcal{C}}[I]_{\mathcal{C},\mathcal{B}} = [I]_{\mathcal{C},\mathcal{C}} = I_n.$$

Since $P_{\mathcal{B},\mathcal{C}}$ is square, this proves that $[I]_{\mathcal{C},\mathcal{B}} = (P_{\mathcal{B},\mathcal{C}})^{-1}$. ■

Example 5.6.7 We consider two bases for $\mathbb{F}_5^3$: the standard basis $\mathcal{E}$ and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}.$$

(The reader should verify that $\mathcal{B}$ is a basis for $\mathbb{F}_5^3$.)

Lemma 5.6.5, p. 145 tells us that the change-of-basis matrix $P_{\mathcal{B},\mathcal{E}}$ is easy to

write down:

$$P_{\mathcal{B},\mathcal{E}} = \begin{bmatrix} 4 & 3 & 0 \\ 2 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}.$$

Then Lemma 5.6.6, p. 145 says that $P_{\mathcal{E},\mathcal{B}} = (P_{\mathcal{B},\mathcal{E}})^{-1}$, so we can find that matrix without too much difficulty as well:

$$P_{\mathcal{E},\mathcal{B}} = \begin{bmatrix} 3 & 1 & 1 \\ 3 & 2 & 2 \\ 4 & 3 & 4 \end{bmatrix}.$$

▲
The final results of this section deal with linear transformations. This theorem relates the coordinate matrix for a linear transformation to the situation in which we want to change bases in the domain and codomain.

Theorem 5.6.8 *Let V and W be finite-dimensional vector spaces, and let $T \in L(V, W)$. Additionally, let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for V , and let $\mathcal{C}_1$ and $\mathcal{C}_2$ be bases for W . Then*

$$[T]_{\mathcal{B}_2, \mathcal{C}_2} = [I]_{\mathcal{C}_1, \mathcal{C}_2} [T]_{\mathcal{B}_1, \mathcal{C}_1} [I]_{\mathcal{B}_2, \mathcal{B}_1}.$$

Proof. We will use Theorem 5.5.15, p. 140:

$$[T]_{\mathcal{B}_2, \mathcal{C}_2} = [ITI]_{\mathcal{B}_2, \mathcal{C}_2} = [I]_{\mathcal{C}_1, \mathcal{C}_2} [T]_{\mathcal{B}_1, \mathcal{C}_1} [I]_{\mathcal{B}_2, \mathcal{B}_1}.$$

■
The most important (and most common) use of this theorem happens when $V = W$, $\mathcal{B}_1 = \mathcal{C}_1$, and $\mathcal{B}_2 = \mathcal{C}_2$.

Corollary 5.6.9 *Let $\mathcal{B}$ and $\mathcal{C}$ be bases for a finite-dimensional vector space V , and let $T \in L(V)$. Then*

$$[T]_{\mathcal{C}} = (P_{\mathcal{C},\mathcal{B}})^{-1} [T]_{\mathcal{B}} P_{\mathcal{C},\mathcal{B}}.$$

Proof. This result is due to Theorem 5.6.8, p. 146 and Lemma 5.6.6, p. 145. ■

We will end this section with an example that takes advantage of Corollary 5.6.9, p. 146.

Example 5.6.10 We consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which is reflection across the line $y = \frac{1}{2}x$. While the action of T is not impossible to write down in the usual coordinate system, it is even easier using the alternate basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

To see why this linear transformation is easier to describe in the $\mathcal{B}$ -coordinates, we recall how easy reflection across the y -axis is to describe relative to the standard basis—simply negate the first coordinate! The $\mathcal{B}$ -basis vectors in this case lie on the axis of reflection and along the line perpendicular to that axis.

We note that $T(\mathbf{v}_1) = \mathbf{v}_1$ and that $T(\mathbf{v}_2) = -\mathbf{v}_2$. This shows that the coordinate matrix of T with respect to $\mathcal{B}$ is

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

(Writing the action of T this way makes it especially easy to see that performing this transformation twice puts us back where we started.) We will use

Corollary 5.6.9, p. 146 to calculate the matrix for T relative to the standard basis. That is, we wish to calculate $[T]_{\mathcal{E}}$.

We first note that the matrix $P_{\mathcal{B},\mathcal{E}}$ is, according to Lemma 5.6.5, p. 145,

$$P_{\mathcal{B},\mathcal{E}} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Then, with the help of Lemma 5.6.6, p. 145, we have

$$P_{\mathcal{E},\mathcal{B}} = (P_{\mathcal{B},\mathcal{E}})^{-1} = \begin{bmatrix} 2/5 & 1/5 \\ -1/5 & 2/5 \end{bmatrix}.$$

We can put these together to find $[T]_{\mathcal{E}}$:

$$[T]_{\mathcal{E}} = P_{\mathcal{B},\mathcal{E}}[T]_{\mathcal{B}}P_{\mathcal{E},\mathcal{B}} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2/5 & 1/5 \\ -1/5 & 2/5 \end{bmatrix} = \begin{bmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{bmatrix}.$$

The action of the transformation, as written in the final line here, is perhaps better understood in words rather than symbols. To reflect across the line $y = \frac{1}{2}x$, first shift from the standard coordinates to the alternate $\mathcal{B}$ -coordinates. (This is accomplished by $P_{\mathcal{E},\mathcal{B}}$.) In this new coordinate system, the action of T is easily described. (Thus, $[T]_{\mathcal{B}}$.) After that action is carried out, then we shift back to the standard coordinate system. (That is the work of $P_{\mathcal{B},\mathcal{E}}$.) From start to finish, this gives us a matrix which carries out the action of T relative to $\mathcal{E}$. ▲

5.6.2 Reading Questions

- Let $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{w}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, and $\mathbf{w}_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$. Consider the bases $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2\}$ of $\mathbb{R}^2$. Find the change-of-basis matrix $P_{\mathcal{B},\mathcal{C}}$.
- Using the definitions of the vectors and bases from the previous reading question, find $P_{\mathcal{C},\mathcal{B}}$.

5.6.3 Exercises

- Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2\}$ be bases of $\mathbb{R}^2$, where

$$\mathbf{v}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \mathbf{w}_1 = \begin{bmatrix} 3 \\ -3 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} -5 \\ -3 \end{bmatrix}.$$

Find the change-of-basis matrices $P_{\mathcal{B},\mathcal{C}}$ and $P_{\mathcal{C},\mathcal{B}}$.

- Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ be bases of $\mathbb{F}_5^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 4 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}.$$

Find the change-of-basis matrices $P_{\mathcal{B},\mathcal{C}}$ and $P_{\mathcal{C},\mathcal{B}}$.

3. Consider the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ of $\mathbb{F}_3^2$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

(a) If $\mathcal{E}$ is the standard basis for $\mathbb{F}_3^2$, find $P_{\mathcal{B}, \mathcal{E}}$ and $P_{\mathcal{E}, \mathcal{B}}$.

(b) Use your work in part (a) to find $[\mathbf{v}]_{\mathcal{B}}$ if $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

4. Let $\mathcal{B} = \{p_1, p_2, p_3\}$ be a basis for P_2 , where

$$p_1 = 3t - 5t^2, \quad p_2 = 3 - 3t + 5t^2, \quad p_3 = -1 - t - t^2.$$

(a) If $\mathcal{E}$ is the standard basis for P_2 , find $P_{\mathcal{B}, \mathcal{E}}$ and $P_{\mathcal{E}, \mathcal{B}}$.

(b) Use your work in part (a) to find $[p]_{\mathcal{B}}$ if $p = -3 + \frac{1}{2}t + \frac{3}{2}t^2$.

(c) If $T : P_2 \rightarrow P_2$ is the linear transformation that takes the derivative, find $[T]_{\mathcal{B}}$.

5. Let $\mathcal{B} = \{p_1, p_2, p_3\}$ be a basis for P_2 , where

$$p_1 = 1 + 5t - 5t^2, \quad p_2 = -2 + t + t^2, \quad p_3 = 5 - t - 4t^2.$$

(a) If $\mathcal{E}$ is the standard basis for P_2 , find $P_{\mathcal{B}, \mathcal{E}}$ and $P_{\mathcal{E}, \mathcal{B}}$.

(b) If $T : P_2 \rightarrow P_2$ is the linear transformation

$$T(p) = p(2)t + p(1)t^2,$$

find $[T]_{\mathcal{B}}$.

6. Let $\mathcal{B} = \{A_1, A_2, A_3, A_4\}$ be a basis for $M_2(\mathbb{F}_5)$, where

$$A_1 = \begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

(a) If $\mathcal{E}$ is the standard basis for $M_2(\mathbb{F}_5)$, find $P_{\mathcal{B}, \mathcal{E}}$ and $P_{\mathcal{E}, \mathcal{B}}$.

(b) If $T : M_2(\mathbb{F}_5) \rightarrow M_2(\mathbb{F}_5)$ is the linear transformation

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} 2a & 4b + c \\ a + d & 3b + d \end{bmatrix},$$

find $[T]_{\mathcal{B}}$.

7. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which is projection onto the line $y = -x$.

(a) Propose a basis $\mathcal{B}$ for $\mathbb{R}^2$ where $[T]_{\mathcal{B}}$ will be easy to determine.

(b) Find $[T]_{\mathcal{B}}$.

(c) If $\mathcal{E}$ is the standard basis for $\mathbb{R}^2$, find $P_{\mathcal{B}, \mathcal{E}}$ and $P_{\mathcal{E}, \mathcal{B}}$.

(d) Using your work in previous parts of this problem, find $[T]_{\mathcal{E}}$.

8. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which is reflection across the line $y = -\frac{1}{4}x$.

(a) Propose a basis $\mathcal{B}$ for $\mathbb{R}^2$ where $[T]_{\mathcal{B}}$ will be easy to determine.

- (b) Find $[T]_{\mathcal{B}}$.
- (c) If $\mathcal{E}$ is the standard basis for $\mathbb{R}^2$, find $P_{\mathcal{B},\mathcal{E}}$ and $P_{\mathcal{E},\mathcal{B}}$.
- (d) Using your work in previous parts of this problem, find $[T]_{\mathcal{E}}$.

Writing Exercises

- 9. Let V be an n -dimensional vector space over $\mathbb{F}$, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V . Prove that the columns of the matrix $P_{\mathcal{B},\mathcal{C}}$ are linearly independent.
- 10. Let V be an n -dimensional vector space over $\mathbb{F}$, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V . Prove that the columns of the matrix $P_{\mathcal{B},\mathcal{C}}$ span $\mathbb{F}^n$.
- 11. Let $A \in M_n(\mathbb{F})$ be invertible. Prove that there exist bases $\mathcal{B}$ and $\mathcal{C}$ for $\mathbb{F}^n$ such that $A = P_{\mathcal{B},\mathcal{C}}$.

Chapter 6

Eigenvalues and Eigenvectors

6.1 Eigenvalues and Eigenvectors

For linear transformations $T : V \rightarrow W$, there isn't often a connection between $\mathbf{v} \in V$ and $T(\mathbf{v}) \in W$ that is easy to describe. These vectors, after all, live in different vector spaces, so they need not have any obvious relationship to each other. When $W = V$, we sometimes have a different story to tell (for some vectors).

Suppose that V is a vector space with U as a subspace. If $T : V \rightarrow V$ is a linear transformation, then we can ask whether or not $T(U) \subseteq U$. If $T(U) \subseteq U$, then U is what is known as an *invariant subspace* under T . Though T may have all sorts of actions on V outside of U , when T is applied to vectors in U they are sent to other vectors in U .

This section is concerned with a version of an invariant subspace. We will shortly be able to analyze (nonzero) vectors $\mathbf{v} \in V$ for which $T(\mathbf{v}) \in \text{Span}\{\mathbf{v}\}$. This is close to satisfying what we might mean by a one-dimensional invariant subspace of V under T .

6.1.1 Defining Eigenvalues and Eigenvectors

We first define the sorts of vectors we alluded to in the previous paragraphs.

Definition 6.1.1 Let V be a vector space over $\mathbb{F}$, and let $T \in L(V)$. A nonzero vector $\mathbf{v} \in V$ is an **eigenvector** for T if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{F}$. A scalar λ is called an **eigenvalue** of T if there is a nontrivial solution to the equation $T(\mathbf{x}) = \lambda\mathbf{x}$. Such a solution is called an **eigenvector for T corresponding to λ** .

If $A \in M_n(\mathbb{F})$, the eigenvectors and eigenvalues of A are the eigenvectors and eigenvalues of the transformation $T \in L(\mathbb{F}^n)$ defined by $T(\mathbf{x}) = A\mathbf{x}$. ♦

Informally, eigenvectors for T are nonzero vectors on which T acts by scalar multiplication. The next example shows that for a T that has eigenvectors, it is not (always) *every* vector in V that has this special property.

Example 6.1.2 When we are given a matrix A and a vector $\mathbf{v}$, it is easy to determine whether or not $\mathbf{v}$ is an eigenvector for A . Consider the following:

$$A = \begin{bmatrix} 3 & 0 \\ 7 & -1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

We take the product $A\mathbf{u}$,

$$\begin{bmatrix} 3 & 0 \\ 7 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 12 \\ 21 \end{bmatrix}.$$

Since $A\mathbf{u} = 3\mathbf{u}$, $\mathbf{u}$ is an eigenvector for A with eigenvalue 3. Also, since

$$A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 7 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ -15 \end{bmatrix},$$

we can see that $\mathbf{v}$ is not an eigenvector for A , because $A\mathbf{v}$ is not a scalar multiple of $\mathbf{v}$. ▲

When $\mathbf{v}$ is an eigenvector of T , then applying T may change the *length* of $\mathbf{v}$ but it will not change the *direction* of $\mathbf{v}$. (To say this we must include “pointing in the exact opposite direction” as being in the same direction.) This is a simplification, because not every vector space has a neat, geometric interpretation.

Example 6.1.3 Let $T : P_2 \rightarrow P_2$ be the following linear transformation:

$$T(a + bt + ct^2) = (4a - b + 6c) + (2a + b + 6c)t + (2a - b + 8c)t^2.$$

If $p = 1 + t + t^2$, it is not difficult to check that

$$T(p) = 9 + 9t + 9t^2 = 9p.$$

Therefore, p is an eigenvector for T with eigenvalue 9. ▲

Example 6.1.4 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which is counterclockwise rotation about the origin by an angle of θ . We can see that T will have an eigenvector if and only if θ is an integer multiple of π radians. If θ is an even integer multiple of π , then every nonzero vector in $\mathbb{R}^2$ is an eigenvector for T with eigenvalue 1, and if θ is an odd integer multiple of π , then every nonzero vector in $\mathbb{R}^2$ is an eigenvector for T with eigenvalue -1 . ▲

We take a slightly different approach in our next example. Instead of verifying that a vector is an eigenvector, we provide the eigenvalue and then search for the eigenvector(s).

Example 6.1.5 We consider the matrix A from Example 6.1.2, p. 150. Let's show that -1 is an eigenvalue of A and find the corresponding eigenvectors.

We know that -1 is an eigenvalue of A if the equation $A\mathbf{x} = -\mathbf{x}$ has a nontrivial solution for some $\mathbf{x} \in \mathbb{R}^2$. This is equivalent to saying that the equation $A\mathbf{x} + \mathbf{x} = \mathbf{0}$ has a nontrivial solution. We can also view $\mathbf{x}$ as $I\mathbf{x}$, so if -1 is an eigenvalue of A , there is a nonzero vector $\mathbf{x}$ which satisfies

$$(A + I)\mathbf{x} = \mathbf{0}. \tag{6.1}$$

Viewed from the correct angle, we have reduced this problem to finding the null space of a matrix.

We will calculate $A + I$:

$$A + I = \begin{bmatrix} 3 & 0 \\ 7 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 7 & 0 \end{bmatrix}.$$

We can see that the columns of $(A + I)$ are linearly dependent, so we know that (6.1) has nontrivial solutions. This proves that -1 is an eigenvalue of A .

In order to find the eigenvectors of A that correspond to $\lambda = -1$, we describe

the null space of the appropriate matrix. We row-reduce $(A + I)$:

$$\begin{bmatrix} 4 & 0 \\ 7 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

This shows that every eigenvector of A corresponding to $\lambda = -1$ has the form $x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, as long as $x_2 \neq 0$. The interested/vigilant reader can check that, for example, $A \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$. ▲

The process we undertook in the previous example showed that there are almost always multiple eigenvectors for a linear transformation which correspond to a specific eigenvalue. In fact, the collection of such vectors forms almost an entire subspace.

Theorem 6.1.6 *Let V be a vector space over $\mathbb{F}$, let $T \in L(V)$, and let $\lambda \in \mathbb{F}$ be an eigenvalue of T . Then the set of all eigenvectors for T corresponding to λ , along with the zero vector, is a subspace of V .*

Proof. We note that a nonzero vector $\mathbf{v} \in V$ is an eigenvector for T corresponding to λ if and only if $T(\mathbf{v}) = \lambda\mathbf{v}$. The vector $\mathbf{v}$ satisfies this equation if and only if $T(\mathbf{v}) - \lambda I(\mathbf{v}) = \mathbf{0}$, which is true exactly when $(T - \lambda I)\mathbf{v} = \mathbf{0}$. This shows that a nonzero $\mathbf{v}$ is an eigenvector for T corresponding to λ if and only if $\mathbf{v} \in \ker(T - \lambda I)$.

Since we already know (Theorem 3.4.2, p.83) that the kernel of a linear transformation is a subspace, this completes the proof of this theorem. ■

Note 6.1.7 The awkwardness in the statement of this theorem regarding the zero vector is only present because the zero vector (by definition) cannot be an eigenvector.

This previous theorem justifies the following definition.

Definition 6.1.8 Let V be a vector space and let $T \in L(V)$. If $\lambda \in \mathbb{F}$ is an eigenvalue of T , then the **eigenspace** of T corresponding to λ is the subspace of V defined by

$$\text{eig}_\lambda(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \lambda\mathbf{v}\}.$$

We may refer to the eigenspace corresponding to λ as the **λ -eigenspace**. ◆

In the following example, we will calculate the eigenspace corresponding to an eigenvalue.

Example 6.1.9 We consider the following matrix A :

$$A = \begin{bmatrix} 4.5 & -2.5 & -2.5 \\ 2.5 & -0.5 & -2.5 \\ 5 & -5 & -3 \end{bmatrix}.$$

Let $T \in L(\mathbb{R}^3)$ be multiplication by A . If we know that $\lambda = 2$ is an eigenvalue for A , we can calculate a basis for $\text{eig}_2(T)$.

We need to form the matrix $A - 2I$ and then find the RREF:

$$A - 2I = \begin{bmatrix} 2.5 & -2.5 & -2.5 \\ 2.5 & -2.5 & -2.5 \\ 5 & -5 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The presence of free variables here confirms that 2 is an eigenvalue of A . If

$\mathbf{x} \in \text{null}(A - 2I)$, then

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 + x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

From this calculation we can see that $\text{eig}_2(T)$ is two-dimensional, with basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \text{ and } \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

▲

6.1.2 Results About Eigenvalues and Eigenvectors

In general, the eigenvalues of a linear transformation are not easy to spot. There are some situations, however, when we can identify eigenvalues at a glance.

Theorem 6.1.10 *The eigenvalues of a triangular matrix $A \in M_n(\mathbb{F})$ are the entries on the main diagonal of A .*

Proof. Suppose $A \in M_n(\mathbb{F})$ is an upper triangular matrix. Then $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} - \lambda & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} - \lambda & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & a_{nn} - \lambda \end{bmatrix}.$$

We can see that λ is an eigenvalue for A if and only if $\text{null}(A - \lambda I) \neq \{\mathbf{0}\}$, and this happens if and only if $A - \lambda I$ has at least one non-pivot column. Because A (and therefore $A - \lambda I$) is upper triangular, $A - \lambda I$ has at least one non-pivot column if and only if at least one of the entries on the main diagonal of $A - \lambda I$ is zero. This happens if and only if λ equals one of the entries on the main diagonal of A .

We have saved the case of a lower triangular matrix for the exercises. ■

Example 6.1.11 If $A \in M_3(\mathbb{F}_5)$ is given by

$$A = \begin{bmatrix} 4 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix},$$

the eigenvalues of A are 4, 2, and 1. The reader might use this opportunity to find the associated eigenvectors/eigenspaces! ▲

Of all possible scalars $\lambda \in \mathbb{F}$, it is especially noteworthy when $\lambda = 0$ is an eigenvalue for $T \in L(V)$. In this situation, there is a nonzero vector $\mathbf{x}$ such that $T(\mathbf{x}) = \mathbf{0}$. In other words, $\mathbf{x}$ is a nonzero vector in $\ker(T)$.

This short argument establishes a connection between the injectivity of T and the presence of $\lambda = 0$ as an eigenvalue for T . Because of previous results, we have the following theorem. (We leave the proof of both statements in this theorem as exercises so the reader might get practice connecting the logic of results in various chapters of the book.)

Theorem 6.1.12 Suppose V is a finite dimensional vector space and $T \in L(V)$. Then 0 is an eigenvalue of T if and only if T is not invertible.

If $A \in M_n(\mathbb{F})$, then 0 is an eigenvalue of A if and only if A is not invertible.

The final result in this section will be useful later, but we have all of the tools we need to prove it now.

Theorem 6.1.13 Suppose that $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors of $T \in L(V)$ corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent.

Proof. We will argue by contradiction. Suppose that the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly dependent. Since $\mathbf{v}_1 \neq \mathbf{0}$ (because eigenvectors cannot be $\mathbf{0}$), we can apply the Linear Dependence Lemma (Theorem 5.1.19, p. 112). Therefore, there is some $j \geq 2$ such that $\mathbf{v}_j \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\}$. There may be multiple subscripts j for which this is true; we use the smallest such j . We therefore have

$$\mathbf{v}_j = a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1}, \quad (6.2)$$

for scalars $a_i \in \mathbb{F}$.

We now apply T to both sides of this equation and use the eigenvector assumptions (as well as the linearity of T) to get

$$\begin{aligned} T(\mathbf{v}_j) &= T(a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1}) \\ T(\mathbf{v}_j) &= a_1T(\mathbf{v}_1) + \dots + a_{j-1}T(\mathbf{v}_{j-1}) \\ \lambda_j\mathbf{v}_j &= a_1\lambda_1\mathbf{v}_1 + \dots + a_{j-1}\lambda_{j-1}\mathbf{v}_{j-1}. \end{aligned} \quad (6.3)$$

If we multiply both sides of (6.2) by λ_j and subtract the result from (6.3), we get

$$\mathbf{0} = a_1(\lambda_1 - \lambda_j)\mathbf{v}_1 + \dots + a_{j-1}(\lambda_{j-1} - \lambda_j)\mathbf{v}_{j-1}.$$

Since $\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\}$ is linearly independent by assumption, we must have $a_i(\lambda_i - \lambda_j) = 0$ for each i , $1 \leq i \leq j-1$. But we assumed that the eigenvalues are all distinct, so this means that $\lambda_i - \lambda_j \neq 0$ for all i , and therefore we must have $a_i = 0$ for all i . But this implies, from (6.2), that $\mathbf{v}_j = \mathbf{0}$, which is a contradiction as $\mathbf{0}$ cannot be an eigenvector.

This contradiction proves that $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ must be linearly independent. ■

6.1.3 Reading Questions

1. Consider the following matrix $A \in M_2(\mathbb{R})$ and the vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$:

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}.$$

- (a) Is $\mathbf{u}$ an eigenvector for A ? How do you know?
 - (b) Is $\mathbf{v}$ an eigenvector for A ? How do you know?
2. Consider the matrix A from the previous reading question. Show that -2 is an eigenvalue of A and find the corresponding eigenvectors. Follow Example 6.1.5, p. 151.

6.1.4 Exercises

1. Let $A \in M_2(\mathbb{R})$ be the matrix

$$A = \begin{bmatrix} -12 & -14 \\ 7 & 9 \end{bmatrix}.$$

- (a) Is $\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ an eigenvector for A ? If so, find the eigenvalue.
- (b) Is $\mathbf{v} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$ an eigenvector for A ? If so, find the eigenvalue.

2. Let $A \in M_3(\mathbb{R})$ be the matrix

$$A = \begin{bmatrix} 1 & -3 & -1 \\ 2 & 6 & 1 \\ -4 & -10 & -1 \end{bmatrix}.$$

- (a) Is 4 an eigenvalue for A ? If so, find at least one eigenvector.
- (b) Is 3 an eigenvalue for A ? If so, find at least one eigenvector.

3. Let $A \in M_3(\mathbb{R})$ be the following matrix

$$A = \begin{bmatrix} -8 & 20 & 10 \\ 4 & -6 & -4 \\ -12 & 24 & 14 \end{bmatrix}.$$

- (a) Show that $\lambda = -4$ is an eigenvalue for A and find a basis for $\text{eig}_{-4}(A)$.
- (b) Show that $\lambda = 2$ is an eigenvalue for A and find a basis for $\text{eig}_2(A)$.

4. Let $A \in M_3(\mathbb{R})$ be the following matrix

$$A = \begin{bmatrix} -3 & 1 & -1 \\ 8 & 4 & 1 \\ 7 & 7 & -2 \end{bmatrix}.$$

- (a) Show that $\lambda = 5$ is an eigenvalue for A and find a basis for $\text{eig}_5(A)$.
- (b) Show that $\lambda = -2$ is an eigenvalue for A and find a basis for $\text{eig}_{-2}(A)$.

5. Let $A \in M_3(\mathbb{F}_5)$ be the following matrix

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 4 & 0 & 3 \\ 1 & 4 & 1 \end{bmatrix}.$$

- (a) Show that $\lambda = 4$ is an eigenvalue for A and find a basis for $\text{eig}_4(A)$.
- (b) Show that $\lambda = 2$ is an eigenvalue for A and find a basis for $\text{eig}_2(A)$.

6. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 5 \\ 1 & 3 & 5 \end{bmatrix}.$$

Find one eigenvalue of A without any calculation. Explain your reasoning.

7. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which is orthogonal projection onto the line $y = 5x$. Find all eigenvalues and all eigenvectors of this transformation.
8. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation which is reflection across the xy -plane. Find all eigenvalues and all eigenvectors of this transformation.
9. Let $A \in M_n(\mathbb{R})$ be the matrix with the number 1 in every entry. Find all eigenvalues and eigenvectors for A .
 - (a) Explain why A is not invertible. Based on this fact, use a result from this section to find an eigenvalue for A .
 - (b) Find the eigenvectors for A that correspond to the eigenvalue found in part a.
 - (c) There is one more eigenvalue of A . Consider the effect multiplication by A has on a vector $\mathbf{v} \in \mathbb{R}^n$. Using your answer, determine the remaining eigenvalue and eigenvector(s) for A . (Some students may find it easier to locate the eigenvector first; some may not!)

Writing Exercises

10. Prove that if A^2 is the zero matrix, then the only eigenvalue of A is 0.
11. Prove that an $n \times n$ matrix A can have at most n distinct eigenvalues.
12. If λ is an eigenvalue for an invertible linear transformation T , prove that λ^{-1} is an eigenvalue for T^{-1} .
13.
 - (a) Let A be an $n \times n$ matrix. Prove that λ is an eigenvalue for A if and only if λ is an eigenvalue for A^T . (Hint: Figure out how $A - \lambda I$ and $A^T - \lambda I$ are related.)
 - (b) Use part (a) to complete the proof of Theorem 6.1.10, p.153 for lower triangular matrices.
14. In a matrix, a *row sum* refers to the sum of all of the entries in a particular row.
 Let A be an $n \times n$ matrix where all of the row sums are equal to the same number k . Show that k is an eigenvalue of A . (Hint: Find an eigenvector.)
15. Suppose that $A \in M_n(\mathbb{F})$ and that for each $j = 1, \dots, n$, $\mathbf{e}_j$ is an eigenvector of A . Prove that A is a diagonal matrix.
16.
 - (a) Suppose that λ is an eigenvalue of $T \in L(V)$ and that $k \in \mathbb{N}$. Prove that λ^k is an eigenvalue of T^k and that $\text{eig}_\lambda(T) \subseteq \text{eig}_{\lambda^k}(T^k)$. (Here T^k means the composition of T with itself k times.)
 - (b) Give an example of a linear transformation $T : V \rightarrow V$ with an eigenvalue λ such that $\text{eig}_\lambda(T) \neq \text{eig}_{\lambda^2}(T^2)$.
17. Complete the proof of both halves of Theorem 6.1.12, p.154.

6.2 The Characteristic Equation

In the previous section we were able to check whether or not a given vector (scalar) was an eigenvector (eigenvalue) for a matrix A . What was missing was a process for finding the eigenvalues and eigenvectors for a matrix from scratch. This section will outline such a process. (Once we find the eigenvalues, we can use the method of Example 6.1.9, p. 152 to find bases for the eigenspaces.)

6.2.1 How to Find Eigenvalues

We recall from Section 6.1, p. 150 that a scalar $\lambda \in \mathbb{F}$ is an eigenvalue for a matrix $A \in M_n(\mathbb{F})$ if there is a nontrivial solution to the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$. Since this happens precisely when $A - \lambda I$ is *not* invertible, by Theorem 5.4.13, p. 133 we can use the determinant to find the eigenvalues for A .

Specifically, λ is an eigenvalue for A if and only if $\det(A - \lambda I) = 0$. If we consider λ as a variable, we can begin the process of finding the eigenvalues for square matrices. We will illustrate this method in the following example before making all of the steps explicit (with the appropriate notation and terminology).

Example 6.2.1 Let $A \in M_2(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 2 & -2 \\ 2 & 7 \end{bmatrix}.$$

A scalar λ is an eigenvalue for A if and only if $\det(A - \lambda I) = 0$. Since $A - \lambda I$ is 2×2 , we can find the determinant with ease:

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -2 \\ 2 & 7 - \lambda \end{vmatrix} = (2 - \lambda)(7 - \lambda) + 4.$$

With some algebra, we can see that

$$\det(A - \lambda I) = \lambda^2 - 9\lambda + 18 = (\lambda - 6)(\lambda - 3).$$

Since λ is an eigenvalue when $\det(A - \lambda I) = 0$, we see that the eigenvalues of A are $\lambda = 6$ and $\lambda = 3$.

Let's verify this method by showing that $\lambda = 6$ and $\lambda = 3$ are eigenvalues for A and finding bases for their eigenspaces. First, if $\lambda = 6$, then

$$A - 6I = \begin{bmatrix} -4 & -2 \\ 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{bmatrix}.$$

From this we see that $\text{eig}_6(A)$ is one-dimensional with basis $\left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$.

Now, when $\lambda = 3$, we have

$$A - 3I = \begin{bmatrix} -1 & -2 \\ 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

Thus $\text{eig}_3(A)$ is also one-dimensional with basis $\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$. ▲

We now give some names and notation to the elements of this method to find the eigenvalues of a matrix.

Definition 6.2.2 If $A \in M_n(\mathbb{F})$, then $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A . $\blacklozenge$

Based on our previous discussion, we can say that λ is an eigenvalue of A if and only if λ satisfies the characteristic equation of A .

It is occasionally useful to have a term and notation for just one piece of the characteristic equation.

Definition 6.2.3 For a matrix $A \in M_n(\mathbb{F})$, $\det(A - \lambda I)$ is a degree n polynomial referred to as the **characteristic polynomial** of A . Sometimes the notation $p_A(\lambda)$ is used for this polynomial. $\blacklozenge$

We will see that the characteristic equation can get quite difficult to solve as the size of the matrix increases. The next example provides a glimpse of this for a 3×3 matrix.

Example 6.2.4 Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 1 & -3 & -1 \\ 2 & 6 & 1 \\ -4 & -10 & -1 \end{bmatrix}.$$

To find the characteristic polynomial of A , we will use cofactor expansion along the first row of $A - \lambda I$:

$$\begin{aligned} \det(A - \lambda I) &= (1 - \lambda) \begin{vmatrix} 6 - \lambda & 1 \\ -10 & -1 - \lambda \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ -4 & -1 - \lambda \end{vmatrix} - 1 \begin{vmatrix} 2 & 6 - \lambda \\ -4 & -10 \end{vmatrix} \\ &= (1 - \lambda)(\lambda^2 - 5\lambda + 4) + 3(2 - 2\lambda) - (4 - 4\lambda) \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6. \end{aligned}$$

For the matrix A given above, we have $p_A(\lambda) = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$. $\blacktriangle$

Since one can solve a quadratic equation easily—either by factoring or using the quadratic formula—some mistakenly think that all polynomial equations are easy to solve. Not so. As we consider the characteristic polynomial for the matrix in the previous example, we find the task of finding the eigenvalues quite challenging.

While there is a formula for solving a cubic polynomial, it is not nearly as nice as the quadratic formula. It is, in fact, something that should rarely see the light of day. It is ugly.

In practice, the eigenvalues of an $n \times n$ matrix are usually found with a computer algebra system when $n \geq 3$. However, there is one way that we could describe an eigenvalue problem and still have a reasonable expectation that the reader could find a solution.

Example 6.2.5 Consider the following matrix $A \in M_3(\mathbb{R})$ (from the previous example):

$$A = \begin{bmatrix} 1 & -3 & -1 \\ 2 & 6 & 1 \\ -4 & -10 & -1 \end{bmatrix}.$$

If we know that $\lambda = 1$ is an eigenvalue for A , we can find all of the eigenvalues.

Our previous work showed that $p_A(\lambda) = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$. If we know that $\lambda = 1$ is an eigenvalue, that means that $(\lambda - 1)$ must be a factor of this polynomial. Using long division of polynomials, we can complete the factorization:

$$p_A(\lambda) = (\lambda - 1)(-\lambda^2 + 5\lambda - 6) = -(\lambda - 1)(\lambda - 2)(\lambda - 3).$$

This shows that the eigenvalues for A are $\lambda = 1, 2, 3$. $\blacktriangle$

When a characteristic polynomial is factored to reveal the eigenvalues, we might see some of those eigenvalues repeated. We now introduce a term to describe this repetition.

Definition 6.2.6 The **algebraic multiplicity** of an eigenvalue is the number of times it appears as a root of the characteristic polynomial for a matrix. We will often refer to this number as an eigenvalue's **multiplicity**. ♦

Example 6.2.7 We consider the matrix from Example 6.1.9, p. 152 in the previous section:

$$A = \begin{bmatrix} 4.5 & -2.5 & -2.5 \\ 2.5 & -0.5 & -2.5 \\ 5 & -5 & -3 \end{bmatrix}.$$

The characteristic polynomial for this matrix is

$$p_A(\lambda) = -\lambda^3 + \lambda^2 + 8\lambda - 12 = -(\lambda - 2)^2(\lambda + 3).$$

From this we see that A has an eigenvalue of 2 with multiplicity 2 and an eigenvalue of -3 with multiplicity 1. ▲

Before we move to the second half of this section, we point out that not every matrix over $\mathbb{F}$ has eigenvalues in $\mathbb{F}$. The following example illustrates this, but we can understand this intuitively by recognizing that (for example) not every polynomial over $\mathbb{R}$ can be completely factored into linear terms.

Example 6.2.8 Let $A \in M_2(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} -5 & -4 \\ 5 & -6 \end{bmatrix}.$$

We can calculate the characteristic polynomial for A :

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} -5 - \lambda & -4 \\ 5 & -6 - \lambda \end{vmatrix} \\ &= (-5 - \lambda)(-6 - \lambda) + 20 \\ &= \lambda^2 + 11\lambda + 50. \end{aligned}$$

This quadratic polynomial has no real roots, as the reader can check by verifying that the discriminant (the expression under the square root in the quadratic formula) is negative.

If A is considered as a matrix over $\mathbb{C}$, then it has two complex eigenvalues. But it has no real eigenvalues. ▲

One of the major differences between $\mathbb{R}$ and $\mathbb{C}$ is that $\mathbb{C}$ is an **algebraically closed field**, while $\mathbb{R}$ is not. The fact that $\mathbb{C}$ is algebraically closed is another way to restate the famous Fundamental Theorem of Algebra, which says that every polynomial with coefficients in $\mathbb{C}$ can be completely reduced to linear factors. There are interesting, deep waters to explore here, but perhaps not in the present text.

6.2.2 Similarity

Earlier in this book we discussed one way matrices can be related—row equivalence. We now introduce another relationship which has connections to the characteristic equation.

Definition 6.2.9 Suppose that $A, B \in M_n(\mathbb{F})$. Then A is **similar to** B if there exists an invertible matrix $P \in M_n(\mathbb{F})$ such that $P^{-1}AP = B$ or $A = PBP^{-1}$.



Note 6.2.10 Since we could write $Q = P^{-1}$ in the previous definition and have $Q^{-1}BQ = A$ when $P^{-1}AP = B$, we see that B is similar to A when A is similar to B . For this reason, we can say that A and B are **similar**.

The next theorem presents one of the most compelling reasons for studying similarity in matrices.

Theorem 6.2.11 *If matrices $A, B \in M_n(\mathbb{F})$ are similar, then they have the same characteristic polynomials and therefore the same eigenvalues (with the same multiplicities).*

Proof. If A and B are similar, then there exists an invertible matrix P such that $B = P^{-1}AP$. When forming $B - \lambda I$, we have

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P.$$

Since the determinant is multiplicative (Theorem 4.2.13, p. 101), we have

$$\begin{aligned} \det(B - \lambda I) &= \det(P^{-1}) \det(A - \lambda I) \det(P) \\ &= \det(P^{-1}) \det(P) \det(A - \lambda I) \\ &= \det(P^{-1}P) \det(A - \lambda I) \\ &= \det(I) \det(A - \lambda I) \\ &= \det(A - \lambda I). \end{aligned}$$

This shows that A and B have the same characteristic polynomials. ■

The next example showcases two similar matrices.

Example 6.2.12 Let A and B be the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} -10 & -5 \\ 24 & 13 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -4 \\ -3 & 2 \end{bmatrix}.$$

We will show that these are similar matrices by considering the matrix P :

$$P = \begin{bmatrix} -4 & -1 \\ -5 & -2 \end{bmatrix}.$$

The reader can check that P is invertible and that $A = P^{-1}BP$. This shows that A and B are similar.

Further calculation shows that $p_B(\lambda) = \lambda^2 - 3\lambda - 10$ and that this is the same as $p_A(\lambda)$. ▲

It is important to note the logical implication that Theorem 6.2.11, p. 160 provides with care. *If* two matrices are similar, *then* they have the same eigenvalues. However, we may have two matrices with the same eigenvalues which are not similar.

Example 6.2.13 Let $A, B \in M_2(\mathbb{R})$ be the following matrices:

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -2 \\ 0 & 5 \end{bmatrix}.$$

Since both A and B are triangular matrices, we can see that they have the same eigenvalue (with the same multiplicity). However, these matrices are *not* similar. In fact, we will show in the exercises that a matrix kI , for $k \in \mathbb{F}$, is only similar to itself. In this example, $A = 5I$. ▲

We end this section with a small note of caution. Similarity and row equivalence are not the same thing, and the distinction between these two relations is important to keep in mind. In particular, elementary row operations usually alter the eigenvalues of a matrix, while we have seen that a similarity transformation does not.

6.2.3 Reading Questions

1. Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} 2 & -1 \\ -6 & -3 \end{bmatrix}.$$

- (a) Write down the characteristic equation for A . (Feel free to use t instead of λ if typing λ is difficult.)
- (b) Determine the two eigenvalues for A .
2. Consider the matrix A from the previous reading question. Write down two separate matrices which are similar to A . Explain how you obtained these matrices. (There are many, many correct answers!)

6.2.4 Exercises

1. Let $A \in M_2(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} -7 & 8 \\ -6 & 7 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A .

2. Let $A \in M_2(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 7 & -3 \\ 6 & -2 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A .

3. Let $A \in M_2(\mathbb{F}_5)$ be the following matrix:

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A .

4. Let $A \in M_2(\mathbb{F}_5)$ be the following matrix:

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A .

5. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -1 & 1 \\ 3 & -1 & 1 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A by hand.

6. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 2 & 6 & 6 \\ 0 & -1 & 0 \\ -3 & -6 & -7 \end{bmatrix}.$$

Find the characteristic polynomial and the eigenvalues of A by hand.

7. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 5 & 8 & -8 \\ -4 & -11 & 16 \\ 0 & -4 & 9 \end{bmatrix}.$$

Find the characteristic polynomial of A by hand. If $\lambda = 1$ is an eigenvalue of A , find the other eigenvalues of A by hand.

8. Let $A \in M_3(\mathbb{R})$ be the following matrix:

$$A = \begin{bmatrix} 1 & -4 & -4 \\ -2 & -1 & -4 \\ 4 & 4 & 9 \end{bmatrix}.$$

Find the characteristic polynomial of A by hand. If $\lambda = 3$ is an eigenvalue of A , find the other eigenvalues of A by hand.

Writing Exercises

9. Let $A \in M_n(\mathbb{R})$ have n real eigenvalues, $\lambda_1, \dots, \lambda_n$, repeated according to their multiplicities, so that

$$\det(A - \lambda I) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda).$$

Prove that $\det(A)$ is the product of the n eigenvalues of A .

10. Prove that similarity is an equivalence relation on the set $M_n(\mathbb{F})$.
 11. Let $I_n \in M_n(\mathbb{F})$ be the identity matrix and let $k \in \mathbb{F}$ be a scalar. If $A = kI_n$, prove that the only matrix A is similar to is itself.
 12. Prove that if A and B are similar, then $\det(A) = \det(B)$.
 13. Let $A \in M_n(\mathbb{F})$. Prove that $\text{rank}(A) = n$ if and only if $p_A(0) \neq 0$.

6.3 Diagonalization

An $n \times n$ matrix A often has a huge number of matrices to which it is similar. In this section, we will focus on matrices that are similar to diagonal matrices, and we do so because of how straightforward the action of a diagonal matrix is.

6.3.1 Diagonalizable Matrices

If A is similar to a diagonal matrix D , then $A = PDP^{-1}$ for some invertible matrix P . Such a factorization of A encodes much of the information about the eigenvalues and eigenvectors of A , and it also allows us to raise A to integer powers rather easily.

If D is diagonal, then powers of D are easy to compute. Consider the following matrices D and D^2 :

$$D = \begin{bmatrix} -3 & 0 \\ 0 & 4 \end{bmatrix}, \quad D^2 = \begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix}.$$

In general, the many zeros in a diagonal matrix make the powers of that matrix easy to calculate. For this specific D , we have, for any integer $k \geq 1$,

$$D^k = \begin{bmatrix} (-3)^k & 0 \\ 0 & 4^k \end{bmatrix}.$$

Given the number of calculations that are usually involved in matrix multiplication, this is a huge savings in computing time.

Now, if A is similar to a diagonal matrix D , we find related behavior. Suppose that $A = PDP^{-1}$. Then

$$\begin{aligned} A^2 &= (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} \\ &= PD(I)DP^{-1} = PD^2P^{-1}. \end{aligned}$$

Since A^3 can be written as AA^2 , we have $A^3 = PD^3P^{-1}$. This pattern holds for any $k \geq 2$:

$$A^k = PD^kP^{-1}.$$

Perhaps we have convinced a skeptical reader that there are some advantages when A is similar to a diagonal matrix. This is worthy of a formal definition.

Definition 6.3.1 A matrix $A \in M_n(\mathbb{F})$ is **diagonalizable** if $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D . ♦

Given this definition, it is natural to ask exactly when a matrix is diagonalizable. That answer comes in the following theorem.

Theorem 6.3.2 The Diagonalization Theorem. *A matrix $A \in M_n(\mathbb{F})$ is diagonalizable if and only if A has n linearly independent eigenvectors.*

More specifically, $A = PDP^{-1}$, for a diagonal matrix D , if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are the eigenvalues of A which correspond, respectively, to the columns of P .

Proof. If P is an $n \times n$ matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if D is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then we have

$$AP = A \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} A\mathbf{v}_1 & \cdots & A\mathbf{v}_n \end{bmatrix}, \quad (6.4)$$

and also

$$PD = \begin{bmatrix} \lambda_1 \mathbf{v}_1 & \cdots & \lambda_n \mathbf{v}_n \end{bmatrix}. \quad (6.5)$$

(If the reader has trouble believing (6.5), thinking of matrix multiplication, in each column of the product, as a linear combination of the columns of P with weights coming from the corresponding column of D , may help!) If A is diagonalizable, then $A = PDP^{-1}$ and $AP = PD$. From (6.4) and (6.5), by equating columns in AP and PD we see that $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ for $1 \leq i \leq n$. Since P is invertible, the columns of P must be linearly independent. Further, since the columns of P cannot be zero vectors, this argument shows that λ_i is an eigenvalue of A with eigenvector $\mathbf{v}_i$, for each i . This proves one direction of the theorem.

If we are given $\mathbf{v}_1, \dots, \mathbf{v}_n$ as eigenvectors of A with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, then we can form the matrices P and D . The argument in the

previous paragraph shows that $AP = PD$. (Note that we have not yet used the linear independence of the eigenvectors!) If the eigenvectors are linearly independent, then P is invertible, and $AP = PD$ implies $A = PDP^{-1}$, making A diagonalizable. ■

Note 6.3.3 Theorem 6.3.2, p. 163 says that A is diagonalizable if and only if there is a basis of $\mathbb{F}^n$ consisting of eigenvectors of A . We call such a basis an **eigenvector basis** of $\mathbb{F}^n$.

6.3.2 How to Diagonalize a Matrix

Using Theorem 6.3.2, p. 163, we see there are four steps to diagonalizing a matrix. We will summarize them in the following algorithm.

Algorithm 6.3.4 If $A \in M_n(\mathbb{F})$, we follow these steps to diagonalize A .

1. Find the eigenvalues of A . This usually involves solving the characteristic equation for A . (If the characteristic polynomial does not factor into linear factors, the matrix is not diagonalizable.)
2. Find n linearly independent eigenvectors for A . (This step may fail, in which case the matrix is not diagonalizable.)
3. Construct the matrix P using the vectors from the previous step. Form P by using the eigenvectors as its columns. The order of these vectors does not matter.
4. Construct D from the eigenvalues. Once the matrix P is formed, the order of these eigenvalues does matter—the eigenvalues must be placed along the diagonal of D in the order corresponding to the columns of P . In other words, entry d_{ii} in D should be the eigenvalue for the vector which is column i in P .

After forming P and D , it is a good idea to check that the process was successful. We may verify the equation $A = PDP^{-1}$, or alternatively we may check that $AP = PD$. (This avoids the need to find P^{-1} .)

Example 6.3.5 We consider the matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} -2 & 2 & -2 \\ -4 & 4 & -2 \\ -6 & 3 & -1 \end{bmatrix}.$$

Though the reader can determine this independently, we will provide the characteristic polynomial to save time and space:

$$p_A(\lambda) = -(\lambda + 3)(\lambda - 2)^2.$$

From this we see that the eigenvalues of A are $\lambda = -3$ and $\lambda = 2$.

We now find bases for the associated eigenspaces. Again, we will suppress all of the calculations since the previous sections have gone through these in some detail. We find that

$$\text{eig}_{-3}(A) = \text{Span} \left\{ \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \right\},$$

and

$$\text{eig}_2(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

Since we have three linearly independent eigenvectors, we know from Theorem 6.3.2, p. 163 that A is diagonalizable.

We now form the matrices P and D according to the algorithm:

$$P = \begin{bmatrix} 2 & 1 & -1 \\ 2 & 2 & 0 \\ 3 & 0 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

We can check that our diagonalization was successful by calculating AP and PD :

$$\begin{aligned} AP &= \begin{bmatrix} -2 & 2 & -2 \\ -4 & 4 & -2 \\ -6 & 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 2 & 2 & 0 \\ 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -6 & 2 & -2 \\ -6 & 4 & 0 \\ -9 & 0 & 4 \end{bmatrix} \\ PD &= \begin{bmatrix} 2 & 1 & -1 \\ 2 & 2 & 0 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -6 & 2 & -2 \\ -6 & 4 & 0 \\ -9 & 0 & 4 \end{bmatrix}. \end{aligned}$$

▲

We now consider another example of a 3×3 matrix.

Example 6.3.6 We consider the matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 0 & -6 & -4 \\ 5 & -11 & -6 \\ -6 & 9 & 4 \end{bmatrix}.$$

The characteristic polynomial for A is $p_A(\lambda) = -(\lambda + 2)^2(\lambda + 3)$. So the eigenvalues of A are $\lambda = -2$ and $\lambda = -3$.

When we look for eigenvectors, we find the following for $A + 2I$:

$$A + 2I = \begin{bmatrix} 2 & -6 & -4 \\ 5 & -9 & -6 \\ -6 & 9 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

This shows that $\text{eig}_{-2}(A)$ is only one-dimensional. Since we need three total linearly independent eigenvectors to diagonalize A , and we will only get one from $\text{eig}_{-2}(A)$, we needed $\text{eig}_{-3}(A)$ to be two-dimensional. This shows that A is not diagonalizable. ▲

The difference between the last two examples shows that diagonalizability is subtle. There are times when we can tell if a matrix is diagonalizable without a lot of work, but sometimes we need to get all the way to the eigenspace calculation before having an answer. The following theorem states a situation in which diagonalizability is easier to confirm.

Theorem 6.3.7 Let $A \in M_n(\mathbb{F})$. If A has n distinct eigenvalues, A is diagonalizable.

Proof. If A has n distinct eigenvalues, let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be eigenvectors which correspond to those eigenvalues. Then, by Theorem 6.1.13, p. 154, the set $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent. Since our vectors are in $\mathbb{F}^n$, by Theorem 6.3.2, p. 163 this basis of eigenvectors of A means that A is diagonalizable. ■

Example 6.3.8 We consider the following matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & -2 & 0 \\ 3 & -1 & 5 \end{bmatrix}.$$

Since A is lower triangular, we can read the eigenvalues off of the main diagonal: $\lambda = 1, -2, 5$. Since there are three distinct eigenvalues for this 3×3 matrix, then Theorem 6.3.7, p. 165 says that A is diagonalizable. $\blacktriangle$

Note 6.3.9 Having n distinct eigenvalues is a *sufficient* condition for a matrix to be diagonalizable, but it is not *necessary*. In other words, a matrix can still be diagonalizable with fewer than n distinct eigenvalues. We have already encountered this in Example 6.3.5, p. 164.

The following theorem collects some facts about the diagonalizability of a matrix. (We omit the proof.)

Theorem 6.3.10 Let $A \in M_n(\mathbb{F})$ have distinct eigenvalues $\lambda_1, \dots, \lambda_p$.

1. For $1 \leq k \leq p$, the dimension of $\text{eig}_{\lambda_k}(A)$ is less than or equal to the (algebraic) multiplicity of λ_k .
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n , and this happens if and only if (i) the characteristic polynomial factors completely into linear factors, and (ii) the dimension of $\text{eig}_{\lambda_k}(A)$ equals the multiplicity of λ_k for each k .
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for $\text{eig}_{\lambda_k}(A)$, then $\mathcal{B} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_p$ is an eigenvector basis for $\mathbb{F}^n$.

In the previous theorem, we used the dimension of an eigenspace several times. This is worthy of a definition.

Definition 6.3.11 The **geometric multiplicity** of an eigenvalue λ of a matrix A is the dimension of the eigenspace $\text{eig}_{\lambda}(A)$. $\blacklozenge$

The reader will now be able to phrase some of the results contained in Theorem 6.3.10, p. 166 in terms of how the (algebraic) multiplicity and geometric multiplicity of the eigenvalues of a matrix compare to each other.

6.3.3 Linear Transformations and Diagonalizability

In Subsection 5.5.2, p. 137 we saw that, for linear transformations between finite-dimensional vector spaces, we could view these transformations as multiplication by a matrix if we were content to handle coordinate vectors. And while we didn't have the current terminology at that point, in Section 5.6, p. 144 we were calculating coordinate matrices for linear transformations using similarity. (See Example 5.6.10, p. 146.)

This means that our discussion of similar matrices has implications for linear transformations broadly. And these implications are, unsurprisingly, related to eigenvalues and eigenvectors.

Definition 6.3.12 Let V be a finite-dimensional vector space and let $T \in L(V)$. Then T is **diagonalizable** if there exists a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ is diagonal. $\blacklozenge$

Based on our discussion thus far in this section, the reader may guess that the vectors in the basis $\mathcal{B}$ referenced in Definition 6.3.12, p. 166 are eigenvectors for T . What is true for matrices is (generally) true in the proper context for linear transformations.

Proposition 6.3.13 *A linear transformation $T \in L(V)$ is diagonalizable if and only if there exists a basis $\mathcal{B}$ of V consisting entirely of eigenvectors of T .*

Example 6.3.14 In Example 5.6.10, p. 146, we considered the linear transformation on $\mathbb{R}^2$ which is reflection across the line $y = \frac{1}{2}x$. In that example, we looked at the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ for $\mathbb{R}^2$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

We saw that the matrix $[T]_{\mathcal{B}}$ was diagonal, and now we know that was because the basis vectors are eigenvectors for T . Since $\mathbf{v}_1$ lies on the line of reflection, it is an eigenvector for T with eigenvalue 1, and since $\mathbf{v}_2$ lies on the line perpendicular to the line of reflection, it is an eigenvector for T with eigenvalue -1 . The matrix $[T]_{\mathcal{B}}$ is

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The following result is basically a restatement of Corollary 5.6.9, p. 146, using the language of similar matrices. ▲

Proposition 6.3.15 *Suppose that V is a finite-dimensional vector space and that $T \in L(V)$. Suppose further that $\mathcal{B}$ and $\mathcal{C}$ are bases for V . Then the coordinate matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$ are similar.*

Proof. Since change-of-basis matrices are invertible, this really is just a restatement of Corollary 5.6.9, p. 146. ■

The final result in this section brings several prior results together, tying the diagonalizability of linear transformations and matrices to each other in a predictable way.

Theorem 6.3.16 *Let V be a finite-dimensional vector space, let $\mathcal{B}$ be a basis for V , and let $T \in L(V)$. Then $[T]_{\mathcal{B}}$ is a diagonalizable matrix if and only if T is a diagonalizable linear transformation.*

This theorem says that a linear transformation T is diagonalizable if there is a basis of V with respect to which the coordinate matrix of T is diagonalizable.

We finish this section with an example which is perhaps a bit contrived but which is also, hopefully, illustrative.

Example 6.3.17 Let $T : P_1 \rightarrow P_1$ be the following linear transformation:

$$T(a + bt) = (a - 4b) + (3a - 6b)t.$$

If $\mathcal{B}$ is the standard basis for P_1 , then $[T]_{\mathcal{B}}$ is

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & -4 \\ 3 & -6 \end{bmatrix}.$$

It is fairly easy to determine that $[T]_{\mathcal{B}}$ is diagonalizable, since the characteristic polynomial is

$$\lambda^2 + 5\lambda + 6 = (\lambda + 2)(\lambda + 3).$$

Since $[T]_{\mathcal{B}}$ is diagonalizable, that means that T is a diagonalizable linear transformation.

Using coordinate vectors, we can also determine the basis $\mathcal{C}$ of P_1 with respect to which T has a diagonal coordinate matrix. (It is a basis of eigenvectors of T !)

Since the eigenvalues of $[T]_{\mathcal{B}}$ are $\lambda = -2, -3$, we can find bases for the related eigenspaces. For ease of notation, let $[T]_{\mathcal{B}} = A$. Now

$$\text{eig}_{-2}(A) = \text{Span} \left\{ \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right\}, \quad \text{eig}_{-3}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

These are the coordinate vectors for the eigenvectors of T with respect to the standard basis. Therefore, an eigenvector basis of P_1 is

$$\mathcal{C} = \{4 + 3t, 1 + t\},$$

and $[T]_{\mathcal{C}}$ is a diagonal matrix with diagonal entries -2 and -3 . ▲

6.3.4 Reading Questions

1. Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}.$$

- (a) Find the characteristic polynomial and the eigenvalues of A . Show your work.
 - (b) Using only part a (this means you should make no additional calculations), explain why A is diagonalizable.
2. Let A be the same matrix as in the first reading question. Following Algorithm 6.3.4, p. 164, diagonalize A .

6.3.5 Exercises

1. Let $P, D \in M_2(\mathbb{R})$ be the following matrices:

$$P = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

If $A = PDP^{-1}$, calculate A^4 .

2. Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} -5 & -3 \\ 6 & 4 \end{bmatrix}.$$

Determine whether or not A is diagonalizable. If it is, diagonalize it. If it isn't, explain why it isn't.

3. Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 5 \end{bmatrix}.$$

Determine whether or not A is diagonalizable. If it is, diagonalize it. If it isn't, explain why it isn't.

4. Consider the following matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 14 & 3 & 12 \\ -2 & 3 & -2 \\ -7 & -1 & -5 \end{bmatrix}.$$

Determine whether or not A is diagonalizable. If it is, diagonalize it. If it isn't, explain why it isn't. (Hint: One of the eigenvalues of A is $\lambda = 5$.)

5. Consider the following matrix $A \in M_3(\mathbb{R})$:

$$A = \begin{bmatrix} -9 & -8 & -16 \\ -4 & -5 & -8 \\ 4 & 4 & 7 \end{bmatrix}.$$

Determine whether or not A is diagonalizable. If it is, diagonalize it. If it isn't, explain why it isn't. (Hint: One of the eigenvalues of A is $\lambda = -1$.)

6. Suppose that $A \in M_4(\mathbb{R})$ has three distinct eigenvalues. One eigenspace is one-dimensional and one of the other eigenspaces is two-dimensional. Is it possible for A *not* to be diagonalizable? Explain.
7. Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 4 & -1 \\ -4 & 4 \end{bmatrix}.$$

Show that a diagonalization $A = PDP^{-1}$ is not unique by finding two pairs of matrices (P, D) which diagonalize A .

Writing Exercises

8. Prove that if A is both invertible and diagonalizable, then so is A^{-1} .
9. Let $A \in M_n(\mathbb{F})$. Prove that if A has n linearly independent eigenvectors, then so does A^T .
10. This problem explores the relationship between invertibility and diagonalizability.
- Construct a nonzero 2×2 matrix which is invertible but not diagonalizable.
 - Construct a nondiagonal 2×2 matrix that is diagonalizable but not invertible.
11. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be projection onto the xy -plane. Prove that T is diagonalizable.
12. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be orthogonal projection onto the line $y = -6x$. Prove that T is diagonalizable.
13. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be counterclockwise rotation by $\frac{\pi}{2}$ radians.
- Prove that T is not diagonalizable.
 - Prove that T^2 is diagonalizable.

6.4 Invariants

One of the fun aspects of linear algebra is that it touches so many different areas of mathematics. In this section we will connect row equivalence and similarity to the topic of *mathematical invariants*, which is used in many mathematical disciplines to determine whether or not two objects are related.

6.4.1 Introduction to Invariants

Many areas of mathematics have a notion of the *relatedness* of the important objects of study. In linear algebra, we have already learned about vector spaces being isomorphic (Theorem 5.3.12, p. 127), and this idea shows up widely. We often prove that mathematical objects are related by defining a function between them and showing that it has all the right properties.

However, showing that two mathematical objects are *not* related is harder when working strictly from the definitions. It is difficult to show that a function with certain properties *cannot exist*—not being able to find or think of such a function is not a sufficient argument! This is where *invariants* enter the picture.

Definition 6.4.1 An **invariant** of a mathematical object is a property of that object that is either the same for all members of an equivalence class or unchanged after some sort of transformation. ♦

Note 6.4.2 In our present context, we will be thinking about equivalence relations. This is a good opportunity to think through the logic of invariants. Suppose we have two objects A and B which are related by some equivalence relation. We will write $A \sim B$. (This tells us that A and B are in the same equivalence class.) If there is an invariant f in view, then $A \sim B$ *implies* the values of the invariant are equal—in other words, $A \sim B$ implies $f(A) = f(B)$.

The contrapositive of this implication is what we end up using with great frequency. If $f(A) \neq f(B)$, then we know that $A \not\sim B$. (In other words, A and B are not related.) Importantly, knowing that $f(A) = f(B)$ *does not* prove that A and B are related. The implication goes in one direction but not the other.

Before this discussion gets too vague, let's look at an example that most readers of this book should recognize.

Example 6.4.3 We say that two geometric objects are **congruent** if they have the same shape and size. Most readers will likely have spent some time considering congruent triangles in a geometry class, where they encountered several congruence theorems for triangles: Side-Side-Side, Side-Angle-Side, etc. (Note that congruent triangles are not the same as similar triangles, though there are some, ahem, similarities.)

In this example, *congruence* provides the relevant equivalence relation on the set of all planar triangles.

One invariant that can be used to distinguish between triangles that are not congruent is the area of the triangle. It is easy to show that if two triangles are congruent, they have the same area. Therefore, if two triangles have different areas, we can conclude that they are not congruent.

To be specific, suppose that triangle A is a 30-60-90 triangle with side lengths $(10, 10\sqrt{3}, 20)$ and triangle B is a 45-45-90 triangle with side lengths $(10\sqrt[4]{3}, 10\sqrt[4]{3}, 10\sqrt{2}\sqrt[4]{3})$. Then since the area of a right triangle is easy to calculate using the $\frac{1}{2}bh$ formula, we can see that both triangles A and B have an area of $50\sqrt{3}$ square units. But these triangles are not congruent—they have different angles, after all.

In summary, the area of a triangle can be used to distinguish between triangles that are *not* congruent. However, area cannot be used to conclude that two triangles are congruent. In this way, area is a typical invariant. ▲

In this text, we have discussed two equivalence relations on sets of matrices for which there exist some interesting invariants. For fixed values of m and n , *row equivalence* is an equivalence relation on the set $M_{m,n}(\mathbb{F})$. (See Exercise 1.2.10, p. 16.) More recently, we have looked at *similarity*, which is

an equivalence relation on the set $M_n(\mathbb{F})$ for fixed values of n . (See Exercise 6.2.4.10, p. 162.)

We have shown in Theorem 6.2.11, p. 160 that the characteristic polynomial is an invariant for similar matrices. If two matrices are similar, they have the same characteristic polynomial, and therefore the same eigenvalues (with the same multiplicities). This means that if we know of two matrices of the same size with different eigenvalues, they cannot be similar.

Additionally—and using much the same argument—the determinant is an invariant for similar matrices. (We asked the reader to prove this in Exercise 6.2.4.12, p. 162.) So, two matrices of the same size with different determinants cannot be similar.

We will summarize these two already-established facts in the following proposition.

Proposition 6.4.4 *Let n be a positive integer and let $\mathbb{F}$ be a field. The determinant and the characteristic polynomial are both invariants on the set $M_n(\mathbb{F})$ with respect to the relation of similarity.*

While the characteristic polynomial and the determinant are useful invariants for similarity, they are far from the only ones. In the second half of this section we will develop two more invariants which respect similarity, one of which is quite surprising.

6.4.2 Rank and Trace as Invariants

We will first state and prove a few results needed to establish rank as an invariant.

Theorem 6.4.5 *Suppose that V and W are finite-dimensional vector spaces and that $T \in L(V, W)$. Then, for any bases $\mathcal{B}$ of V and $\mathcal{C}$ of W , we have*

$$\text{rank}(T) = \text{rank}([T]_{\mathcal{B}, \mathcal{C}}) \quad \text{and} \quad \dim(\ker(T)) = \dim(\text{null}([T]_{\mathcal{B}, \mathcal{C}})).$$

Proof. Suppose that $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and let $A = [T]_{\mathcal{B}, \mathcal{C}}$. Since $V = \text{Span}(\mathcal{B})$, we have $\text{im}(T) = \text{Span}\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$. By Theorem 5.5.3, p. 135 we know that $\text{Span}\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is isomorphic to

$$\text{Span}\{[T(\mathbf{v}_1)]_{\mathcal{C}}, \dots, [T(\mathbf{v}_n)]_{\mathcal{C}}\}.$$

The vectors $[T(\mathbf{v}_i)]_{\mathcal{C}}$ are exactly the columns of A , so

$$\text{rank}(T) = \dim(\text{im}(T)) = \dim(\text{col}(A)) = \text{rank}(A).$$

For the second equality, we refer to The Rank-Nullity Theorem, p. 132. If $\dim(V) = n$ and $\dim(W) = m$, then A is $m \times n$. We have proved that $\text{rank}(T) = \text{rank}(A)$, and comparing the two equations in Theorem 5.4.10, p. 132, we are left with $\dim(\text{null}(A)) = \dim(\ker(T))$, as desired. ■

Proposition 6.4.6 *Suppose that V is an n -dimensional vector space over $\mathbb{F}$, that $\mathcal{B}$ is a basis of V , and that $T \in L(V)$. Suppose further that A and B are similar matrices in $M_n(\mathbb{F})$ and that $A = [T]_{\mathcal{B}}$. Then there exists a basis $\mathcal{C}$ of V such that $B = [T]_{\mathcal{C}}$.*

Proof. Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and suppose that $B = SAS^{-1}$. We define $Q = S^{-1}$ and, for each j , we define $\mathbf{w}_j$ by

$$\mathbf{w}_j = \sum_{i=1}^n q_{ij} \mathbf{v}_i,$$

so that $\mathbf{q}_j$ is the coordinate vector $[\mathbf{w}_j]_{\mathcal{B}}$. Since Q is invertible, then $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ is a basis of $\mathbb{F}^n$, so $\mathcal{C} = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a basis of V . (This is again due to the coordinate mapping being an isomorphism.) We also have

$$P_{\mathcal{C}, \mathcal{B}} \mathbf{e}_j = P_{\mathcal{C}, \mathcal{B}} [\mathbf{w}_j]_{\mathcal{C}} = [\mathbf{w}_j]_{\mathcal{B}} = \mathbf{q}_j$$

for each j , so $P_{\mathcal{C}, \mathcal{B}} = Q$. Therefore, by Lemma 5.6.6, p. 145, $S = P_{\mathcal{B}, \mathcal{C}}$, and so by Corollary 5.6.9, p. 146, $[T]_{\mathcal{C}} = SAS^{-1} = B$. ■

Note 6.4.7 Proposition 6.4.6, p. 171, along with Proposition 6.3.15, p. 167 show us that two matrices are similar if and only if they represent the same linear transformation with respect to different bases.

With these two results in hand, we can state the first invariant we have been pointing toward.

Corollary 6.4.8 *If $A, B \in M_n(\mathbb{F})$ are similar, then $\text{rank}(A) = \text{rank}(B)$.*

Proof. This follows from Theorem 6.4.5, p. 171 (in the case where $V = W$ and $\mathcal{B} = \mathcal{C}$) as well as Proposition 6.4.6, p. 171. ■

In the following example, we put this invariant to work.

Example 6.4.9 Let $A, B \in M_3(\mathbb{R})$ be the following matrices:

$$A = \begin{bmatrix} -3.5 & 5 & -4.5 \\ 3.5 & -3 & -1.5 \\ -3 & 5 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 & 4 \\ -2 & -4 & 0 \\ 3.5 & -5 & -2 \end{bmatrix}.$$

Some quick row reduction shows that $\text{rank}(A) = 2$ while $\text{rank}(B) = 3$, so we know that A and B are not similar. ▲

We now move into the development and discussion of our final invariant, the *trace* of a matrix. Of all the invariants we discuss, the trace is *by far* the easiest to calculate!

Definition 6.4.10 Let $A \in M_n(\mathbb{F})$. The **trace** of A is the sum of the numbers along the main diagonal of A . In other words,

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

◆

Example 6.4.11 Here are two matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 0 \\ 3.5 & -2 & -3 \\ 4 & -2.5 & -2.5 \end{bmatrix}.$$

By summing the entries along the main diagonal, we see that $\text{tr}(A) = 1$ and $\text{tr}(B) = -2.5$. ▲

We need surprisingly few additional results in order to establish the trace as an invariant.

Proposition 6.4.12 *If $A \in M_{m,n}(\mathbb{F})$ and $B \in M_{n,m}(\mathbb{F})$, then*

$$\text{tr}(AB) = \text{tr}(BA).$$

Proof. We begin by using the definitions of the trace and matrix multiplication:

$$\operatorname{tr}(AB) = \sum_{i=1}^m [AB]_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji}.$$

Since multiplication and addition in a field are both commutative, we can switch the order of summation and multiplication. The calculation is finished using the definitions of matrix multiplication and the trace again:

$$\operatorname{tr}(AB) = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \sum_{j=1}^n [BA]_{jj} = \operatorname{tr}(BA).$$

■

This next result finishes our argument.

Corollary 6.4.13 *If A and B are similar matrices, then $\operatorname{tr}(A) = \operatorname{tr}(B)$.*

Proof. Suppose that $B = PAP^{-1}$. By the associativity of matrix multiplication and Proposition 6.4.12, p. 172, we have

$$\begin{aligned} \operatorname{tr}(B) &= \operatorname{tr}(PAP^{-1}) = \operatorname{tr}((PA)P^{-1}) = \operatorname{tr}(P^{-1}(PA)) \\ &= \operatorname{tr}((P^{-1}P)A) = \operatorname{tr}(IA) = \operatorname{tr}(A). \end{aligned}$$

■

We can now use the trace as an invariant.

Example 6.4.14 Let A and B be the following matrices over $\mathbb{F}_5$:

$$A = \begin{bmatrix} 4 & 1 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 2 \\ 3 & 0 \end{bmatrix}.$$

Since $\operatorname{tr}(A) = 1$ and $\operatorname{tr}(B) = 2$, we know that A and B are not similar matrices. ▲

One consequence of this invariant is that it allows us to define the trace of a linear transformation.

Corollary 6.4.15 *Let V be a finite-dimensional vector space and let $T \in L(V)$. If $\mathcal{B}$ and $\mathcal{C}$ are two bases for V , then $\operatorname{tr}([T]_{\mathcal{B}}) = \operatorname{tr}([T]_{\mathcal{C}})$.*

Proof. This is a direct and immediate consequence of Proposition 6.3.15, p. 167 and Corollary 6.4.13, p. 173. ■

Definition 6.4.16 Let V be a finite-dimensional vector space and let $T \in L(V)$. Then the **trace** of T is $\operatorname{tr}(T) = \operatorname{tr}([T]_{\mathcal{B}})$, where $\mathcal{B}$ is any basis of V . ◆

Example 6.4.17 We end this section with an example of a linear transformation $T : P_1 \rightarrow P_1$ which was defined in Example 6.3.17, p. 167. In that example, we saw that

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & -4 \\ 3 & -6 \end{bmatrix},$$

so we can see that $\operatorname{tr}(T) = -5$. ▲

6.4.3 Reading Questions

1. Use only the rank invariant to answer the following questions. If that invariant does not give you enough information to answer the question,

explain why that is.

$$A = \begin{bmatrix} 1.5 & 3 & -1.5 \\ -0.5 & 0 & 1.5 \\ 0.5 & -3 & -4.5 \end{bmatrix}, \quad B = \begin{bmatrix} 3.5 & -4.5 & -4 \\ 3.5 & -0.5 & 4 \\ -3.5 & -1.5 & 3.5 \end{bmatrix},$$

$$C = \begin{bmatrix} 4 & -2 & -6 \\ 2.5 & 5 & 2.5 \\ -2.5 & -5 & -2.5 \end{bmatrix}.$$

- (a) Are A and B similar? Explain.
 - (b) Are B and C similar? Explain.
 - (c) Are A and C similar? Explain.
2. Use only the trace invariant to answer the following questions. If that invariant does not give you enough information to answer the question, explain why that is. (The matrices for this question are the same as for the previous reading question.)
- (a) Are A and B similar? Explain.
 - (b) Are B and C similar? Explain.
 - (c) Are A and C similar? Explain.

6.4.4 Exercises

1. Consider the following two matrices in $M_3(\mathbb{R})$:

$$A = \begin{bmatrix} -5 & 4.5 & 1.5 \\ -1.5 & 0 & 4.5 \\ 1 & -1.5 & 1.5 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} -3.5 & 3.5 & -3.5 \\ -4.5 & 3.5 & -2.5 \\ -4 & 0.5 & 3 \end{bmatrix}.$$

- (a) Explain why both determinant and rank are *not* useful invariants to say that A and B are not similar.
 - (b) Prove that A and B are not similar.
2. Consider the following two matrices in $M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 1 & -6 & 2 \\ 1 & 2 & 1 \\ 0 & 6 & -1 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 10 & 13 & -5 \\ -3 & -6 & -1 \\ 3 & 3 & -2 \end{bmatrix}.$$

- (a) Explain why both trace and rank are *not* useful invariants to say that A and B are not similar.
 - (b) Prove that A and B are not similar.
3. Consider the following two matrices in $M_3(\mathbb{R})$:

$$A = \begin{bmatrix} -10 & -20 & -2 \\ 7 & 14 & 1 \\ -5 & -10 & -1 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 & 1 \\ 2 & -2 & 2 \\ 2 & -3 & 3 \end{bmatrix}.$$

- (a) Explain why the trace, rank, and determinant are *not* useful invariants to say that A and B are not similar.
- (b) Prove that A and B are not similar.

4. Consider the following two matrices in $M_2(\mathbb{F}_7)$:

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 4 & 4 \\ 5 & 3 \end{bmatrix}.$$

- (a) Explain why the trace and rank are *not* useful invariants to say that A and B are not similar.
- (b) Prove that A and B are not similar.

5. Consider the following two matrices in $M_3(\mathbb{F}_7)$:

$$A = \begin{bmatrix} 4 & 6 & 2 \\ 1 & 5 & 6 \\ 1 & 5 & 5 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 2 & 2 & 2 \\ 4 & 4 & 1 \\ 5 & 5 & 1 \end{bmatrix}.$$

- (a) Explain why the trace, rank, and determinant are *not* useful invariants to say that A and B are not similar.
- (b) Prove that A and B are not similar.

6. Consider the following two matrices in $M_3(\mathbb{F}_7)$:

$$A = \begin{bmatrix} 2 & 2 & 4 \\ 5 & 3 & 3 \\ 4 & 0 & 0 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 2 & 0 & 0 \\ 5 & 4 & 3 \\ 2 & 5 & 6 \end{bmatrix}.$$

- (a) Explain why the trace, rank, determinant, and characteristic polynomial are *not* useful invariants to say that A and B are not similar.
- (b) Prove that A and B are not similar.

7. There are 16 elements of the set $M_2(\mathbb{F}_2)$. How many equivalence classes are there for this set under the similarity equivalence relation? Use as many of the invariants as you can to distinguish between matrices that are not similar to each other.

Writing Exercises

8. Let $A \in M_2(\mathbb{F})$ be this 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

- (a) Write out the characteristic polynomial of A , substituting $\det(A)$ and $\text{tr}(A)$ where appropriate.
- (b) Argue that, for 2×2 matrices, the fact that the characteristic polynomial is an invariant (for similarity) implies that both the trace and determinant are invariants.
9. Fix a positive integer $n \geq 2$. Is diagonalizability an invariant (for similarity) on the set $M_n(\mathbb{F})$? In other words, for $A, B \in M_n(\mathbb{F})$, if A and B are similar, must it be true that either (a) they are both diagonalizable, or (b) they are both not diagonalizable?
10. Suppose that $A \in M_n(\mathbb{F})$.
- (a) Prove that if A has n distinct eigenvalues, then $\text{tr}(A)$ is the sum of the eigenvalues of A .

- (b) Is the result from part (a) true if A does not have n distinct eigenvalues? For example, if $A \in M_3(\mathbb{R})$ has only two distinct eigenvalues, must $\text{tr}(A)$ be the sum of the eigenvalues (with multiplicity)?

11.

- (a) Prove that if $A, B, C \in M_n(\mathbb{F})$, then

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

- (b) Give an example of three matrices $A, B, C \in M_n(\mathbb{F})$ such that $\text{tr}(ABC) \neq \text{tr}(ACB)$.

- 12.** For a matrix $A \in M_n(\mathbb{F})$, define the function $S(A)$ as the sum of all of the entries of A . Prove or disprove that S is an invariant for similar matrices.
- 13.** Is rank an invariant for row equivalence? Justify your answer.
- 14.** Is the trace an invariant for row equivalence? Justify your answer.
- 15.** Is the determinant an invariant for row equivalence? Justify your answer.
- 16.** Is invertibility an invariant for row equivalence? Justify your answer.
- 17.** Is invertibility an invariant for similarity? Justify your answer.

Chapter 7

Inner Products

7.1 Inner Products

A general vector space need not have any relevant geometry, and in most of our work up to this point, geometric notions did not play a central role. In this chapter, however, we will begin to take advantage of the geometry present in *some* vector spaces.

7.1.1 The Dot Product

In Euclidean geometry, we are introduced to the *dot product* quite early. The dot product in $\mathbb{R}^n$ is essential to our understandings of length and distance.

Definition 7.1.1 For two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we have the **dot product** of $\mathbf{x}$ and $\mathbf{y}$ given by

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i,$$

where $\mathbf{x} = [x_i]$ and $\mathbf{y} = [y_i]$. ◆

Example 7.1.2 Suppose that $\mathbf{x}$ and $\mathbf{y}$ are the following two vectors in $\mathbb{R}^3$:

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}.$$

Then $\mathbf{x} \cdot \mathbf{y} = (-1)(0) + 2(1) + (-2)(-3) = 8$. ▲

Note 7.1.3 Having facility with matrix multiplication now, the observant reader will notice that $\mathbf{x} \cdot \mathbf{y} = \mathbf{y}^T \mathbf{x}$.

Definition 7.1.4 The **length** or **norm** of a vector $\mathbf{x} \in \mathbb{R}^n$ is the nonnegative scalar $\|\mathbf{x}\|$ defined by

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Example 7.1.5 If $\mathbf{x} \in \mathbb{R}^3$ is the same as in Example 7.1.2, p. 177, then ◆

$$\|\mathbf{x}\| = \sqrt{(-1)^2 + 2^2 + (-2)^2} = \sqrt{9} = 3.$$

▲

Of special relevance for us is the fact that the dot product gives us a notion of angles and perpendicularity.

Definition 7.1.6 Two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ are **orthogonal** if $\mathbf{x} \cdot \mathbf{y} = 0$. $\blacklozenge$

Note 7.1.7 The word “orthogonal” is another way of saying “perpendicular,” but “orthogonal” is used much more frequently in linear algebra.

Example 7.1.8 Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be the following vectors in $\mathbb{R}^2$:

$$\mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

We can see that $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal, since $\mathbf{u} \cdot \mathbf{v} = 10$. However, $\mathbf{u}$ and $\mathbf{w}$ are orthogonal, as $\mathbf{u} \cdot \mathbf{w} = 0$. $\blacktriangle$

As this chapter continues, the reader will see just how important orthogonality is. For now, we note that all of the vectors in the standard basis of $\mathbb{R}^n$, $\mathbf{e}_1, \dots, \mathbf{e}_n$, are orthogonal to each other. That is, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ whenever $i \neq j$.

A consequence of this last fact is stated in the following proposition.

Proposition 7.1.9 If $\mathbf{v} \in \mathbb{R}^n$ is given by $\mathbf{v} = [v_i]$, then $v_i = \mathbf{v} \cdot \mathbf{e}_i$ for each i , $1 \leq i \leq n$.

Proof. By definition of the dot product, we have

$$\mathbf{v} \cdot \mathbf{e}_i = \sum_{j=1}^n v_j (\mathbf{e}_i)_j = v_i,$$

since the only nonzero entry of $\mathbf{e}_i$ is $(\mathbf{e}_i)_i = 1$. $\blacksquare$

This has been a very brief review/introduction to the dot product. As we generalize this function in what follows, we will remind the reader of important facts and properties as we need them.

7.1.2 The Inner Product

In the same way that vectors in $\mathbb{R}^n$ gave us the intuition to consider a general vector space, the dot product in $\mathbb{R}^n$ points us toward a more general function on vector spaces. Our generalization of the dot product is called the *inner product*.

Note 7.1.10 Before this definition we need a quick reminder. For a complex number $z = a + bi$, recall that the **complex conjugate** of z is defined by $\bar{z} = a - bi$. This will be used in the following definition.

Definition 7.1.11 Let V be a vector space over a field $\mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. An **inner product** on V is a function that associates to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V an element of the field $\langle \mathbf{u}, \mathbf{v} \rangle$ satisfying all of the following axioms. For all $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V , and all $c \in \mathbb{F}$:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$;
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c \langle \mathbf{u}, \mathbf{v} \rangle$; and
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

A vector space together with an inner product is called an **inner product space**. $\blacklozenge$

Note 7.1.12 If the field we have in mind is $\mathbb{R}$ instead of $\mathbb{C}$, then the first property listed in the definition is just $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$. (If $x \in \mathbb{R}$, then $\bar{x} = x$.) Also, if our field is $\mathbb{C}$, we still require $\langle \mathbf{u}, \mathbf{u} \rangle$ to be a real number, as this is implicit in the fourth property where $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$.

Before we introduce examples, we should address why the only fields we allow for inner product spaces are $\mathbb{R}$ and $\mathbb{C}$. The inner product requires that a notion of *order* be present in any field over which a vector space is defined. This is inherent in the fourth property listed in the definition of an inner product, where we must have $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ for all $\mathbf{u} \in V$. We do not have this sort of ordering in a finite field like $\mathbb{F}_5$.

As we discuss in Appendix A, p. 200, each field $\mathbb{F}_p$ is really a set of equivalence classes of $\mathbb{Z}$ under the equivalence relation congruence mod p . So when we write 2 as an element of $\mathbb{F}_5$, we're referring to $[2]$, the equivalence class of all integers congruent to 2 mod 5. And although our convention is to use the integers $0, 1, \dots, p-1$ as the equivalence class representatives for the elements of $\mathbb{F}_p$, this is not a requirement. So, 7 and 12 and -3 could all be used as the representative of $[2] \in \mathbb{F}_5$. This means that we cannot in any coherent way say that $[2] \in \mathbb{F}_5$ is "greater than or equal to 0." (Since 0 in $\mathbb{F}_5$ is $[0]$, what we mean is that " $[2] \geq [0]$ " has no meaning.) Because of this lack of ordering, finite fields do not have the geometric properties that we require for an inner product space. We must bid a fond farewell to these dear friends for now, knowing that we will cross paths with them again in our mathematical futures.

Example 7.1.13 All real vector spaces $\mathbb{R}^n$ with the dot product are inner product spaces. (Once again, we would be particularly bad at generalizing if the motivating case were not an example of the general situation!) ▲

Example 7.1.14 For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, the *standard inner product* is defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i \bar{v}_i,$$

where $\mathbf{u} = [u_i]$ and $\mathbf{v} = [v_i]$.

As an example calculation, we consider the following two vectors in $\mathbb{C}^2$:

$$\mathbf{u} = \begin{bmatrix} 1+i \\ -2i \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2-i \\ 3+4i \end{bmatrix}.$$

Then we have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= (1+i)\overline{(2-i)} + (-2i)\overline{(3+4i)} \\ &= (1+i)(2+i) + (-2i)(3-4i) \\ &= -7-3i. \end{aligned}$$

We will leave for the exercises the proof that the inner product axioms hold for this function. ▲

Example 7.1.15 Let $C([0, 1])$ denote the vector space of continuous real-valued functions on the interval $[0, 1]$. (See Example 2.3.5, p. 47 for a discussion of vector spaces like this one.) We can study an inner product on this space defined by the following:

$$\langle f, g \rangle = \int_0^1 f(x)g(x) \, dx.$$

Again, we provide an example of a calculation. If $f(x) = 2x$ and $g(x) = x^2 - 4$,

then

$$\langle f, g \rangle = \int_0^1 2x(x^2 - 4) dx = \int_0^1 (2x^3 - 8x) dx = -\frac{7}{2}.$$

Proving that the inner product axioms hold requires recalling a few facts from calculus. We leave this to the exercises. $\blacktriangle$

Example 7.1.16 We consider an inner product on P_2 , the vector space of all real-valued polynomials of degree at most two. For $p, q \in P_2$, we define the function

$$\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2).$$

To become familiar with this function, we can calculate the inner product of $p = t - 2t^2$ and $q = 3 + 4t$. Calculating $\langle p, q \rangle$ only involves evaluating these polynomials at $t = 0$, $t = 1$, and $t = 2$ and then finding the sum of the products. We find that

$$\begin{aligned} \langle p, q \rangle &= p(0)q(0) + p(1)q(1) + p(2)q(2) \\ &= (0)(3) + (-1)(7) + (-6)(11) = -73. \end{aligned}$$

The first inner product property holds since multiplication in the real numbers is commutative. The second and third properties hold by the definitions of vector addition and scalar multiplication in P_2 . The first part of the third property holds because the sum of squared real numbers must always be non-negative. The final part of the third property holds by an important fact about polynomials: any polynomial of degree n which has $n + 1$ zeros must be the zero polynomial. (This is why we must take *three* evaluation points for this function to be an inner product in P_2 .) $\blacktriangle$

The following properties flow fairly quickly from the definition of an inner product.

Proposition 7.1.17 *Suppose that V is an inner product space. Then the following statements are all true.*

1. For each $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.
2. For each $\mathbf{u}, \mathbf{v} \in V$ and each $c \in \mathbb{F}$, $\langle \mathbf{u}, c\mathbf{v} \rangle = \bar{c} \langle \mathbf{u}, \mathbf{v} \rangle$.
3. For each $\mathbf{u} \in V$, $\langle \mathbf{0}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{0} \rangle = 0$.
4. If $\mathbf{u} \in V$ and $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for every $\mathbf{v} \in V$, then $\mathbf{u} = \mathbf{0}$.

Proof. We will prove the second property and leave the others for the exercises. Using the first and third axioms from the definition of the inner product, we have

$$\langle \mathbf{u}, c\mathbf{v} \rangle = \overline{\langle c\mathbf{v}, \mathbf{u} \rangle} = \overline{c \langle \mathbf{v}, \mathbf{u} \rangle} = \bar{c} \overline{\langle \mathbf{v}, \mathbf{u} \rangle} = \bar{c} \langle \mathbf{u}, \mathbf{v} \rangle.$$

$\blacksquare$

The presence of an inner product gives us a good way to define the length of a vector.

Definition 7.1.18 Let V be an inner product space and let $\mathbf{v} \in V$. Then the **norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

If $\|\mathbf{v}\| = 1$, then $\mathbf{v}$ is called a **unit vector**. $\blacklozenge$

In the following examples we calculate the norm of a few vectors in different vector spaces.

Example 7.1.19 We consider the following vector in $\mathbb{C}^3$:

$$\mathbf{v} = \begin{bmatrix} 2 + 4i \\ -2 + 4i \\ 2i \end{bmatrix}.$$

Using the standard inner product on $\mathbb{C}^3$, we have

$$\|\mathbf{v}\| = \sqrt{(4+16) + (4+16) + 4} = \sqrt{44}.$$

▲

Example 7.1.20 Returning to the vector space $C([0, 1])$ with the inner product defined in Example 7.1.15, p. 179, we can find the norm of $f(x) = 2 + x$:

$$\int_0^1 (2+x)^2 dx = \int_0^1 (4+4x+x^2) = \frac{19}{3}.$$

This means that $\|f\| = \sqrt{\frac{19}{3}}$.

▲

Using the definition of the norm, we can examine what happens to the “length” of a vector when it is multiplied by a scalar:

$$\|c\mathbf{v}\| = \sqrt{\langle c\mathbf{v}, c\mathbf{v} \rangle} = \sqrt{|c|^2 \langle \mathbf{v}, \mathbf{v} \rangle} = |c| \|\mathbf{v}\|.$$

(Note that when $\mathbb{C}$ is our field, $|c|$ for a scalar $c = a + bi$ is $|c| = \sqrt{a^2 + b^2}$.) From this calculation we can see that when a vector is multiplied by a scalar, its length is multiplied by that same scalar, in a way. (We can make the most geometric sense of this when $\mathbb{R}$ is our field and when c is positive.)

Example 7.1.21 Often we will want a *unit vector* that points in the same direction as a given vector. We accomplish this by dividing a vector by its length in order to form a vector of length 1.

If we consider the vector $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ in $\mathbb{R}^2$ with the dot product, then we have

$$\|\mathbf{v}\| = \sqrt{1+16} = \sqrt{17}.$$

Therefore, a unit vector in the direction of $\mathbf{v}$ would be

$$\frac{1}{\sqrt{17}}\mathbf{v} = \begin{bmatrix} -\frac{1}{\sqrt{17}} \\ \frac{4}{\sqrt{17}} \end{bmatrix}.$$

▲

7.1.3 Orthogonality

In the same way that we used the dot product to define orthogonality in $\mathbb{R}^n$, we can now extend that definition to our more general setting.

Definition 7.1.22 Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is orthogonal if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ whenever $i \neq j$. ◆

One of the ways that orthogonality is used is through the following result.

Proposition 7.1.23 *An orthogonal set of nonzero vectors in an inner product space V is linearly independent.*

Proof. Let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthogonal set of vectors in V . Suppose that

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

for some scalars $c_1, \dots, c_n \in \mathbb{F}$. We want to show that all the scalars must be zero. Then, for each k , we have

$$0 = \langle \mathbf{0}, \mathbf{v}_k \rangle = \left\langle \sum_{i=1}^n c_i \mathbf{v}_i, \mathbf{v}_k \right\rangle = \sum_{i=1}^n c_i \langle \mathbf{v}_i, \mathbf{v}_k \rangle = c_k \langle \mathbf{v}_k, \mathbf{v}_k \rangle = c_k \|\mathbf{v}_k\|^2.$$

Since $c_k \|\mathbf{v}_k\|^2 = 0$ but $\mathbf{v}_k \neq \mathbf{0}$, we know that $\|\mathbf{v}_k\|^2 \neq 0$, so $c_k = 0$. This is true for each k , $1 \leq k \leq n$, so V' is linearly independent. ■

The next result is sometimes referred to as the Pythagorean Theorem for general inner product spaces. When there are only two orthogonal vectors, the reader will recognize the reference to the Pythagorean Theorem.

Theorem 7.1.24 *If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthogonal set of vectors in an inner product space V , then*

$$\|\mathbf{v}_1 + \dots + \mathbf{v}_n\|^2 = \|\mathbf{v}_1\|^2 + \dots + \|\mathbf{v}_n\|^2.$$

7.1.4 Results for Inner Product Spaces

The property of orthogonality is so powerful that we will occasionally want to call upon it even when it is not already on the scene.

Lemma 7.1.25 *Let V be an inner product space and let $\mathbf{u}, \mathbf{v} \in V$. Then $\mathbf{u}$ can be written as*

$$\mathbf{u} = c\mathbf{v} + \mathbf{w}, \tag{7.1}$$

where $c \in \mathbb{F}$ and $\mathbf{w}$ is orthogonal to $\mathbf{v}$. Specifically, if $\mathbf{v} \neq \mathbf{0}$, then

$$c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}, \text{ and } \mathbf{w} = \mathbf{u} - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v}.$$

Proof. If $\mathbf{v} = \mathbf{0}$, then we can take $\mathbf{w} = \mathbf{u}$ and $c = 1$, as every vector is orthogonal to $\mathbf{0}$. So, we now suppose that $\mathbf{v} \neq \mathbf{0}$.

If there exists $c \in \mathbb{F}$ such that $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ with $\mathbf{w}$ orthogonal to $\mathbf{v}$, then we must have

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle c\mathbf{v} + \mathbf{w}, \mathbf{v} \rangle = c \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle = c \|\mathbf{v}\|^2.$$

This shows that the only possibility for c is $c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$.

Once c has been determined, then the choice of $\mathbf{w}$ is determined by (7.1)—we must have $\mathbf{w} = \mathbf{u} - c\mathbf{v}$. Now it is easy to check that, with these values, we indeed have $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ and that (7.1) holds. ■

Example 7.1.26 We consider two vectors in $\mathbb{R}^2$ to understand the relationship in this lemma:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The lemma specifies our calculations:

$$c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} = \frac{4}{5}, \quad \mathbf{w} = \mathbf{u} - c\mathbf{v} = \begin{bmatrix} \frac{14}{5} \\ \frac{7}{5} \end{bmatrix}.$$

We will verify the properties specified by the lemma. Since

$$\mathbf{v} \cdot \mathbf{w} = -\frac{14}{5} + \frac{14}{5} = 0,$$

we have $\langle \mathbf{v}, \mathbf{w} \rangle = 0$. Additionally, since

$$c\mathbf{v} + \mathbf{w} = \begin{bmatrix} -\frac{4}{5} \\ \frac{8}{5} \end{bmatrix} + \begin{bmatrix} \frac{14}{5} \\ \frac{7}{5} \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

we see that $\mathbf{u} = c\mathbf{v} + \mathbf{w}$. ▲

There are two famous results which involve the norm in an inner product space. We present them without proof.

Theorem 7.1.27 The Cauchy-Schwarz Inequality. *For every pair of vectors $\mathbf{u}, \mathbf{v}$ in an inner product space V , we have*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

with equality holding if and only if one vector is a scalar multiple of the other.

Theorem 7.1.28 The Triangle Inequality. *For every pair of vectors $\mathbf{u}, \mathbf{v}$ in an inner product space V , we have*

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

We end this section with one final example of an inner product space.

Example 7.1.29 We consider the vector space $\mathbb{R}^2$ with a modified inner product:

$$\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + u_2v_2.$$

The only change from the dot product in $\mathbb{R}^2$ is the coefficient 2 on the first term. It is not difficult to verify that this is an inner product.

Since an inner product provides a way to measure distance and length (as well as angles), it is instructive to consider how this inner product changes our experience of $\mathbb{R}^2$. Just to take one example, if we think of the “unit circle” as the collection of all unit vectors in $\mathbb{R}^2$, then using this inner product we no longer have a circle but an ellipse. The radii of this ellipse would be $\frac{1}{\sqrt{2}}$ horizontally and 1 vertically. ▲

7.1.5 Reading Questions

1. Consider Example 7.1.29, p.183 and the inner product on $\mathbb{R}^2$ defined there.

(a) If $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, calculate $\|\mathbf{u}\|$.

(b) Describe all of the vectors in $\mathbb{R}^2$ which are orthogonal to $\mathbf{u}$ using this inner product. All of these vectors fall on a line through the origin—what is that line?

2. Consider the following function on P_1 . For polynomials p and q , define $\langle p, q \rangle$ by

$$\langle p, q \rangle = p(0)q(0) - p(1)q(1).$$

Explain why this function is not an inner product on P_1 . (You must show why one of the inner product axioms fails, and to do this you should use an example.)

7.1.6 Exercises

1. Consider the following inner product on P_2 . For $p, q \in P_2$,

$$\langle p, q \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1).$$

(You do not need to prove that this is an inner product.)

- (a) Calculate $\langle p, q \rangle$ where $p = 3 - t$ and $q = 2 + 2t^2$.
 - (b) Find a nonzero vector $r \in P_2$ which is orthogonal to the vector p from part (a).
 - (c) Calculate $\|p\|$ and $\|q\|$ for p, q from part (a).
2. Use Proposition 7.1.23, p. 181 to prove that the following set of vectors in $\mathbb{C}^3$ is linearly independent: $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 - i \\ 3 + 2i \\ 4 + i \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 8 + 38i \\ 5 - 25i \\ 13 + 13i \end{bmatrix}.$$

3. Consider the following inner product on $\mathbb{R}^2$:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2.$$

- (a) Give an example of two vectors in $\mathbb{R}^2$ which are orthogonal with respect to the dot product but which are not orthogonal with respect to this inner product.
 - (b) Give an example of two vectors in $\mathbb{R}^2$ which are orthogonal with respect to this inner product but which are not orthogonal with respect to the dot product.
4. Let A be the following matrix over $\mathbb{R}$:

$$A = \begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix}.$$

Define a function on $\mathbb{R}^2$ by

$$\langle \mathbf{u}, \mathbf{v} \rangle = (A\mathbf{u}) \cdot (A\mathbf{v}),$$

where the right side of the equals sign uses the standard dot product in $\mathbb{R}^2$. (This function defines an inner product, but you do not need to prove this right now.)

- (a) Let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Calculate $\langle \mathbf{u}, \mathbf{v} \rangle$ using this inner product.

- (b) Calculate $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$ for the vectors $\mathbf{u}$ and $\mathbf{v}$ given in part a.
- (c) Find a vector $\mathbf{w} \in \mathbb{R}^2$ which is orthogonal to the vector $\mathbf{u}$ (given in part a.) with respect to this inner product.

5. Define the following function on $M_2(\mathbb{R})$:

$$\langle A, B \rangle = \text{tr}(A^T B).$$

(This function defines an inner product, but you do not need to prove this right now.)

- (a) Let A and B be the following matrices:

$$A = \begin{bmatrix} 2 & 0 \\ 1 & -3 \end{bmatrix}, \text{ and } B = \begin{bmatrix} -4 & 1 \\ 1 & 2 \end{bmatrix}.$$

Calculate $\langle A, B \rangle$ using this inner product.

- (b) Calculate $\|A\|$ and $\|B\|$ for the matrices A and B given in part a.

- (c) Find a matrix $C \in M_2(\mathbb{R})$ which is orthogonal to the matrix A (given in part a.) with respect to this inner product.

6. Consider the following function defined on $M_2(\mathbb{R})$:

$$\langle A, B \rangle = \det(A) \cdot \det(B).$$

Show that this function is not an inner product.

7. Consider the following function defined on P_2 :

$$\langle p, q \rangle = p(-1)q(-1) + p(2)q(2).$$

Show that this function is not an inner product.

Writing Exercises

8. Consider the following function defined on $\mathbb{R}^2$:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_2 + u_2 v_1.$$

Prove or disprove: this function is an inner product.

9. Prove that the function given in Exercise 7.1.6.4, p. 184 is an inner product.
10. Prove that the function given in Exercise 7.1.6.5, p. 185 is an inner product.
11. Prove that the standard inner product on $\mathbb{C}^n$, defined in Example 7.1.14, p. 179, is an inner product.
12. Prove that the inner product defined in Example 7.1.15, p. 179 is an inner product.
13. Suppose that V is a vector space, W is an inner product space, and that $T \in L(V, W)$ is injective. For $\mathbf{v}_1, \mathbf{v}_2 \in V$, define $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle_T$ by

$$\langle \mathbf{v}_1, \mathbf{v}_2 \rangle_T = \langle T(\mathbf{v}_1), T(\mathbf{v}_2) \rangle,$$

where the right-hand side is the inner product on W . Prove that this defines an inner product on V .

14. Prove properties 1, 3, and 4 of Proposition 7.1.17, p. 180.

7.2 Orthonormal Bases

While the previous section extended the dot product in $\mathbb{R}^n$ to the notion of an inner product in V , in this section we focus on bases for a vector space. One remarkably nice feature of the standard basis in $\mathbb{R}^n$ is that, with respect to the dot product, it is an orthogonal set in which every vector is a unit vector. In this section we will see that such a basis exists in every inner product space.

7.2.1 Orthonormality

Our first definition is the most important in the section, as we generalize the important characteristics of the standard basis in $\mathbb{R}^n$.

Definition 7.2.1 Let V be an inner product space, and let $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a subset of V . Then V' is an **orthonormal** set if it is an orthogonal set and $\|\mathbf{v}_i\| = 1$ for all $\mathbf{v}_i \in V'$. If V' is a basis for V , then it is called an **orthonormal basis**. $\blacklozenge$

Example 7.2.2 The standard basis is an orthonormal basis of either $\mathbb{R}^n$ or $\mathbb{C}^n$. $\blacktriangle$

Example 7.2.3 The following is an orthonormal basis of either $\mathbb{R}^2$ or $\mathbb{C}^2$:

$$\mathcal{B} = \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

Example 7.2.4 For any fixed $\theta \in \mathbb{R}$, the following is an orthonormal basis of $\mathbb{R}^2$ or $\mathbb{C}^2$:

$$\mathcal{B} = \left\{ \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix} \right\}.$$

Example 7.2.5 Because of the requirement that each vector be a unit vector, the vectors in an orthonormal basis aren't the prettiest to behold. Here is an orthonormal basis of $\mathbb{R}^3$:

$$\mathcal{B} = \left\{ \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{\sqrt{2}}{\sqrt{15}} \\ \frac{1}{\sqrt{30}} \\ -\frac{\sqrt{5}}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} -\frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} \right\}.$$

Example 7.2.6 Consider the following inner product on the space P_1 :

$$\langle p, q \rangle = p(0)q(0) + p(1)q(1).$$

The set $\mathcal{B} = \{t, 1-t\}$ is an orthonormal basis for P_1 . We first verify that this is an orthogonal set:

$$\langle t, 1-t \rangle = (0)(1) + (1)(0) = 0.$$

We can also see that each of these are unit vectors:

$$\langle t, t \rangle = 0^2 + 1^2 = 1$$

$$\langle 1-t, 1-t \rangle = 1^2 + 0^2 = 1.$$

Thus, $\mathcal{B}$ is an orthonormal basis for P_1 with this inner product. $\blacktriangle$

7.2.2 Coordinates in Orthonormal Bases

Having an orthonormal basis makes some tasks easier than they would be otherwise. In particular, when we need to find coordinates of vectors with respect to an orthonormal basis, the path is fairly gentle to walk.

Theorem 7.2.7 *Let V and W be inner product spaces, and let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthonormal basis for V and $\mathcal{C} = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ be an orthonormal basis for W . For every $\mathbf{v} \in V$ and every $T \in L(V, W)$, we have*

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{v}_1 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{v}_n \rangle \end{bmatrix} \quad (7.2)$$

and

$$[[T]_{\mathcal{B}, \mathcal{C}}]_{jk} = \langle T(\mathbf{v}_k), \mathbf{w}_j \rangle. \quad (7.3)$$

Proof. For $\mathbf{v} \in V$, we have

$$\mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i$$

for some $c_i \in \mathbb{F}$ since $\mathcal{B}$ is a basis of V . If we take the inner product of both sides of this equation with $\mathbf{v}_k$ and use both the linearity of the inner product and the orthonormality of $\mathcal{B}$, we have

$$\langle \mathbf{v}, \mathbf{v}_k \rangle = \left\langle \sum_{i=1}^n c_i \mathbf{v}_i, \mathbf{v}_k \right\rangle = \sum_{i=1}^n c_i \langle \mathbf{v}_i, \mathbf{v}_k \rangle = c_k.$$

This means that

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{v}_i \rangle \mathbf{v}_i,$$

which is the same as (7.2).

We proceed similarly for the second statement in the theorem. For any $\mathbf{w} \in W$, we have

$$\mathbf{w} = \sum_{i=1}^m \langle \mathbf{w}, \mathbf{w}_i \rangle \mathbf{w}_i.$$

Since $T(\mathbf{v}_i) \in W$, this must also be true for each $T(\mathbf{v}_i)$:

$$T(\mathbf{v}_i) = \sum_{i=1}^m \langle T(\mathbf{v}_i), \mathbf{w}_i \rangle \mathbf{w}_i.$$

This is the same as (7.3). ■

We can illustrate one part of this theorem with an example.

Example 7.2.8 In $\mathbb{R}^2$ with the dot product, we consider the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ from Example 7.2.3, p.186. Then the coordinate vector of $\mathbf{v} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$ with respect to $\mathcal{B}$ is

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{v}_1 \rangle \\ \langle \mathbf{v}, \mathbf{v}_2 \rangle \end{bmatrix} = \begin{bmatrix} -\frac{3}{\sqrt{2}} \\ \frac{7}{\sqrt{2}} \end{bmatrix}.$$

▲

The previous theorem has an important consequence. The inner product of any two vectors in an inner product space is the same as the usual inner

product of their coordinate vectors in $\mathbb{C}^n$.

Theorem 7.2.9 *Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthonormal basis for an inner product space V . Then, for any vectors $\mathbf{u}, \mathbf{v} \in V$, we have*

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n \langle \mathbf{u}, \mathbf{v}_i \rangle \overline{\langle \mathbf{v}, \mathbf{v}_i \rangle}$$

and

$$\|\mathbf{v}\|^2 = \sum_{i=1}^n |\langle \mathbf{v}, \mathbf{v}_i \rangle|^2.$$

Proof. By Theorem 7.2.7, p. 187, we have

$$\mathbf{u} = \sum_{i=1}^n \langle \mathbf{u}, \mathbf{v}_i \rangle \mathbf{v}_i \quad \text{and} \quad \mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{v}_i \rangle \mathbf{v}_i.$$

So, using the properties of the inner product and the orthonormality of $\mathcal{B}$, we have

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= \left\langle \sum_{i=1}^n \langle \mathbf{u}, \mathbf{v}_i \rangle \mathbf{v}_i, \sum_{j=1}^n \langle \mathbf{v}, \mathbf{v}_j \rangle \mathbf{v}_j \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n \langle \mathbf{u}, \mathbf{v}_i \rangle \overline{\langle \mathbf{v}, \mathbf{v}_j \rangle} \langle \mathbf{v}_i, \mathbf{v}_j \rangle \\ &= \sum_{i=1}^n \langle \mathbf{u}, \mathbf{v}_i \rangle \overline{\langle \mathbf{v}, \mathbf{v}_i \rangle}. \end{aligned}$$

■

7.2.3 The Gram-Schmidt Process

Orthonormal bases are quite useful, but they do not appear around every corner. In this section we will describe a reliable process to produce an orthonormal basis of an inner product space.

Algorithm 7.2.10 The Gram-Schmidt Process. *This process results in an orthonormal basis for any finite-dimensional inner product space V .*

1. Start with a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V .
2. Define $\mathbf{e}_1 = \frac{1}{\|\mathbf{v}_1\|} \mathbf{v}_1$.
3. For $j = 2, \dots, n$, define $\mathbf{e}_j$ recursively by

$$\mathbf{e}'_j = \mathbf{v}_j - \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k \rangle \mathbf{e}_k$$

and

$$\mathbf{e}_j = \frac{1}{\|\mathbf{e}'_j\|} \mathbf{e}'_j.$$

Then $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis of V . Additionally, for each $j = 1, \dots, n$, we have $\text{Span}\{\mathbf{e}_1, \dots, \mathbf{e}_j\} = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}$.

Note 7.2.11 We omit the proof that the Gram-Schmidt process does what it claims to do, as the proof is on the long and technical side. However, it may be helpful for readers to have a non-technical description of the process. Starting with the original basis, for each vector we strip away the parts of the vector that point in the direction of previous basis vectors. (We understand that “direction” only makes geometric sense in $\mathbb{R}^n$, but perhaps the reader will allow this use of analogy.) What remains gets normalized to be a unit vector and then added to the growing orthonormal basis.

To get a good handle on this algorithm, we now present some examples.

Example 7.2.12 Consider the vector space P_1 with the inner product introduced in Example 7.2.6, p. 186. The standard basis for P_1 is $\{1, t\}$, but this is not an orthonormal basis. We will apply the Gram-Schmidt process to this basis.

First, we calculate that $\langle 1, 1 \rangle = \sqrt{2}$, so our first vector in the orthonormal basis should be $\mathbf{e}_1 = \frac{1}{\sqrt{2}}\mathbf{v}_1 = \frac{1}{\sqrt{2}}$. Then, for the second calculation we need

$$\langle t, \mathbf{e}_1 \rangle = (0) \left(\frac{1}{\sqrt{2}} \right) + (1) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}}.$$

So, we have

$$\mathbf{e}'_2 = t - \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} = t - \frac{1}{2}.$$

Our final step is to normalize $\mathbf{e}'_2$. We find that $\|\mathbf{e}'_2\| = \frac{1}{\sqrt{2}}$, meaning that

$$\mathbf{e}_2 = \sqrt{2}\mathbf{e}'_2 = \sqrt{2}t - \frac{\sqrt{2}}{2}.$$

The reader may want to verify that $\{\mathbf{e}_1, \mathbf{e}_2\}$ is indeed an orthonormal set in P_1 with this inner product. ▲

Example 7.2.13 We will carry out the Gram-Schmidt process on a basis for $\mathbb{R}^3$ where we will use the standard dot product in $\mathbb{R}^3$. Consider the set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}.$$

It is easily checked that $\mathcal{B}$ is a basis for $\mathbb{R}^3$. Now $\|\mathbf{v}_1\| = \sqrt{2}$, so we have

$$\mathbf{e}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

We need $\mathbf{v}_2 \cdot \mathbf{e}_1 = \frac{2}{\sqrt{2}}$ for the next calculation:

$$\mathbf{e}'_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{e}_1)\mathbf{e}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Since $\|\mathbf{e}'_2\| = \sqrt{3}$, we have

$$\mathbf{e}_2 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}.$$

Finally, we will calculate $\mathbf{e}'_3$ and $\mathbf{e}_3$. We need these two dot product calculations first:

$$\mathbf{v}_3 \cdot \mathbf{e}_1 = -\frac{2}{\sqrt{2}}, \quad \mathbf{v}_3 \cdot \mathbf{e}_2 = \frac{1}{\sqrt{3}}.$$

Then we have

$$\begin{aligned} \mathbf{e}'_3 &= \mathbf{v}_3 - (\mathbf{v}_3 \cdot \mathbf{e}_1)\mathbf{e}_1 - (\mathbf{v}_3 \cdot \mathbf{e}_2)\mathbf{e}_2 \\ &= \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ -\frac{1}{3} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}. \end{aligned}$$

Since $\|\mathbf{e}'_3\| = \frac{\sqrt{6}}{3}$, we have

$$\mathbf{e}_3 = \frac{3}{\sqrt{6}}\mathbf{e}'_3 = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}.$$

The set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is an orthonormal basis of $\mathbb{R}^3$. ▲

Example 7.2.14 In our final example we will consider $\mathbb{R}^2$ with the alternative inner product introduced in Example 7.1.29, p. 183. We consider a basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for $\mathbb{R}^2$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

For the Gram-Schmidt process, we first need to calculate $\|\mathbf{v}_1\| = \sqrt{3}$. So, we have

$$\mathbf{e}_1 = \frac{1}{\sqrt{3}}\mathbf{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}.$$

Then we need $\langle \mathbf{v}_2, \mathbf{e}_1 \rangle = \frac{6}{\sqrt{3}}$ for the next step of the process:

$$\mathbf{e}'_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{e}_1 \rangle \mathbf{e}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Finally, we normalize $\mathbf{e}'_2$ in order to get $\mathbf{e}_2$. We find that $\|\mathbf{e}'_2\| = \sqrt{6}$, so

$$\mathbf{e}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{bmatrix}. \quad \blacktriangle$$

While the Gram-Schmidt process has obvious computational implications, it also has some theoretical consequences.

Corollary 7.2.15 *Every finite-dimensional inner product space V has an orthonormal basis.*

Proof. Since we know that every finite-dimensional vector space has a basis (Corollary 5.2.13, p. 119), we can apply the Gram-Schmidt process to that basis.

This proves that such a V always has an orthonormal basis. ■

Corollary 7.2.16 *Suppose that $V' = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an orthonormal set in a finite-dimensional inner product space V . Then V' can be extended to an orthonormal basis of V .*

Proof. By Theorem 5.3.16, p.127, V' can be extended to a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V . We can apply the Gram-Schmidt process to $\mathcal{B}$, producing an orthonormal basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of V . Since V' is an orthonormal set, the Gram-Schmidt process will produce vectors $\mathbf{e}_i$ such that $\mathbf{e}_j = \mathbf{v}_j$ for $j = 1, \dots, k$, meaning that V' is a subset of $\mathcal{B}$. This justifies the claim that $\mathcal{B}$ is an extension of V' . ■

7.2.4 Reading Questions

1. Define the following inner product on P_1 :

$$\langle p, q \rangle = p(-2)q(-2) + p(2)q(2).$$

(You do not need to prove that this is an inner product.) Let $p_1 = \frac{1}{4}t + \frac{1}{2}$ and $p_2 = \frac{1}{4}t - \frac{1}{2}$. Prove or disprove that $\{p_1, p_2\}$ is an orthonormal basis of P_1 . Show your work.

2. Consider the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ of vectors in $\mathbb{R}^2$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Using the standard dot product in $\mathbb{R}^2$, use the Gram-Schmidt process on this basis to produce an orthonormal basis for $\mathbb{R}^2$.

7.2.5 Exercises

1. Consider the following basis for $\mathbb{R}^2$:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}.$$

- (a) Use the Gram-Schmidt process to create an orthonormal basis from $\mathcal{B}$. (Use the standard dot product on $\mathbb{R}^2$.)

- (b) Let $\mathbf{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. Use Theorem 7.2.7, p. 187 to find the coordinate vector of $\mathbf{v}$ with respect to the orthonormal basis of $\mathbb{R}^2$ you created in part (a).

2. Consider the basis $\mathcal{B} = \{p_1, p_2\}$ for P_1 , where

$$p_1 = 2 + t, \quad p_2 = 1 - 2t.$$

Consider the inner product on P_1 defined by

$$\langle p, q \rangle = 2p(0)q(0) + p(1)q(1).$$

(You do not need to prove that this is an inner product.)

- (a) Use the Gram-Schmidt process to create an orthonormal basis from $\mathcal{B}$.

- (b) Let $p = 1 + t$. Use Theorem 7.2.7, p.187 to find the coordinate

vector of p with respect to the orthonormal basis of P_1 you created in part (a).

3. Carry out the Gram-Schmidt process on the following set of vectors in $\mathbb{C}^3$. Use the standard dot product in $\mathbb{C}^3$:

$$\left\{ \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ i \end{bmatrix}, \begin{bmatrix} i \\ 0 \\ 1 \end{bmatrix} \right\}.$$

4. Consider the standard basis of the vector space P_2 . Carry out the Gram-Schmidt process on this basis with respect to the following inner product on P_2 :

$$\langle p, q \rangle = \int_0^1 p(t)q(t) dt.$$

5. Let A be the following matrix in $M_3(\mathbb{R})$:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Consider the inner product on $\mathbb{R}^3$ defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = (A\mathbf{u}) \cdot (A\mathbf{v}),$$

where the standard dot product is in view on the right side of this equation. Find an orthonormal basis of $\mathbb{R}^3$ with respect to this inner product.

Writing Exercises

6. Suppose that the matrix of $T \in L(V)$ with respect to some basis $\mathcal{B}$ is upper triangular. Show that if $\mathcal{C}$ is the orthonormal basis obtained by applying the Gram-Schmidt process to $\mathcal{B}$, then $[T]_{\mathcal{C}}$ is also upper triangular.

7.3 Orthogonal Projections

An inner product provides the tool to decompose vectors into useful components. We have already seen this in Lemma 7.1.25, p.182, but in this section we will expand our discussion. The process of orthogonal projection opens the door to many applications.

7.3.1 Orthogonal Complements

In an inner product space, we can collect all vectors orthogonal to any given set of vectors. In particular, we can do this with a subspace.

Definition 7.3.1 Let U be a subspace of an inner product space V . Then the **orthogonal complement** of U , denoted $U^\perp$, is defined as

$$U^\perp = \{\mathbf{v} \in V \mid \langle \mathbf{u}, \mathbf{v} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

◆

It is relatively easy to verify that $U^\perp$ is itself a subspace of V . We will leave that proof as an exercise.

Proposition 7.3.2 *If U is a subspace of an inner product space V , then $U^\perp$ is also a subspace of V .*

The easiest examples of orthogonal complements to visualize are in $\mathbb{R}^2$ and $\mathbb{R}^3$. If L is a line through the origin in $\mathbb{R}^2$, then $L^\perp$ is the line perpendicular to L which passes through the origin. If P is a plane through the origin in $\mathbb{R}^3$, then $P^\perp$ is the line through the origin which is perpendicular to P .

In an inner product space, any vector can be uniquely decomposed with reference to a subspace and its orthogonal complement.

Theorem 7.3.3 *Let V be an inner product space and let U be a finite-dimensional subspace of V . Then every vector $\mathbf{v} \in V$ can be uniquely written in the form*

$$\mathbf{v} = \mathbf{u} + \mathbf{w},$$

where $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$.

Proof. Since U is finite-dimensional, there is an orthonormal basis for U , $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$. For any $\mathbf{v} \in V$, we define $\mathbf{u}$ by

$$\mathbf{u} = \sum_{i=1}^m \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i,$$

and we let $\mathbf{w} = \mathbf{v} - \mathbf{u}$. Then we have $\mathbf{v} = \mathbf{u} + \mathbf{w}$ and $\mathbf{u} \in U$. For each k , we have

$$\langle \mathbf{u}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle,$$

so

$$\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \langle \mathbf{u}, \mathbf{e}_k \rangle = 0.$$

Since $\mathbf{w}$ is orthogonal to each element of the orthonormal basis of U , we have $\mathbf{w} \in U^\perp$.

We now need to prove that $\mathbf{u}$ and $\mathbf{w}$ are unique. Suppose that $\mathbf{u}_1, \mathbf{u}_2 \in U$ and $\mathbf{w}_1, \mathbf{w}_2 \in U^\perp$ such that

$$\mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}_2.$$

We consider the vector $\mathbf{x}$,

$$\mathbf{x} = \mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1.$$

Since U and $U^\perp$ are subspaces, we have $\mathbf{x} \in U$ and $\mathbf{x} \in U^\perp$, which means that $\langle \mathbf{x}, \mathbf{x} \rangle = 0$. This means that $\mathbf{x} = \mathbf{0}$, so that $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$. ■

7.3.2 Orthogonal Projections

Once we have the sort of decomposition that Theorem 7.3.3, p. 193 provides, we can properly talk about orthogonal projections.

Definition 7.3.4 Let U be a subspace of an inner product space V . The **orthogonal projection** onto U is the function $\text{proj}_U : V \rightarrow V$ given by $\text{proj}_U(\mathbf{v}) = \mathbf{u}$, where $\mathbf{v} = \mathbf{u} + \mathbf{w}$ for $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. ♦

Orthogonal projection has some important properties which we now collect in the following theorem.

Theorem 7.3.5 *Let U be a finite-dimensional subspace of an inner product space V .*

1. *The function proj_U is a linear transformation.*

2. If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis of U , then

$$\text{proj}_U(\mathbf{v}) = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$$

for each $\mathbf{v} \in V$.

3. For each $\mathbf{v} \in V$, $\mathbf{v} - \text{proj}_U(\mathbf{v}) \in U^\perp$.

4. For each $\mathbf{v}, \mathbf{w} \in V$,

$$\langle \text{proj}_U(\mathbf{v}), \mathbf{w} \rangle = \langle \text{proj}_U(\mathbf{v}), \text{proj}_U(\mathbf{w}) \rangle = \langle \mathbf{v}, \text{proj}_U(\mathbf{w}) \rangle.$$

5. If $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ is an orthonormal basis for V , and $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis of U (with $n \leq m$), then $[\text{proj}_U]_{\mathcal{B}}$ is a diagonal matrix with the first n diagonal entries being 1 and the remaining diagonal entries being 0.

6. We have $\text{im}(\text{proj}_U) = U$ and $\text{proj}_U(\mathbf{u}) = \mathbf{u}$ for all $\mathbf{u} \in U$.

7. We have $\ker(\text{proj}_U) = U^\perp$.

8. If V is finite-dimensional, then $\text{proj}_{U^\perp} = I - \text{proj}_U$.

9. We have $(\text{proj}_U)^2 = \text{proj}_U$.

Proof. We will prove property 1. Let $\mathbf{v}_1, \mathbf{v}_2 \in V$, so we can write $\mathbf{v}_1 = \mathbf{u}_1 + \mathbf{w}_1$ and $\mathbf{v}_2 = \mathbf{u}_2 + \mathbf{w}_2$, for $\mathbf{u}_1, \mathbf{u}_2 \in U$ and $\mathbf{w}_1, \mathbf{w}_2 \in U^\perp$. Then

$$\mathbf{v}_1 + \mathbf{v}_2 = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2),$$

which tells us that

$$\text{proj}_U(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{u}_1 + \mathbf{u}_2 = \text{proj}_U(\mathbf{v}_1) + \text{proj}_U(\mathbf{v}_2).$$

We now let $c \in \mathbb{F}$ and $\mathbf{v} \in V$. We write $\mathbf{v} = \mathbf{u} + \mathbf{w}$, with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. We note that $c\mathbf{u} \in U$ and $c\mathbf{w} \in U^\perp$ since U and $U^\perp$ are subspaces. Then

$$\text{proj}(c\mathbf{v}) = \text{proj}(c\mathbf{u} + c\mathbf{w}) = c\mathbf{u} = c\text{proj}_U(\mathbf{v}).$$

This proves that proj_U is a linear transformation.

We will also prove property 3. Let $\mathbf{v} \in V$ and write $\mathbf{v} = \mathbf{u} + \mathbf{w}$, with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. Then

$$\mathbf{v} - \text{proj}_U(\mathbf{v}) = \mathbf{v} - \mathbf{u} = \mathbf{w} \in U^\perp.$$

We leave the proof of the other properties to the exercises. ■

We can use part of this theorem to describe the matrix of proj_U explicitly.

Proposition 7.3.6 *Let U be a subspace of $\mathbb{R}^n$ or $\mathbb{C}^n$ with orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$. Then the matrix of proj_U with respect to the standard basis $\mathcal{E}$ is*

$$[\text{proj}_U]_{\mathcal{E}} = \sum_{i=1}^m \mathbf{v}_i \mathbf{v}_i^*,$$

where A^* denotes the conjugate transpose of a matrix.

Proof. This fact follows from part 2 of Theorem 7.3.5, p. 193 and the fact that the standard inner product in $\mathbb{C}^n$ can be written as

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{u},$$

where matrix multiplication is in view on the right side of the equals sign. ■

Lest this endeavor become purely speculative, we now carry out an example.

Example 7.3.7 We consider the plane through the origin in $\mathbb{R}^3$ defined by $x + 2y - z = 0$. This is a subspace of $\mathbb{R}^3$, let's call it U , and we can identify the following basis:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

We use the Gram-Schmidt process on this basis to produce this orthonormal basis of U :

$$\left\{ \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{5}{\sqrt{30}} \end{bmatrix} \right\}.$$

Using this orthonormal basis of U , we can write the matrix of proj_U with respect to the standard basis. We have

$$\begin{aligned} [\text{proj}_U]_{\mathcal{E}} &= \mathbf{e}_1 \mathbf{e}_1^* + \mathbf{e}_2 \mathbf{e}_2^* \\ &= \begin{bmatrix} \frac{4}{5} & -\frac{2}{5} & 0 \\ -\frac{2}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{30} & \frac{1}{15} & \frac{1}{6} \\ \frac{1}{15} & \frac{2}{15} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} \\ &= \begin{bmatrix} \frac{5}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix}. \end{aligned}$$

To finish this example, we will decompose a specific vector $\mathbf{v}$ into the pieces specified by Theorem 7.3.3, p. 193. Let

$$\mathbf{v} = \begin{bmatrix} 3 \\ -6 \\ 2 \end{bmatrix}.$$

We note that $\mathbf{v} \notin U$. Now we calculate $\text{proj}_U(\mathbf{v})$:

$$\text{proj}_U(\mathbf{v}) = \begin{bmatrix} \frac{5}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} \begin{bmatrix} 3 \\ -6 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{29}{6} \\ -\frac{7}{3} \\ \frac{1}{6} \end{bmatrix}.$$

Then $\mathbf{w} = \mathbf{v} - \text{proj}_U(\mathbf{v})$, so

$$\mathbf{w} = \begin{bmatrix} -\frac{11}{6} \\ -\frac{11}{3} \\ \frac{11}{6} \end{bmatrix}.$$

This completes the decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$. ▲

Proposition 7.3.6, p. 194 depended on having an orthonormal basis for the

subspace U . We can always find such a basis through the Gram-Schmidt process, but there is an alternative way to produce the matrix for orthogonal projection.

Proposition 7.3.8 *Let U be a subspace of $\mathbb{R}^n$ or $\mathbb{C}^n$ with basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Let A be the $n \times k$ matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_k$. Then*

$$[\text{proj}_U]_{\mathcal{E}} = A(A^*A)^{-1}A^*.$$

Proof. We note that $\text{proj}_U(\mathbf{x})$ is an element of U , so it can be written as a linear combination of the columns of A . In other words, there is a vector $\mathbf{x}'$ which satisfies $\text{proj}_U(\mathbf{x}) = A\mathbf{x}'$. By part 3 of Theorem 7.3.5, p. 193,

$$\mathbf{x} - \text{proj}_U(\mathbf{x}) = \mathbf{x} - A\mathbf{x}' \in U^\perp.$$

Specifically, we have

$$\langle \mathbf{x} - A\mathbf{x}', \mathbf{v}_j \rangle = \mathbf{v}_j^*(\mathbf{x} - A\mathbf{x}') = 0$$

for each $\mathbf{v}_j$. If we rewrite these k equations in matrix form, we have

$$A^*(\mathbf{x} - A\mathbf{x}') = \mathbf{0}$$

or $A^*A\mathbf{x}' = A^*\mathbf{x}$. If A^*A is invertible, then we can multiply both sides of this equation by $A(A^*A)^{-1}$, and we get

$$A\mathbf{x}' = A(A^*A)^{-1}A^*\mathbf{x}.$$

This completes the proof, since $A\mathbf{x}' = \text{proj}_U(\mathbf{x})$.

In the last paragraph we assumed that A^*A was invertible, so we now prove that fact. We can do this by proving that the null space of A^*A is trivial. Suppose that $A^*A\mathbf{x} = \mathbf{0}$, so we have

$$0 = \langle A^*A\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^*(A^*A\mathbf{x}) = (A\mathbf{x})^*(A\mathbf{x}) = \|A\mathbf{x}\|^2.$$

But since the columns of A are linearly independent (they are basis vectors for U), A must have rank k . By the Rank-Nullity Theorem (Theorem 5.4.10, p. 132), this means that $\dim(\text{null}(A)) = 0$, and since $\|A\mathbf{x}\|^2 = 0$ means $A\mathbf{x} = \mathbf{0}$, we must have $\mathbf{x} = \mathbf{0}$. This proves that A^*A is invertible. ■

In the following theorem we capture two important geometric properties of orthogonal projections.

Theorem 7.3.9 *Let U be a finite-dimensional subspace of an inner product space V .*

1. *For each $\mathbf{v} \in V$, $\|\text{proj}_U(\mathbf{v})\| \leq \|\mathbf{v}\|$, with equality if and only if $\mathbf{v} \in U$.*
2. *For each $\mathbf{v} \in V$ and each $\mathbf{u} \in U$, we have*

$$\|\mathbf{v} - \text{proj}_U(\mathbf{v})\| \leq \|\mathbf{v} - \mathbf{u}\|,$$

with equality if and only if $\mathbf{u} = \text{proj}_U(\mathbf{v})$.

Proof. Since $\mathbf{v} - \text{proj}_U(\mathbf{v}) \in U^\perp$ by part 3 of Theorem 7.3.5, p. 193, $\mathbf{v} - \text{proj}_U(\mathbf{v})$ is orthogonal to $\text{proj}_U(\mathbf{v})$, so, using Theorem 7.1.24, p. 182 we have

$$\begin{aligned} \|\mathbf{v}\|^2 &= \|\text{proj}_U(\mathbf{v}) + (\mathbf{v} - \text{proj}_U(\mathbf{v}))\|^2 \\ &= \|\text{proj}_U(\mathbf{v})\|^2 + \|\mathbf{v} - \text{proj}_U(\mathbf{v})\|^2 \end{aligned}$$

$$\geq \|\text{proj}_U(\mathbf{v})\|^2.$$

Equality holds here if and only if $\|\mathbf{v} - \text{proj}_U(\mathbf{v})\|^2 = 0$, which is true if and only if $\mathbf{v} = \text{proj}_U(\mathbf{v})$. This only happens if $\mathbf{v} \in U$.

We now move on to the second part of the theorem. We know that $\mathbf{v} - \text{proj}_U(\mathbf{v}) \in U^\perp$ and that $\text{proj}_U(\mathbf{v}) - \mathbf{u} \in U$, so

$$\begin{aligned} \|\mathbf{v} - \mathbf{u}\|^2 &= \|(\mathbf{v} - \text{proj}_U(\mathbf{v})) + (\text{proj}_U(\mathbf{v}) - \mathbf{u})\|^2 \\ &= \|\mathbf{v} - \text{proj}_U(\mathbf{v})\|^2 + \|\text{proj}_U(\mathbf{v}) - \mathbf{u}\|^2 \\ &\geq \|\mathbf{v} - \text{proj}_U(\mathbf{v})\|^2. \end{aligned}$$

We have equality here if and only if $\|\text{proj}_U(\mathbf{v}) - \mathbf{u}\|^2 = 0$, which happens if and only if $\text{proj}_U(\mathbf{v}) = \mathbf{u}$; that is, if and only if $\mathbf{v} \in U$. ■

Note 7.3.10 This theorem says that, first, orthogonal projections result in a *shorter* vector. That is, orthogonal projection is a type of *contraction*. Secondly, $\text{proj}_U(\mathbf{v})$ is the closest vector in U to the vector $\mathbf{v}$.

Finding the closest vector to $\mathbf{v}$ in a subspace U can be thought of as giving the best approximation of $\mathbf{v}$ by elements of U . This leads to our application of least squares approximation.

7.3.3 Least Squares Approximation

We consider a set of points $\{(x_i, y_i)\}$ in $\mathbb{R}^2$; in practice, these are usually the result of data collection, perhaps a sample of two numeric variables from a population. A graph of such points is called a *scatterplot*, and we often want to find the “line of best fit” for these data. There are many ways to measure “best fit,” and our method here will be the least squares linear regression technique.

Define a subspace U of $\mathbb{R}^n$ in the following way, where the x_i in the definition are the x -coordinates of the data:

$$U = \{[mx_i + b] \in \mathbb{R}^n \mid m, b \in \mathbb{R}\}.$$

If we let $\mathbf{1}$ denote the $n \times 1$ vector where each entry is 1, then

$$U = \{m\mathbf{x} + b\mathbf{1} \mid m, b \in \mathbb{R}\},$$

where $\mathbf{x}$ is the vector of all of the first coordinates in our data set.

The points (x_i, y_i) all lie on a single line $y = mx + b$ if and only if the vector $\mathbf{y}$ of second coordinates of our data lies in U . This does not happen often, as U is only a two-dimensional subspace of $\mathbb{R}^n$. So, we want to find the closest point in U to $\mathbf{y}$ —by Theorem 7.3.9, p. 196, we can find this through orthogonal projection. When we find $\text{proj}_U(\mathbf{y})$, the m and b will give us our equation of the regression line.

This is called a “least squares” regression, because minimizing the distance from $\mathbf{y}$ to U involves minimizing a distance. This distance in $\mathbb{R}^n$, when using the dot product, looks like a sum of squares.

Example 7.3.11 Consider the following set of five points in $\mathbb{R}^2$:

$$\{(2, 1), (1, 0), (4, 4), (4, 5), (3, 2)\}.$$

Our subspace $U \subseteq \mathbb{R}^5$ is spanned by $\mathbf{x}$ and $\mathbf{1}$, where $\mathbf{x}$ is the vector of first

coordinates

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 4 \\ 3 \end{bmatrix}.$$

We form the 5×2 matrix A with columns $\mathbf{x}$ and $\mathbf{1}$. Then, by Proposition 7.3.8, p.196, we have

$$[\text{proj}_U]_{\mathcal{E}} \mathbf{y} = A(A^*A)^{-1}A^*\mathbf{y},$$

where $\mathbf{y}$ is the 5×1 vector of the y -coordinates of our data.

We calculate the following:

$$A^*A = \begin{bmatrix} 46 & 14 \\ 14 & 5 \end{bmatrix}, \quad (A^*A)^{-1} = \begin{bmatrix} \frac{5}{34} & -\frac{7}{17} \\ -\frac{7}{17} & \frac{23}{17} \end{bmatrix}.$$

Now, we don't actually want $\text{proj}_U(\mathbf{y})$, because that is a vector in $\mathbb{R}^5$. We want to know the coefficients m and b in the linear combination of the column vectors of A which produce $\text{proj}_U(\mathbf{y})$. In other words, we want the vector

$$\mathbf{w} = (A^*A)^{-1}A^*\mathbf{y} = \begin{bmatrix} \frac{26}{17} \\ -\frac{32}{17} \end{bmatrix}.$$

Since $\text{proj}_U(\mathbf{y}) = A\mathbf{w}$, this means that $m = \frac{26}{17}$ and $b = -\frac{32}{17}$. We can see that this is a believable solution by looking at the graph below which contains the five points as well as the line $y = \frac{26}{17}x - \frac{32}{17}$.

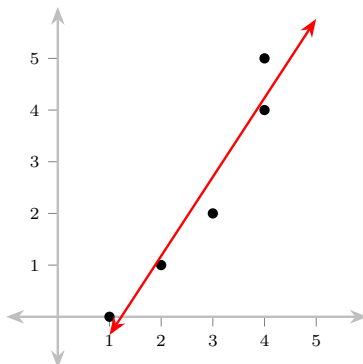


Figure 7.3.12 Graph of least-squares regression line with data



7.3.4 Reading Questions

1. Let L be the line $y = \frac{1}{2}x$ in $\mathbb{R}^2$. (This is a subspace of $\mathbb{R}^2$.)
 - (a) Calculate an orthonormal basis for L . (We are considering $\mathbb{R}^2$ with the usual dot product.)
 - (b) Let $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Using part 2 of Theorem 7.3.5, p.193, calculate $\text{proj}_L(\mathbf{v})$.
2. Consider the same situation as in the first reading question. Using Proposition 7.3.6, p.194, find the matrix of proj_L with respect to the standard

basis $\mathcal{E}$ of $\mathbb{R}^2$.

7.3.5 Exercises

1. Let L be the line $y = \frac{3}{5}x$ in $\mathbb{R}^2$. Write the vector $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ as the sum of a vector in L and a vector in $L^\perp$. (Use the standard dot product as the inner product in $\mathbb{R}^2$.)
2. Let $U = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$$

- (a) Find the matrix $[\text{proj}_U]_{\mathcal{E}}$. (Use the standard dot product as the inner product in $\mathbb{R}^3$.)
- (b) Using your work from part (a), find the vector in U which is closest to $\mathbf{v}$, if

$$\mathbf{v} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}.$$

3. Consider the following inner product on P_2 :

$$\langle p, q \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1).$$

Let $U = \text{Span}\{t - t^2, 1 + 2t\}$. If $p = 2 - t + 2t^2$, write p as the sum of a vector in U and a vector in $U^\perp$.

4. Consider the following four points in $\mathbb{R}^2$:

$$(-1, -1), (1, 2), (2, 0.5), (-0.75, 1).$$

Find the least-squares regression line for these points.

Writing Exercises

5. Let U be a subspace of an inner product space V . Prove that $U^\perp$ is a subspace of V .
6. Let $A \in M_{m,n}(\mathbb{R})$. Prove that $(\text{row}(A))^\perp = \text{null}(A)$.

Appendix A

The Integers Modulo n

In this short appendix we will define modular arithmetic. There are other books and sources that develop these ideas in a more thorough, formal manner. However, we are primarily aiming for a streamlined approach that will serve our discussion of finite fields in Section 2.1, p. 34.

The Division Algorithm is essential to what follows.

Theorem A.0.1 The Division Algorithm. *Let a be an integer and let b be a positive integer. Then there exist unique integers q and r for which*

$$a = qb + r,$$

and $0 \leq r < b$.

In this statement of the Division Algorithm, the reader should think of dividing a by b , where q is the quotient and r is the remainder. For example,

$$17 = 5(3) + 2$$

captures the information that dividing 17 by 3 leaves a remainder of 2. Similarly, the equation

$$-10 = -2(7) + 4$$

tells us that dividing -10 by 7 gives a remainder of 4. It is crucial for uniqueness in the Division Algorithm that we insist the remainder r is in the specified range. (In other words, it is *not* appropriate to write $-10 = -1(7) + (-3)$ as an application of the Division Algorithm to $a = -10$ and $b = 7$.)

The Division Algorithm provides a necessary link between the definition of congruence and what follows.

Definition A.0.2 Let a and b be integers and let n be a natural number. Then we say that a and b are **congruent modulo n** if $n \mid (a - b)$. We write this as $a \equiv b \pmod{n}$. ◆

While this is the definition of congruence mod n , there is an equivalent way we can understand the meaning of two numbers being congruent in this way.

Proposition A.0.3 *Let a and b be integers and let n be a natural number. Then $a \equiv b \pmod{n}$ if and only if a and b have the same remainder when divided by n .*

We now define a relation on $\mathbb{Z}$. Let n be a natural number. If a and b are integers, then we say that a is related to b when $a \equiv b \pmod{n}$. (Proposition A.0.3, p. 200 gives the more intuitive way to understand this relation—two integers are related if they have the same remainder when divided by n .) It is

a good (and not terribly difficult) exercise to show that this is an *equivalence relation* on $\mathbb{Z}$. For an integer m , we will use the notation $[m]$ to denote the equivalence class of m under this relation.

Definition A.0.4 Let $n \in \mathbb{N}$. Then the equivalence classes of the *congruence mod n* equivalence relation are $[0], [1], \dots, [n-1]$. The **integers modulo n** is the set

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}.$$

Addition and multiplication can be defined on this set in the following way. For any $[a], [b] \in \mathbb{Z}_n$,

$$\begin{aligned}[a] + [b] &:= [a + b] \\ [a] \cdot [b] &:= [ab].\end{aligned}$$

◆

It takes a little bit of work (but not too much!) to show that these operations are well-defined.

When performing calculations in $\mathbb{Z}_n$, we will prefer to use one of the numbers $0, 1, \dots, n-1$ as the equivalence class representative. For example, though it is correct to write $[5] + [6] = [11]$ in $\mathbb{Z}_7$, we will prefer to write $[5] + [6] = [4]$. (Note that knowing the integer n is *essential* in these calculations!)

Example A.0.5 Let's carry out some basic arithmetic within $\mathbb{Z}_8$.

1. $[3] + [5] = [0]$
2. $[6] + [4] = [2]$
3. $[2] \cdot [7] = [6]$
4. $[3] \cdot [5] = [7]$
5. $[3]([4] + [5]) = [3]$

▲

At this point the reader may see that, for each $n \in \mathbb{N}$, $\mathbb{Z}_n$ is its own mathematical universe, just like $\mathbb{Z}$ or $\mathbb{R}$, with its own calculations and quirks. In particular, it is illuminating to think about what properties of addition and multiplication $\mathbb{Z}_n$ shares with $\mathbb{R}$. (Also: In which cases does the value of n affect whether or not these properties hold?)

While the full impact of those questions is best pursued in an abstract algebra class, one question has immediate relevance for our current subject: For which $n \in \mathbb{N}$ do all nonzero elements in $\mathbb{Z}_n$ have multiplicative inverses?

In Example A.0.5, p. 201 we carried out calculations in $\mathbb{Z}_8$, so let's examine that set again. Importantly, the element $[1]$ is the *multiplicative identity* in $\mathbb{Z}_8$, meaning that multiplying each $[m] \in \mathbb{Z}_8$ by $[1]$ leaves $[m]$ unchanged. By multiplying every element in $\mathbb{Z}_8$ by $[2]$, we can see that $[1]$ is never the result. This shows that $[2]$ has no multiplicative inverse in $\mathbb{Z}_8$. Further, it is illuminating to note that the elements $[1]$, $[3]$, $[5]$, and $[7]$ have multiplicative inverses in $\mathbb{Z}_8$ while the elements $[2]$, $[4]$, and $[6]$ do not. However, the issue is deeper than the parity of the elements of the equivalence class; the key ingredient is relative primeness with n .

We hope this discussion makes the following theorem believable.

Theorem A.0.6 *The set $\mathbb{Z}_n$ has the property that all nonzero elements have a multiplicative inverse if and only if n is prime.*

In the context of Section 2.1, p. 34, Theorem A.0.6, p. 201 leads to the result

that $\mathbb{Z}_n$ is a field if and only if n is prime. In this case, we will use the notation $\mathbb{F}_n$ in place of $\mathbb{Z}_n$. We will also often drop the square-bracket notation and, for example, refer to 2 instead of $[2]$ as an element of $\mathbb{F}_3$.

Appendix B

Hints, Answers, and Solutions to Exercises

1 · Solving Systems of Linear Equations

1.1 · Systems of Linear Equations

· Exercises

1.1.2. Answer. The linear system to solve is

$$a - b + c = 6$$

$$a + b + c = 0$$

$$9a + 3b + c = -2.$$

1.1.5. Answer.

(a) No.

(b) No.

(c) No.

1.1.6. Answer.

(a) The solution set is $\{(8/9, -11/9)\}$.

(b) The solution set is empty.

(c) The solution set is empty.

(d) The solution set is $\{(x, y) \mid y = \frac{1}{2}x + \frac{1}{4}\}$.

(e) The solution set is $\{(5/13, 15/13)\}$.

Writing Exercises

1.1.7. Answer.

(a) This system has no solution because the lines are parallel:

$$-2x + y = 0$$

$$-2x + y = 1.$$

- (b) This system has one solution because the lines only intersect at the origin:

$$\begin{aligned} -2x + y &= 0 \\ 2x + y &= 0. \end{aligned}$$

- (c) This system has infinitely many solutions because the equations describe the same line:

$$\begin{aligned} -2x + y &= 0 \\ 2x - y &= 0. \end{aligned}$$

1.1.8. Answer. This is true if and only if the system is a *homogeneous* linear system.

1.2 • Matrices

• Exercises

1.2.4. Answer.

- (a) Neither
- (b) REF
- (c) REF
- (d) Both

1.2.5. Answer. Two possibilities (there are many!):

$$B = \begin{bmatrix} -2 & 3 \\ 0 & 1/2 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 1 & -3/2 \\ 0 & 1/2 \end{bmatrix}.$$

1.2.6. Answer.

- (a)
- (b) $\begin{bmatrix} 1 & 0 & 31/10 \\ 0 & 1 & -3/10 \end{bmatrix}$
- (c)
- (d) $\begin{bmatrix} 1 & 0 & 21/23 \\ 0 & 1 & -5/23 \\ 0 & 0 & 0 \end{bmatrix}$

1.2.7. Answer.

- (a) There are 4 possible RREFs.
- (b) There are 7 possible RREFs.
- (c) There are 4 possible RREFs.

1.3 • Solving Linear Systems With Matrices

• Exercises

1.3.1. Answer.

- (a) The only solution is $(-4, 5, 0)$.

(b) The solutions can be written parametrically as

$$\begin{cases} x_1 = 2x_2 + 2 \\ x_2 \text{ is free} \\ x_3 = -3. \end{cases}$$

(c) There are no solutions.

(d) The solutions can be written parametrically as

$$\begin{cases} x_1 = -2x_3 + 4x_4 + 7 \\ x_2 = -9x_3 + x_4 - 4 \\ x_3 \text{ is free} \\ x_4 \text{ is free} \\ x_5 = 4. \end{cases}$$

1.3.5. Answer. This system is consistent if and only if $3b_1 + 2b_2 + b_3 = 0$.

1.3.8. Answer. The function is $f(x) = \frac{1}{2}x^2 - 3x + \frac{5}{2}$.

1.3.9. Answer. The values are $a = \frac{5}{3}$, $b = 0$, $c = -\frac{2}{3}$, and $d = 3$.

Writing Exercises

1.3.10. Hint. It is easy to find at least one solution to every homogeneous system. Why?

1.3.13. Answer. Yes, this system is consistent. (Why?)

1.4 · Vectors

1.4.4 · Exercises

1.4.4.1. Answer. Here is the linear system:

$$\begin{aligned} 2x &= 7 \\ x - 3y &= -1 \\ -2x + y &= -1. \end{aligned}$$

1.4.4.2. Answer. Here is the vector equation:

$$x \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix} + z \begin{bmatrix} -5 \\ 1 \\ -7 \end{bmatrix} = \begin{bmatrix} 9 \\ -1 \\ 0 \end{bmatrix}.$$

1.4.4.3. Answer. There are an infinite number of correct answers! Three vectors that are in the span are $\mathbf{u} = 1\mathbf{u} + 0\mathbf{v}$, $\mathbf{v} = 0\mathbf{u} + 1\mathbf{v}$, and $\mathbf{0} = 0\mathbf{u} + 0\mathbf{v}$. Here are two additional vectors in the span:

$$\begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} = \mathbf{u} + \mathbf{v}, \text{ and } \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix} = \mathbf{u} - \mathbf{v}.$$

1.4.4.4. Answer. If $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, then $\mathbf{x}$ can be written as

$$\mathbf{x} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w},$$

where $a = \frac{1}{3} - \frac{1}{3}c$, $b = \frac{5}{3} + \frac{10}{3}c$, and c can be any real number.

1.4.4.7. Answer. The vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ as long as the following equation is satisfied:

$$b_1 + \frac{3}{5}b_2 - \frac{7}{5}b_3 = 0.$$

Writing Exercises

1.4.4.9. Hint. One might try to prove that $\mathbf{u} = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} b \\ d \end{bmatrix}$ are collinear if and only if $ad - bc = 0$. Then an exercise from section 1.3 can be invoked.

1.4.4.12. Solution. We must prove that for all real numbers c and all $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ we have

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}.$$

Let $\mathbf{u}$ and $\mathbf{v}$ be denoted in the following way:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

Then we have

$$c(\mathbf{u} + \mathbf{v}) = c \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix} = \begin{bmatrix} c(u_1 + v_1) \\ \vdots \\ c(u_n + v_n) \end{bmatrix}.$$

Now, since the distributive law holds for real numbers, we have $c(u_i + v_i) = cu_i + cv_i$ for all $i = 1, \dots, n$. This means that

$$\begin{aligned} c(\mathbf{u} + \mathbf{v}) &= \begin{bmatrix} cu_1 + cv_1 \\ \vdots \\ cu_n + cv_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix} + \begin{bmatrix} cv_1 \\ \vdots \\ cv_n \end{bmatrix} \\ &= c \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + c \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = c\mathbf{u} + c\mathbf{v}. \end{aligned}$$

This completes the proof.

2 · Fields and Vector Spaces

2.1 · Fields

· Exercises

Writing Exercises

2.1.6. Answer.

- No, $\mathbb{Z}[x]$ is not a field. All of the integers are contained in $\mathbb{Z}[x]$, and since (for example) $2 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$, it will not have a multiplicative inverse in $\mathbb{Z}[x]$.
- No, $\mathbb{R}[x]$ is not a field. The element $x \in \mathbb{R}[x]$ has no multiplicative inverse. We can argue this by contradiction. If $b = a_0 + a_1x + a_2x^2 + \dots$ was the multiplicative inverse of x , then

$$1 = xb = a_0x + a_1x^2 + a_2x^3 + \dots$$

But since the constant term on the right side of this equation is 0 and the constant term on the left side is 1, and since those are not equal, we have a contradiction.

2.1.8. Answer.

(a)

- (b) Yes, this is a field. The most difficult axiom to check is the one about multiplicative inverses. It turns out that the multiplicative inverse of $a + b\alpha$ is $c + d\alpha$ where

$$c = \frac{1}{a^2 - 2b^2}, \text{ and } d = -\frac{b}{a^2 - 2b^2}.$$

One needs to check that for all non-zero elements $a + b\alpha$ of $\mathbb{F}_5[\alpha]$ we have $a^2 - 2b^2 \neq 0$. But this is relatively easy to verify.

2.1.10. Answer. Yes, this is a field. Some of the axioms are quite tedious to check. It is perhaps worth noting that the multiplicative inverse of (a, b) is

$$\left(\frac{a}{a^2 + 2b^2}, -\frac{b}{a^2 + 2b^2} \right).$$

2.1.12. Solution. Let $b \in \mathbb{F}$ be a non-zero element with multiplicative inverse b' . Suppose that the element $a \in \mathbb{F}$ also has the properties of a multiplicative inverse of b . This means that $ab = 1$. If we multiply both sides of this equation by b' , we have

$$\begin{aligned} (ab)b' &= b' \\ a(bb') &= b' \\ a(1) &= b' \\ a &= b'. \end{aligned}$$

This proves that $a = b'$, so multiplicative inverses are unique.

2.2 · Solving Linear Systems Over Fields

· Exercises

2.2.1. Answer. This system has the solution $x = 1$, $y = 0$, $z = 0$.

2.2.2. Answer. This linear system is inconsistent.

2.2.5. Answer. This system is inconsistent.

2.2.6. Answer. The unique solution is $x = 2$, $y = 4$, and $z = 1$.

Writing Exercises

2.2.9. Answer.

- (a) The same argument as in Exercise 1.3.11, p. 24 works here, recognizing that $2 = -1$ in $\mathbb{F}_3$.
- (b) The correct inequality is $ad + (p-1)bc \neq 0$.

2.2.10. Answer.

- (a) If all of the coefficients and constants begin as integers, all of the elementary row operations will produce coefficients and constants within $\mathbb{Q}$. None of the elementary row operations can produce an irrational number when beginning with integers.

- (b) If there is a unique solution, there are no free variables, so whether one considers the solution in $\mathbb{Q}^n$ or $\mathbb{R}^n$, there cannot be another solution. Thus, the numbers in the solution set must be rational numbers.

2.3 · Vector Spaces

· Exercises

2.3.1. Answer.

- (a) No, this is not a vector space. For example, it is not closed under scalar multiplication. We can see that $-2(1, 1) = (-2, -2) \notin V$.
- (b) No, this is not a vector space. For example, it is not closed under addition. We can see that $(1, 4) + (-2, -1) = (-1, 3) \notin V$.
- (c) Yes, this is a vector space. All of the vector space axioms hold.

2.3.3. Answer. The only axiom that fails is axiom 7. We can see, for example, that $1 \cdot (1, 1) = (0, 1) \neq (1, 1)$, so this axiom does not hold.

2.3.4. Answer. We calculate that

$$2p_1 - 3p_2 - 2p_3 = -5t^3 + 6t^2 - 18t + 10.$$

2.3.6. Answer.

- (a) This vector space contains $2^3 = 8$ distinct vectors.
- (b) The set $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ contains four distinct vectors: $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$. These are all distinct.

2.3.9. Answer. No, $\mathbf{w}$ is not in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$. When we form the matrix with $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ as its columns, it reduces to the following:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since there is a pivot in the final column, the corresponding vector equation $x_1\mathbf{u} + x_2\mathbf{v} = \mathbf{w}$ has no solutions.

Writing Exercises

2.3.12. Answer. No, $\mathbb{Q}$ is not a vector space over $\mathbb{R}$. It is not closed under scalar multiplication. For example, we see that $\pi(1, 1) = (\pi, \pi) \notin \mathbb{Q}^2$.

2.3.15. Solution. Suppose that $\mathbf{0}$ is a zero vector in V and that $\mathbf{z}$ also has the properties of a zero vector. Since $\mathbf{0}$ is a zero vector, we have $\mathbf{0} + \mathbf{z} = \mathbf{z}$. Since $\mathbf{z}$ has the properties of a zero vector, we have $\mathbf{0} + \mathbf{z} = \mathbf{0}$. Therefore, we have

$$\mathbf{z} = \mathbf{0} + \mathbf{z} = \mathbf{0},$$

so $\mathbf{z} = \mathbf{0}$. This proves that the zero vector is unique.

2.4 · Subspaces

· Exercises

2.4.2. Answer.

- (a) Yes, W is a subspace.
- (b) Yes, W is a subspace.

- (c) No, W is not a subspace. It is not closed under scalar multiplication.
- (d) Yes, W is a subspace.
- (e) Yes, W is a subspace.

2.4.4. Answer.

- (a) Yes, W is a subspace as it can be written as the span of two vectors.
- (b) No, W is not a subspace. It is not closed under scalar multiplication, for example.
- (c) Yes, W is a subspace as it can be written as the span of four vectors.
- (d) No, W is not a subspace as it does not contain the zero vector.

Writing Exercises

2.4.6. Answer. Let V denote the set of functions described in the statement of the problem. We first observe that the zero function is in V since it integrates to zero over any interval $[a, b]$. If $f, g \in V$, then

$$\begin{aligned}\int_a^b (f+g)(x) \, dx &= \int_a^b (f(x) + g(x)) \, dx \\ &= \int_a^b f(x) \, dx + \int_a^b g(x) \, dx = 0 + 0 = 0,\end{aligned}$$

which proves that $f+g \in V$. Finally, if $c \in \mathbb{R}$ and $f \in V$, we have

$$\int_a^b (cf)(x) \, dx = \int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx = c \cdot 0 = 0,$$

which shows that $cf \in V$. (These last two calculations rely on the *linear* properties of the definite integral.) This completes the proof that V is a subspace.

2.4.9. Answer. No. Let W_1 and W_2 be the following two subspaces of $\mathbb{R}^2$:

$$W_1 = \{(x, 2x) \mid x \in \mathbb{R}\}, \quad W_2 = \{(x, 3x) \mid x \in \mathbb{R}\}.$$

Then $(1, 2) \in W_1$ and $(1, 3) \in W_2$, so both of these elements are in $W_1 \cup W_2$. However, $(1, 2) + (1, 3) = (2, 5)$, and $(2, 5) \notin W_1 \cup W_2$. This shows that $W_1 \cup W_2$ is not closed under addition, so it is not a subspace.

3 · Linear Transformations**3.1 · Linear Transformations****3.1.7 · Exercises**

3.1.7.2. Answer. Yes, T is a linear transformation.

3.1.7.4. Answer. No, T is not a linear transformation. Consider the following example:

$$\begin{aligned}T(2(1, 1, 0)) &= T(2, 2, 0) = (6 + 2 - 0, -4) \equiv (3, 1) \\ 2T(1, 1, 0) &= 2(3 + 1 - 0, -1) \equiv 2(4, 4) \equiv (3, 3).\end{aligned}$$

This shows that the scalar multiplication property does not hold, as $2T(1, 1, 0) \neq T(2(1, 1, 0))$.

3.1.7.6. Answer.

(a) $A\mathbf{v} = (0, 1, 0)$

(b) $A\mathbf{v} = (1, 2, 2)$

(c) $A\mathbf{v} = (0, 1, 0)$

3.1.7.8. Answer.

(a) No, since

$$\left[\begin{array}{cc|c} 3 & -2 & 1 \\ 1 & 4 & 1 \\ -1 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

The pivot in the final column shows that there is no solution to the matrix-vector equation $A\mathbf{x} = \mathbf{v}$, where $\mathbf{v} = (1, 1, 1)$.

(b) No. Since in part a we identified a vector which is not in the image of T , this means that T cannot be surjective.

Writing Exercises

3.1.7.11. Answer. Let $V = C[0, \infty)$ and let $f, g \in V$. Then we have

$$\begin{aligned} T(f+g)(x) &= \int_0^x (f+g)(y) \, dy = \int_0^x (f(y) + g(y)) \, dy \\ &= \int_0^x f(y) \, dy + \int_0^x g(y) \, dy = T(f)(x) + T(g)(x). \end{aligned}$$

We now let $c \in \mathbb{R}$ and $f \in V$. Then

$$\begin{aligned} T(cf)(x) &= \int_0^x (cf)(y) \, dy = \int_0^x cf(y) \, dy \\ &= c \int_0^x f(y) \, dy = cT(f)(x). \end{aligned}$$

Both of these calculations rely on the *linear* properties of the definite integral.

3.1.7.12. Solution. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in U . Then, since S and T are both linear transformations, we have

$$\begin{aligned} (S \circ T)(\mathbf{u} + \mathbf{v}) &= S(T(\mathbf{u} + \mathbf{v})) = S(T(\mathbf{u}) + T(\mathbf{v})) \\ &= S(T(\mathbf{u})) + S(T(\mathbf{v})) = (S \circ T)(\mathbf{u}) + (S \circ T)(\mathbf{v}). \end{aligned}$$

This proves that $S \circ T$ has the first property of a linear transformation.

We now let $c \in \mathbb{F}$ and $\mathbf{u} \in U$. Then, since S and T are both linear transformations, we have

$$\begin{aligned} (S \circ T)(c\mathbf{u}) &= S(T(c\mathbf{u})) = S(cT(\mathbf{u})) \\ &= cS(T(\mathbf{u})) = c(S \circ T)(\mathbf{u}). \end{aligned}$$

This shows that $S \circ T$ is a linear transformation.

3.1.7.13. Solution.

(a) We assume that $S \circ T$ is injective, and we let $\mathbf{u}_1$ and $\mathbf{u}_2$ be vectors in U with $T(\mathbf{u}_1) = T(\mathbf{u}_2)$. Then, since $S \circ T$ is injective, we have

$$S(T(\mathbf{u}_1)) = S(T(\mathbf{u}_2))$$

$$\mathbf{u}_1 = \mathbf{u}_2.$$

This proves that T is injective.

- (b) We assume that $S \circ T$ is surjective, and we let $\mathbf{w} \in W$. Since $S \circ T$ is surjective, there is a vector $\mathbf{u} \in U$ such that $(S \circ T)(\mathbf{u}) = \mathbf{w}$. But this means that $S(T(\mathbf{u})) = \mathbf{w}$; in other words, S sends $T(\mathbf{u})$ to $\mathbf{w}$. This proves that S is surjective, since we have found a vector $\mathbf{v} = T(\mathbf{u}) \in V$ such that $S(\mathbf{v}) = \mathbf{w}$.

3.2 · The Matrix of a Linear Transformation

3.2.5 · Exercises

3.2.5.3. Answer. The matrix for T is $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$.

3.2.5.5. Answer. The matrix for T is $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

3.3 · Inverting a Matrix

3.3.5 · Exercises

3.3.5.2. Answer.

- (a) The RREF of A is I_3 and it takes nine elementary row operations to reduce A .
- (b) The RREF of A is I_3 and it takes six elementary row operations to reduce A .

3.3.5.4. Answer.

- (a) The matrix A is invertible and the inverse of A is

$$\begin{bmatrix} 0 & 2 & 1 \\ 2 & 4 & 2 \\ 4 & 0 & 2 \end{bmatrix}.$$

- (b) The matrix A is not invertible since its RREF is

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

3.3.5.6. Answer. If $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$, then the inverse over $\mathbb{F}_5$ is $A^{-1} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$.

Then we have

$$\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Writing Exercises

3.3.5.12. Solution. Let $A \in M_n(\mathbb{F})$ be invertible. This means that A^{-1} exists and that $AA^{-1} = A^{-1}A = I_n$. These equalities involving A and A^{-1} show that A^{-1} is invertible and that A is a matrix which has the properties of the inverse of A^{-1} . Since the inverse of a matrix is unique, it must be that $(A^{-1})^{-1} = A$.

3.3.5.14. Answer.

- (a) We suppose that $(B - C)D = 0$ where D is invertible. Then we have

$$\begin{aligned}(B - C)D &= 0 \\ (B - C)DD^{-1} &= 0D^{-1} \\ B - C &= 0 \\ B &= C.\end{aligned}$$

- (b) Suppose that $B = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, and $D = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Then, it is true that D is not invertible. We obviously have $B \neq C$. But we do have $B - C = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix}$ and then

$$(B - C)D = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

3.3.5.16. Answer. Suppose that A is invertible and upper triangular. Then we can use elementary row operations to reduce $[A \mid I]$ to $[I \mid A^{-1}]$. But the elementary row operations that are needed for this reduction will be only the scaling operation and the replacement operation, where we add a multiple of one row to another higher up in the matrix. (Specifically, we will only need to add a multiple of a row i to a row j where $i > j$.) The elementary matrices that correspond to these specific elementary row operations are all upper triangular. And since the product of upper triangular matrices is upper triangular (this should probably be proved), this shows that A^{-1} will be upper triangular.

3.4 · Subspaces and Linear Transformations**3.4.4 · Exercises****3.4.4.1. Answer.**

- (a) Yes, we calculate $A\mathbf{x} = \mathbf{0}$.
- (b) No, we calculate $A\mathbf{x} = (6, 6, 5)$, where we have written the vector horizontally out of convenience.

3.4.4.4. Answer.

- (a) It is easy to calculate $A\mathbf{x}$ and see that $A\mathbf{x} = \mathbf{0}$. To show that $\mathbf{x}$ is not in $\text{col}(A)$, we form the matrix $[A \mid \mathbf{x}]$ and row reduce:

$$\left[\begin{array}{cc|c} 3 & -1 & 2 \\ -9 & 3 & 6 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -1/3 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

The pivot in the last column shows that $\mathbf{x} \notin \text{col}(A)$.

- (b) Since

$$A \sim \begin{bmatrix} 1 & -1/3 \\ 0 & 0 \end{bmatrix},$$

we can see that any vector $\mathbf{v}$ in the null space of A must be a multiple of $\mathbf{x}$. And since $\mathbf{x}$ is not in $\text{col}(A)$, no non-zero multiple of $\mathbf{x}$ can be in $\text{col}(A)$. This shows that the only vector in $\text{null}(A) \cap \text{col}(A)$ is the zero vector.

- (c) Yes, this is possible. Consider the matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$. The vector

$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is in the null space of A . (This can easily be checked.) Also, since $\mathbf{v}$ is a column of A , it is in $\text{col}(A)$.

3.4.4.7. Solution.

- (a) This linear transformation is injective. Observe that if T sends the vector (a, b, c) to the zero polynomial, then we have the following linear system in which each equation must hold:

$$\begin{aligned} a + b &= 0 \\ b + c &= 0 \\ a + b + c &= 0. \end{aligned}$$

We can solve this using methods from earlier in this book. We find that

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This shows that the only solution to this linear system is the trivial one—that is, where $a = b = c = 0$. But this means that the only vector that T sends to the zero polynomial is the zero vector.

This proves that the kernel of T contains only the zero vector, so T is injective.

- (b) We claim that T is surjective. Let $p = d + et + ft^2$ be an arbitrary element of P_2 . In order to demonstrate that there is an element $\mathbf{x} \in \mathbb{R}^3$ such that $T(\mathbf{x}) = p$, we need to solve the following linear system:

$$\begin{aligned} a + b &= d \\ b + c &= e \\ a + b + c &= f. \end{aligned}$$

However, the RREF of the coefficient matrix for this system looks like this:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The fact that there is a pivot in each row of this RREF means that a combination of a , b , and c can be found such that $T(\mathbf{x}) = p$. This proves that T is surjective.

3.4.4.9. Solution.

- (a) This linear transformation is not injective. We can calculate that $T(-\frac{2}{3} - \frac{1}{3}t + t^2) = \mathbf{0}$. Since this shows that T has a nonzero vector in the kernel, T is not injective.

- (b) This linear transformation is surjective. Let $\mathbf{x} = \begin{bmatrix} d \\ e \end{bmatrix}$ be an element of $\mathbb{R}^2$. We claim that we can find an element $p \in P_2$ such that $T(p) = \mathbf{x}$. If $p = a + bt + ct^2$, we will need to solve the following system of equations to find the coefficients of p :

$$\begin{aligned} a - 2b &= d \\ 3b + c &= e. \end{aligned}$$

But the coefficient matrix of this system has the following RREF:

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2/3 \\ 0 & 1 & 1/3 \end{bmatrix}.$$

The important thing to notice here is that there is a pivot in each row of this RREF. This means that the linear system always has a solution, so we can always find values of a , b , and c such that $T(p) = \mathbf{x}$. This proves that T is surjective.

Writing Exercises

3.4.4.11. Solution. Since U' is a subspace of U , we must have $\mathbf{0} \in U'$. Further, we know that $T(\mathbf{0}) = \mathbf{0}$ for all linear transformations. This proves that $\mathbf{0} \in T(U')$.

We now let $\mathbf{v}_1, \mathbf{v}_2 \in T(U')$. This means that there exist $\mathbf{u}_1, \mathbf{u}_2 \in U'$ with $T(\mathbf{u}_1) = \mathbf{v}_1$ and $T(\mathbf{u}_2) = \mathbf{v}_2$. Since U' is a subspace of U , we know that $\mathbf{u}_1 + \mathbf{u}_2 \in U'$, so $T(\mathbf{u}_1 + \mathbf{u}_2) \in T(U')$. However, because of the properties of a linear transformation, we have

$$T(\mathbf{u}_1 + \mathbf{u}_2) = T(\mathbf{u}_1) + T(\mathbf{u}_2) = \mathbf{v}_1 + \mathbf{v}_2.$$

This proves that $\mathbf{v}_1 + \mathbf{v}_2 \in T(U')$, which shows that $T(U')$ is closed under addition.

Finally, we let $\mathbf{v} \in T(U')$ and $c \in \mathbb{F}$. This means that there exists $\mathbf{u} \in U'$ such that $T(\mathbf{u}) = \mathbf{v}$. Since U' is a subspace of U , we know that $c\mathbf{u} \in U'$, so $T(c\mathbf{u}) \in T(U')$. Using the properties of a linear transformation, we see that

$$T(c\mathbf{u}) = cT(\mathbf{u}) = c\mathbf{v}.$$

This proves that $c\mathbf{v} \in T(U')$, showing that $T(U')$ is closed under scalar multiplication. This completes the proof that $T(U')$ is a subspace of V .

4 • Determinants

4.1 • Defining the Determinant

4.1.4 • Exercises

4.1.4.2. Answer.

(a) 6

(b) 66

4.1.4.4. Answer.

(a) The elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and $\det(E) = -3$.

(b) The elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

and $\det(E) = -1$.

(c) The elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{bmatrix},$$

and $\det(E) = 1$.

4.1.4.6. Solution. We note that $kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$, and

$$\det(kA) = kakd - kbbc = k^2ad - k^2bc = k^2(ad - bc) = k^2\det(A).$$

4.1.4.8. Answer. We must have $c = -\frac{13}{2}$ in order to have $\det(A) = 0$.

4.1.4.9. Answer. The values are $x = 1$ and $x = 6$.

Writing Exercises

4.1.4.12. Answer. We let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and we consider the case of adding k times row 1 to row 2. We will call the resulting matrix B , and it has this appearance:

$$B = \begin{bmatrix} a & b \\ ka + c & kb + d \end{bmatrix}.$$

We know that $\det(A) = ad - bc$, and we will now calculate $\det(B)$:

$$\begin{aligned} \det(B) &= a(kb + d) - b(ka + c) \\ &= kab + ad - kab - bc \\ &= ad - bc = \det(A). \end{aligned}$$

Technically we have only proved one of the two necessary cases to complete this proof.

4.2 • Properties of the Determinant

4.2.5 • Exercises

4.2.5.2. Answer.

(a) -12

(b) -3

4.2.5.4. Answer. -10

4.2.5.6. Answer. 27

Writing Exercises

4.2.5.8. Solution. If A has identical columns, then A^T has identical rows. We can use an elementary row operation to add -1 times one of these rows to the other, producing a row of zeros in this matrix we will call B . Since we used the replace row operation to go from A^T to B , we have $\det(A^T) = \det(B)$ by Theorem 4.2.5, p. 98. Since B has a row of zeros, we know that $\det(B) = 0$ by Proposition 4.2.1, p. 96. This means that $\det(A^T) = 0$, and since we have $\det(A) = \det(A^T)$ by Theorem 4.1.9, p. 93, this means that $\det(A) = 0$, as desired.

4.2.5.11. Solution. If $A, B \in M_n(\mathbb{F})$, we have

$$\det(AB) = \det(A)\det(B) = \det(B)\det(A) = \det(BA).$$

This string of equations uses Theorem 4.2.13, p. 101 twice as well as the fact that the determinant of a matrix is an element of $\mathbb{F}$, and elements of $\mathbb{F}$ commute via multiplication.

5 • The Dimension of a Vector Space

5.1 • Linear Independence

• Exercises

5.1.3. Answer. The set is linearly dependent if and only if $c = 20$.

5.1.4. Answer. 4

5.1.5. Answer.

- (a) False.
- (b) False.
- (c) False.
- (d) True.

Writing Exercises

5.1.9. Solution. Suppose that

$$V_1 = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$$

and

$$V_2 = \{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}.$$

Since V_1 is linearly dependent, we know that there exist scalars $c_1, \dots, c_k$, not all of which are zero, such that

$$c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}.$$

Then it is easy to produce a linear dependence relation for the set V_2 :

$$c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k + 0\mathbf{v}_{k+1} + \dots + 0\mathbf{v}_n = \mathbf{0}.$$

This proves that V_2 is linearly dependent.

5.1.10. Solution.

- (a) If $A \in M_n(\mathbb{F})$, then $\text{null}(A) = \{\mathbf{0}\}$ if and only if the RREF of A is I_n . It is always true that the null space is $\{\mathbf{0}\}$ if and only if we have a pivot in each column of the RREF; when A is square, this means that the RREF must be I_n .

We also know that $\text{col}(A) = \mathbb{F}^n$ if and only if the RREF of A is I_n . It is always true that the column space is $\mathbb{F}^n$ if and only if we have a pivot in each row of the RREF; when A is square, this means that the RREF must be I_n .

Putting these two paragraphs together, we conclude that $\text{null}(A) = \{\mathbf{0}\}$ if and only if $\text{col}(A) = \mathbb{F}^n$.

- (b) If T is a linear transformation $\mathbb{F}^n \rightarrow \mathbb{F}^n$, then by Theorem 3.2.2, p. 69

there is a matrix $B \in M_n(\mathbb{F})$ such that T is multiplication by B . Then the kernel of T is the null space of B and the range of T is the column space of B . So, T is injective if and only if $\text{null}(B) = \{\mathbf{0}\}$, and T is surjective if and only if $\text{col}(B) = \mathbb{F}^n$. Then the result from part a completes the argument.

5.2 • Basis of a Vector Space

5.2.4 • Exercises

5.2.4.2. Answer.

(a) No.

(b) No.

(c) No.

5.2.4.4. Answer. Any matrix that has two pivot columns and two non-pivot columns will work as an answer here. This is one of many such matrices:

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Of course, any matrix row equivalent to a matrix like this one would also work as an answer.

5.2.4.6. Answer. A basis is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

5.2.4.7. Answer. We will argue using the (contrapositive of the) Linear Dependence Lemma. Since $\sin(t)$ is not the zero function, then $\{\sin(t)\}$ is a linearly independent set. Also, since $\sin(2t)$ is not a scalar multiple of $\sin(t)$ (this can be verified by comparing graphs of the two functions), the set $\{\sin(t), \sin(2t)\}$ is also linearly independent. However, there is a trig identity which says that $\sin(2t) = 2\sin(t)\cos(t)$, meaning that the set $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$ is linearly dependent (as one vector is a multiple of another within this set). Therefore, one basis for H is $\{\sin(t), \sin(2t)\}$.

Writing Exercises

5.2.4.12. Solution. We let $B = \{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_{n-1} + \mathbf{v}_n, \mathbf{v}_n\}$. We will first show that B is linearly independent.

We first suppose that $c_1, \dots, c_n$ are scalars such that

$$c_1(\mathbf{v}_1 + \mathbf{v}_2) + \dots + c_{n-1}(\mathbf{v}_{n-1} + \mathbf{v}_n) + c_n\mathbf{v}_n = \mathbf{0}.$$

Rearranging this equation, we see that it is equivalent to

$$c_1\mathbf{v}_1 + (c_1 + c_2)\mathbf{v}_2 + \dots + (c_{n-1} + c_n)\mathbf{v}_n = \mathbf{0}.$$

However, since we were given that $B' = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V , this means that the coefficients in this last equation must all be zero, since B' is linearly independent. This means that $c_1 = 0$, and then since we must also have $c_1 + c_2 = 0$, we have $c_2 = 0$, and so on. The result is that $c_i = 0$ for all i , $i = 1, \dots, n$. This proves that B is linearly independent.

We will now show that B spans V . Let $\mathbf{v} \in V$. We want to argue that $\mathbf{v}$ can be written as a linear combination of the vectors in B . Since B' is a basis for V , there exist scalars $d_1, \dots, d_n$ such that

$$\mathbf{v} = d_1\mathbf{v}_1 + \dots + d_n\mathbf{v}_n. \quad (\text{B.1})$$

We want to argue that we can always find scalars $c_1, \dots, c_n$ such that

$$\mathbf{v} = c_1(\mathbf{v}_1 + \mathbf{v}_2) + \cdots + c_n \mathbf{v}_n.$$

This equation can be rewritten as

$$\mathbf{v} = c_1 \mathbf{v}_1 + (c_1 + c_2) \mathbf{v}_2 + \cdots + (c_{n-1} + c_n) \mathbf{v}_n, \quad (\text{B.2})$$

and by The Unique Representation Theorem (Theorem 5.2.11, p. 118), we know that the coefficients on the right sides of (B.1) and (B.2) must be equal. Immediately we see that $c_1 = d_1$ and then since we must have $c_1 + c_2 = d_2$, we conclude $c_2 = d_2 - d_1$. We can continue on in this way, eventually producing an expression for each c_i in terms of the d_i coefficients.

This proves that B spans V which concludes the proof that B is a basis of V .

5.3 · Dimension

5.3.4 · Exercises

5.3.4.1. Answer. The dimension is 2.

5.3.4.4. Answer.

- (a) This is true by Lemma 5.3.1, p. 123.
- (b) This is false, as $\mathbb{R}^2$ is not even a subset of $\mathbb{R}^3$.
- (c) This is false. The vector space $\mathbb{R}^2$ is a counter-example, as we know that $\dim(\mathbb{R}^2) = 2$ but it is spanned by an infinite set (the entire vector space).

5.3.4.6. Answer. Since $\dim(P_3) = 4$ and this is a set of four polynomials, we only need to argue that this set is linearly independent (see Theorem 5.3.17, p. 127). If we label the polynomials as $p_1 = 1$, $p_2 = 2t$, $p_3 = -2 + 4t^2$, and $p_4 = -12t + 8t^3$, then we can argue that the set $\{p_1, p_2, p_3, p_4\}$ is linearly independent by the contrapositive of the Linear Dependence Lemma. The set containing only p_1 is linearly independent since $p_1 \neq 0$. Then the set $\{p_1, p_2\}$ is linearly independent since neither polynomial is a scalar multiple of the other. Then since p_3 cannot be a linear combination of p_1 and p_2 for degree reasons, $\{p_1, p_2, p_3\}$ is linearly independent. Similarly, since p_4 cannot be a linear combination of p_1 , p_2 , and p_3 for degree reasons, $\{p_1, p_2, p_3, p_4\}$ is linearly independent.

Writing Exercises

5.3.4.8. Answer.

- (a) The null space of a matrix has a basis vector for each column in the RREF which does not contain a pivot. Therefore, the dimension of the null space is the number of non-pivot columns.
- (b) By Algorithm 5.2.14, p. 119, we see that there is a vector in the basis for $\text{col}(A)$ for each pivot in the RREF of A . Therefore, $\dim(\text{col}(A))$ is the number of pivot columns of A .

5.3.4.9. Answer. Since V contains the vector space of all polynomials, and since the vector space of all polynomials is infinite-dimensional (see Example 5.3.3, p. 123), then V must be infinite-dimensional.

5.4 • Rank and Nullity

5.4.4 • Exercises

5.4.4.4. Answer. This is not possible. A 9×10 linear system has a 9×10 coefficient matrix which we will call A . This situation can be reinterpreted as a linear transformation from $\mathbb{F}^{10} \rightarrow \mathbb{F}^9$ which is multiplication by A . The information given in the problem tells us that this transformation is surjective, meaning that the rank of A is as large as possible, which is 9 in this case. Therefore, the Rank-Nullity Theorem says that $\dim(\text{null}(A)) = 1$. This means that every nonzero vector in the null space of A is a scalar multiple of a single basis vector. Therefore, the associated homogeneous linear system only has nonzero solutions which are multiples of each other.

Writing Exercises

5.4.4.7. Answer.

- (a) If T is surjective, then $\text{im}(T) = W$, which means that $\dim(\text{im}(T)) = \dim(W)$. This implies that $\text{rank}(T) = \dim(W)$. On the other hand, if $\text{rank}(T) = \dim(W)$, then $\dim(\text{im}(T)) = \dim(W)$. Since $\text{im}(T)$ is a subspace of W , Theorem 5.3.18, p. 128 implies that $\text{im}(T) = W$, which means that T is surjective.
- (b) We will be able to argue both implications at once. A linear transformation T is injective if and only if $\ker(T) = \{\mathbf{0}\}$. This occurs if and only if $\dim(\ker(T)) = 0$. The Rank-Nullity Theorem then says that $\dim(\ker(T)) = 0$ if and only if $\text{rank}(T) = \dim(V)$. This proves the result.

5.4.4.8. Answer. The given information means there is no pivot in the last column of the augmented matrix but that there is at least one non-pivot column among the first six columns. The coefficient matrix A then has at least one non-pivot column, meaning that $\text{rank}(A) \leq 5$. But this means that the linear transformation $T : \mathbb{F}^6 \rightarrow \mathbb{F}^6$ which is multiplication by A cannot be surjective, since the rank of A is less than six. Therefore, we can find a vector $\mathbf{c} \in \mathbb{F}^6$ such that $A\mathbf{x} = \mathbf{c}$ is inconsistent.

5.4.4.10. Answer. We know that $\text{rank}(A)$ is the number of pivots in the RREF of A , but the number of pivots must be less than both the number of columns (n) and the number of rows (m) of A . Then if $\text{rank}(A) \leq m$ and $\text{rank}(A) \leq n$, we have $\text{rank}(A) \leq \min\{m, n\}$.

5.5 • Coordinates

5.5.4 • Exercises

5.5.4.1. Answer.

$$(a) \mathbf{v} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$$

$$(b) \mathbf{v} = \begin{bmatrix} -19 \\ -2 \end{bmatrix}$$

5.5.4.3. Answer.

$$(a) [\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 2 \\ -1 \end{bmatrix}$$

$$(b) [\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

5.5.4.6. Answer.

(a) This set of vectors spans P_2 .

(b) This set of vectors does not span P_2 .

5.5.4.7. Answer.

$$(a) [T]_{\mathcal{B}, \mathcal{E}} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & -3 \end{bmatrix}$$

$$(b) T(-10 + 3t^2) = [T]_{\mathcal{B}, \mathcal{E}}[p]_{\mathcal{B}} = \begin{bmatrix} -17 \\ -9 \end{bmatrix}$$

Writing Exercises

5.5.4.13. Solution. Let V be an n -dimensional vector space over $\mathbb{F}$, and let

$$\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$$

be a basis for V . Let $C_{\mathcal{B}} : V \rightarrow \mathbb{F}^n$ be the coordinate mapping. We will prove that $C_{\mathcal{B}}$ is injective by showing that it has a trivial kernel.

Suppose that $\mathbf{v} \in \ker(C_{\mathcal{B}})$. This means that $C_{\mathcal{B}}(\mathbf{v}) = \mathbf{0} \in \mathbb{F}^n$, so $[\mathbf{v}]_{\mathcal{B}} = \mathbf{0}$. Since this is the coordinate vector of $\mathbf{v}$, this tells us that

$$\mathbf{v} = 0\mathbf{v}_1 + \dots + 0\mathbf{v}_n.$$

This proves that $\mathbf{v} = \mathbf{0} \in V$, and therefore $\ker(C_{\mathcal{B}}) = \{\mathbf{0}\}$. This proves that the coordinate mapping is injective.

5.6 • Change of Basis

5.6.3 • Exercises

5.6.3.1. Answer. Here are the change-of-basis matrices:

$$P_{\mathcal{B}, \mathcal{C}} = \begin{bmatrix} 5/24 & 1/12 \\ -7/8 & 1/4 \end{bmatrix}, \text{ and } P_{\mathcal{C}, \mathcal{B}} = \begin{bmatrix} 2 & -2/3 \\ 7 & 5/3 \end{bmatrix}.$$

5.6.3.3. Answer.

(a) Here are the change-of-basis matrices:

$$P_{\mathcal{B}, \mathcal{E}} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}, \text{ and } P_{\mathcal{E}, \mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}.$$

(Yes, they are the same!)

(b) We find that

$$[\mathbf{v}]_{\mathcal{B}} = P_{\mathcal{E}, \mathcal{B}}[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

5.6.3.8. Answer.

(a) We will use the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \text{ and } \mathbf{v}_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

$$(b) [T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

(c) We find that

$$P_{\mathcal{B},\mathcal{E}} = \begin{bmatrix} 4 & 1 \\ -1 & 4 \end{bmatrix}, \text{ and } P_{\mathcal{E},\mathcal{B}} = \begin{bmatrix} 4/17 & -1/17 \\ 1/17 & 4/17 \end{bmatrix}.$$

$$(d) [T]_{\mathcal{E}} = \begin{bmatrix} 15/17 & -8/17 \\ -8/17 & -15/17 \end{bmatrix}$$

Writing Exercises

5.6.3.10. Solution. The i th column of $P_{\mathcal{B},\mathcal{C}}$ is $[\mathbf{v}_i]_{\mathcal{C}}$. Or, stated differently, if $C_{\mathcal{C}} : V \rightarrow \mathbb{F}^n$ is the coordinate mapping, then the i th column of $P_{\mathcal{B},\mathcal{C}}$ is $C_{\mathcal{C}}(\mathbf{v}_i)$.

We know (by Theorem 5.5.3, p. 135) that the coordinate mapping is an isomorphism. Since $\mathcal{B}$ is a basis for V , $\mathcal{B}$ is a spanning set for V . But then the set containing the columns of $P_{\mathcal{B},\mathcal{C}}$ is $C_{\mathcal{C}}(\mathcal{B})$, and since $\mathcal{B}$ spans V we know that $C_{\mathcal{C}}(\mathcal{B})$ will span $\mathbb{F}^n$.

6 · Eigenvalues and Eigenvectors

6.1 · Eigenvalues and Eigenvectors

6.1.4 · Exercises

6.1.4.3. Answer.

- (a) The RREF of $A+4I$ has one non-pivot column, so $\lambda = -4$ is an eigenvalue for A . A basis for $\text{eig}_{-4}(A)$ is

$$\left\{ \begin{bmatrix} 5 \\ -2 \\ 6 \end{bmatrix} \right\}.$$

- (b) The RREF of $A-2I$ has two non-pivot columns, so $\lambda = 2$ is an eigenvalue for A . A basis for $\text{eig}_2(A)$ is

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

6.1.4.5. Answer.

- (a) The RREF of $A-4I$ has two non-pivot columns, so $\lambda = 4$ is an eigenvalue for A . A basis for $\text{eig}_4(A)$ is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

- (b) The RREF of $A-2I$ has one non-pivot column, so $\lambda = 2$ is an eigenvalue for A . A basis for $\text{eig}_2(A)$ is

$$\left\{ \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

6.1.4.9. Answer. Since A is clearly not invertible—its columns are linearly dependent—it has 0 as an eigenvalue by Theorem 6.1.12, p. 154. A basis for $\text{eig}_0(A)$ is

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} -1 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The dimension of $\text{eig}_0(A)$ is $n - 1$.

The other eigenvalue is $\lambda = n$, and a basis for $\text{eig}_n(A)$ is

$$\left\{ \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right\}.$$

Writing Exercises

6.1.4.10. Solution. Suppose that λ is an eigenvalue of A . Then there exists a nonzero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. But then

$$A^2\mathbf{v} = A(\lambda\mathbf{v}) = \lambda \cdot A\mathbf{v} = \lambda^2\mathbf{v}.$$

If $A^2 = 0$, then $A^2\mathbf{v} = \mathbf{0}$, so we have $\lambda^2\mathbf{v} = \mathbf{0}$. But since $\mathbf{v}$ is nonzero, and since $\mathbb{F}$ is a field, we must have $\lambda = 0$. Therefore, the only eigenvalue of A is 0.

6.1.4.11. Solution. We will argue by contradiction. Suppose that an $n \times n$ matrix A has $n + 1$ distinct eigenvalues. This means that there are corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_{n+1}$. By Theorem 6.1.13, p. 154, we know that the set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_{n+1}\}$ is linearly independent. However, this gives us a linearly independent set of $n + 1$ vectors in $\mathbb{F}^n$, which is a contradiction by Corollary 5.1.13, p. 110. This proves that A can have at most n distinct eigenvalues.

6.2 • The Characteristic Equation

6.2.4 • Exercises

6.2.4.2. Answer. The characteristic polynomial is $p_A(\lambda) = \lambda^2 - 5\lambda + 4$. The eigenvalues are $\lambda = 1$ and $\lambda = 4$.

6.2.4.4. Answer. The characteristic polynomial is $p_A(\lambda) = \lambda^2 + \lambda + 3$, and the eigenvalues are $\lambda = 1$ and $\lambda = 3$.

6.2.4.6. Answer. The characteristic polynomial is $p_A(\lambda) = -\lambda^3 - 6\lambda^2 - 9\lambda - 4$, but it is easier in this problem NOT to expand the calculation to this form. The characteristic polynomial as calculated is $p_A(\lambda) = (-1 - \lambda)(\lambda^2 + 5\lambda + 4)$. The eigenvalues are $\lambda = -4$ and $\lambda = -1$.

6.2.4.8. Answer. The characteristic polynomial is $p_A(\lambda) = -\lambda^3 + 9\lambda^2 - 23\lambda + 15$. The eigenvalues are $\lambda = 1$, $\lambda = 3$, and $\lambda = 5$.

Writing Exercises

6.2.4.10. Solution. For the purposes of notation, we will say that $A \sim B$ if A is similar to B . We first note that a matrix $A \in M_n(\mathbb{F})$ is similar to itself since $A = IPI^{-1}$, where I is the $n \times n$ identity matrix. This proves that the

relation is reflexive.

We now suppose that $A, B \in M_n(\mathbb{F})$ with $A \sim B$. This means that there is an invertible matrix P such that $A = PBP^{-1}$. We can manipulate this equation to give $B = P^{-1}AP$, and since P^{-1} is invertible (and since $(P^{-1})^{-1} = P$), this proves that $B \sim A$. This proves that the relation is symmetric.

We now suppose that $A, B, C \in M_n(\mathbb{F})$ with $A \sim B$ and $B \sim C$. This means that we have invertible matrices P and Q such that $A = PBP^{-1}$ and $B = QCQ^{-1}$. Putting these equations together, we have

$$A = P(QCQ^{-1})P^{-1} = (PQ)C(Q^{-1}P^{-1}) = (PQ)C(PQ)^{-1}.$$

In this derivation we have used Proposition 3.3.3, p. 76. Since PQ is invertible (again by Proposition 3.3.3, p. 76), this proves that $A \sim C$, which shows that the relation is transitive.

We have shown that the relation of similarity on the set $M_n(\mathbb{F})$ is reflexive, symmetric, and transitive, so it is an equivalence relation.

6.3 • Diagonalization

6.3.5 • Exercises

6.3.5.3. Answer. This matrix is not diagonalizable. There is only one eigenvalue, and the dimension of the eigenspace is one.

6.3.5.5. Answer. This matrix is diagonalizable using the following matrices P and D :

$$P = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -5 \end{bmatrix}.$$

6.3.5.6. Answer. No, that is not possible. The matrix A must be diagonalizable. The dimension of the eigenspace that is not yet specified must be at least one. The sum of all dimensions of the eigenspaces must be at most four (since A is 4×4), and the given information tells us that this sum will be exactly four. This means that A is diagonalizable.

6.3.5.7. Answer. Here is one pair of matrices that diagonalizes A :

$$P = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}.$$

We can make subtle manipulations to these matrices to find another pair which diagonalizes A :

$$P = \begin{bmatrix} -1 & 1 \\ 2 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}.$$

There are many, many other pairs of matrices that diagonalize A , mainly because the matrix P can have infinitely many different columns.

Writing Exercises

6.3.5.9. Solution. If A has n linearly independent eigenvectors, then it is diagonalizable by Theorem 6.3.2, p. 163. This means there exist matrices P and D such that $A = PDP^{-1}$. If we take the transpose of both sides of this equation, we get

$$A^T = (PDP^{-1})^T = (P^{-1})^T D^T P^T.$$

Since D is diagonal, $D^T = D$. Also, Exercise 3.3.5.8, p. 82 tells us that

$(P^{-1})^T = (P^T)^{-1}$. So, we have

$$A^T = (P^T)^{-1}DP^T.$$

This proves that A^T is diagonalizable, and Theorem 6.3.2, p. 163 allows us to conclude that A^T has n linearly independent eigenvectors.

6.4 • Invariants

6.4.4 • Exercises

6.4.4.2. Answer.

- (a) Since $\text{rank}(A) = \text{rank}(B) = 3$ and $\text{tr}(A) = \text{tr}(B) = 2$, neither the rank nor the trace can tell us whether or not A and B are similar matrices.
- (b) We can calculate that $\det(A) = -2$ while $\det(B) = -12$, so this means that A and B are not similar.

6.4.4.3. Answer.

- (a) We find that $\text{tr}(A) = \text{tr}(B) = 3$, $\text{rank}(A) = \text{rank}(B) = 2$, and $\det(A) = \det(B) = 0$, so the trace, rank, and determinant do not give us enough information to determine whether or not A and B are similar.
- (b) Some calculation shows that the eigenvalues for A are $\lambda = 0, -1, 4$, while the eigenvalues for B are $\lambda = 0, 1, 2$. Since these are different (and therefore the characteristic polynomials are different), we see that A and B are not similar.

6.4.4.6. Answer.

- (a) We find that $\text{rank}(A) = \text{rank}(B) = 3$, $\text{tr}(A) = \text{tr}(B) = 5$, and $\det(A) = \det(B) = 4$. Further, the eigenvalues for both A and B are $\lambda = 1, 2, 2$. These four invariants do not provide enough information to determine whether or not A and B are similar.
- (b) We can show that A is not diagonalizable but B is. Since diagonalizability is an invariant for similarity, this proves that A and B are not similar.

Writing Exercises

6.4.4.14. Answer. No, trace is not an invariant for row equivalence. Consider the following matrix $A \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

The matrix A is invertible, so it is row equivalent to I_2 . However, $\text{tr}(A) = 3$ and $\text{tr}(I_2) = 2$. This proves that trace is not an invariant for row equivalence.

6.4.4.17. Answer. Yes, invertibility is an invariant for similarity. This argument relies on the fact that the determinant is an invariant for similarity.

We will denote the similarity relation by $A \sim B$. Suppose that $A \sim B$ and that A is invertible. This means that $\det(A) \neq 0$, and since $A \sim B$, this means that $\det(B) \neq 0$ also. But the fact that $\det(B) \neq 0$ means that B is invertible.

We now suppose that $A \sim B$ and that A is not invertible. This means that $\det(A) = 0$, and since $A \sim B$, this means that $\det(B) = 0$. But $\det(B) = 0$ means that B is not invertible.

This proves that invertibility is an invariant for similarity.

7 · Inner Products

7.1 · Inner Products

7.1.6 · Exercises

7.1.6.2. Answer. Calculation shows that the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is orthogonal, so then it is linearly independent.

7.1.6.3. Answer.

(a) We let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Then it is easy to see that $\mathbf{u} \cdot \mathbf{v} = 0$, but $\langle \mathbf{u}, \mathbf{v} \rangle = 4$.

(b) We let $\mathbf{u}$ and $\mathbf{v}$ be the following vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}.$$

Then it is easy to see that $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, but $\mathbf{u} \cdot \mathbf{v} = -4$.

7.1.6.6. Answer. Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$. It is fairly easy to see that $\det(A) = 0$, so we have

$$\langle A, A \rangle = 0 \cdot 0 = 0.$$

However, since A is not the zero matrix (i.e., the zero vector for the vector space $M_2(\mathbb{R})$), the fourth axiom of the inner product does not hold for this function.

Writing Exercises

7.1.6.8. Solution. This function is not an inner product. Consider the vector $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. We have

$$\langle \mathbf{u}, \mathbf{u} \rangle = (1)(0) + (0)(1) = 0.$$

However, since $\mathbf{u}$ is not the zero vector in $\mathbb{R}^2$, the fourth axiom of the inner product does not hold.

7.1.6.12. Solution. We will show that all four of the inner product axioms hold. For the first axiom, we note that everything here is real-valued, so we do not need to worry about any complex conjugates. Since the order of the functions within a definite integral can be switched, we have

$$\begin{aligned} \langle g, f \rangle &= \int_0^1 g(x)f(x) \, dx \\ &= \int_0^1 f(x)g(x) \, dx = \langle f, g \rangle. \end{aligned}$$

This shows that the first axiom holds.

For the second axiom, we let $f, g, h \in C([0, 1])$. The definite integral is linear with respect to the sum of functions, so we have

$$\langle f + g, h \rangle = \int_0^1 (f(x) + g(x))h(x) \, dx$$

$$\begin{aligned}
&= \int_0^1 (f(x)h(x) + g(x)h(x)) \, dx \\
&= \int_0^1 f(x)h(x) \, dx + \int_0^1 g(x)h(x) \, dx \\
&= \langle f, h \rangle + \langle g, h \rangle.
\end{aligned}$$

This proves that the second axiom holds.

Let $f, g \in C([0, 1])$ and let $c \in \mathbb{R}$. The definite integral is linear with respect to scalar multiplication by a real number, so we have

$$\begin{aligned}
\langle cf, g \rangle &= \int_0^1 (cf(x))g(x) \, dx \\
&= c \int_0^1 f(x)g(x) \, dx \\
&= c \langle f, g \rangle.
\end{aligned}$$

This proves that the third axiom holds.

Let $f \in C([0, 1])$. We observe that $f(x)^2$ is a function with values that are always non-negative (since each value of this function is a real number squared). Since the definite integral can be interpreted as calculating signed area between the graph of a function and the x -axis, we know that

$$\langle f, f \rangle = \int_0^1 f(x)^2 \, dx \geq 0.$$

Finally, using this same signed area interpretation of the definite integral, the only way a non-negative function could produce a zero value for the definite integral is if the function was uniformly the zero function. This means that if $\langle f, f \rangle = 0$, we must have $f(x)^2 = 0$ and therefore $f(x) = 0$. This proves that the fourth axiom holds.

7.2 • Orthonormal Bases

7.2.5 • Exercises

7.2.5.2. Answer.

(a) The orthonormal basis we obtain is

$$\left\{ \frac{1}{\sqrt{17}}(2+t), \frac{1}{\sqrt{34}}(3-7t) \right\}.$$

(b) The coordinate vector is

$$\begin{bmatrix} 10/\sqrt{17} \\ -2/\sqrt{34} \end{bmatrix}.$$

7.2.5.3. Answer. The orthonormal basis we obtain is

$$\left\{ \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} i/\sqrt{6} \\ 1/\sqrt{6} \\ 2i/\sqrt{6} \end{bmatrix}, \frac{\sqrt{3}}{2\sqrt{2}} \begin{bmatrix} -\frac{1}{3} + \frac{4}{3}i \\ -\frac{2}{3} + \frac{1}{3}i \\ \frac{1}{3} - \frac{1}{3}i \end{bmatrix} \right\}.$$

7.3 • Orthogonal Projections

7.3.5 • Exercises

7.3.5.1. Answer. If we let $\mathbf{w}$ and $\mathbf{w}'$ be the following vectors,

$$\mathbf{w} = \begin{bmatrix} 35/34 \\ 21/34 \end{bmatrix}, \text{ and } \mathbf{w}' = \begin{bmatrix} -69/34 \\ 115/34 \end{bmatrix},$$

then $\mathbf{v} = \mathbf{w} + \mathbf{w}'$ where $\mathbf{w} \in L$ and $\mathbf{w}' \in L^\perp$.

7.3.5.2. Answer.

(a) Our matrix is

$$[\text{proj}_U]_{\mathcal{E}} = \begin{bmatrix} \frac{1}{11} & -\frac{3}{11} & \frac{1}{11} \\ -\frac{3}{11} & \frac{101}{110} & \frac{3}{110} \\ \frac{1}{11} & \frac{3}{110} & \frac{109}{110} \end{bmatrix}.$$

(b) The vector in U which is closest to $\mathbf{v}$ is

$$\begin{bmatrix} \frac{3}{11} \\ \frac{15}{22} \\ \frac{105}{22} \end{bmatrix}.$$

7.3.5.3. Answer. Let q and q' be the following polynomials:

$$q = \frac{11}{10} - \frac{17}{20}t + \frac{61}{20}t^2, \text{ and } q' = \frac{9}{10} - \frac{3}{20}t - \frac{21}{20}t^2.$$

Then $p = q + q'$, where $q \in U$ and $q' \in U^\perp$.

Writing Exercises

7.3.5.5. Solution. First, we know that $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ for all $\mathbf{u} \in U$ by Proposition 7.1.17, p. 180. This proves that $\mathbf{0} \in U^\perp$.

Next, we let $\mathbf{w}_1, \mathbf{w}_2 \in U^\perp$ and $\mathbf{u} \in U$. Then, by the properties of the inner product, we have

$$\langle \mathbf{w}_1 + \mathbf{w}_2, \mathbf{u} \rangle = \langle \mathbf{w}_1, \mathbf{u} \rangle + \langle \mathbf{w}_2, \mathbf{u} \rangle = 0 + 0 = 0.$$

This shows that $\mathbf{w}_1 + \mathbf{w}_2 \in U^\perp$, meaning that $U^\perp$ is closed under addition.

Finally, we let $\mathbf{w} \in U^\perp$ and $c \in \mathbb{F}$. (Here, $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.) We also let $\mathbf{u} \in U$. Then, by the properties of the inner product, we have

$$\langle c\mathbf{w}, \mathbf{u} \rangle = c \langle \mathbf{w}, \mathbf{u} \rangle = c \cdot 0 = 0.$$

This proves that $c\mathbf{w} \in U^\perp$ so that $U^\perp$ is closed under scalar multiplication.

Since $U^\perp$ contains the zero vector and is closed under addition and scalar multiplication, $U^\perp$ is a subspace of V .

7.3.5.6. Solution. We will prove that each of these sets is a subset of the other. First, we let $\mathbf{x} \in \text{null}(A)$, so that $A\mathbf{x} = \mathbf{0}$. The fact that $A\mathbf{x} = \mathbf{0}$ means that $\mathbf{r} \cdot \mathbf{x} = 0$ for all rows $\mathbf{r}$ of A . (Entry i in $A\mathbf{x}$ is the dot product of the i th row of A with $\mathbf{x}$.) Since the rows of A span $\text{row}(A)$, the fact that $\mathbf{r} \cdot \mathbf{x} = 0$ for each row $\mathbf{r}$ of A means that $\mathbf{x} \cdot \mathbf{v} = 0$ for all $\mathbf{v} \in \text{row}(A)$. This proves that $\mathbf{x} \in (\text{row}(A))^\perp$, so $\text{null}(A) \subseteq (\text{row}(A))^\perp$.

We now let $\mathbf{x} \in (\text{row}(A))^\perp$. We want to show that $\mathbf{x} \in \text{null}(A)$. Since $\mathbf{x} \in (\text{row}(A))^\perp$, we know that $\mathbf{x} \cdot \mathbf{r} = 0$ for each row $\mathbf{r}$ of A . This shows that $A\mathbf{x} = \mathbf{0}$, which proves that $\mathbf{x} \in \text{null}(A)$. Therefore, $(\text{row}(A))^\perp \subseteq \text{null}(A)$.

Since we have shown that $\text{null}(A) \subseteq (\text{row}(A))^\perp$ and $(\text{row}(A))^\perp \subseteq \text{null}(A)$,

we can conclude that $(\text{row}(A))^\perp = \text{null}(A)$, as desired.

Colophon

This book was authored in PreTeXt.